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Foreword

Publishing information

This British Standard is published by BSI Standards Limited, under licence from The British Standards Institution, and came into effect on 31 October 2025. It was prepared by Subcommittee TPR/1/8, *Technical Product Realization - BS 8888 Technical product specification*, under the authority of Technical Committee TPR/1, *Technical product realization*. A list of organizations represented on these committees can be obtained on request to the committee manager.

Supersession

This British Standard supersedes BS 8888:2020, which is withdrawn.

Relationship with other publications

The function of BS 8888 is to draw together, in an easily accessible manner, the full complement of international standards relevant to the preparation of technical product specifications. However, it is not the intention for BS 8888 to be a standalone standard. TPR/1 is responsible for a suite of related national standards, including the various parts of BS 8887 and PD 68888, a new training document.

Information about this document

This is a full revision of the document, and introduces relevant international standards published since the 2020 edition. It also incorporates more fully than previous editions some of the fundamental requirements of the key international standards relevant to the preparation of technical product specifications, such as BS EN ISO 1101:2017 and BS EN ISO 5459. It is hoped that UK industry will find this edition of BS 8888 more user-friendly than previous editions. It aims to help organizations better understand and implement the full complement of international standards developed by ISO/TC 213, *Geometrical product specifications and verification*, and ISO/TC 10, *Technical product documentation*.

Copyright is claimed on Figure 72, Figure 85, Figure 97, Figure 131, Figure B.1, Figure B.3, Figure B.7, Figure B.9, Figure C.1, Figure H.1, Figure H.2, Figure H.3, Table 18, Table G.6. Copyright holder is Iain Macleod Associated Ltd, 29 Clay Lane, Hale, Altrincham, Cheshire, WA15 8PJ, United Kingdom.

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Where websites and webpages have been cited, they are provided for ease of reference and are correct at the time of publication. The location of a webpage or website, or its contents, cannot be guaranteed.

Presentational conventions

The provisions of this document are presented in roman (i.e. upright) type. Its requirements are expressed in sentences in which the principal auxiliary verb is “shall”.

Commentary, explanation and general informative material is presented in smaller italic type, and does not constitute a normative element.

All dimensions shown in the figures in this British Standard are in millimetres.

Where words have alternative spellings, the preferred spelling of the *Shorter Oxford English Dictionary* is used (e.g. “organization” rather than “organisation”).

Contractual and legal considerations

This publication has been prepared in good faith, however no representation, warranty, assurance or undertaking (express or implied) is or will be made, and no responsibility or liability is or will be accepted by BSI in relation to the adequacy, accuracy, completeness or reasonableness of this publication. All and any such responsibility and liability is expressly disclaimed to the full extent permitted by the law.

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The recipient is advised to consider seeking professional guidance with respect to its use of this publication.

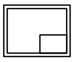
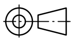
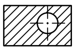
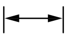
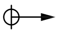
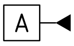
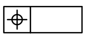
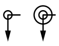
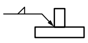
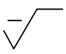
This publication is not intended to constitute a contract. Users are responsible for its correct application.

Compliance with a British Standard cannot confer immunity from legal obligations.

Introduction

Table 1 is a visual index and an innovation to improve the useability of the standard. It is not exhaustive, but a first attempt to improve the accessibility of the content.

Table 1 — *Visual index*

Symbol	Explanation	Relevant clause
Ⓕ	Non-rigid parts	**5.2.11**
Ⓔ	Envelope requirement	**5.2.13.2**
	Drawing sheets	**6.5**
	Views and projections	**6.11.2**
	Sections	**6.13**
	Dimensions	Clause **7**
	Symbols for dimensions	**7.2.7**
	Running dimensioning	**7.2.14.4**
±	Dimensional tolerances	**7.4.3**
	Datums	Clause **8**
	Datum targets	**8.8**
	Geometrical specifications	Clause **9**
Ⓜ	Maximum material requirement	**9.10.1**
Ⓛ	Least material requirement	**9.10.4**
Ⓡ	Reciprocity requirement	**9.10.7**
Ⓟ	Projected tolerance zones	**9.11.1**
	All-around/over	Clause **10**
A ←→ B	Between	**10.1**
UF	United feature	**10.4**
CZ/SZ	Feature patterns	**10.5**
4× A/2× B ⌖ ⌀0,2 SZ CZ A	Nested patterns	**10.5.3**
	Plane and feature indicators	**10.6**
	Statistical tolerancing	Clause **12**
	Edge conditions	Clause **13**
	Welding symbols	Clause **14**
	Surface texture	Clause **15**
	Document handling	Clause **18**

The annexes have been updated to reflect the changes in the content:

- Annex A: Datums and datum systems: Definitions and explanations;
- Annex B: Associations;
- Annex C: Implementation in CAD;
- Annex D: Document security – Enhanced;
- Annex E: Tolerancing system: Former practice;
- Annex F: Selected ISO fits: Hole and shaft basis;
- Annex G: Filtration;
- Annex H: Dimensioning of slots;
- Annex I: Guidance for parts produced by additive manufacture (AM);
- Annex J: Castings;
- Annex K: Examples of profile and areal surface texture specifications, and summary of changes from BS EN ISO 1302 (withdrawn).

1 Scope

This British Standard specifies requirements for technical product specification and documentation, principally for manufacturing industries and organizations associated with engineering disciplines, such as mechanical, electrical, nuclear, automotive and aerospace.

The needs of building, architectural, civil, and structural engineering and construction services industries continue to be covered by the BS EN ISO 19650 series.

Most of this British Standard consists of an implementation of the ISO system for technical product specification and documentation. The ISO system is defined in a large number of interlinked and related international standards which are referenced in this British Standard.

The purpose of this British Standard is to facilitate the use of the ISO system by providing:

a) an index to the international standards which make up the ISO system, referencing them according to their area of application;

b) key elements of the ISO standards to facilitate their application;

c) references to additional British and European standards where they provide information or guidance over and above that provided by ISO standards; and

d) commentary and recommendations on the application of the standards where this is deemed useful.

The requirements refer to international and European standards which have been implemented as British Standards, either in the BS EN, BS EN ISO, BS ISO series or as international standards renumbered as British Standards.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes provisions, or limits the application, of this document[1]. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

BS 1916-1, *Limits and fits for engineering – Part 1: Guide to limits and tolerances*

BS 1916-2, *Limits and fits for engineering – Part 2: Guide to the selection of fits in BS 1916-1*

BS 1916-3, *Limits and fits for engineering – Part 3: Guide to tolerances, limits and fits for large diameters*

BS 3238-1, *Graphical symbols for components of servo mechanisms – Part 1: Transductors and magnetic amplifiers*

BS 3238-2, *Graphical symbols for components of servo mechanisms – Part 2: General servo-mechanisms*

BS 3643-1:2007, *ISO metric screw threads – Part 1: Principles and basic data*

BS 3643-2:2007, *ISO metric screw threads – Part 2: Specification for selected limits of size*

BS 4500, *Limits and fits – Guidance for system of cone (taper) fits and tolerances for cones from C = 1:3 to 1:500, lengths from 6 mm to 630 mm and diameters up to 500 mm*

BS 4827:1972, *Specification for ISO miniature screw threads – Metric series*

BS EN 2066, *Aerospace series – Extruded section in aluminium alloys – General tolerances*

[1] Documents that are referred to solely in an informative manner are listed in the Bibliography.

BS EN 10058, *Hot rolled flat steel bars for general purposes – Dimensions and tolerances on shape and dimensions*

BS EN 22768-1:1993, *General tolerances – Part 1: Tolerances for linear and angular dimensions without individual tolerance indications*

BS EN 80000-13, *Quantities and units – Part 13: Information science and technology*

BS EN 80000-14, *Quantities and units – Part 14: Telebiometrics related to human physiology*

BS EN ISO 128-2:2022, *Technical product documentation (TPD) – General principles of representation – Part 2: Basic conventions for lines*

BS EN ISO 129-1, *Technical product documentation (TPD) – Indications of dimensions and tolerances – Part 1: General principles*

BS EN ISO 286-1:2010, *Geometrical product specifications (GPS) – ISO code system for tolerances on linear sizes – Part 1: Basis of tolerances, deviations and fits*

BS EN ISO 286-2:2010, *Geometrical product specifications (GPS) – ISO code system for tolerances on linear sizes – Part 2: Tables of standard tolerance classes and limit deviations for holes and shafts*

BS EN ISO 1101:2017, *Geometrical product specifications (GPS) – Geometrical tolerancing – Tolerances of form, orientation, location and run-out*

BS EN ISO 1119:2011, *Geometrical product specifications (GPS) – Series of conical tapers and taper angles*

BS EN ISO 1660, *Geometrical product specifications (GPS) – Geometrical tolerancing – Profile tolerancing*

BS EN ISO 2162-1, *Technical product documentation – Springs – Part 1: Simplified representation*

BS EN ISO 2162-2, *Technical product documentation – Springs – Part 2: Presentation of data for cylindrical helical compression springs*

BS EN ISO 2203, *Technical drawings – Conventional representation of gears*

BS EN ISO 2553, *Welding and allied processes – Symbolic representation on drawings – Welded joints*

BS EN ISO 2692, *Geometrical product specifications (GPS) – Geometrical tolerancing – Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR)*

BS EN ISO 3040, *Geometrical product specifications (GPS) – Dimensioning and tolerancing – Cones*

BS EN ISO 3098 (all parts), *Technical product documentation – Lettering*

BS EN ISO 4063, *Welding, brazing, soldering and cutting – Nomenclature of processes and reference numbers*

BS EN ISO 5261:1999, *Technical drawings – Simplified representation of bars and profile sections*

BS EN ISO 5456-2, *Technical drawings – Projection methods – Part 2: Orthographic representations*

BS EN ISO 5456-3, *Technical drawings – Projection methods – Part 3: Axonometric representations*

BS EN ISO 5457, *Technical product documentation – Sizes and layout of drawing sheets*

BS EN ISO 5458, *Geometrical product specifications (GPS) – Geometrical tolerancing – Positional tolerancing*

BS EN ISO 5459, *Geometrical product specifications (GPS) – Geometrical tolerancing – Datums and datum systems*

BS EN ISO 5817, *Fusion-welded joints in steel, nickel, titanium and their alloys (beam welding excluded) – Quality levels for imperfections*

BS EN ISO 5845-1, *Technical drawings – Simplified representation of the assembly of parts with fasteners – Part 1: General principles*

BS EN ISO 6410-1, *Technical drawings – Screw threads and threaded parts – Part 1: General conventions*

BS EN ISO 6410-2, *Technical drawings – Screw threads and threaded parts – Part 2: Screw thread inserts*

BS EN ISO 6411, *Technical drawings – Simplified representation of centre holes*

BS EN ISO 6412-1, *Technical drawings – Simplified representation of pipelines – Part 1: General rules and orthogonal representation*

BS EN ISO 6412-2, *Technical drawings – Simplified representation of pipelines – Part 2: Isometric projection*

BS EN ISO 6412-3, *Technical drawings – Simplified representation of pipelines – Part 3: Terminal features of ventilation and drainage systems*

BS EN ISO 6413, *Technical drawings – Representation of splines and serrations*

BS EN ISO 6428, *Technical drawings – Requirements for microcopying*

BS EN ISO 6947, *Welding and allied processes – Welding positions*

BS EN ISO 7083, *Technical product documentation – Symbols used in technical product documentation – Proportions and dimensions*

BS EN ISO 7200, *Technical product documentation – Data fields in title blocks and document headers*

BS EN ISO 8015, *Geometrical product specifications (GPS) – Fundamentals – Concepts, principles and rules*

BS EN ISO 8062-1, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 1: Vocabulary*

BS EN ISO 8062-3, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 3: General dimensional and geometrical tolerances and machine allowances for castings using ± tolerances for indicated dimensions*

BS EN ISO 8826-1, *Technical drawings – Roller bearings – Part 1: General simplified representation*

BS EN ISO 8826-2, *Technical drawings – Roller bearings – Part 2: Detailed simplified representation*

BS EN ISO 9013, *Thermal cutting – Classification of thermal cuts – Geometrical product specification and quality tolerances*

BS EN ISO 9222-1, *Technical drawings – Seals for dynamic application – Part 1: General simplified representation*

BS EN ISO 9222-2, *Technical drawings – Seals for dynamic application – Part 2: Detailed simplified representation*

BS EN ISO 10042, *Welding – Arc-welded joints in aluminium and its alloys – Quality levels for imperfections*

BS EN ISO 10135, *Geometrical product specifications (GPS) – Drawing indications for moulded parts in technical product documentation (TPD)*

BS EN ISO 10209:2022, *Technical product documentation – Vocabulary – Terms relating to technical drawings, product definition and related documentation*

BS EN ISO 10579, *Geometrical product specifications (GPS) – Dimensioning and tolerancing – Non-rigid parts*

BS EN ISO 11442, *Technical product documentation – Document management*

BS EN ISO 12180-1, *Geometrical product specifications (GPS) – Cylindricity – Part 1: Vocabulary and parameters of cylindrical form*

BS EN ISO 12180-2, *Geometrical product specifications (GPS) – Cylindricity – Part 2: Specification operators*

BS EN ISO 12181-1, *Geometrical product specifications (GPS) – Roundness – Part 1: Vocabulary and parameters of roundness*

BS EN ISO 12181-2, *Geometrical product specifications (GPS) – Roundness – Part 2: Specification operators*

BS EN ISO 12780-1, *Geometrical product specifications (GPS) – Straightness – Part 1: Vocabulary and parameters of straightness*

BS EN ISO 12780-2, *Geometrical product specifications (GPS) – Straightness – Part 2: Specification operators*

BS EN ISO 12781-1, *Geometrical product specifications (GPS) – Flatness – Part 1: Vocabulary and parameters of flatness*

BS EN ISO 12781-2, *Geometrical product specifications (GPS) – Flatness – Part 2: Specification operators*

BS EN ISO 13715, *Technical product documentation – Edges of undefined shape – Indication and dimensioning*

BS EN ISO 13919 (all parts), *Welding – Electron and laser beam welded joints – Guidance on quality levels for imperfections*

BS EN ISO 13920, *Welding – General tolerances for welded constructions – Dimensions for lengths and angles, shape and position*

BS EN ISO 14171, *Welding consumables – Solid wire electrodes, tubular cored electrodes and electrode/flux combinations for submerged arc welding of non alloy and fine grain steels – Classification*

BS EN ISO 14253 (all parts), *Geometrical product specifications (GPS) – Inspection by measurement of workpieces and measuring equipment*

BS EN ISO 14341, *Welding consumables – Wire electrodes and weld deposits for gas shielded metal arc welding of non alloy and fine grain steels – Classification*

BS EN ISO 14405-1, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 1: Linear sizes*

BS EN ISO 14405-2, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 2: Dimensions other than linear or angular sizes*

BS EN ISO 14405-3, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 3: Angular sizes*

BS EN ISO 15785, *Technical drawings – Symbolic presentation and indication of adhesive, fold and pressed joints*

BS EN ISO 16610 (all parts), *Geometrical product specifications (GPS) – Filtration*

BS EN ISO 17295:2023, *Additive manufacturing – General principles – Part positioning, coordinates and orientation*

BS EN ISO 17450 (all parts), *Geometrical product specifications (GPS) – General concepts*

BS EN ISO 21920 (all parts), *Geometrical product specifications (GPS) – Surface texture: Profile*

BS EN ISO 22432, *Geometrical product specifications (GPS) – Features used in specification and verification*

BS EN ISO 25178 (all parts), *Geometrical product specifications (GPS) – Surface texture: Areal*

BS EN ISO 26909, *Springs – Vocabulary*

BS EN ISO 80000-1, *Quantities and units – Part 1: General*

BS EN ISO 80000-2, *Quantities and units – Part 2: Mathematics*

BS EN ISO 80000-3, *Quantities and units – Part 3: Space and time*

BS EN ISO 80000-4, *Quantities and units – Part 4: Mechanics*

BS EN ISO 80000-5, *Quantities and units – Part 5: Thermodynamics*

BS EN ISO 80000-8, *Quantities and units – Part 8: Acoustics*

BS EN ISO 80000-9, *Quantities and units – Part 9: Physical chemistry and molecular physics*

BS EN ISO 80000-10, *Quantities and units – Part 10: Atomic and nuclear physics*

BS EN ISO 80000-11, *Quantities and units – Part 11: Characteristic numbers*

BS EN ISO 80000-12, *Quantities and units – Part 12: Condensed matter physics*

BS ISO 128 (all parts), *Technical product documentation (TPD) – General principles of presentation*

BS ISO 261, *ISO general purpose metric screw threads – General plan*

BS ISO 262, *ISO general purpose metric screw threads – Selected sizes for screws, bolts and nuts*

BS ISO 965-1, *ISO general purpose metric screw threads – Tolerances – Part 1: Principles and basic data*

BS ISO 1219-1, *Fluid power systems and components – Graphical symbols and circuit diagrams – Part 1: Graphical symbols for conventional use and data-processing applications*

BS ISO 3302-1, *Rubber – Tolerances for products – Part 1: Dimensional tolerances*

BS ISO 3302-2, *Rubber – Tolerances for products – Part 2: Geometrical tolerances*

BS ISO 5456-4, *Technical drawings – Projection methods – Part 4: Central projection*

BS ISO 5623, *Plastics – Joining of thermoplastic moulded components – Specification for quality levels for imperfections*

BS ISO 7573, *Technical product documentation – Parts lists*

BS ISO 8062-4, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 4: Rules and general tolerances for castings using profile tolerancing in a general datum system*

BS ISO 13583-1, *Centrifugally cast steel and alloy products – Part 1: General testing and tolerances*

BS ISO 14617-1, *Graphical symbols for diagrams – Part 1: General rules*

BS ISO 14617-2, *Graphical symbols for diagrams – Part 2: Graphical symbols*

BS ISO 16016, *Technical product documentation – Protection notices for restricting the use of documents and products*

BS ISO 16792, *Technical product documentation – Digital product definition data practices*

BS EN ISO 19650 (all parts), *Organization and digitization of information about buildings and civil engineering works, including building information modelling (BIM) – Information management using building information modelling*

BS ISO 20457, *Plastics moulded parts – Tolerances and acceptance conditions*

BS ISO 29845, *Technical product documentation – Document types*

BS ISO 80000-7, *Quantities and units – Part 7: Light and radiation*

ISO/IEC Guide 98-3, *Uncertainty of measurement – Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*

ISO/IEC Guide 99, *International vocabulary of metrology – Basic and general concepts and associated terms (VIM)*

PD CEN ISO/TS 8062-2, *Geometrical Product Specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 2: Rules*

3 Terms and definitions

For the purposes of this British Standard, the terms and definitions given in BS EN ISO 10209, BS EN ISO 17450 (all parts) and BS EN ISO 25378 apply, together with the following.

3.1 association

feature operation used to fit ideal feature(s) to non-ideal feature(s) according to a criterion

NOTE This is sometimes referred to as a "best fit" operation.

[SOURCE: BS EN ISO 17450-1:2011, **3.4.1.4**]

3.2 associated feature

ideal feature established from a non-ideal surface model or from a real feature through an association operation

[SOURCE: BS EN ISO 17450-1:2011, **3.3.8**]

3.3 date of issue

point in time at which the technical product specification is officially released for its intended use

NOTE The date of issue is important for legal reasons, e.g. patent rights, traceability.

3.4 derived feature

geometrical feature, which does not exist physically on the real surface of the workpiece and which is not natively a nominal integral feature

NOTE The following are derived features:

- *centre plane (or median plane or median surface);*
- *a centre line (or axis or median line); and*
- *a centre point.*

[SOURCE: BS EN ISO 17450-1:2011, **3.3.6**, modified – notes 1 and 2 and examples 1 to 3 omitted and replaced with new note]

3.5 extracted feature

geometrical feature defining a set of a finite number of points

NOTE The set of data obtained by sampling a real feature in a measurement operation is an extracted feature. The set of data representing a derived feature (e.g. a median surface, median line or median point) which has been derived from data extracted from a real feature is also known as an extracted feature.

[SOURCE: BS EN ISO 17450-1:2011, **3.3.7**, modified – notes 1 to 3 omitted and replaced with a new note]

3.6 extracted surface

finite set of data points sampled from the primary surface

3.7 free state

condition of a part subjected only to the force of gravity

[SOURCE: BS EN ISO 10579:2013, **3.2**]

3.8 geometrical product specification and verification

system developed in ISO by technical committee ISO/TC213 for the definition and verification of geometrical requirements for mechanical parts

NOTE This is also known as "Geometrical Product Specification" and "ISO GPS".

3.9 integral feature

geometrical feature belonging to the real surface of the workpiece or to a surface model

NOTE An integral feature is continuous (i.e. composed of an infinite number of points).

[SOURCE: BS EN ISO 17450-1:2011, **3.3.5**, modified – notes 1 to 3 omitted and replaced with a new note]

3.10 nesting index

N_{is}, N_{ic}, N_{if}

number or set of numbers indicating the relative level of nesting for a particular primary mathematical model

NOTE 1 For some types of filters, this might be referred to as a "cut-off".

NOTE 2 The cut-off wavelength for the Gaussian filter is an example of a nesting index.

NOTE 3 Using the different nesting indices, specific lateral scale components of a scale-limited profile or surface are extracted.

[SOURCE: BS EN ISO 16610-1:2015, **3.2.1**, modified – definition and notes revised]

3.11 nominal feature

ideal feature defined in the technical product documentation by the product designer

[SOURCE: BS EN ISO 17450-1:2011, **3.3.3**, modified – notes and example omitted]

3.12 real feature

geometrical feature corresponding to a part of the workpiece real surface

[SOURCE: BS EN ISO 17450-1:2011, **3.3.4**]

3.13 skin model
non-ideal surface model

<of a workpiece> model of the physical interface of the workpiece with its environment

[SOURCE: BS EN ISO 17450-1:2011, **3.2.2**]

3.14 specification

expression of permissible limits on a characteristic

NOTE Some specifications are expressed as limits for a value (i.e. a tolerance on a dimension). Some specifications are expressed as a space limited by one or more ideal features within which a non-ideal real feature is contained (i.e. a tolerance applied to flatness).

[SOURCE: BS EN ISO 17450-1:2011, **3.6**, modified – note added]

3.15 surface texture terms

3.15.1 surface texture

geometrical irregularities contained in a scale-limited profile

NOTE Surface texture does not include geometrical irregularities contributing to the form or shape of the profile.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.2**]

3.15.2 areal surface texture

surface texture defined and evaluated over a portion of a surface

NOTE 1 This is covered by the BS EN ISO 25178 series of standards.

NOTE 2 Areal surface texture can be considered as 3D surface texture.

3.15.3 electromagnetic profile

profile obtained by the electromagnetic interaction with the skin model of a workpiece

NOTE The electromagnetic profile is the profile trace obtained with an optical measuring instrument.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.7**, modified – notes replaced]

3.15.4 electromagnetic surface

surface obtained by the electromagnetic interaction with the skin model of a workpiece

NOTE The electromagnetic surface is the surface with which an optical measurement system interacts.

[SOURCE: BS EN ISO 14406:2010, **3.1.2**, modified – notes replaced]

3.15.5 evaluation area

A $\tilde{A}$

portion of the scale-limited surface for specifying the area under evaluation

NOTE The symbol A is used for the numerical value of the evaluation area and the symbol Ã for the domain (of integration or definition).

3.15.6 evaluation length

l_e

length in the direction of the x-axis used for identifying the geometrical structures characterizing the scale-limited profile

NOTE The traverse length is longer than the evaluation length.

3.15.7 feature parameter

parameter defined from a subset of predefined topographic features from the scale-limited surface, such as S_{pd}, S_{vd} or $S10_z$

[SOURCE: BS EN ISO 21920-2:2022, **3.2.2**, modified – text added]

3.15.8 field parameter

parameter defined from all the points on a scale-limited surface, such as S_a, S_z or S_p

[SOURCE: BS EN ISO 21920-2:2022, **3.2.1**, modified – text added]

3.15.9 mechanical profile

boundary of the mathematical erosion, by a circular disc of radius *r*, of the locus of the centre of an ideal tactile sphere, also with radius *r*, rolled along a trace over the skin model of a workpiece

NOTE The mechanical profile is the profile trace obtained with a mechanical contact measuring instrument.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.5**, modified – notes replaced]

3.15.10 mechanical surface

boundary of the mathematical erosion, by a sphere of radius *r*, of the locus of the centre of an ideal tactile sphere, also with radius *r*, rolled over the skin model of a workpiece

NOTE The mechanical surface is the surface with which a contact measurement system interacts.

[SOURCE: BS EN ISO 14406:2010, **3.1.1**, modified – notes replaced]

3.15.11 nesting index

$\boldsymbol{N}_{\mathrm{is}}$, $\boldsymbol{N}_{\mathrm{ic}}$, $\boldsymbol{N}_{\mathrm{if}}$

number or set of numbers indicating the relative level of nesting for a particular primary mathematical model

NOTE 1 For some types of filters, this might be referred to as a "cut-off".

NOTE 2 The cut-off wavelength for the Gaussian filter is an example of a nesting index.

NOTE 3 Using the different nesting indices, specific lateral scale components of a scale-limited profile or surface are extracted.

[SOURCE: BS EN ISO 16610-1:2015, **3.2.1**, modified – definition and notes revised]

3.15.12 number of sections

$\boldsymbol{n}_{\mathrm{sc}}$

integer number used to obtain section length parameters

NOTE Default values of n_{sc} are contained in BS EN ISO 21920-3.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.18**]

3.15.13 primary profile
P-profile

scale-limited profile at any position *x* derived from the primary surface profile by removing the form using a profile F-operation with nesting index N_{if}

NOTE The primary profile is the basis for evaluation of the P-parameters.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.14.1**, modified – notes 1, 3 and 4 omitted]

3.15.14 primary surface

surface portion obtained when a surface portion is represented as a specified primary mathematical model with specified nesting index

NOTE An S-filter is used to derive the primary surface.

3.15.15 primary surface profile

surface profile trace obtained when a surface profile trace is represented as a specified primary mathematical model with specified nesting index N_{is}

NOTE In the BS EN ISO 21920 series, a profile S-filter is used to derive the primary surface profile from a profile trace (e.g. mechanical profile).

[SOURCE: BS EN ISO 21920-2:2022, **3.1.12**, modified – notes 2 and 3 omitted]

3.15.16 profile F-operation

operation which removes form from a profile or surface

[SOURCE: BS EN ISO 21920-2:2022, **3.1.13.3**, modified]

3.15.17 profile L-filter

profile filter which removes large lateral scale (long wavelength) components from a profile or surface

[SOURCE: BS EN ISO 21920-2:2022, **3.1.13.2**, modified]

3.15.18 profile S-filter

profile filter which removes small lateral scale (short wavelength) components from a profile or surface

[SOURCE: BS EN ISO 21920-2:2022, **3.1.13.1**, modified]

3.15.19 profile surface texture

surface texture defined and evaluated in a plane perpendicular to the surface with a specified orientation

NOTE 1 This is covered by the BS EN ISO 21920 series of standards.

NOTE 2 Profile surface texture can be considered as 2D surface texture.

3.15.20 reference line

line corresponding to a specific large lateral scale component

NOTE 1 The x-axis of the specification coordinate system coincides with the reference line of the assessed profile and the z-axis is oriented in an outward direction (from the material to the surrounding medium).

NOTE 2 The reference line for the primary profile and waviness profile are the large lateral scale component of primary surface profile removed by the profile F-operation.

NOTE 3 The reference line for the roughness profile is the component of the primary profile removed by the profile L-filter.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.15**]

3.15.21 reference surface

<surface texture> surface associated to the scale-limited surface according to a criterion

NOTE This reference surface is used as the origin of heights for surface texture parameters.

3.15.22 restrained state

condition of a part subjected to forces in addition to gravity

3.15.23 roughness profile
R-profile

scale-limited profile at any position x derived from the primary profile by removing large-scale lateral components by a specific type of profile L-filter with a nesting index N_{ic}

NOTE The roughness profile is the basis for evaluation of the R-parameters.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.14.3**]

3.15.24 scale-limited profile

profile structure scale components between specified nesting indices

EXAMPLE: A profile is scale-limited after applying a profile filter with a specified nesting index.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.14**]

3.15.25 scale-limited surface

S-F surface or S-L surface

3.15.26 section length

l_{sc}

length in the direction of the *x*-axis used to obtain section length parameters such as Rz, Rp and Rv

NOTE Default values of l_{sc} are found in BS EN ISO 21920-3.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.17**]

3.15.27 setting class

Scn

identifier to label default settings

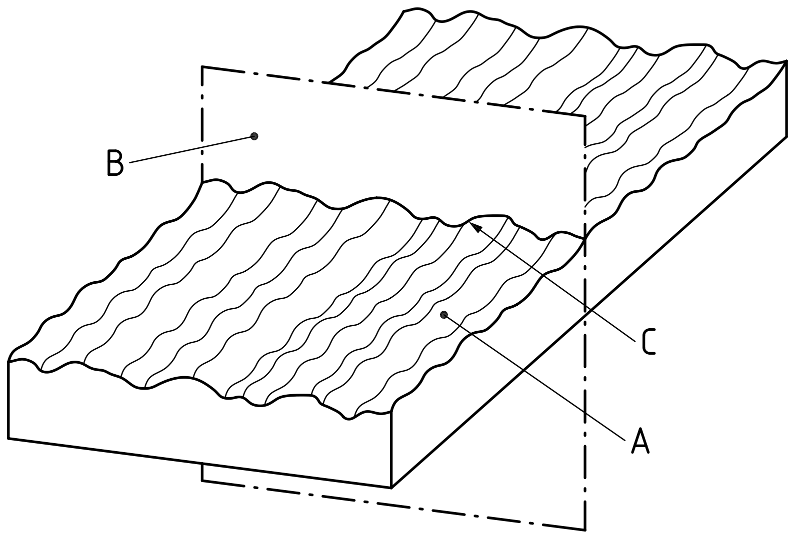
3.15.28 surface profile trace

intersection of the skin model by an intersection plane perpendicular to the skin model and in a specified direction

NOTE See Figure 1.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.4**, modified]

Figure 1 — *Surface profile trace*



Key

A Skin model

B Intersection plane

C Profile trace

NOTE SOURCE: BS EN ISO 21920-2:2022, Figure 2.

3.15.29 S-parameter

field parameter or feature parameter that is not a V-parameter

3.15.30 S-F surface

surface derived from the primary surface by removing the form using an F-operation

3.15.31 S-L surface

surface derived from the S-F surface by removing the large-scale components using an L-filter

3.15.32 V-parameter

material volume or void volume field parameter

3.15.33 waviness profile
W-profile

scale-limited profile at any position *x* derived from the primary profile by removing small-scale lateral components by a specific type of profile S-filter with a nesting index N_{ic}

NOTE The waviness profile is the basis for evaluation of the W-parameters.

[SOURCE: BS EN ISO 21920-2:2022, **3.1.14.2**, modified – notes 2 and 3 omitted]

3.16 technical product documentation (TPD)

means of conveying all or part of a design definition or specification of a product

[SOURCE: BS EN ISO 10209:2022, **3.10.163**]

3.17 technical product specification (TPS)

technical product documentation comprising the complete design definition and specification of a product for manufacturing and verification purposes

NOTE A TPS (which may contain drawings, 3-D models, parts lists or other documents forming an integral part of the specification, in whatever format they are presented) may consist of one or more TPDs.

[SOURCE: BS EN ISO 10209:2022, **3.10.164**]

4 Standards underpinning BS 8888

The following documents shall be applied as “global” standards in support of BS 8888.

- ISO/IEC Guide 98-3, *Uncertainty of measurement – Part 3: Guide to the expression of uncertainty in measurement (GUM:1995)*.
- ISO/IEC Guide 99, *International vocabulary of metrology – Basic and general concepts and associated terms (VIM)*.

The principles in Clause **5** shall always be applied where conformity with BS 8888 is claimed.

5 Technical product specification (TPS)

5.1 Types of specification

*COMMENTARY ON **5.1***

Specifications relating to mechanical engineering products can be broadly categorized as functional, manufacturing or verification specifications. Functional specifications specify design intent which is imperative to the safe function and performance of products. Product functionality is paramount and thus differentiation between requirement types is necessary to prevent functional requirements from being unintentionally or unnecessarily affected or obscured by changes in manufacturing or verification processes. Differentiation between specification types also facilitates communication between stakeholders and provides clarity on ownership of requirements throughout the supply chain. This clause provides a structure of GPS specifications in accordance with PD ISO/TS 21619:2018 to facilitate differentiation between functional, manufacturing and verification specifications. Each of these specification types can be at the assembly, sub-assembly or component/part level.

"The functional specification (FUN-SPEC) of the assembly is the master specification for the functional sub-assembly specification(s) (FUN-SPEC-SUB) and the functional component specification(s) (FUNSPEC-COM). All functional specifications in the three levels are master specifications for the manufacturing specification and the verification specification."

[SOURCE: PD ISO/TS 21619:2018, **4.1**]

NOTE All specifications may be documents other than drawings.

5.1.1 Functional specifications

*COMMENTARY ON **5.1.1***

"A functional specification is a document stating functional requirements."

*[SOURCE: PD ISO/TS 21619:2018, **3.8**]*

Functional specifications are typically the responsibility of the designer and shall not contain manufacturing or verification requirements.

NOTE 1 Functional specifications may be used to manufacture or verify products.

Functional specifications shall be manufacturing and verification process agnostic.

NOTE 2 Where features of the manufacturing process need to be taken into account at a functional level, they may be included in the functional specification. This can be due, for example, to split lines from moulding/casting, surface roughness lay from machining, welds from fabrication or anisotropic material properties due to additive manufacturing processes.

As illustrated in Figure 2, functional specifications may be used to derive manufacturing and verification specifications, but manufacturing and verification specifications shall not be used to express functional requirements.

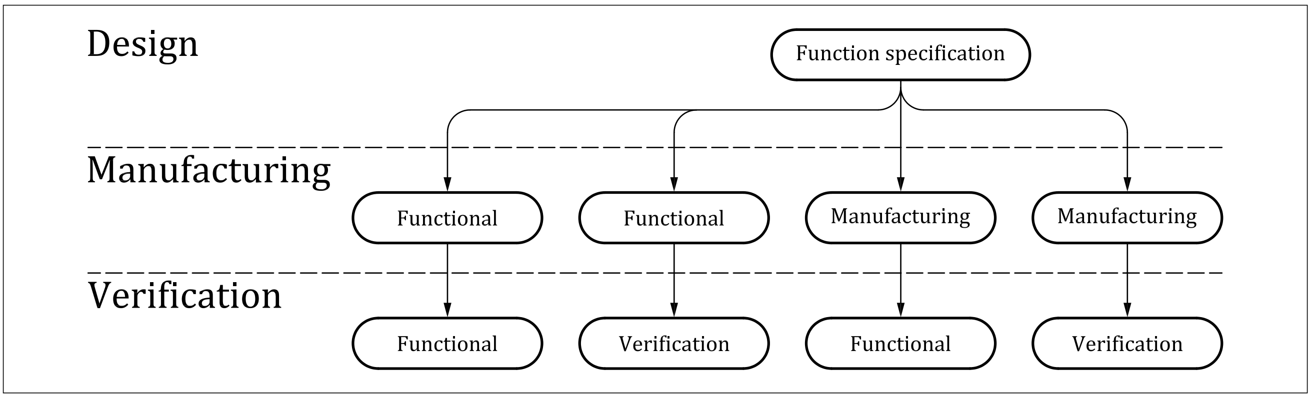
Since manufacturing and verification specifications are more frequently subject to change due to non-functional factors, such as manufacturing locations, manufacturing or verification equipment and contractual agreements, removal of these requirements from functional specifications shall be used to facilitate:

a) better differentiation between types of requirements;

b) better communication; and

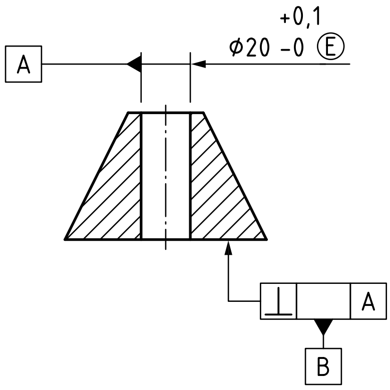
c) greater clarity in contractual obligations (the designer is not responsible for how a part/sub-assembly/assembly is manufactured or verified).

Figure 2 — *Hierarchy of specification types*



NOTE 3 An example of a functional specification is shown in Figure 3.

Figure 3 — *Example of functional specification*



NOTE The primary datum feature has been selected in accordance with the function of the part, e.g. this is a tapered roller where the 20 ID bore is mounted to an axle in the higher assembly.

5.1.2 Manufacturing specifications

COMMENTARY ON ***5.1.2***

"A manufacturing specification is a document stating manufacturing process-related requirements."

[SOURCE: PD ISO/TS 21619:2018, ***3.9****]*

The manufacturer or manufacturing engineering function is typically responsible for manufacturing specifications.

Manufacturing specifications shall be derived from functional specifications.

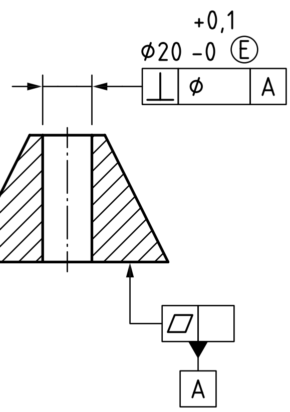
NOTE 1 One or more manufacturing specifications may be derived from a single functional specification. The transcription of the requirements between the functional and manufacturing specifications is the responsibility of the party doing the transformation.

Measures shall be implemented to prevent incorrect or ambiguous transmission of information between the specifications and to manage change where derived specifications exist.

NOTE 2 An example of a manufacturing specification is shown in Figure 4.

NOTE 3 Conformity to one or more individual manufacturing specifications does not guarantee conformity to the functional specification.

Figure 4 — *Example of manufacturing specification*



NOTE The manufacturing specification has been derived from the functional specification shown in Figure 3. The datum system has been modified to reflect the requirements of the manufacturing process, e.g. the internal diameter is drilled relative to the bottom surface, datum feature A.

5.1.3 Verification specifications

*COMMENTARY ON **5.1.3***

"A verification specification is a document stating verification related requirements."

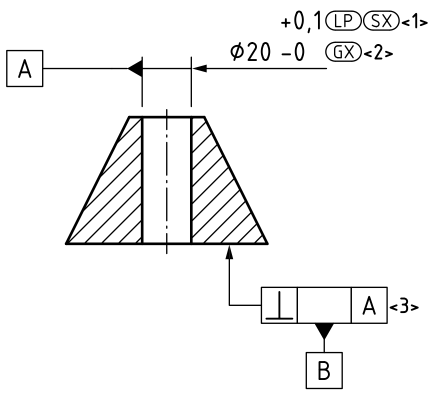
*[SOURCE: PD ISO/TS 21619:2018, **3.10**]*

The manufacturer, verifier or quality function is typically responsible for verification specifications. Verification specifications may relate to and be derived from either functional or manufacturing specifications. One or more verification specifications may be derived from functional or manufacturing specifications. The transcription of the requirements from the functional or manufacturing specifications to verification specifications is the responsibility of the party doing the transformation.

Measures shall be implemented to prevent incorrect or ambiguous transmission of information between the specifications and to manage change where derived specifications exist.

NOTE 1 An example of a verification specification is shown in Figure 5.

Figure 5 — *Example of verification specification*



NOTE 1 The verification specification has been derived from the functional specification in Figure 3. Verification specifications may detail specific requirements relating to verification processes. Figure 5 translates the special specification operator (envelope requirement) expressed into two verification operations (LP and GX) (see BS EN ISO 17450-2) in accordance with the duality principle (see BS EN ISO 8015). The GPS [functional] specification does not dictate which verification operators are acceptable. The acceptability of a verification operator is evaluated using the measurement uncertainty and any ambiguity of the specification (see BS EN ISO 8015).

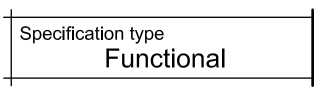
*NOTE 2 Verification specifications (labelled <1>, <2> and <3>) may contain special indicators (see **6.11**) to provide traceability in related documents.*

NOTE 2 If the verification specification(s) has been correctly derived and cover all the requirements contained in a functional specification, conformity to the verification specification(s) should be interpreted as demonstrating conformity to the functional specification.

Throughout the derivation and translation of dependent requirements, traceability shall be verified and documented such that conformity to the master (functional) specifications is proven.

NOTE 3 The type of specification may be indicated in technical product documentation (TPD). The field "Specification Type" can be added to title blocks as shown in Figure 6.

Figure 6 — *Indication of specification type in TPD*



5.2 Principles of specification

5.2.1 Properties

COMMENTARY ON ***5.2.1***

A TPS defines various properties and requirements. Some properties or requirements may be defined for an entire workpiece (e.g. a material specification), some for part of a workpiece (e.g. a particular surface treatment), and some for individual features on a workpiece (e.g. a size requirement). Properties or requirements may also be defined for a process or procedure. Properties which may be specified include:

- *size;*
- *location;*
- *orientation;*
- *form;*
- *surface texture;*
- *surface imperfections;*
- *properties of edges;*
- *material;*
- *coatings and finishes;*
- *hardness;*
- *chemical composition;*
- *treatments which are to be applied to the workpiece;*
- *processing requirements;*
- *recycling requirements;*
- *packaging requirements; and*
- *handling requirements.*

*Properties of location and orientation may be defined solely with toleranced dimensions (see Clause **7**), but this results in ambiguous specifications which are open to a wide range of interpretations.*

For an unambiguous definition of the size, location, orientation and form of features on the workpiece, the ISO system of Geometrical Product Specification (GPS) shall be used.

*NOTE This utilizes datums and geometrical specifications to minimize ambiguity (see Clause **8** and Clause **9**).*

5.2.2 Interpretation

A TPS shall clearly indicate which standards, or systems of standards, apply to its interpretation.

If different versions of standards apply to changes or updates to a TPS, this shall be clearly indicated.

When the format of data in a TPS conforms to the British and ISO standards identified in BS 8888, a reference to BS 8888 itself shall be used to indicate that these standards apply to the specification. Such a reference shall take the following form:

CONFORMS TO BS 8888

NOTE 1 This note may be placed in the title block, in a drawing note or elsewhere within the drawing frame.

NOTE 2 The phrase "format of data" refers to 2D and 3D TPS, and includes aspects such as format of attributes, layout of drawing borders and title blocks, layout of views and projections, format of letters, numbers, dimensions and tolerances, and use of different line types and thicknesses.

When the geometry of a workpiece is defined according to the requirements of ISO Geometrical Product Specification and Verification (ISO GPS) which are documented in standards such as BS EN ISO 8015, BS EN ISO 5459 and BS EN ISO 1101:2017, a reference to BS EN ISO 8015 shall be used to indicate that all geometrical product specification and verification standards apply to the interpretation of that specification. Such a reference shall take the following form:

TOLERANCING ISO 8015

NOTE 3 This note may be placed in the title block, in a drawing note or elsewhere within the drawing frame. This note is required in addition to any reference to BS 8888. This marking is a BS 8888 requirement, and not a requirement of BS EN ISO 8015 and is necessary to avoid possible misinterpretation of which system of standards govern the interpretation of a TPS.

NOTE 4 Reference to BS EN ISO 8015 is not just important to confirm which standards interpretation method is required for the TPS. Reference to BS EN ISO 8015 also enables the invocation principle in BS EN ISO 8015 and, as such, no further reference to other ISO GPS standards is required; for instance reference to BS EN ISO 5459 for Datums is not required, reference to BS EN ISO 1101:2017 for Geometrical Tolerancing is not required, etc.

5.2.3 ISO Geometrical product specification (ISO GPS) invocation principle

"Once a portion of the GPS and verification system is invoked on a mechanical engineering product specification, the entire GPS and verification system shall be invoked, unless otherwise indicated on the specification e.g. by reference to a relevant document.

"Unless otherwise indicated on the specification" means that it is indicated on the specification that it has been prepared in accordance with a regional, national or company standard, then that standard, and not the GPS and verification system, shall be used to interpret those elements of the specification that are covered by that standard.

NOTE 1 The "entire geometrical product specification and verification system is invoked" means that fundamental and general GPS standards apply and, consequently that e.g. the reference temperature given in BS EN ISO 129-1 and the decision rules given in BS EN ISO 14253-1 apply, unless otherwise indicated. The purpose of the invocation principle is to provide the formal traceability for these GPS standards and rules."

[SOURCE: BS EN ISO 8015:2011, **5.1**, modified – "Tolerancing ISO 8015" can optionally be indicated in or near the title block for information, but is not required to invoke the GPS and verification system, note removed]

NOTE 2 The GPS and verification system is defined in a hierarchy of standards that includes the following types of standards in the given order:

- *fundamental GPS standards;*
- *general GPS standards; and*
- *complementary GPS standards.*

See BS EN ISO 14638 for further details.

"The rules given in standards at a higher level in the hierarchy apply in all cases, unless rules in standards at lower levels in the hierarchy specifically give other rules."

[SOURCE: BS EN ISO 8015:2011, **5.2**]

All rules given in fundamental GPS standards shall apply, in addition to those specifically given in general GPS standards, e.g. BS EN ISO 1101:2017, except where the rules in the general GPS standard are explicitly different from the rules given in fundamental GPS standards and unless the rules in a specific complementary GPS standard give other rules that apply within its scope.

All rules given in fundamental and general GPS standards shall apply in addition to the rules specifically given in complementary GPS standards, including BS EN 22768-1, except where the rules in the complementary GPS standard are explicitly different from the rules given in fundamental and general GPS standards.

5.2.4 Level of detail in a specification

COMMENTARY ON ***5.2.4***

When a specification leaves certain requirements undefined, or open to more than one interpretation, it is ambiguous.

The more ambiguity which exists in a specification, the greater the risk that a manufactured product conforms to that specification but is still unfit for purpose. Conversely, there is also greater risk of rejecting a workpiece which is fit for purpose.

The less ambiguity which exists in a specification, the greater the likelihood that the conforming manufactured product is fit for purpose.

It is not always necessary to minimize or eliminate the ambiguity in a specification. A high level of ambiguity can be acceptable when the specifier is willing to rely on the skill, craftsmanship and knowledge of the manufacturer or fitter to achieve a satisfactory result. The benefit is a very simple specification, but the consequences could include a high degree of variation from one part to the next, a lack of interchangeability and inconsistent quality.

Higher levels of ambiguity also mean much greater probability of disputes over whether a manufactured product is acceptable or not.

The use of datums, geometrical specifications, surface texture requirements and edge tolerances is not obligatory. However, not using these specification elements results in greater ambiguity.

The specifier shall determine what level of detail is required in a specification and define the workpiece in an appropriate manner (see Table 2).

Table 2 — *Level of detail in a TPS*

Specification elements included	**Level of detail**			
	Sketch	**Technical model/drawing**	**Technical model/drawing with geometrical tolerancing**	**GPS and verification**
3D/2D geometry	✓	✓	✓	✓
Dimensions	—	✓	✓	✓
+/− Tolerances	—	✓	✓	✓
Datums	—	—	✓	✓
Geometrical specifications	—	—	✓	✓
Surface texture requirements	—	—	—	✓
Edge tolerances	—	—	—	✓
Level of ambiguity in the specification	**Very high**	**High**	**Low**	**Very low**

5.2.5 Independency principle

"By default, every GPS requirement for a feature or relation between features shall be fulfilled independently of other requirements, except when it is stated in a standard or by special indication (e.g. modifiers Ⓜ or Ⓛ in accordance with BS EN ISO 2692 or CZ or CZR in accordance with BS EN ISO 5458) as part of the actual specification."

[SOURCE: BS EN ISO 8015:2011, **5.5**]

5.2.6 Definitive specification principle

The specification is definite: requirements of the specification shall apply to the finished state of the workpiece unless otherwise indicated. Requirements which apply to an intermediate state of the workpiece shall be identified as such.

NOTE Where geometrical or dimensional properties are affected by a finishing process such as plating or heat treatment, there is often confusion over whether specification requirements apply before or after the process. In such cases, a statement can be used such as "all dimensions and tolerances apply after plating", unless otherwise specified.

5.2.7 Feature principle

*COMMENTARY ON **5.2.7***

The GPS and verification defines several different types of features. The main types of features are:

a) integral features: features which consist of, or represent, surfaces on the workpiece;

b) features of size: integral features which have a size characteristic and include (but are not limited to) the following types of feature:

1) two parallel opposed planes;

2) a cylinder;

3) a sphere;

4) two non-parallel planes (a wedge);

5) a cone;

6) the two rounded ends of a slot or protrusion (see Annex H); and

7) a torus;

c) derived features: theoretical features such as axes or median lines, which are derived from a feature of size (axis or median line, median plane or median surface and centre points).

The GPS and verification also defines a number of different "worlds" or "states" in which features can exist. The features can exist in an ideal form, such as in the specification, while in other states they exist in a non-ideal form, such as surfaces on the manufactured workpiece (see Figure 7).

In the specification features are represented in an ideal state. These features have ideal form, and ideal relationships with each other. These are known as nominal features.

On the manufactured workpiece, features are non ideal and are known as real features.

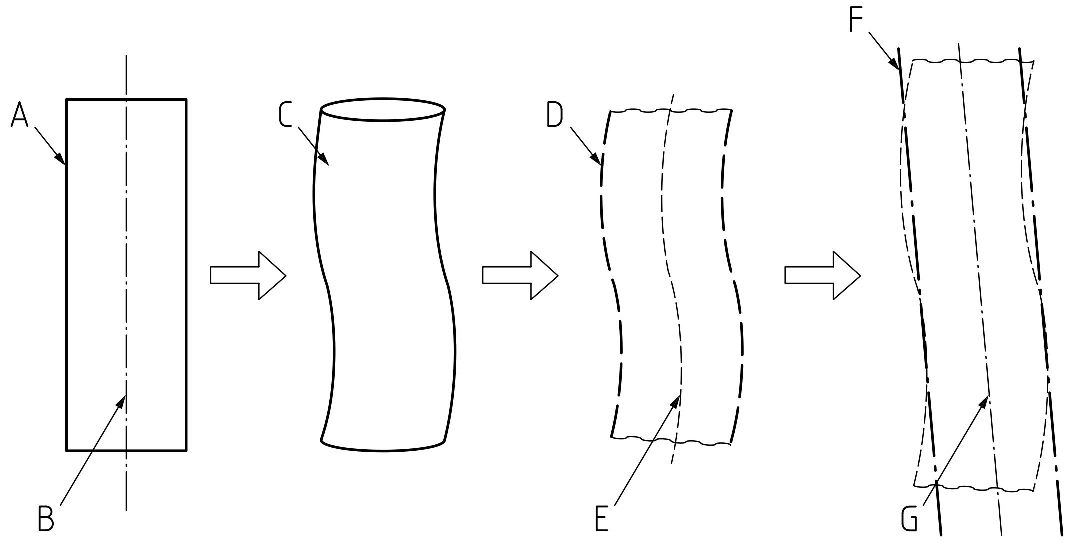
In verification, a feature is represented by a set of data obtained by sampling the workpiece with measuring instruments. This set of data represents the non-ideal real feature and is known as an extracted feature.

The extracted feature is used when determining whether a tolerance requirement has been met.

In verification, an ideal feature may be "fitted" to, or associated with, either the real feature or the extracted feature. This ideal feature is known as an associated feature.

The associated feature is used when determining the orientation of size measurements, or when defining a datum. A range of different association methods may be used.

Figure 7 — *Interrelationship of the geometrical feature definitions*



Key

A	Nominal integral feature	D	Extracted integral feature	G	Associated derived feature
B	Nominal derived feature	E	Extracted derived feature		
C	Real feature	F	Associated integral feature		

A workpiece shall be made up of a number of features limited by natural boundaries. By default, every specification requirement for a feature or relation between features shall apply to the entire feature and each specification requirement shall apply only to one feature or one relation between features.

NOTE This default can only be overridden by explicit indications on the drawing.

5.2.8 Decimal principle

Non-indicated decimals shall be taken as zeros.

NOTE 1 0,2 is the same as 0,200 000 000 000 000 000 000 000 000 000 ... etc.

NOTE 2 10 is the same as 10,000 000 000 000 000 000 000 000 000 000 ... etc.

5.2.9 Reference conditions

Unless otherwise stated, all specification requirements shall apply at 20 °C with the workpiece free from contaminants.

NOTE For more guidance on the conditions, see BS EN ISO 129-1 and BS EN ISO 8015.

Any additional or alternative conditions that apply, e.g. humidity conditions, shall be defined in the specification.

5.2.10 Rigid workpiece principle

Unless otherwise stated, a workpiece shall be treated as having infinite stiffness. All specification requirements shall apply in the free state, with the workpiece undeformed by any external forces (including gravity). Any additional or alternative conditions that apply shall be defined in the specification.

5.2.11 Non-rigid workpieces

Where a workpiece is not to be treated as infinitely stiff, this shall be indicated with the statement "ISO 10579-NR" placed in or near the title block or in the notes for the TPS.

Specification requirements shall be applied either to a free state condition or to a restrained state condition.

NOTE 1 A specification may contain solely free state requirements or solely restrained state requirements or a combination of both.

For free-state requirements, the TPS shall define the direction in which gravity applies and the manner in which the workpiece is supported.

NOTE 2 Free state requirements are identified with the Ⓕ symbol.

For constrained-state requirements, the TPS shall define the restrained condition. When ISO 10579-NR is invoked, specification requirements shall apply in the restrained condition unless otherwise indicated (e.g. with the Ⓕ symbol).

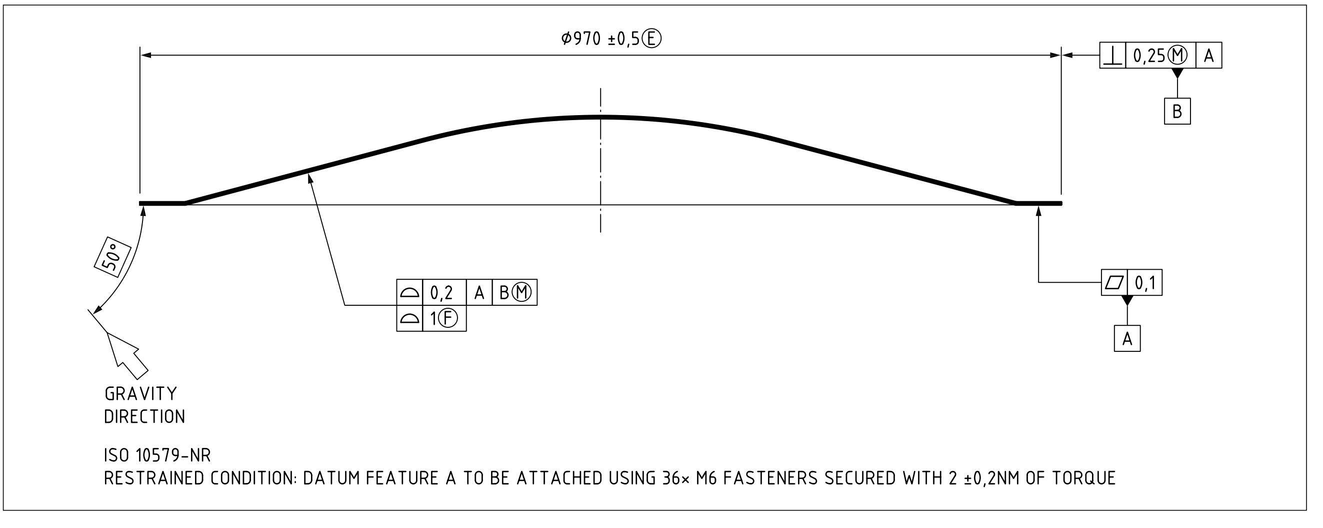
Where more than one free state or restrained state is required, flag notes shall be used to indicate to which state each specification requirement applies.

NOTE 3 The restrained state is usually intended to replicate the assembled condition of the workpiece.

NOTE 4 The restrained state can be identified directly on the TPS with notes, or indirectly with a reference to another document.

NOTE 5 Figure 8 shows an example of ISO 10579-NR non-rigid body applied to a workpiece, the free state, restraining condition specification and tolerance indicator usage.

Figure 8 — *Non-rigid specification applied to a workpiece, the free state, restraining condition specification and tolerance indicator usage*



5.2.12 Size settings in 3D CAD data

*COMMENTARY ON **5.2.12***

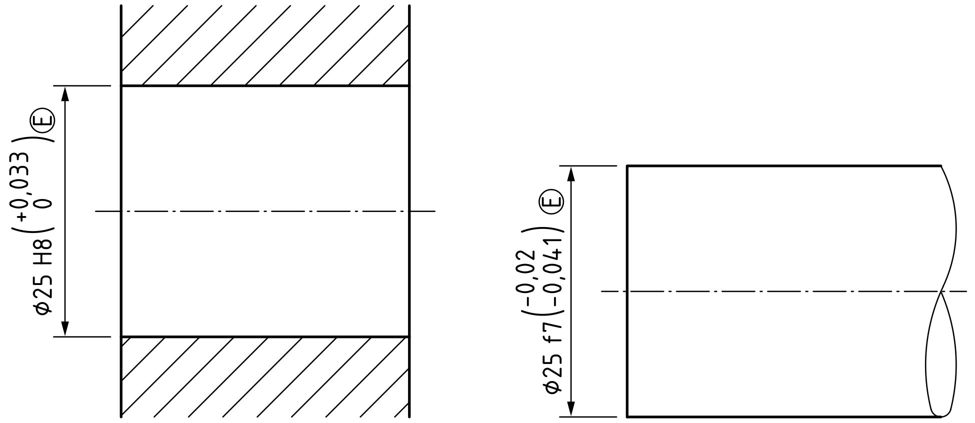
See BS ISO 16792 for requirements for the preparation, revision and presentation of data sets in support of 3D modelling.

Where the 3D CAD geometry includes features of size, two different practices are widely used. These are to model the feature at "nominal" size or to model the feature at "mid-limit" size (see Figure 9).

The "nominal" size is the dimension value to which the tolerance is applied (dimensions which are shown as upper and lower limits do not have a nominal value). The "mid-limit" size is the value half-way through the tolerance range.

Where symmetrical size tolerances are used, "nominal" and "mid-limit" are the same. When non-symmetrical tolerances are used, such as limit and fit codes, they are different.

Figure 9 — *Hole and shaft sizes*



Hole

Nominal size: Ø25

Maximum material size: Ø25

Mid-limit size: Ø25,0165

Least material size: Ø25,033

Shaft

Nominal size: Ø25

Maximum material size: Ø24,98

Mid-limit size: Ø24,9695

Least material size: Ø24,959

When a nominal size is provided by means of a specification or dimension, all features of size shall be modelled to nominal size or to mid-limit value of the tolerance limits.

Where a nominal size is not provided, all features of size shall be modelled to the mid-limit value of the tolerance limits, unless otherwise stated.

NOTE Table 3 shows example mid-limit and nominal values for a 3D CAD model.

Within a 3D CAD model, wherever non-symmetrical size tolerances are used, one practice shall be consistently followed.

Table 3 — *Mid-limit and nominal values for 3D CAD model*

Dimension	Nominal value	Value to be modelled when working to nominal	Mid-limit value	Value to be modelled when working to "mid-limit"
⌀15 ⌀13	Not stated	⌀14	⌀14	⌀14
20,4 19,6	Not stated	20	20	20
10 ±1	10	10	10	10
⌀25 H8Ⓔ	⌀25	⌀25	⌀25,016 5	⌀25,016 5
⌀25 f7Ⓔ	⌀25	⌀25	⌀24,969 5	⌀24,969 5
⌀13,5 $^{+1{,}5}_{-0{,}5}$	⌀13,5	⌀13,5	⌀14	⌀14
⌀14 $^{+1}_{0}$	⌀14	⌀14	⌀14,5	⌀14,5
R0,5 max.	R0,5	R0,5	R0,25 A)	R0,25

A) Min. value = 0.

5.2.13 Interpretations of limits of size for a feature of size

5.2.13.1 Relevant standards

Limits of size for a feature of size shall be interpreted according to the principles and rules defined in the following standards.

- BS EN ISO 8015, *Geometrical product specifications (GPS) – Fundamentals – Concepts, principles and rules*.
- BS EN ISO 14405-1, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 1: Linear sizes*.
- BS EN ISO 14405-2, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 2: Dimensions other than linear or angular sizes*.
- BS EN ISO 14405-3, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 3: Angular sizes*.
- BS EN ISO 17450-1, *Geometrical product specifications (GPS) – General concepts – Part 1: Model for geometrical specification and verification*.
- BS EN ISO 17450-2, *Geometrical product specifications (GPS) – General concepts – Part 2: Basic tenets, specifications, operators, uncertainties and ambiguities*.
- BS EN ISO 17450-3, *Geometrical product specifications (GPS) – General concepts – Part 3: Toleranced features*.
- BS EN ISO 22432, *Geometrical product specifications (GPS) – Features used in specification and verification*.

5.2.13.2 Interpretations of limits of size for a linear feature of size

A linear size specification shall be interpreted as a two-point local size requirement unless otherwise indicated (see BS EN ISO 14405-1).

When a different specification operator is required, it shall be invoked using the appropriate symbol applied to the individual size specification, or by defining a different default linear size specification operator for the entire TPS (see BS EN ISO 14405-1).

If the default size specification operator is to be changed for an entire TPS, the following indication shall appear in or near the title block of each drawing:

LINEAR SIZE ISO 14405 X

where X is the symbol representing the required operator.

EXAMPLE

If the envelope requirement is to be made the default interpretation of linear size specifications for the entire TPS, the indication is:

LINEAR SIZE ISO 14405 Ⓔ

NOTE 1 A additional explanatory note should be used on the drawing as this indication might be overlooked or misunderstood.

NOTE 2 The consequence of using the default "two-point" definition of a linear size tolerance is that size limits only apply to any "two-point" size measurement of a feature. Therefore, the size limits do not control the form deviations of the feature. For example:

a) the size limits on a cylindrical feature do not control its roundness or straightness; and

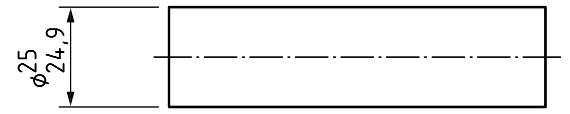
b) the size limits on a feature consisting of two parallel, opposed planes do not control the flatness of the two planes.

Form deviations can be controlled by:

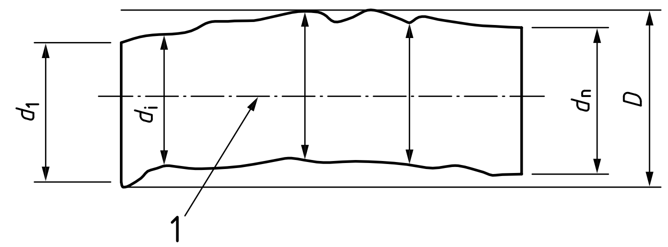
1) *individually specified geometrical specifications;*
2) *general geometrical specifications;*
3) *the use of the envelope requirement, Ⓔ (where the maximum material limit of size defines an envelope of perfect form for the relevant surfaces, see BS EN ISO 14405-1); or*
4) *the use of another global size characteristic (see BS EN ISO 14405-1).*

NOTE 3 Figure 10 gives examples of how size limits could be interpreted where no form control is defined and the specification is incomplete.

Figure 10 — *Interpretations of size limits where no form control is defined and the specification is incomplete*



a) Drawing presentation



b) Size limit being met, but no control over the form of the feature

NOTE Local sizes d_i are contained within the range Ø25 to Ø24,9, where d_1, d_i and d_n are the numerous individual two-point size measurements taken on the real component (see BS EN ISO 17450-3).

Key

1 Axis of the least-squares associated cylinder

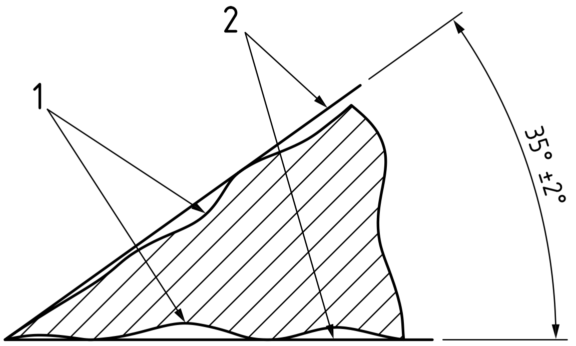
D Undefined overall size

5.2.13.3 Interpretations of limits of size for an angular feature of size

An angular size specification shall be interpreted as a two-line angular size requirement, with minimax association, unless otherwise indicated.

NOTE 1 For angular size, the angle applies to an ideal line or plane fitted (associated) to the manufactured surface. "Minimax" association (also known as a "Chebyshev" association) is a method used for fitting an ideal feature to a manufactured surface (see Figure 11). The minimax association works by minimizing the maximum deviations between the ideal feature and the manufactured feature. It can be further modified by applying a material constraint such as "outside the material" or "inside the material", but this makes no difference to the orientation of the ideal feature, and no material constraint is specified when evaluating angular sizes.

Figure 11 — *Two-line angular size: Second association with minimax criterion*



Key

1 Extracted lines

2 Associated lines

When a different specification operator is required, it shall be invoked using the appropriate modifier applied to the individual angular size specification, or by defining a different default angular size specification operator for the entire TPS (see BS EN ISO 14405-3).

If the default size specification operator is to be changed for an entire TPS, the following indication shall appear in or near the title block of each drawing:

ANGULAR SIZE ISO 14405 X

where X is the modifier representing the required operator.

EXAMPLE

If the global angular size with least-squares association is to be made the default interpretation of angular size specifications for the entire TPS, the indication is:

ANGULAR SIZE ISO 14405 (LG)

NOTE 2 A note on the drawing should be used in addition to the ISO 14405-3 reference, for clarity.

5.2.14 Identification and classification of specification requirements

5.2.14.1 General

When an identification or a classification system is used on a TPS, it shall contain a description of indication, criteria for classification and method of identification.

NOTE 1 Identification and classification may be combined or applied separately using methods and systems used or adopted by the company responsible for product specification.

NOTE 2 The use of identification and/or classification for specification is optional.

5.2.14.2 Identification

*COMMENTARY ON **5.2.14.2***

The use of identification is beneficial to the design, manufacture and verification of a product. Identification provides the framework for specification to be traceable throughout the product life cycle, e.g. links to sources of functional requirements, tolerance sharing and allocation, impact assessment of specification changes. Additionally, identification enables the physical verification of specification to be easily associated to the source requirement, e.g. via a control plan, inspection report or non-conformity.

Where specification is to be uniquely identified, it shall have indication associated to a feature, dimension or tolerance in a TPD, e.g. via a label, tag, flag, note, model-based attribute.

NOTE 1 The format and content of the identification system is not standardized and is specified according to company or industry standards.

NOTE 2 Identification of TPS may conform to BS EN ISO 7533.

5.2.14.3 Classification

*COMMENTARY ON **5.2.14.3***

The use of TPS classification is beneficial to the design, manufacture and verification of the product. By identifying what is important, over-specifying is avoided. Classification allows product characteristics to be categorized based on a functional analysis of the product, e.g. risk analysis. Classification therefore enables efficient product manufacture and verification while managing product risk appropriately. Classification enables product families, common characteristics, manufacturing processes and tolerance capabilities to be collectively monitored, understood and improved for existing and future products.

Where a classification system is used, product characteristics shall be risk assessed and classified according to the severity and susceptibility of specification non-conformity affecting product safety, compliance with regulations, fit, function, performance or subsequent processing.

NOTE 1 The format and coding for a classification system is not standardized and is specified according to company or industry standards.

NOTE 2 Classification of TPS may conform to BS EN ISO 24096-1.

NOTE 3 A collection of features, dimensions or tolerances may be identified as belonging to a single class or category with a similar level of risk and sharing one or more common characteristics such as geometry, material type, tolerance values, physical characteristics or other properties.

NOTE 4 Classification might have implications for the method of manufacture, verification and control of the product's individual or population characteristic to which the specification applies.

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6 Technical product documentation (TPD)

6.1 Formatting of TPD

6.1.1 General

A TPD shall conform to one of the five classifications identified in BS ISO 16792. Data management for TPDs shall conform to BS EN ISO 11442.

6.1.2 Product definition classification code 1 (2D drawing only) – Drawing with optional data set

*COMMENTARY ON **6.1.2***

The TPD consists of a 2D drawing in hardcopy or digital format.

The drawing(s) shall provide all the necessary views, sections, datums, dimensions, tolerances and general notation to provide suitable and sufficient information to define the workpiece.

NOTE A workpiece can consist of a component, fabrication or assembly.

6.1.3 Product definition classification code 2 (2D drawing with supplementary 3D CAD model) – Data set with design model and drawing containing specifications

The drawing(s) shall provide all the necessary views, sections, datums, dimensions, tolerances and general notation to provide suitable and sufficient information to define the workpiece.

NOTE The 3D CAD model can be used as supplementary information but is not required for complete definition of the workpiece.

6.1.4 Product definition classification code 3 (simplified 2D drawing with 3D CAD model) – Data set with design model or annotated model and simplified drawing

The drawing and CAD model together shall provide all the necessary views, sections, datums, dimensions, tolerances and general notation to provide suitable and sufficient information to define the workpiece.

NOTE 1 Neither drawing nor CAD model is sufficient on its own to fully define the workpiece.

In the absence of any dimension on the drawing, values shall be obtained from an interrogation of the 3D model geometry.

NOTE 2 Annotation may appear in either or both of the 2D or 3D documents.

There shall be no discrepancy between the 2D and the 3D documents, except where required for the simplified representation of machine elements such as screw threads.

6.1.5 Product definition classification code 4 (2D drawing with 3D CAD model) – Data set with annotated model and drawing

The drawing and CAD model shall each provide all the necessary views, details, sections, datums, dimensions, tolerances and general notation to provide suitable and sufficient information to define the workpiece.

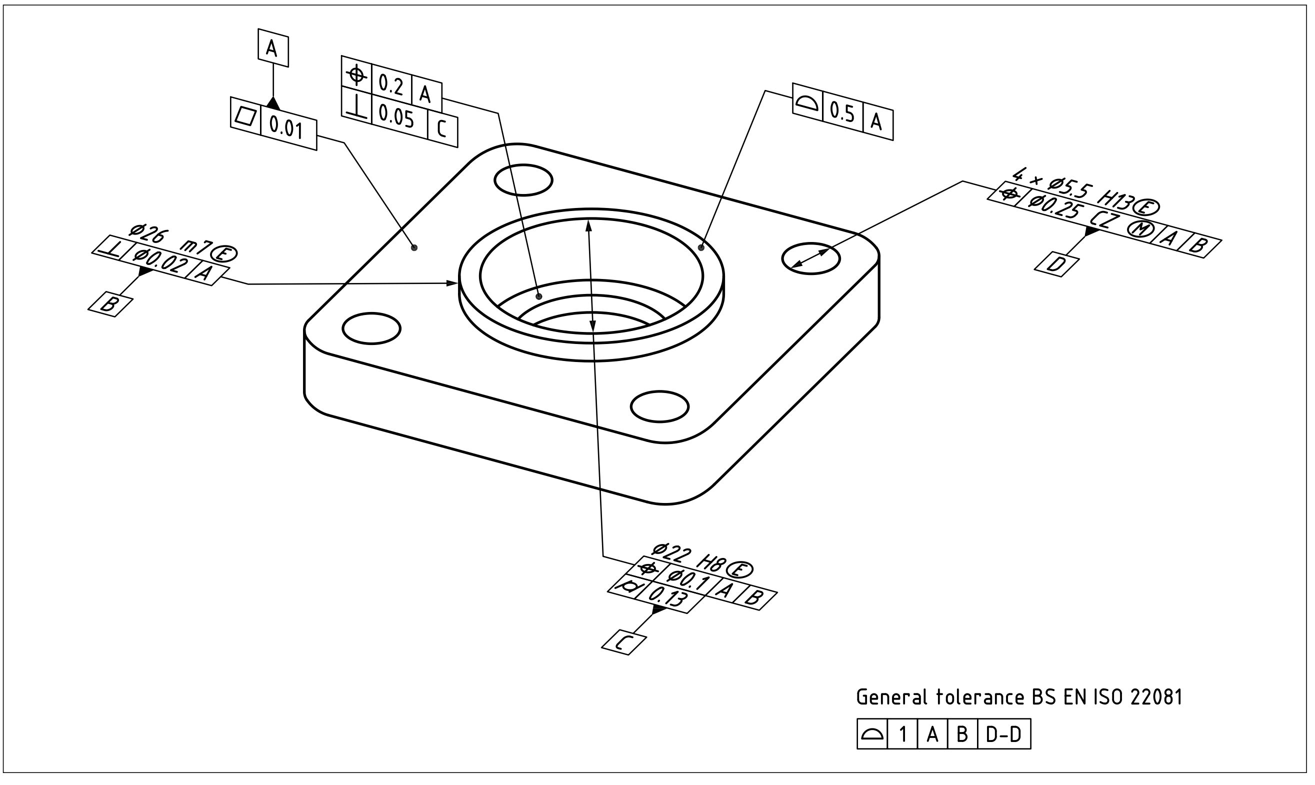
NOTE Either the drawing or the CAD model is sufficient to fully define the workpiece on its own.

6.1.6 Product definition classification code 5 (annotated 3D CAD model) – Data set with annotated model

The data set shall be an annotated model without a supporting drawing.

NOTE 1 An example annotated model is shown in Figure 12.

Figure 12 — *Data set with annotated model*



NOTE 2 The TPD consists of a 3D CAD model.

The 3D data set shall provide all the necessary details, sections, datums, dimensions, tolerances and general notation to provide suitable and sufficient information to define the workpiece.

NOTE 3 No 2D drawing is used as part of the specification, but ancillary 2D documents may be used to express or share date which are derived from the 3D TPD. The use of this type of TPD is sometimes referred to as Model Based Definition (MBD) or Product and Manufacturing Information (PMI).

6.2 3D CAD data used in a TPD

The ability to query the model shall be available.

All model values and resolved dimensions shall be obtained from the model (although they can be displayed on a drawing).

Dimensions displayed on the model, or obtained through a query of the model, shall be rounded to the number of decimal places specified in the model precision.

Values obtained from the model for any feature(s), which are not one of the following items, shall be auxiliary dimensions:

a) values not covered by any tolerance;

b) values which specify the location of a datum target; and

c) values which specify the location and extent of a restricted portion of a feature.

When a 3D model is used as the whole/part of the definition its dimensional condition shall be stated with the maximum material, least material, nominal size or mid-limit. Where the model has a combination of these this shall be clearly defined.

NOTE 1 A standardized approach for modelling/dimensioning holes and shafts should also be in place.

NOTE 2 Additional requirements can be found in BS ISO 16792.

6.3 Document types used in a TPD

*COMMENTARY ON **6.3***

*Various types of documents are used to provide technical information to define the features and requirements for a component or assembly (see Clause **5**). A parts list can also be included on the drawing. See BS ISO 29845 for further document types.*

An assembly drawing (see Figure 13) represents the relative position and/or shape of a group of assembled parts. It depicts the constituents of a parts list.

The list of parts is normally provided in a separate document, i.e. a parts list, but can be included within the drawing, with the individual parts annotated with ballooned numbers matching the item numbers in the parts list.

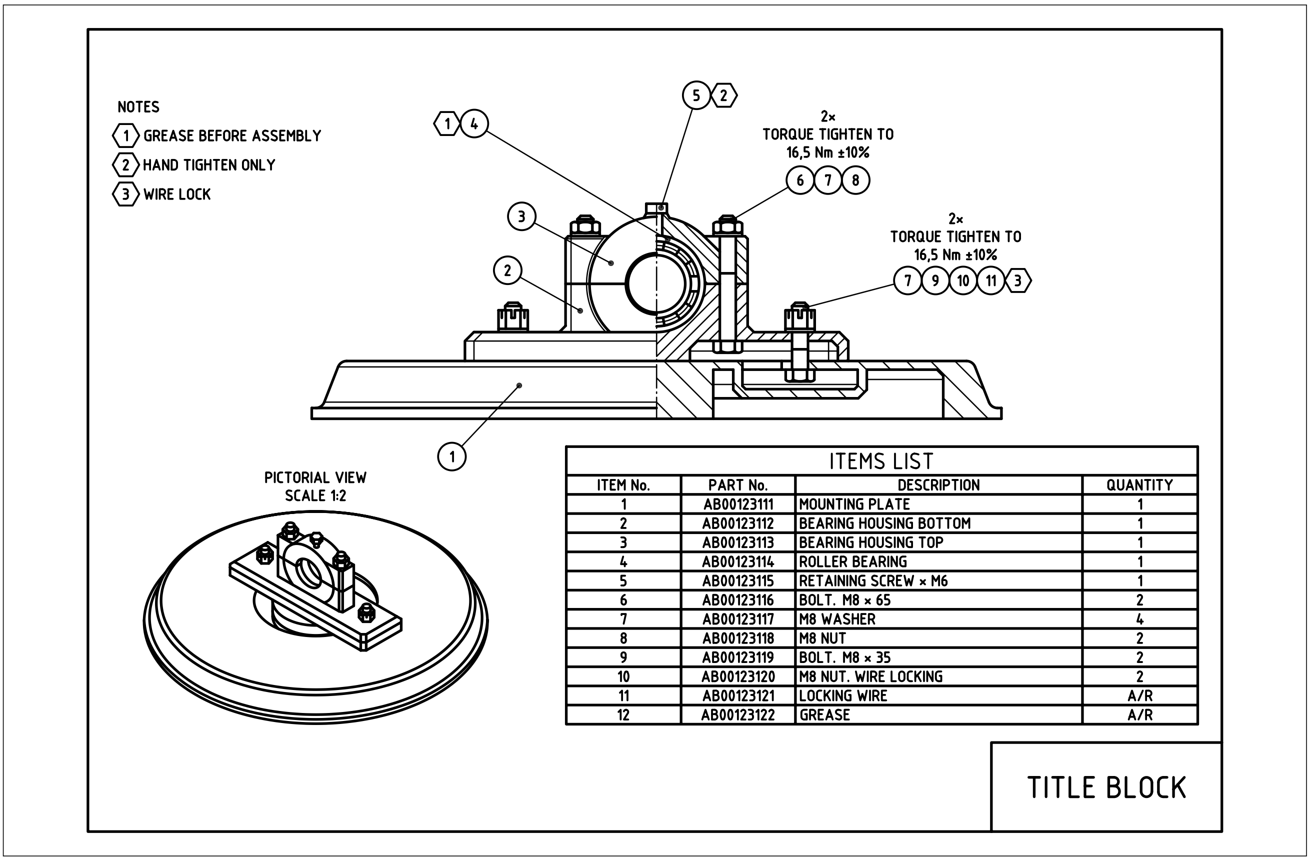
Relevant information can also be added, torque settings can either be shown adjacent to the item balloon or in a torque table, and other information, e.g. lubricating grease, thread adhesive, can be shown alongside corresponding flag notes. Overall and key dimensions can also be included, along with mass and centre of gravity.

A part drawing includes all the necessary information required for the definition of the part, e.g. material properties, dimensions, tolerances, surface texture (see Figure 14).

A fabrication drawing depicts a workpiece which is permanently joined together by means of welding, brazing, adhesion or other method. The constituents can be fully specified (see Figure 15).

The drawing types used in a TPD shall be in accordance with BS EN ISO 5457.

Figure 13 — *Example of assembly drawing*



ITEMS LIST

ITEM No.	PART No.	DESCRIPTION	QUANTITY
1	AB00123111	MOUNTING PLATE	1
2	AB00123112	BEARING HOUSING BOTTOM	1
3	AB00123113	BEARING HOUSING TOP	1
4	AB00123114	ROLLER BEARING	1
5	AB00123115	RETAINING SCREW × M6	1
6	AB00123116	BOLT. M8 × 65	2
7	AB00123117	M8 WASHER	4
8	AB00123118	M8 NUT	2
9	AB00123119	BOLT. M8 × 35	2
10	AB00123120	M8 NUT. WIRE LOCKING	2
11	AB00123121	LOCKING WIRE	A/R
12	AB00123122	GREASE	A/R

Figure 14 — *Example of part drawing*

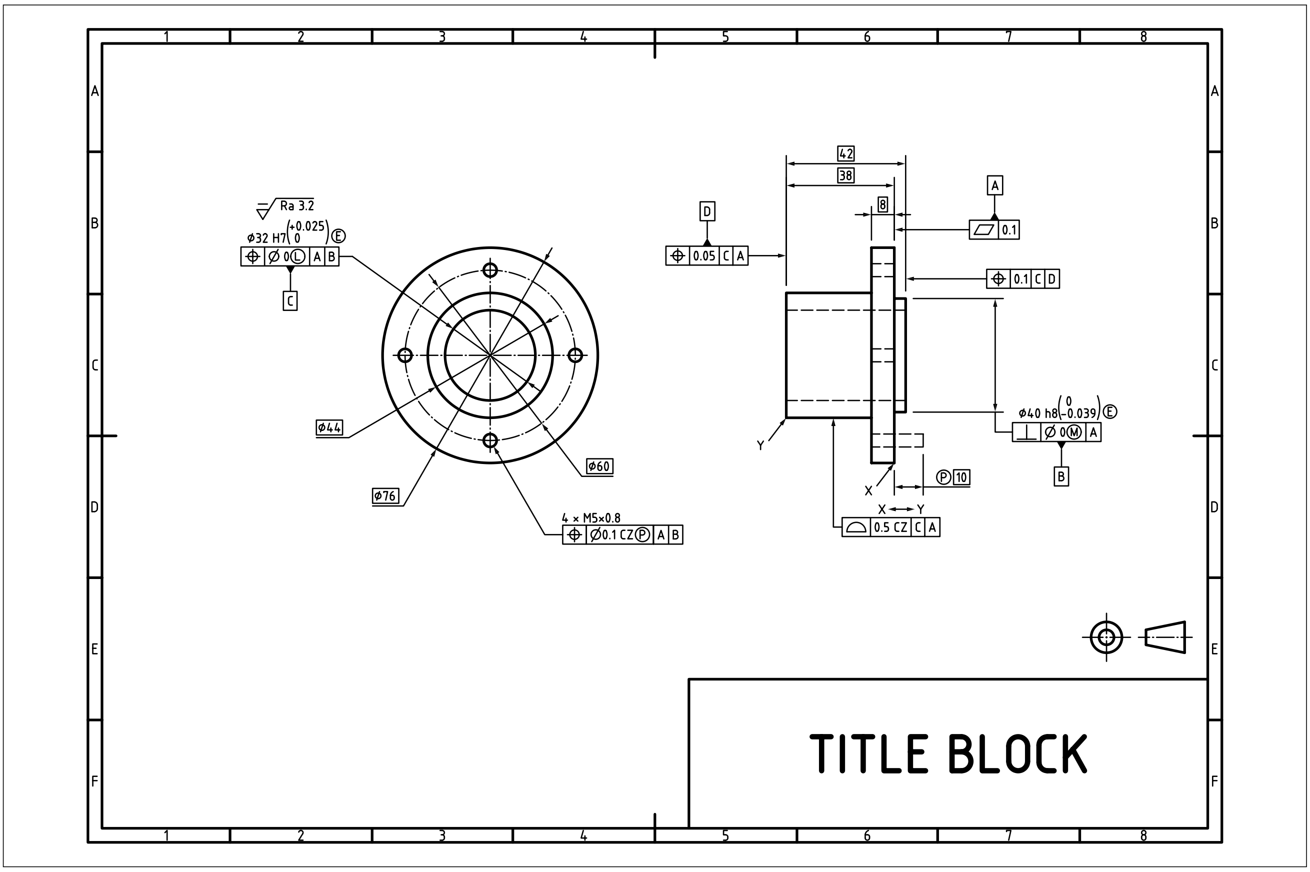
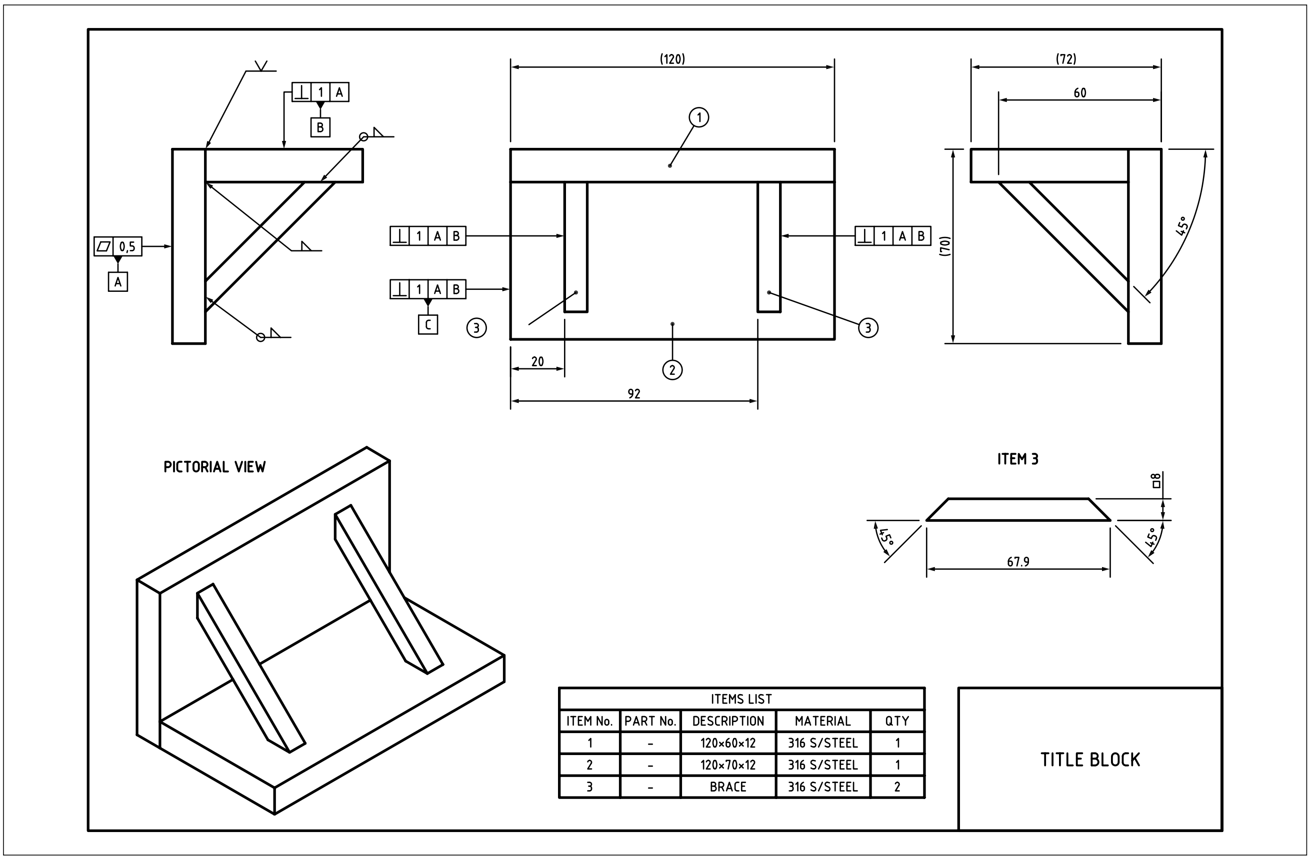


Figure 15 — *Example of fabrication drawing with title block*



6.4 Graphical representation and annotation of 3D data (3D modelling output)

Graphical representation and annotation of 3D models shall conform to the following standard:

BS ISO 16792, *Technical product documentation – Digital product definition data practices.*

NOTE See Clause ***5****.*

6.5 Drawing sheets

COMMENTARY ON ***6.5***

The complete drawing sheet comprises the drawing space, title block, border and frame.

6.5.1 General

The drawing sheet shall conform to BS EN ISO 5457 and BS EN ISO 7200.

6.5.2 Sizes

In accordance with BS EN ISO 5457, the original drawing shall be made on the smallest sheet permitting the necessary clarity and resolution.

NOTE 1 The preferred sizes of the trimmed and untrimmed sheets, as well as the drawing space, are given in Table 4.

NOTE 2 The range of sheet sizes chosen could be rationalized through the use of scales (see ***6.7****).*

Table 4 — *Sizes of trimmed and untrimmed sheets and the drawing space*

Dimensions in millimetres

Designation	Trimmed sheet (T)		Drawing space		Untrimmed sheet (U)	
	a_1 A)	b_1 A)	a_2 ±0,5	b_2 ±0,5	a_3 ±2	b_3 ±2
A0 B)	841	1 189	821	1 159	880	1 230
A1	594	841	574	811	625	880
A2	420	594	400	564	450	625
A3	297	420	277	390	330	450
A4 landscape	210	297	190	267	240	330
A4 portrait	297	210	277	180	330	240

A) For tolerances see BS EN ISO 216.

B) For sizes >A0 see BS EN ISO 216.

6.5.3 Graphical features

6.5.3.1 Title block

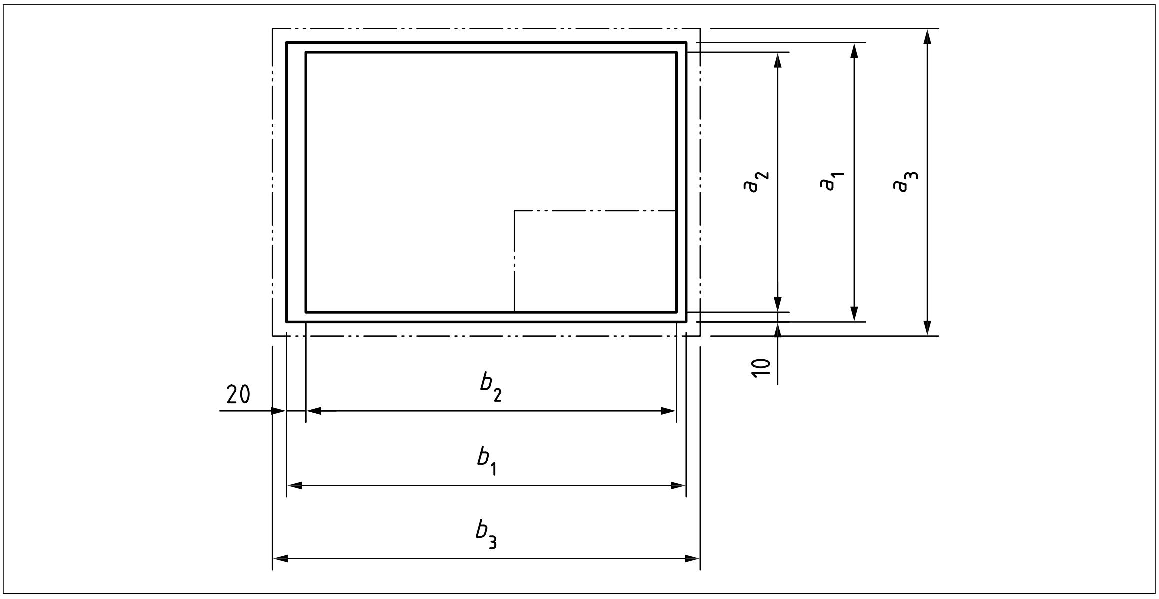
NOTE 1 For the dimensions and layout of title blocks, see BS EN ISO 7200.

Sheet sizes A0 to A3 shall be used in landscape orientation-only (see Figure 16) and the title block shall be located in the bottom right-hand corner of the drawing space.

NOTE 2 A4 sheets can be used in landscape or portrait orientation.

For sheet size A4, the title block shall be situated in the bottom right-hand corner when used in landscape orientation, or the shorter (bottom) part of the drawing space when used in portrait orientation.

Figure 16 — *Size A4 to A0*



The total width of the title block shall be 180 mm to fit an A4 portrait sheet where the left margin is 20 mm and the right margin 10 mm. The same title block shall be used for all paper sizes (see Figure 17).

NOTE 3 The title block of a TPD should contain at least the eight mandatory data fields listed in Table 5. Additional data fields for items such as revision index, or sheet no., should also be taken into account. A complete list of mandatory and optional data fields, along with their recommended character lengths, can be found in BS EN ISO 7200.

Figure 17 — *Title block in compact form*

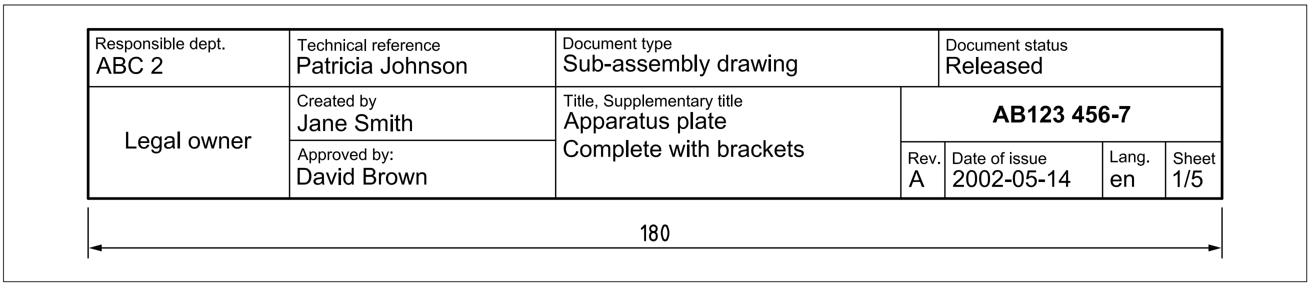


Table 5 — *Identifying, descriptive and administrative data fields in the title block*

Field name	Language-dependent	Recommended number of characters	Field type
Legal owner	—	Unspecified	Identifying
Identification number	No	16	Identifying
Date of issue	No	10	Identifying
Segment/sheet number	No	4	Identifying
Title	Yes	25/30[A)]	Descriptive
Approval person	No/Yes[B)]	20	Administrative
Creator	No/Yes[B)]	20	Administrative
Document type	Yes	20	Administrative

A) 30 to support two-byte-character languages such as Japanese or Chinese.

B) "Yes" to support presentation in different types of alphabet.

6.5.3.2 Borders and frame

Borders enclosed by the edges of the trimmed sheet and the frame limiting the drawing space shall be provided with all sizes. The border shall be 20 mm wide on the left edge, including the frame, and it can be used as a filing margin. All other borders shall be 10 mm wide.

The frame for limiting the drawing space shall be executed with continuous wide lines (see **6.6**).

6.5.3.3 Grid reference system

The sheets shall be divided into fields to permit easy location of details, additions, revisions, etc., on the drawing. The individual fields shall be referenced using capital letters (I and O shall not be used) in the left and right borders, and numerals in the top and bottom borders. For the size A4, they shall be located only at the top and the right side. The start of the ascending sequences of letters and numerals shall begin at the sheet origin, which can either be determined manually or defined by the CAD tool.

The size of letters and characters shall be 3.5 mm. The length of the fields shall be 50 mm, starting either at the sheet origin or from the centring mark.

NOTE The number of fields depends on sheet size (see Table 6).

The letters and numerals shall be placed in the grid reference border and shall be written in vertical characters in accordance with BS EN ISO 3098-2.

The number of fields shall be extended as required for non-standard sheet sizes.

The grid reference system lines shall be executed with continuous narrow lines.

Table 6 — *Number of fields*

Designation	A0	A1	A2	A3	A4
Long side	24	16	12	8	6
Short side	16	12	8	6	4

6.6 Line types and line widths

6.6.1 Types of line

Lines shall conform to BS EN ISO 128-2:2022.

NOTE Examples of the most common line type and width combinations for engineering drawings are given in Table 7.

Table 7 — *Basic line types*

No.	Representation	Description	Application
01.1		Continuous narrow line	Dimension lines, extension lines, leader lines, hatching, roots of screw threads, termination of interrupted views
01.2		Continuous wide line	Visible edges, visible outlines, crests of threads, limit of thread length, section arrows
02.1		Dashed narrow line	Hidden edges, hidden outlines
02.2		Dashed wide line	Permissible areas of surface treatment
04.1		Long-dashed-dotted narrow line	Centre lines, lines and planes of symmetry, pitch circle of gears or holes, spread of surface-hardened areas
04.2		Long-dashed-dotted wide line	Position of cutting planes, restricted area for surface treatment or application of tolerance requirement
05.1		Long-dashed-double-dotted narrow line	Outline of adjacent parts, extreme positions of moveable parts, initial outline prior to forming, projected tolerance zones and outline of datum target areas

*NOTE 1 Example applications in the use of line types can be found in Figure 13 (**6.3**), Figure 14 (**6.3**), Figure 16 (**6.5.3.1**) and Figure 23 to Figure 26 (**6.12.2** and **6.13**).*

NOTE 2 SOURCE: BS EN ISO 128-2:2022, Table 1 and Table 2, modified – updates in descriptions and new column added.

6.6.2 Line width

In accordance with BS EN ISO 128-2:2022, the width, *d*, of all types of line shall be one of the following, depending on the type and size of drawing:

0,13 mm, 0,18 mm, 0,25 mm, 0,35 mm, 0,5 mm, 0,7 mm, 1 mm, 1,4 mm, 2 mm.

NOTE 1 The widths of extra wide, wide and narrow lines are in the ratio 4:2:1.

NOTE 2 For mechanical engineering drawings, two-line thicknesses (typically 0,7 mm and 0,35 mm or 0,5 mm with 0,25 mm) are sufficient for most purposes.

The line width of any one line shall be constant throughout the whole line (see also **6.10.2**, Table 8).

6.6.3 Colours

In accordance with BS EN ISO 128-2:2022, lines shall be drawn in black or white depending on the colour of the background.

NOTE 1 By default, lines are black on a white background.

Other standardized colours may also be used for technical drawing standardized lines, if necessary, in which case, the meaning of the colours shall be explained.

NOTE 2 The use of colour should take into account issues such as readability, reproducibility, accessibility and other user needs.

6.7 Scales

*COMMENTARY ON **6.7***

Traditionally, recommended scales were used, based on ratios of 2, 5 and 10 (and multiples thereof). The use of CAD and the ability to view drawings electronically at any size has made these largely redundant. There is no requirement to use these formerly-recommended scales.

The scale for a drawing shall be chosen according to the complexity of the object to be depicted and the purpose of the representation. In all cases, the selected scale shall be large enough to permit easy and clear interpretation of the information depicted. The scale and the size of the object, in turn, shall determine the size of the drawing. Where no scaleable views are presented, for example a chart drawing or item list, the term N/A ("Not applicable") shall be indicated in the title block scale field.

Details that are too small for complete dimensioning in the main representation shall be shown adjacent to the main representation in a separate detail view (or section) which is drawn to a larger scale.

3D models produced on CAD systems shall always be produced at 1:1.

NOTE For more information on scales, see BS EN ISO 5455.

6.8 Lines

6.8.1 Lines and terminators

Lines shall conform to BS EN ISO 128-2:2022.

6.8.2 Lines, terminators and origin indicators

Arrows and terminators composed of lines shall conform to BS EN ISO 129-1.

6.9 Number formats

6.9.1 General

Numbers used in dimensions and tolerances shall conform to **6.9.2**. Where a "thousands" separator is required, a space shall be used; this can be used with metric and imperial numbers, and both before and after the decimal marker.

6.9.2 Metric values

The decimal marker shall be a comma.

Numbers smaller than 1 shall have a zero in front of the decimal marker, e.g. 0,25 not ,25.

Non-indicated decimals shall be taken as zeros, i.e. 0,25 is the same as 0,250 000 000 000 000 000 …. and not rounded up to 0,25.

Trailing zeros shall be omitted, e.g. 16 not 16,0.

NOTE ISO standards make no provision for the use of imperial units, so when they are used, it is recommended that the American ASME Y14.5 [1] standard conventions are used. ASME Y14.5 [1] specifies the following for imperial (inch) dimension and tolerance values:

- *the decimal marker is to be a full stop;*
- *values smaller than one are not to have a zero in front of the decimal marker, e.g.: Ø.25 not Ø0.25; and*
- *trailing zeros are to be added where necessary, so that the dimension and tolerance have the same number of decimal places, e.g.: .500±.005.*

6.10 Lettering

6.10.1 General

Lettering shall conform to the following standards, as appropriate.

- BS EN ISO 3098-1, *Technical Product Documentation – Lettering – Part 1: General requirements*.
- BS EN ISO 3098-2, *Technical product documentation – Lettering – Part 2: Latin alphabet, numerals and marks*.
- BS EN ISO 3098-3, *Technical product documentation – Lettering – Part 3: Greek alphabet*.
- BS EN ISO 3098-4, *Technical product documentation – Lettering – Part 4: Diacritical and particular marks for the Latin alphabet*.
- BS EN ISO 3098-5, *Technical product documentation – Lettering – Part 5: CAD lettering of the Latin alphabet, numerals and marks*.
- BS EN ISO 3098-6, *Technical product documentation – Lettering – Part 6: Cyrillic alphabet*.

6.10.2 Lettering size

Lettering heights and line widths shall be appropriate for the sheet size and method of reproduction.

NOTE Two types of lettering are described in BS EN ISO 3098-1, type A and type B. Upper case characters of type B have a height of 10 times the line width. Type A take the same character width as type B but with the height increased by a factor of $\sqrt{2}$. BS EN ISO 129-1 recommends the use of type B lettering. Preferred sizes are summarized in Table 8.

Table 8 — *Preferred line widths and lettering heights for drawing sheet sizes*

Sheet size designation	**A0**	**A1**	**A2**	**A3**	**A4**
Line group (wide line)	0,7	0,7	0,7	0,5	0,5
Narrow line	0,35	0,35	0,35	0,25	0,25
Type B lettering height	3,5	3,5	3,5	2,5	2,5
Type B lower case height	2,5	2,5	2,5	1,75	1,75
Type B alternative height (line width)	5,0 (0,5)	5,0 (0,5)	5,0 (0,5)	3,5 (0,35)	3,5 (0,35)

6.10.3 Notes

Notes of a general nature shall, wherever practicable, be grouped together and not distributed over the drawing. Notes relating to specific details shall appear near the relevant feature, but not so near as to crowd the view.

Where emphasis is required, larger characters shall be used.

NOTE 1 Underlining of notes is not recommended.

NOTE 2 Notes may be displayed in capital letters or in lower case (i.e. "sentence case").

NOTE 3 Flag notes can be used (see BS EN ISO 129-1).

6.11 Projections

6.11.1 General

Projections shall conform to one of the following standards.

- BS EN ISO 5456-2, *Technical drawings – Projection methods – Part 2: Orthographic representations*.
- BS EN ISO 5456-3, *Technical drawings – Projection methods – Part 3: Axonometric representations*.
- BS ISO 5456-4, *Technical drawings – Projection methods – Part 4: Central projection*.
- BS EN ISO 10209, *Technical product documentation – Vocabulary – Terms relating to technical drawings, product definition and related documentation*.

NOTE BS EN ISO 5456-1 contains a survey of the various projection methods.

6.11.2 Conventions for arrangement of views on a TPD

6.11.2.1 General

Three main conventions shall be used for arranging the views on a TPD:

a) labelled views (see **6.11.2.3**);

b) first angle orthographic projection (see **6.11.2.4**); and

c) third angle orthographic projection (see **6.11.2.5**).

NOTE 1 The order of this list is not meant to indicate a preference.

NOTE 2 Other projection methods exist (see BS EN ISO 5456 series).

6.11.2.2 Choice of views

When views (including sections and sectional views) are needed, these shall be selected in accordance with the following principles.

a) The number of views (and sections and sectional views) shall be limited to the minimum necessary but shall be sufficient to fully delineate the object without ambiguity.

b) The need for hidden outlines and edges shall be avoided.

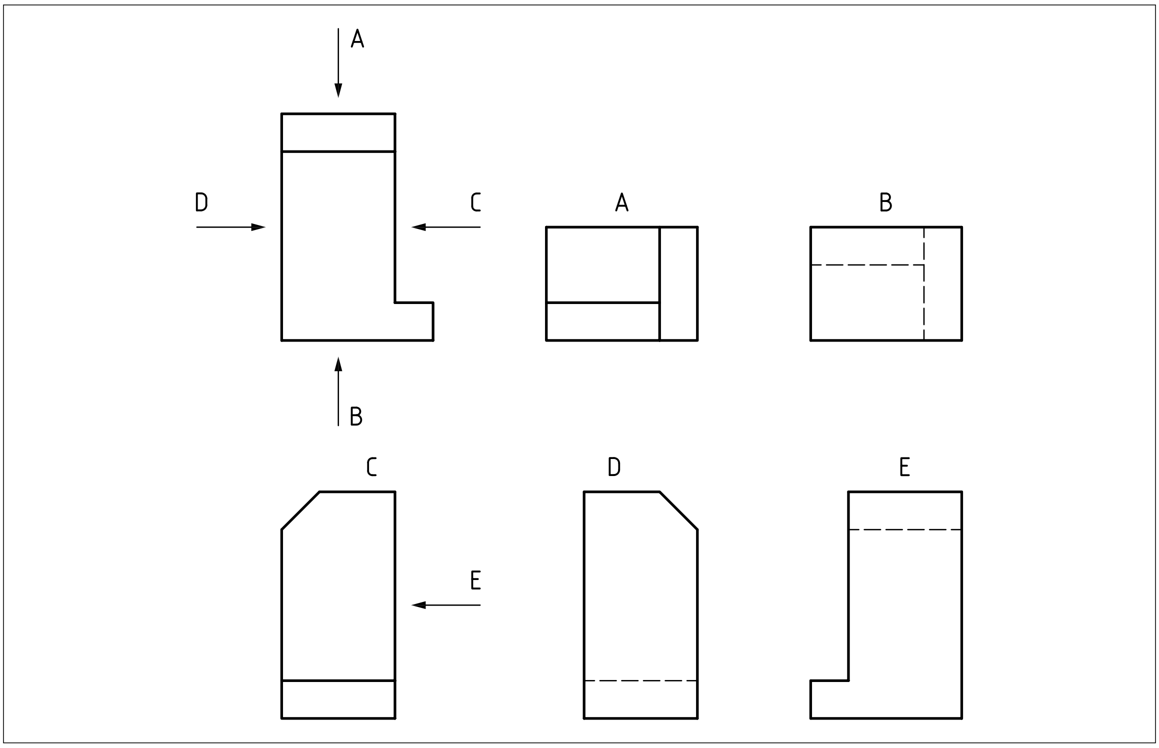
c) The unnecessary repetition of a detail shall be avoided.

6.11.2.3 Labelled view method

As required by BS ISO 128-30, the most informative view of an object shall be used as the front or principal figure, taking account of, for example, its functioning position, position of manufacturing or mounting. Each view, with the exception of the front or principal figure (view, plan, principal figure), shall be given clear identification with a capital letter, repeated near the reference arrow used to indicate the direction of the viewing for the relevant view. Whatever the direction of viewing, the capital letter shall always be positioned in normal relation to the direction of reading and be indicated either above or on the right side of the reference arrow.

The capital letters identifying the referenced views shall be placed immediately above the relevant views (see Figure 18).

Figure 18 — *Labelled view method*



6.11.2.4 First angle projection method

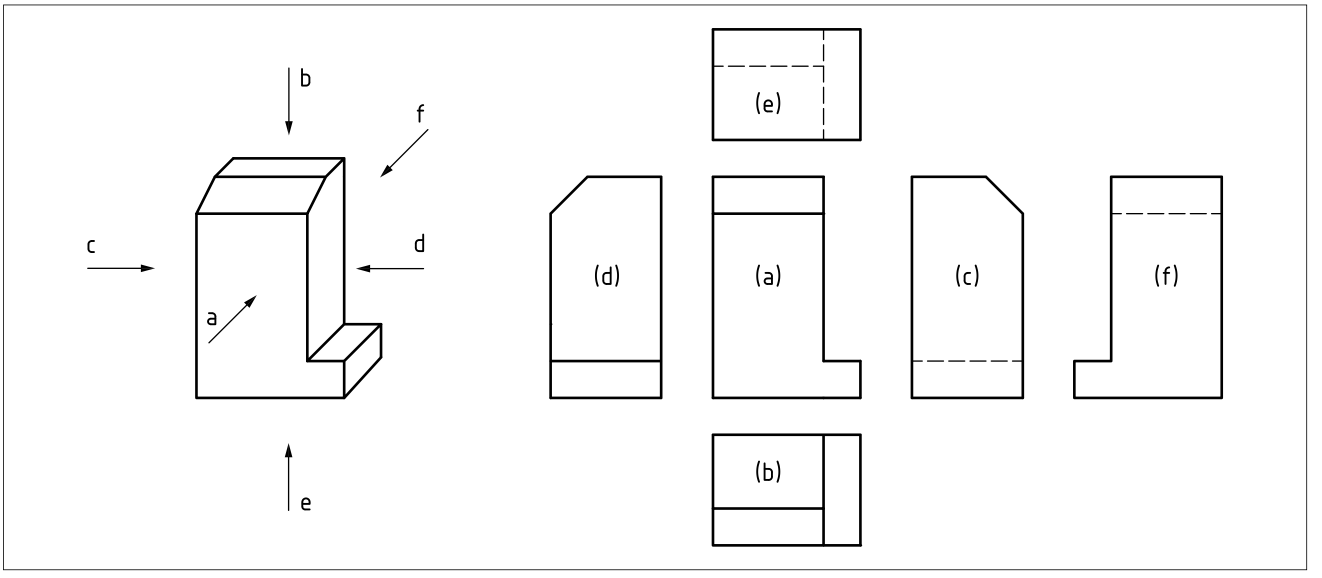
6.11.2.4.1 General

NOTE A more detailed description of the first angle projection method is given in BS ISO 128-30 and BS EN ISO 5456-2.

With reference to the front view (a), the other views shall be arranged as follows (see Figure 19).

- The view from above (b) shall be placed underneath.
- The view from below (e) shall be placed above.
- The view from the left (c) shall be placed on the right.
- The view from the right (d) shall be placed on the left.
- The view from the rear (f) shall be placed on the left or right, as convenient.

Figure 19 — *First angle projection method*

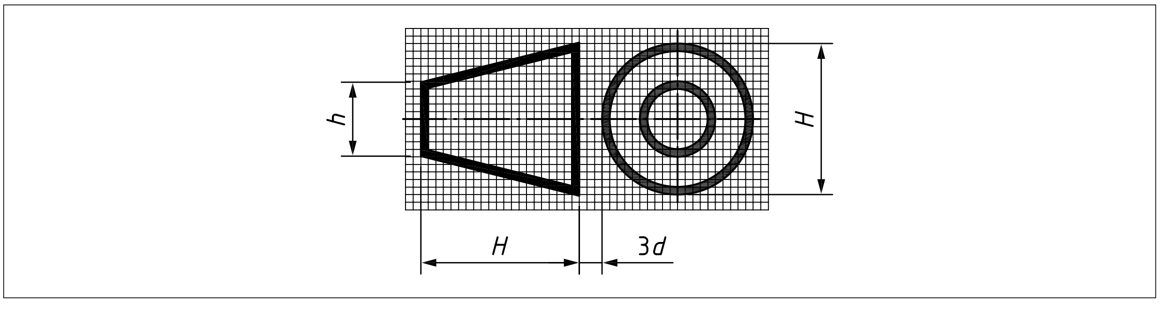


6.11.2.4.2 First angle projection method – Graphical symbol

As specified by BS ISO 128-30, the graphical symbol for the first angle projection method shall be as shown in Figure 20.

The proportions and dimensions of this graphical symbol shall be as specified in BS ISO 128-30.

Figure 20 — *First angle projection method: Graphical symbol*



6.11.2.5 Third angle projection method

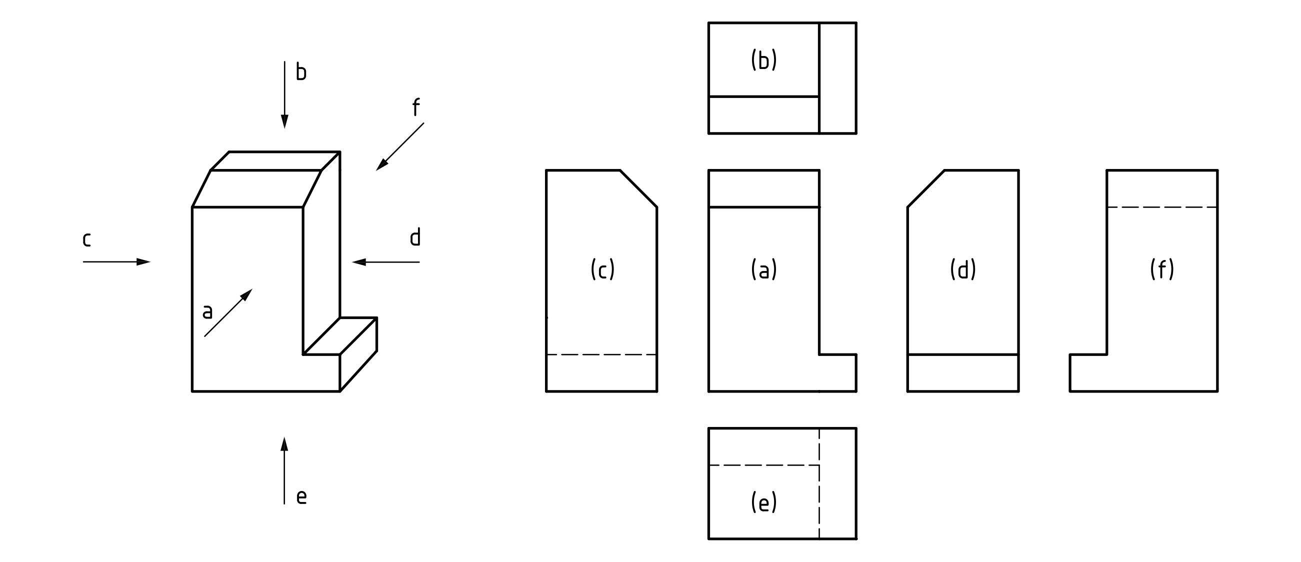
NOTE A more detailed description of the third angle projection method is given in BS ISO 128-30 and BS EN ISO 5456-2.

6.11.2.5.1 General

With reference to the front view (a), the other views shall be arranged as follows (see Figure 21).

- The view from above (b) shall be placed above.
- The view from below (e) shall be placed underneath.
- The view from the left (c) shall be placed on the left.
- The view from the right (d) shall be placed on the right.
- The view from the rear (f) shall be placed on the left or right, as convenient.

Figure 21 — *Third angle projection method*

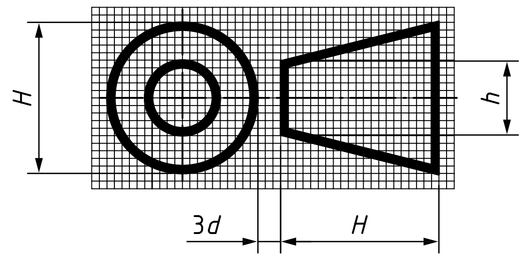


6.11.2.5.2 Third angle projection method – Graphical symbol

As specified by BS ISO 128-30, the graphical symbol for the third angle projection method shall be as shown in Figure 22.

The proportions and dimensions of this graphical symbol shall be as specified in BS ISO 128-30.

Figure 22 — *Third angle projection method: Graphical symbol*



6.12 Views

6.12.1 General

Views shall conform to the following standards.

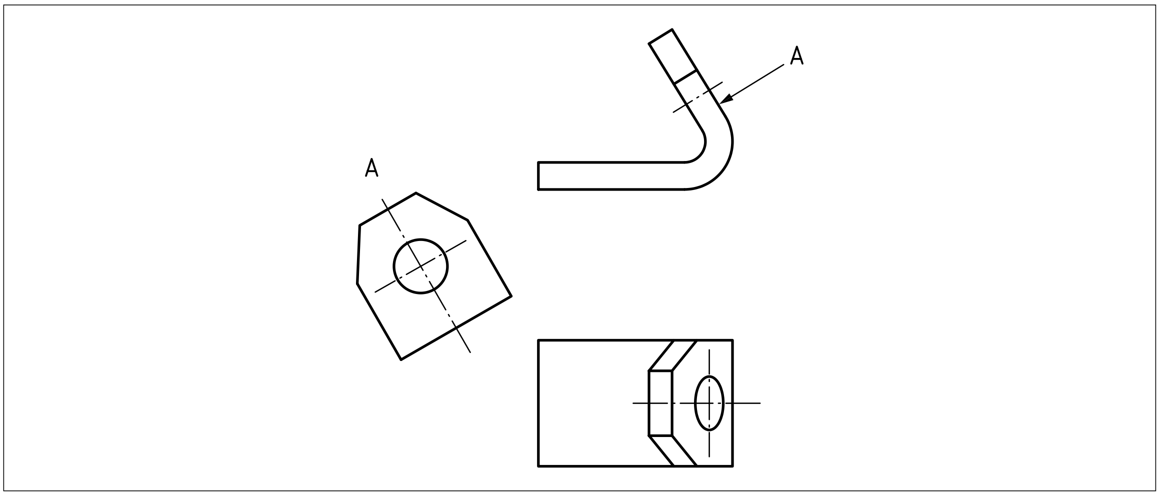
- BS ISO 128-30, *Technical drawings – General principles of presentation – Part 30: Basic conventions for views.*
- BS ISO 128-34, *Technical drawings – General principles of presentation – Part 34: Views on mechanical engineering drawings.*

6.12.2 Auxiliary views

Where true representation of features is necessary, but cannot be achieved on the orthographic views, the features shall be shown in projected auxiliary views.

NOTE An example is shown in Figure 23.

Figure 23 — *Auxiliary view showing true shape of inclined surface*



6.13 Sections

Sections shall conform to the following standards.

- BS ISO 128-40, *Technical drawings – General principles of presentation – Part 40: Basic conventions for cuts and sections*.
- BS ISO 128-44, *Technical drawings – General principles of presentation – Part 44: Sections on mechanical engineering drawings*.
- BS ISO 128-50, *Technical drawings – General principles of presentation – Part 50: Basic conventions for representing areas on cuts and sections*.

NOTE 1 Some figures in BS ISO 128-44 and BS ISO 128-50 contain presentational defects (e.g. line types, line thickness, terminators and letter heights). It is stressed that the text of these standards is technically correct and users should, therefore, regard the figures as illustrations only.

As specified by BS ISO 128-44, ribs, fasteners, shafts, spokes of wheels, etc. shall not be cut in longitudinal sectional views and shall therefore not be represented as sections.

NOTE 2 In practice, it is not normally possible to avoid cutting ribs and spokes of wheels.

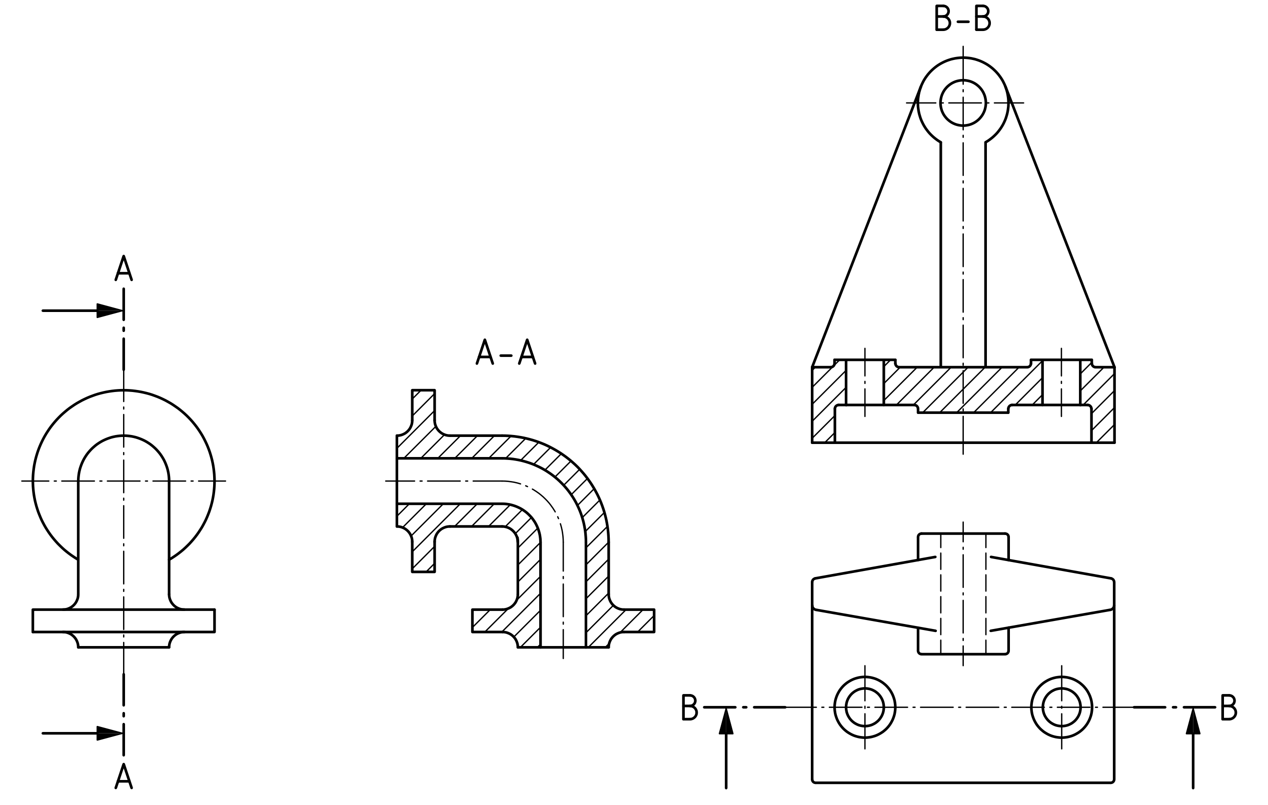
NOTE 3 Like views, sections might be shown in a position other than that indicated by the arrows for the direction of their viewing.

NOTE 4 A section in one plane is shown in Figure 24.

NOTE 5 A section in two parallel planes is shown in Figure 25.

NOTE 6 A section in three contiguous planes is shown in Figure 26.

Figure 24 — *Examples of section view in one plane*



a) Section view in one plane

b) Section view in one plane

Figure 25 — *Section view in two parallel planes*

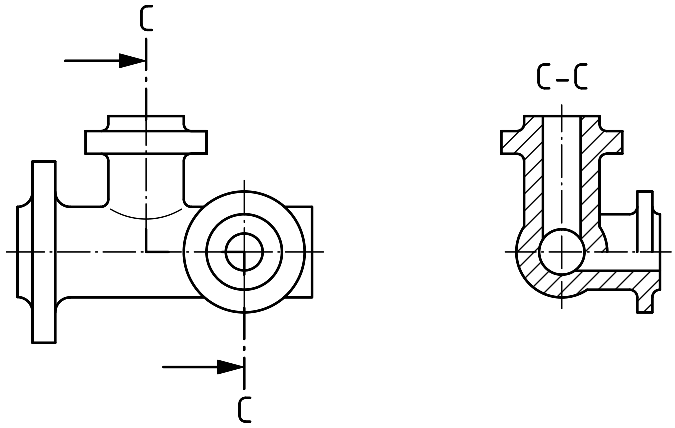
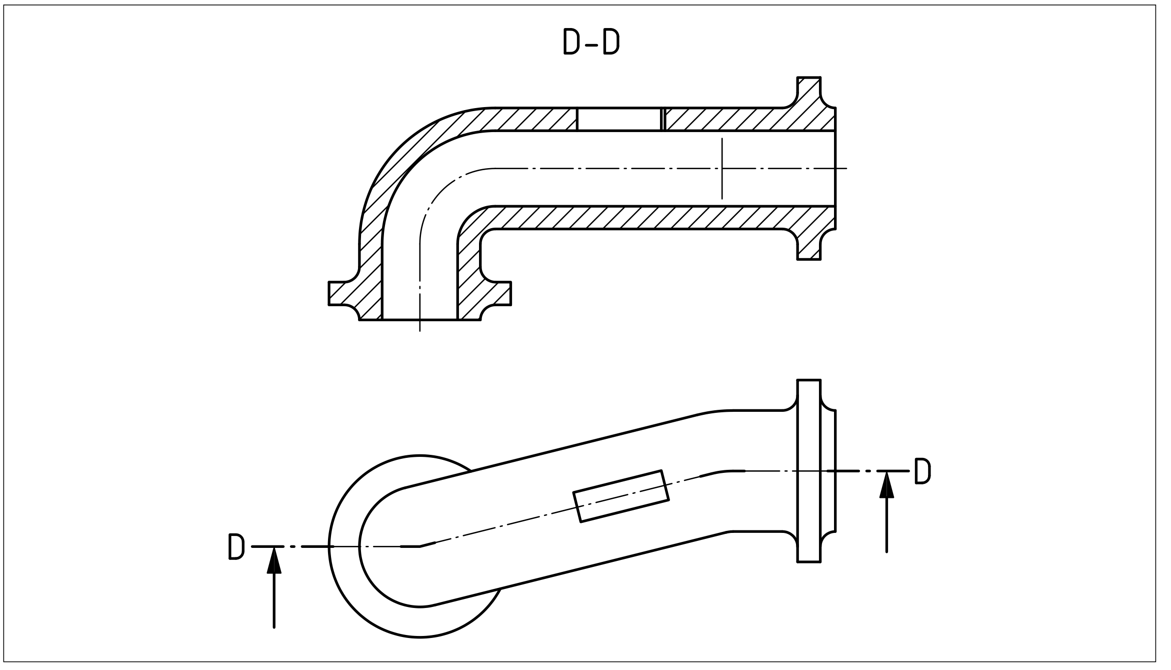


Figure 26 — *Section view in three contiguous planes showing true shape of inclined surface*



6.14 Representation of components

6.14.1 General

Conventions used for the representation of components shall conform to the following standards, as appropriate.

- BS EN ISO 2162-1, *Technical product documentation – Springs – Part 1: Simplified representation*.
- BS EN ISO 2162-2, *Technical product documentation – Springs – Part 2: Presentation of data for cylindrical helical compression springs*.
- BS EN ISO 26909, *Springs – Vocabulary*.
- BS EN ISO 2203, *Technical drawings – Conventional representation of gears*.
- BS ISO 1219-1, *Fluid power systems and components – Graphical symbols and circuit diagrams – Part 1: Graphical symbols for conventional use and data processing applications*.
- BS 3238-1, *Graphical symbols for components of servo mechanisms – Part 1: Transductors and magnetic amplifiers*.
- BS 3238-2, *Graphical symbols for components of servo mechanisms – Part 2: General servo-mechanisms*.
- BS EN ISO 13715, *Technical drawings – Edges of unidentified shape – Vocabulary and indications*.
- BS EN ISO 2553, *Welded and allied processes – Symbolic representation on drawings – Welded joints*.
- BS EN ISO 4063, *Welding, brazing, soldering and cutting – Nomenclature of processes and reference numbers*.
- BS EN ISO 5261, *Technical drawings – Simplified representation of bars and profile sections*.
- BS EN ISO 5845-1, *Technical drawings – Simplified representation of the assembly of parts with fasteners – Part 1: General principles*.
- BS EN ISO 6410-1, *Technical drawings – Screw threads and threaded parts – Part 1: General conventions*.

- BS EN ISO 6410-2, *Technical drawings – Screw threads and threaded parts – Part 2: Screw thread inserts*.
- BS EN ISO 6411, *Technical drawings – Simplified representation of centre holes*.
- BS EN ISO 6412-1, *Technical product documentation – Simplified representation of pipelines – Part 1: General rules and orthogonal representation*.
- BS EN ISO 6412-2, *Technical product documentation – Simplified representation of pipelines – Part 2: Isometric projection*.
- BS EN ISO 6412-3, *Technical product documentation – Simplified representation of pipelines – Part 3: Terminal features of ventilation and drainage systems*.
- BS EN ISO 6413, *Technical product documentation – Representation of splines and serrations*.
- BS EN ISO 8826-1, *Technical drawings – Roller bearings – Part 1: General simplified representation*.
- BS EN ISO 8826-2, *Technical drawings – Roller bearings – Part 2: Detailed simplified representation*.
- BS EN ISO 9222-1, *Technical drawings – Seals for dynamic application – Part 1: General simplified representation*.
- BS EN ISO 9222-2, *Technical drawings – Seals for dynamic application – Part 2: Detailed simplified representation*.
- BS EN ISO 15785, *Technical drawings – Symbolic presentation and indication of adhesive, fold and pressed joints*.

NOTE 1 The BS ISO 128 series of standards covers the general subject of component representation.

NOTE 2 The BS EN ISO 6410-3 (withdrawn) gives the simplified representation of the drawings.

6.14.2 Representation of moulded, cast and forged components

Dimensional tolerancing for metal and metal alloy castings shall conform to the following standards, as appropriate.

- BS EN ISO 8062-1, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 1: Vocabulary*.
- PD CEN ISO/TS 8062-2, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 2: Rules*.
- BS EN ISO 8062-3, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 3: General dimensional and geometrical tolerances and machine allowances for castings using ± tolerances for indicated dimensions*.

6.14.3 Representation of moulded components

Indications for parting lines, draft angles and other elements of moulded components shall conform to the following standard.

- BS EN ISO 10135, *Geometrical product specifications (GPS) – Drawing indications for moulded parts in technical product documentation (TPD)*.

7 Dimensioning

7.1 General

Dimensioning and tolerancing shall conform to the following standards, as appropriate.

- BS EN ISO 129-1, *Technical product documentation (TPD) – Presentation of dimensions and tolerances – Part 1: General principles*.
- BS EN ISO 14405-2, *Geometrical product specifications (GPS) – Dimensional tolerancing – Part 2: Dimensions other than linear sizes*.
- BS EN ISO 1119, *Geometrical product specifications (GPS) – Series of conical tapers and taper angles*.
- BS EN ISO 1660, *Geometrical product specifications (GPS) – Geometrical tolerancing – Profile tolerancing*.
- BS 1916-1, *Limits and fits for engineering – Part 1: Guide to limits and tolerances*.
- BS 1916-2, *Limits and fits for engineering – Part 2: Guide to the selection of fits in BS 1916-1*.
- BS 1916-3, *Limits and fits for engineering – Part 3: Guide to tolerances, limits and fits for large diameters*.
- BS EN ISO 3040, *Geometrical product specifications (GPS) – Dimensioning and tolerancing – Cones*.
- BS 3734-1, *Rubber – Tolerances for products – Part 1: Dimensional tolerances*.
- BS 4500, *Limits and fits – Guidance for system of cone (taper) fits and tolerances for cones from C = 1:3 to 1:500, lengths from 6 mm to 630 mm and diameters up to 500 mm*.
- BS EN ISO 5458, *Geometrical product specifications (GPS) – Geometrical tolerancing – Pattern and combined geometrical specification*.
- BS EN ISO 6410-1, *Technical drawings – Screw threads and threaded parts – Part 1: General conventions*.
- BS EN ISO 7083, *Technical product documentation – Symbols used in technical product documentation – Proportions and dimensions*.
- BS EN ISO 8015, *Geometrical product specifications (GPS) – Fundamentals – Concepts, principles and rules*.
- BS EN ISO 13920, *Welding – General tolerances for welded constructions – Dimensions for lengths and angles, shape and position*.
- BS EN ISO 8062-1, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 1: Vocabulary*.
- PD CEN ISO/TS 8062-2, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 2: Rules*.
- BS EN ISO 8062-3, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 3: General dimensional and geometrical tolerances and machining allowances for castings using ± tolerances for indicated dimensions*.
- BS EN ISO 286-1, *Geometrical product specifications (GPS) – ISO code system for tolerances on linear sizes – Part 1: Basis of tolerances, deviations and fits*.
- BS EN ISO 286-2, *Geometrical product specifications (GPS) – ISO code system for tolerances on linear sizes – Part 2: Tables of standard tolerance classes and limit deviations for holes and shafts*.

7.2 Dimensioning methods

7.2.1 Presentation rules

7.2.1.1 Dimensioning shall conform to BS EN ISO 129-1.

NOTE The dimensioning of shipbuilding drawings is specified in BS ISO 129-4.

7.2.1.2 Only the dimensions which are necessary to unambiguously define the nominal geometry (nominal model) shall be presented. Dimensions shall be used with dimensional tolerances or geometrical specifications as required. Each dimension used with a tolerance shall be presented only once by using a dimension line, a dimension value preceded, and if necessary, by a property indicator.

NOTE 1 Tolerances can be indicated directly on the specification, attached to the dimension or feature concerned, or indirectly through the use of general tolerances.

NOTE 2 When there is a need to repeat the presentation of a dimension, auxiliary dimensions may be used.

NOTE 3 The dimensions shown should be for the purposes of describing the function, production or verification of the product. Auxiliary dimensions may also be included, but do not form part of the specification.

NOTE 4 Dimensions alone are not sufficient to fully define the geometry of a product. Dimensions are typically used with other specification elements such as dimensional tolerances, geometrical specifications and surface texture indications, as necessary, to reduce the ambiguity in a specification.

The specification of geometrical specifications and surface texture requirements shall be in accordance with Clause **9** and Clause **15**.

7.2.1.3 Unless otherwise specified, dimensions shall be indicated for the finished state of the dimensioned feature.

NOTE However, it might be necessary to give additional dimensions at intermediate stages of production if they are shown on the same drawing (e.g. the size of a feature prior to carburizing and finishing).

7.2.1.4 All dimensions, graphical symbols and annotations shall be indicated such that they are legible from the bottom or right-hand side (main reading directions) of the drawing. Dimensioning and annotations of 3D models shall be in accordance with BS ISO 16792.

7.2.1.5 Lettering on drawings shall be in accordance with BS EN ISO 3098 (all parts).

7.2.1.6 Dimension and tolerance indications in a TPS shall be shown in the same character height.

7.2.1.7 All dimensions shall be toleranced, either via a general tolerance or by direct indication of tolerance or limit indications, except for the following cases:

- min. (see **7.4.3**);
- max. (see **7.4.3**);
- auxiliary dimension (see **7.2.8**); and
- theoretically exact dimension (TED) (see **9.9**).

7.2.2 Presentation of decimals

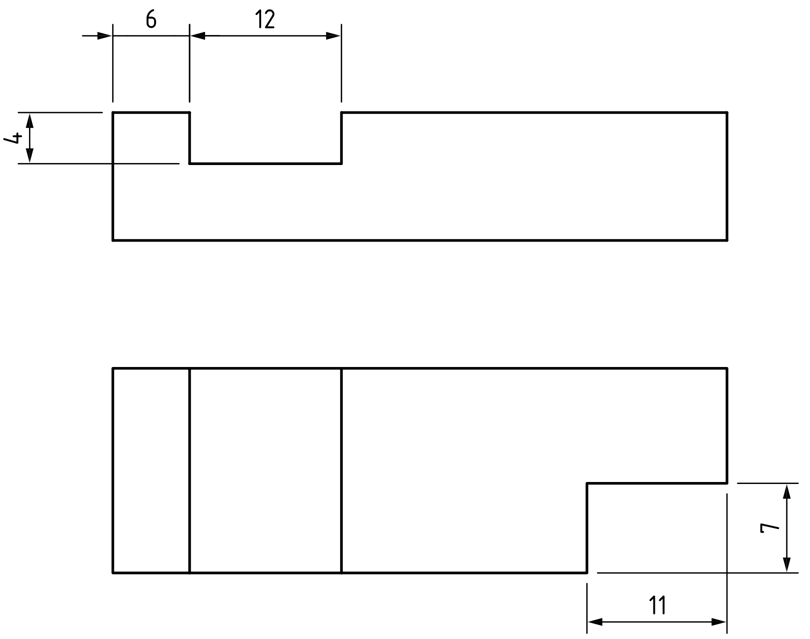
Dimension and tolerance values indicated in decimal notation shall use a comma as the decimal marker.

NOTE Each group of three digits, counting from the decimal marker to the left and to the right, can be separated from other digits by a space as a thousands separator (e.g. 12 345,067 8) if required. A comma or a point cannot be used as a thousands separator. A semi-colon should be used to separate items in a list rather than a comma.

7.2.3 Positioning of dimensions

Dimensions shall be placed in the view or section which shows the relevant feature(s) most clearly (see Figure 27).

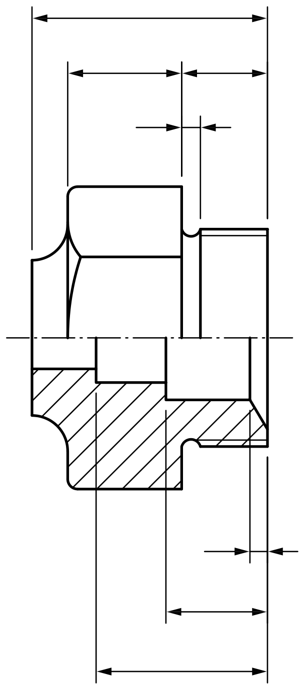
Figure 27 — *Positioning of dimensions*



NOTE The dimensions conform to BS EN ISO 129-1, but do not fully conform to ISO GPS rules, see BS EN ISO 14405-2.

Dimensions for internal features and dimensions for external features shall, where possible, be arranged and indicated in separate groups of dimensions to improve readability (see Figure 28).

Figure 28 — *Arrangement and indication of dimensions internal and external features*



Whenever possible, dimensions shall not be placed within the contour of the depicted item.

7.2.4 Units

The units for dimensions and tolerances shall be specified on the TPS. Dimensional tolerances shall be in the same units as the dimensions to which they apply.

NOTE For linear units, the predominant unit on a drawing may be specified on the TPD (typically in the title block) or in an associated document and the unit omitted from the individual dimensions. Any dimensions expressed in a different unit of measure shall indicate that unit of measure.

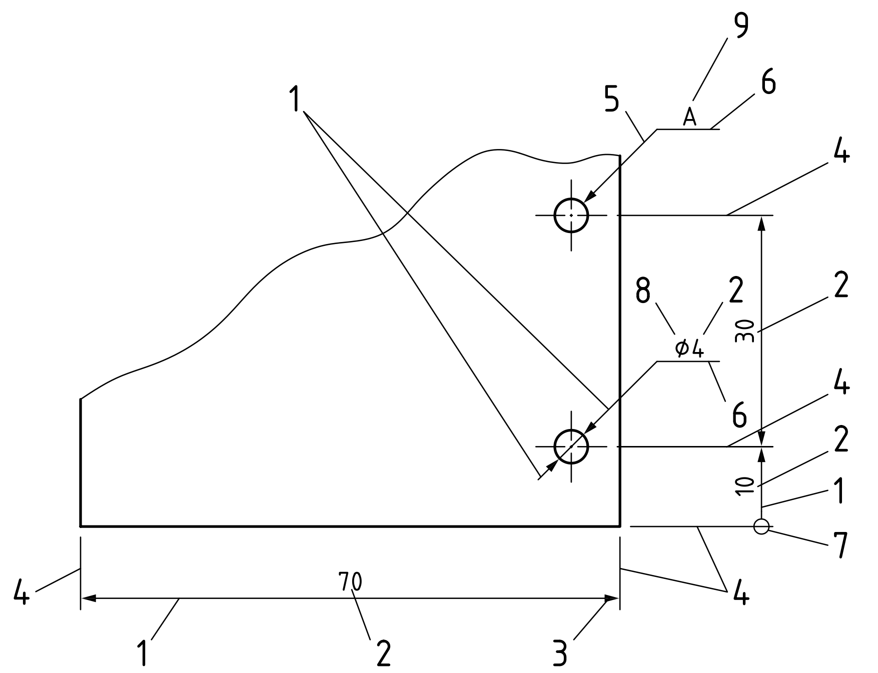
For angular dimensions, the units shall always be specified with the individual dimensions.

7.2.5 Elements of dimensioning: usage

7.2.5.1 General

The various elements of dimensioning shall be indicated as illustrated in Figure 29.

Figure 29 — *Elements of dimensioning*



Key

1 Dimension line
2 Nominal dimensional value
3 Terminator (in this case an arrowhead)
4 Extension line
5 Leader line
6 Reference line
7 Origin symbol
8 Property indicator
9 Reference letter

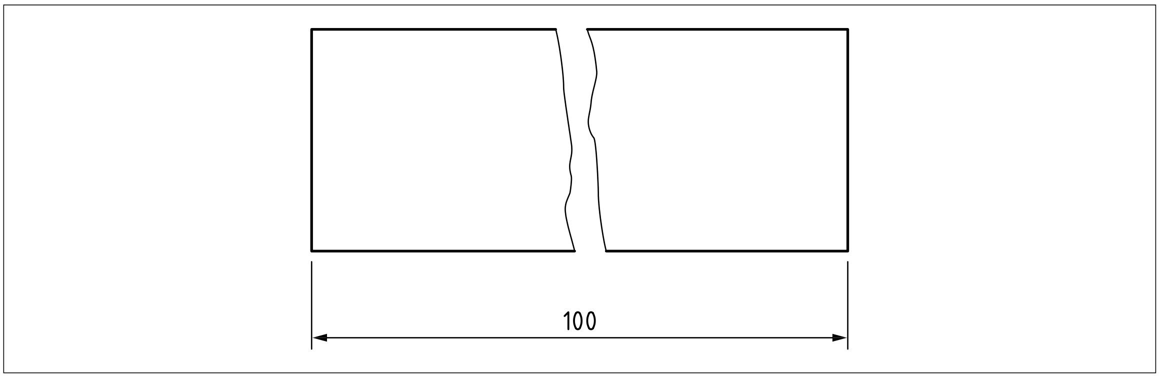
NOTE It is common practice to dimension holes with a single arrow dimension line in accordance with BS EN ISO 129-1:2019+A1, ***5.3****.*

7.2.5.2 Dimension line

Dimension lines shall be indicated as continuous narrow lines in accordance with BS EN ISO 128-2:2022.

Where the feature is shown broken, the corresponding dimension line shall be shown unbroken (see Figure 30).

Figure 30 — *Dimension line of feature that is broken*



Intersection of dimension lines with any other line shall be avoided where possible, but where intersection is unavoidable they shall be shown unbroken.

7.2.5.3 Extension lines

Extension lines shall be drawn as continuous narrow lines in accordance with BS EN ISO 128-2:2022.

Extension lines shall not be drawn between views and shall not be drawn parallel to the direction of hatching.

Extension lines shall extend beyond dimension line by an amount approximately eight times the line width.

NOTE 1 A gap of a similar size may be left between the extension line and the outline of the feature.

Extension lines shall be drawn perpendicular to the dimension line (see Figure 31, Figure 33 and Figure 34), except when:

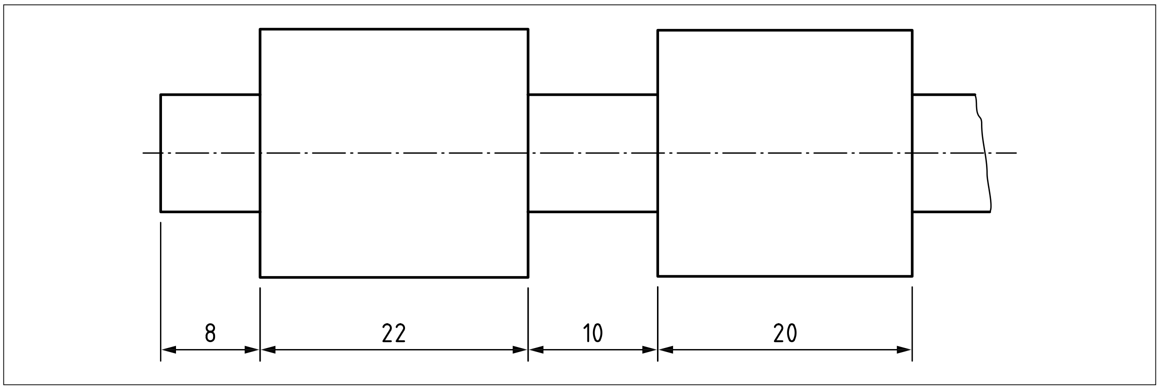
a) they are arranged obliquely (see Figure 32); or

b) they are staggered for clarity on the drawing view.

For circular features the extension line shall be drawn as a continuation of the feature shape (see Figure 36).

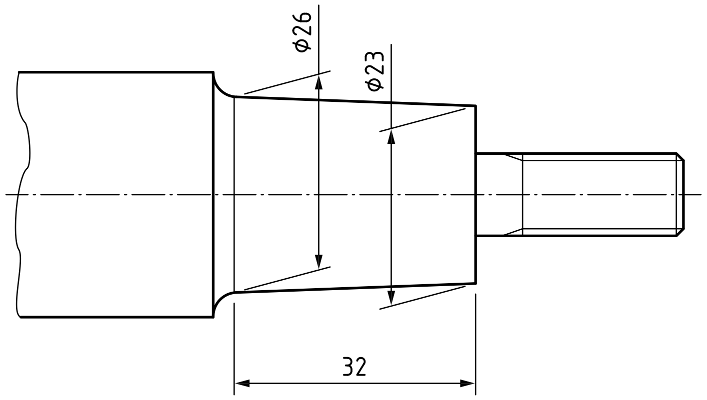
NOTE 2 A gap should exist between the feature and the beginning of the extension line.

Figure 31 — *Extension lines*



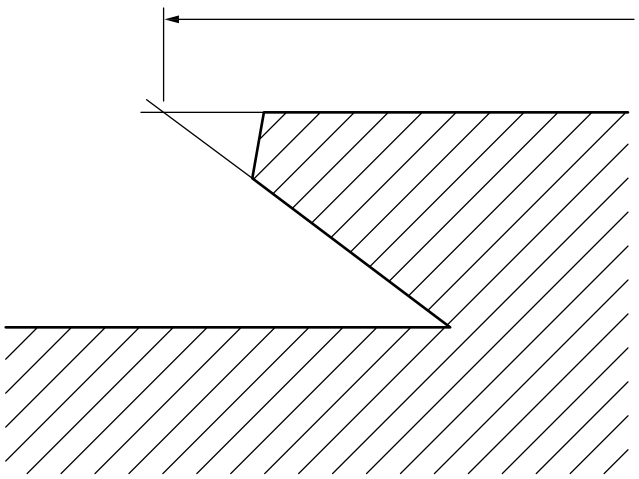
The extension lines may be drawn oblique to the feature but shall be parallel to each other (see Figure 32).

Figure 32 — *Oblique extension lines*



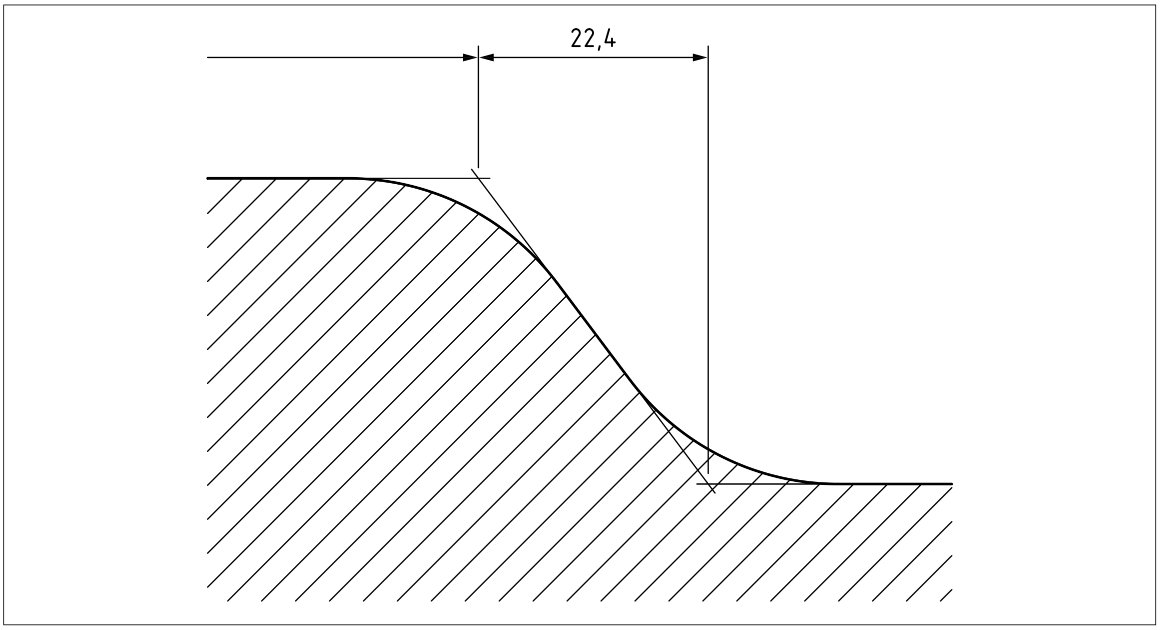
Intersecting projected contours of outlines shall extend approximately eight times the line width beyond the point of intersection (see Figure 33).

Figure 33 — *Intersection of projected contours of outlines*



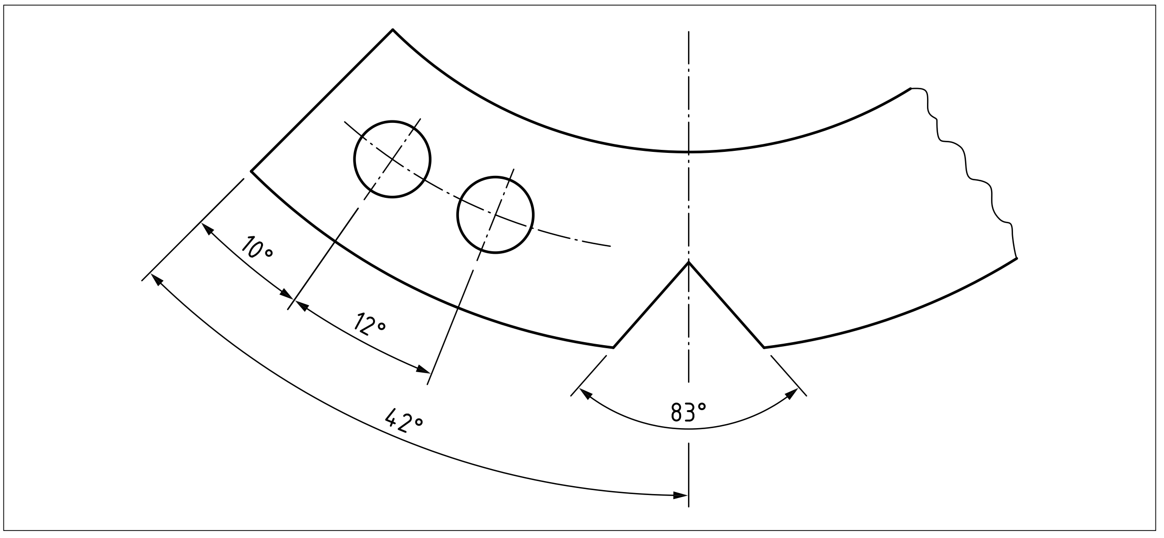
In the case of projected contours of transitions and similar features, the extension lines shall apply at the point of intersection of the projection lines (see Figure 34).

Figure 34 — *Intersection of projected contours of transitions and similar features*



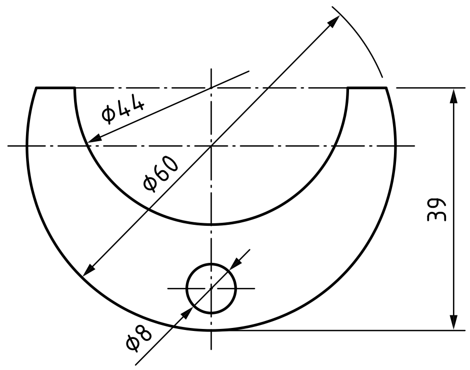
In the case of angular dimensions, the extension lines shall be the extensions of the angle legs (see Figure 35).

Figure 35 — *Extension lines of angular dimensions*



For circular features the extension line shall be drawn as a continuation of the feature shape (see Figure 36).

Figure 36 — *Extension line for circular features*



NOTE SOURCE: BS EN ISO 129-1:2019+A1, Figure 43.

NOTE 3 A gap should exist between the feature and the beginning of the extension line.

7.2.5.4 Leader line

Leader lines shall be drawn in accordance with BS EN ISO 128-2:2022.

7.2.6 Dimensional values

7.2.6.1 Indication

Letters and numbers used for dimensions and dimensional tolerances shall be in accordance with **6.10**.

7.2.6.2 Positions of dimensional values

Dimensional values shall be placed parallel to their dimension line and near the middle of and slightly above that line (see Figure 37 and Figure 38).

NOTE The position of the value can be adapted if needed to achieve readability.

Dimensional values shall be placed in such a way that they are not crossed or separated by any line.

Figure 37 — *Position of dimensional values – 1*

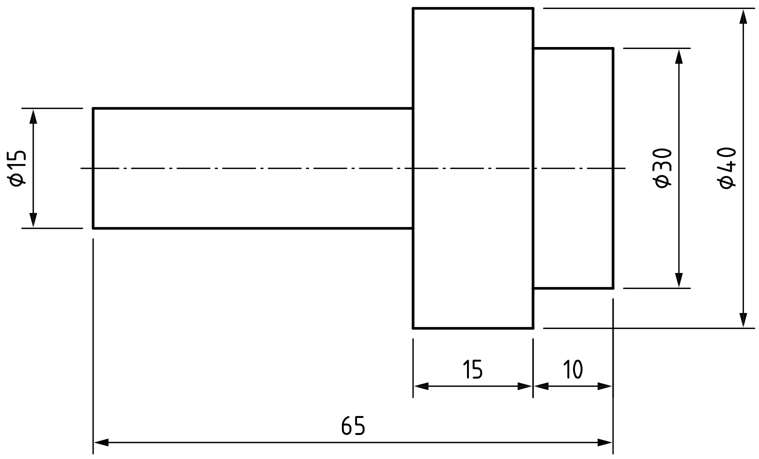
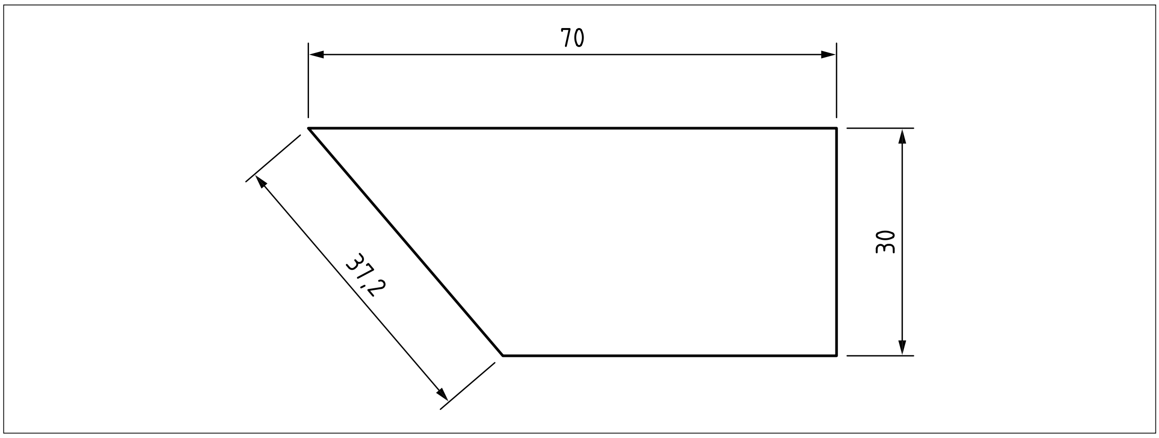


Figure 38 — *Position of dimensional values – 2*



Values of linear dimensions on oblique dimension lines shall be oriented as shown in Figure 39. The values shall be indicated such that they are legible from the bottom or right-hand side of the drawing.

Values of angular dimensions shall be oriented as shown in Figure 40. Angular dimensions shall be placed on top of the dimension line and follow the same rule as linear dimensions (see Figure 39).

Figure 39 — *Orientation of linear dimensions*

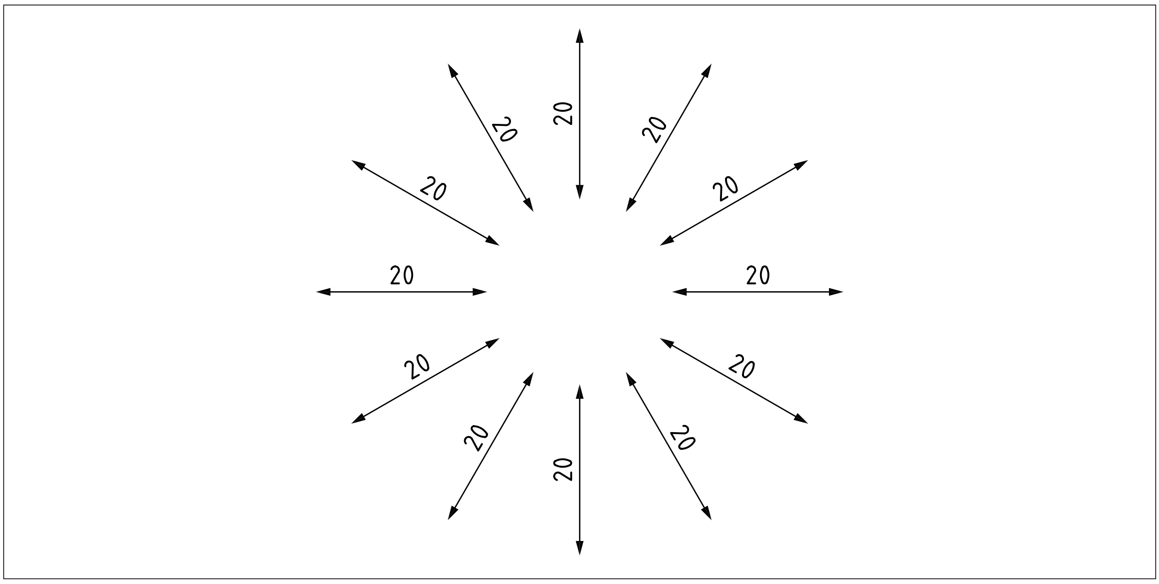
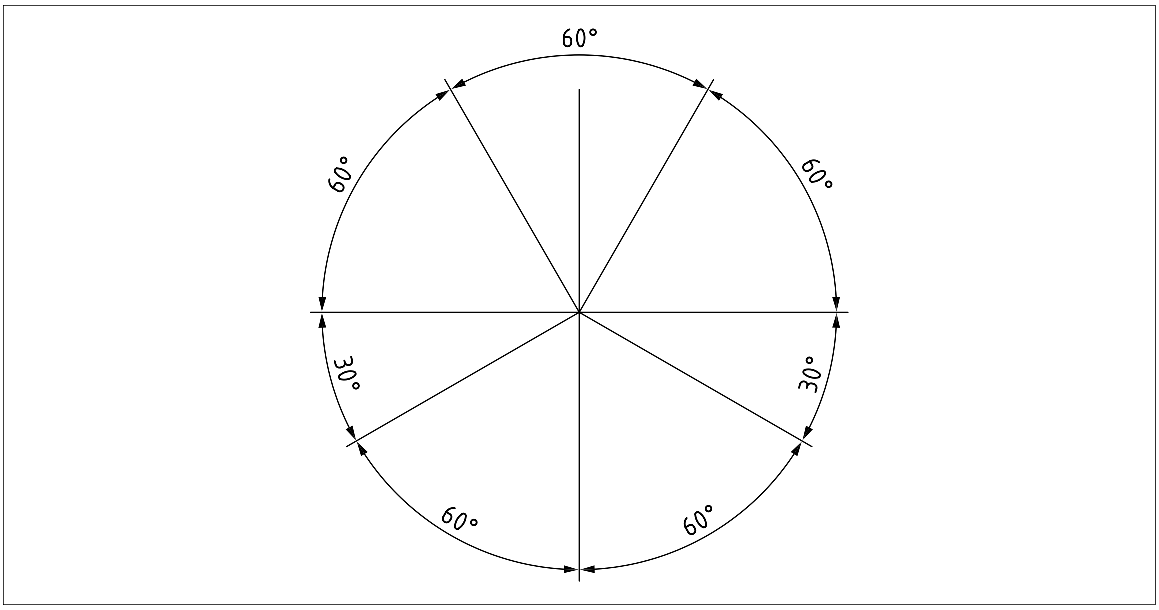


Figure 40 — *Orientation of angular dimensions*



7.2.7 Indications of special dimensions

The symbols shown in Table 9 shall be used with dimensions to identify the feature characteristics. The symbols shall directly precede the dimensional value without space (see Figure 41 to Figure 47).

Table 9 — *Special dimensions*

Property indicator symbol	Description	Associated property	Example of presentation
⌀	Diameter	Cylindrical feature or circular feature described by its diameter	Figure 46
R	Radius	Cylindrical feature or circular feature described by its radius	Figure 41
□	Square	Square feature with four equal angles and four equal sides described by its side dimension	Figure 42
S⌀	Spherical diameter	Spherical feature described by its diameter	Figure 43
SR	Spherical radius	Spherical feature described by its radius	Figure 44
⌒	Arc length	Curvilinear dimension of non-flat feature (e.g. arc length)	Figure 150
t=	Thickness (of thin objects)	Two offset features defined by its thickness	Figure 45
↧	Depth	Depth of a hole or internal feature	Figure 46b) and Figure 46d)
⌴	Cylindrical counterbore	Cylindrical hole with a flat bottom described by its diameter and depth	Figure 46b) and Figure 46d)
⌵	Countersink	Circular chamfer described by a diameter and angle	Figure 47b)
	Developed length	Length of feature prior to bending or forming	—
◄—►	Between	Indication of the extent of a restricted area, used in conjunction with reference letters	Figure 149 and Figure 150

NOTE SOURCE: BS EN ISO 129-1:2019+A1, Table 1 and Table 2.

Figure 41 — *Indications of special dimensions: Radius*

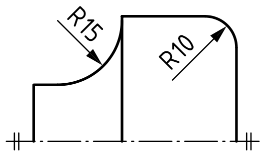


Figure 42 — *Indications of special dimensions: Square*

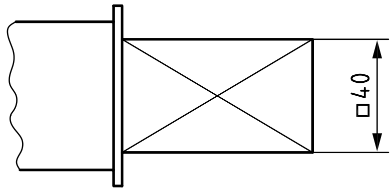


Figure 43 — *Indications of special dimensions: Spherical diameter*

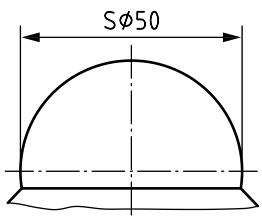


Figure 44 — *Indications of special dimensions: Spherical radius*

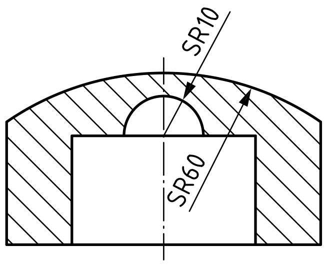
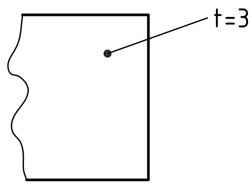
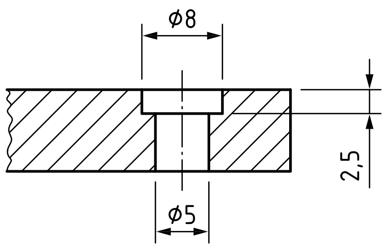


Figure 45 — *Indications of special dimensions: Thickness*

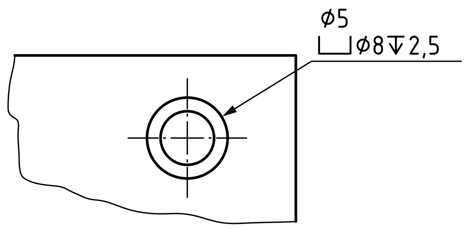


When counterbore is indicated, the diameter, and depth if needed, of the counterbore shall be specified below the dimension value of the hole (see Figure 46). When countersink is indicated, the size and inclusive angle of the countersink shall be specified below the dimensional value of the hole (see Figure 47).

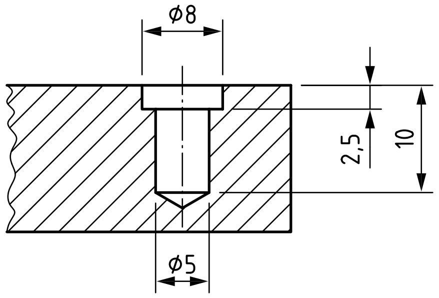
Figure 46 — *Diameter and depth of holes and counterbore*



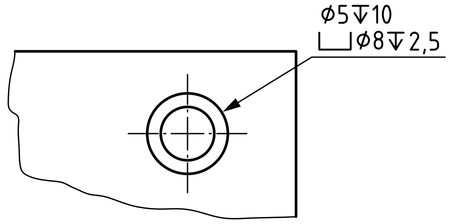
a) Possible indication



b) Equivalent alternative indication

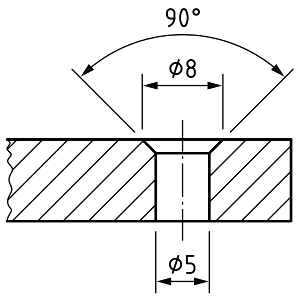


c) Possible indication

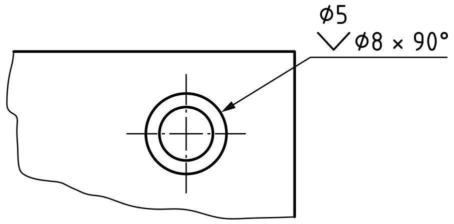


d) Equivalent alternative indication

Figure 47 — *Size and angle of countersink*



a) Possible indication



b) Equivalent alternative indication

NOTE For more details on countersink, see also ***7.2.11****.*

7.2.8 Auxiliary dimensions

Auxiliary dimensions shall be indicated within parentheses and do not constitute an integral part of the specification or requirement (see Figure 48 and Figure 49). Auxiliary dimensions have no tolerances and shall not be used for manufacturing or verification purposes.

NOTE Auxiliary dimensions in drawings are for information only and not for manufacturing and verification purposes. Auxiliary dimensions should have limited use.

7.2.9 Repeated and equally-spaced features

7.2.9.1 Repeated features

NOTE 1 Where clarity is not impaired, features having the same dimensional value can be indicated by that value preceded by the number of features and the symbol "×". The multiplication symbol is "×" is not the same as a lower case "x", but the lower case "x" is normally regarded as an acceptable alternative.

The number of features shall directly precede the symbol "×" without a space and the dimensional value shall be preceded by a space after the symbol "×" (see Figure 48 and Figure 49).

NOTE 2 Figure 48 to Figure 51 conform to BS EN ISO 129-1:2019, but they do not fully conform to ISO GPS rules; see BS EN ISO 14405-2.

Figure 48 — *Indication of features having the same dimensional value – 1*

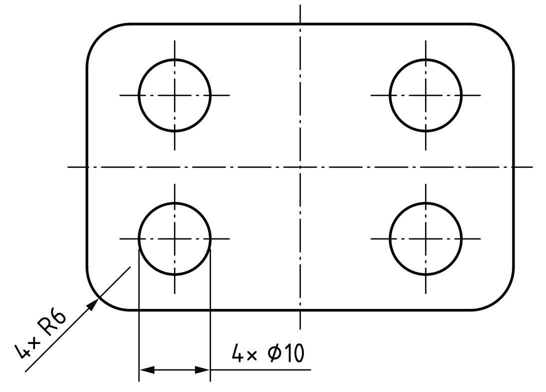
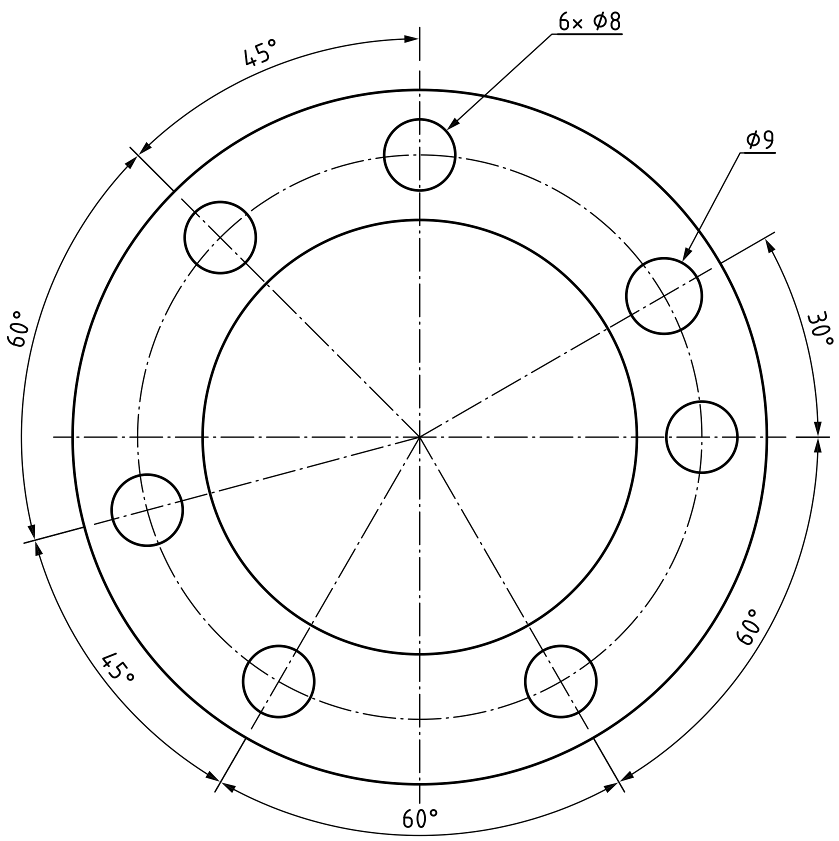


Figure 49 — *Indication of features having the same dimensional value – 2*



7.2.9.2 Equally-spaced features

Features having equal spacing shall be indicated in one of the two following ways:

a) a dimension for a single spacing, with the dimension value preceded by the number of spacings and the multiplication symbol: the number of spacings shall directly precede the symbol "×" without a space and the dimensional value shall be preceded by a space, e.g. 17× 18; or

b) a single dimension to indicate the overall span of repeated features followed by an "=" sign, with the number of spacing followed by the symbol "×" followed by the dimension value for a single spacing preceded by a space: the dimension value for a single spacing shall then be shown in parentheses, e.g. 306= 17× (18).

The sum of linear or angular spacings shall be shown as an auxiliary value, if it is included (see Figure 50 and Figure 51). If there is any risk of confusion between the length of the space and the number of spacings, one space shall additionally be dimensioned as an auxiliary dimension (see Figure 50).

Figure 50 — *Features with equal linear spacing*

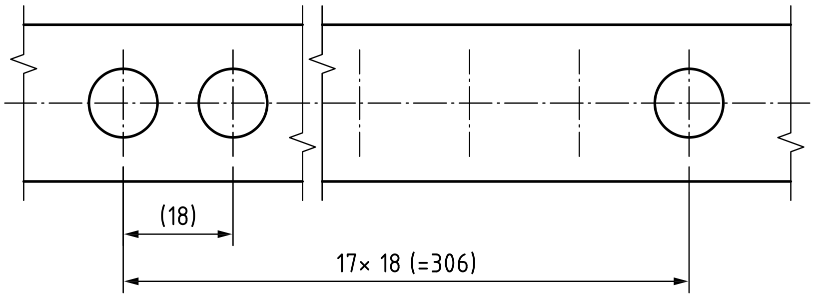
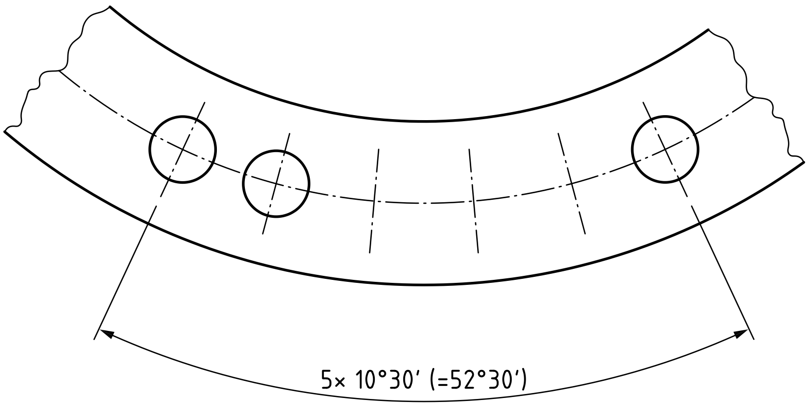


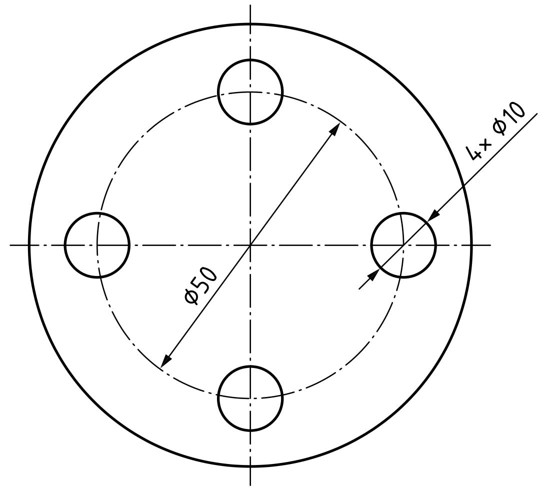
Figure 51 — *Features with equal angular spacing*



NOTE The angles of the spacings can be omitted where spacings are self-evident and the indication does not lead to confusion (see Figure 52).

NOTE Where features are equally-spaced on a pitch diameter, this may be implied rather than stated directly (see Figure 52).

Figure 52 — *Features equally spaced on a pitch diameter*

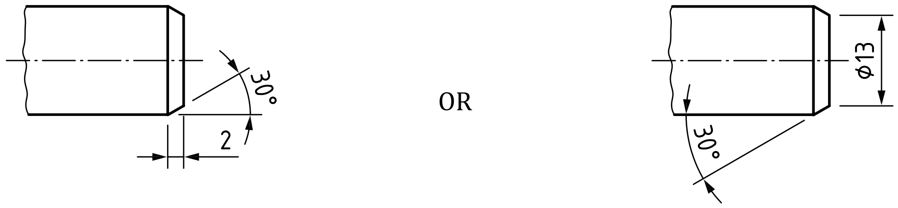


7.2.10 Chamfers

7.2.10.1 External chamfer

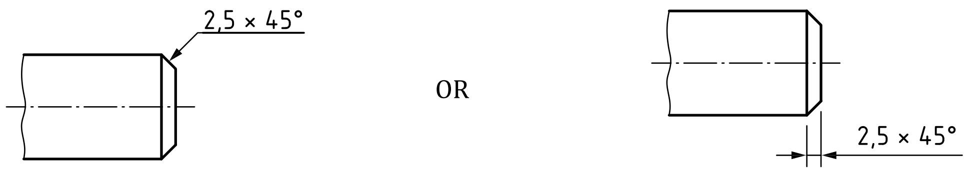
An external chamfer shall be dimensioned conventionally if the angle of the chamfer is not equal to 45° (see Figure 53).

Figure 53 — *Dimension of external chamfer not equal to 45°*



NOTE If the chamfer angle is 45°, the indication can be simplified (see Figure 54).

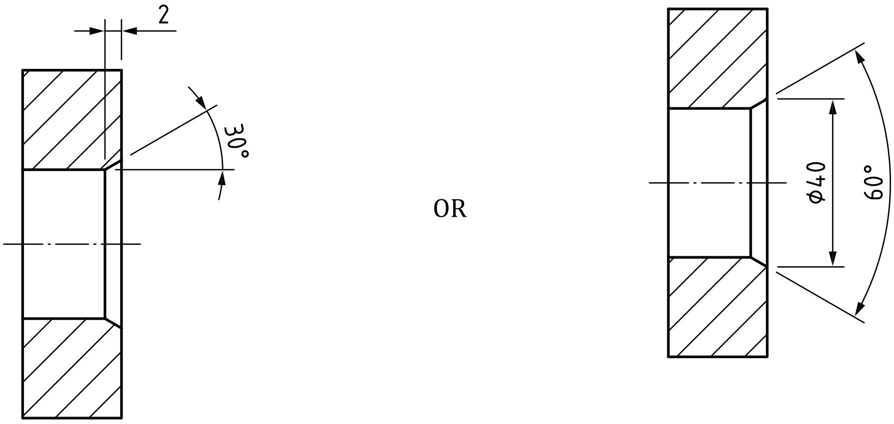
Figure 54 — *Dimension of external chamfer equal to 45°*



7.2.10.2 Internal chamfer

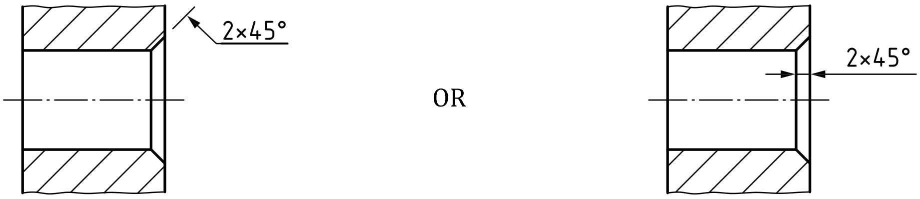
An internal chamfer shall be dimensioned conventionally if the angle of the chamfer is not equal to 45° (see Figure 55).

Figure 55 — *Dimension of internal chamfer not equal to 45°*



NOTE If the chamfer angle is 45°, the indication should be simplified (see Figure 56).

Figure 56 — *Dimension of internal chamfer equal to 45°*

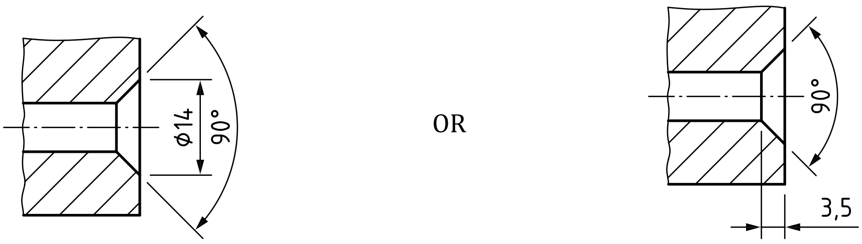


7.2.11 Countersink

A countersink shall be dimensioned by showing either the required diametrical dimension at the surface and the included angle, or the depth and the included angle (see Figure 57).

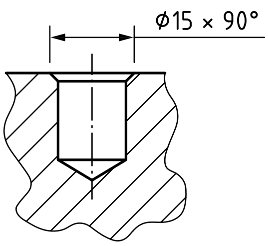
NOTE For the size and angle of countersink, see Figure 47.

Figure 57 — *Dimension of countersink*



NOTE If the included angle is 90°, the indication can be simplified in accordance with BS ISO 15786 (see Figure 58).

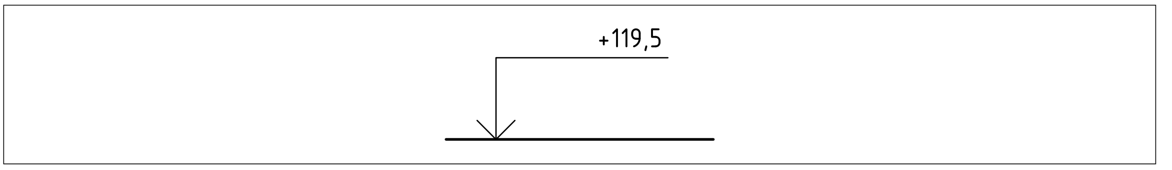
Figure 58 — *Dimension of countersink: simplification*



7.2.12 Indication of levels

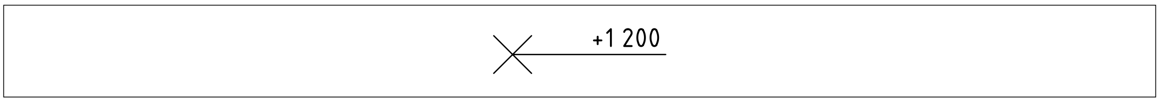
Levels on vertical views, sections and cuts shall be indicated by an open 90° arrowhead connected with a vertical line and horizontal line above which the numerical value of level is placed (see Figure 59).

Figure 59 — *Indication of level on vertical view*



Levels for specified points on horizontal (planes) views and sections shall be indicated by a numerical value of the level placed above a line connected to the point indicated by "X" (see Figure 60).

Figure 60 — *Indication of level on horizontal view or section*



7.2.13 Dimensioning of curved features

7.2.13.1 Curved feature defined by radii

Where appropriate, a curved feature shall be defined by radii, as shown in Figure 61 and Figure 62.

Figure 61 — *Curved feature defined by radii – 1*

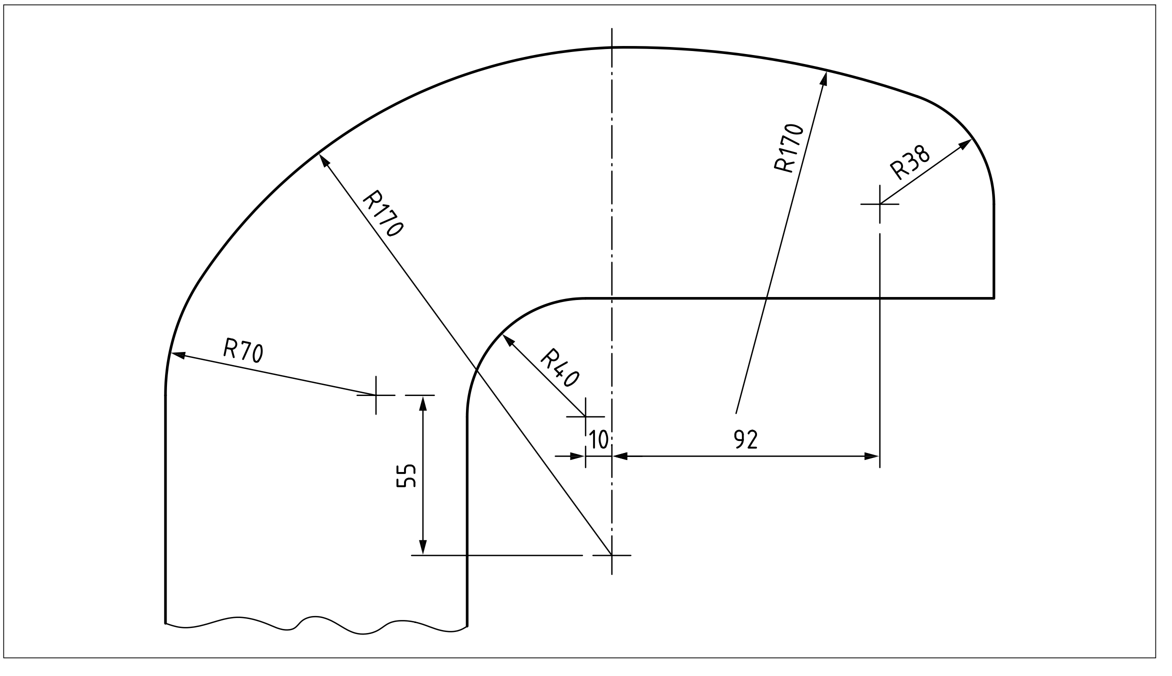
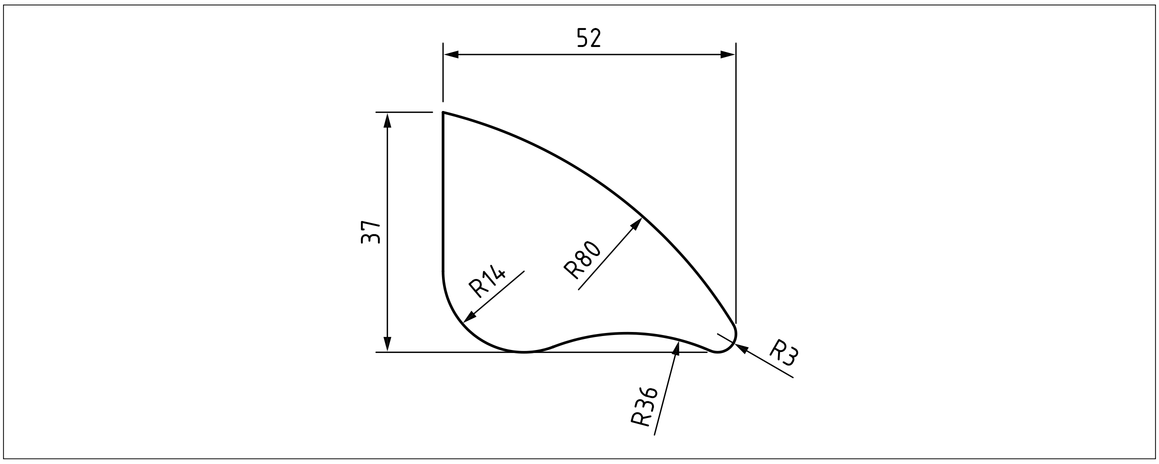


Figure 62 — *Curved feature defined by radii – 2*



7.2.13.2 Curved feature defined by coordinate dimensions

Dimensioning of curved features using linear or polar coordinates shall be indicated by dimensions to points on the profile (see Figure 63 and Figure 64).

Figure 63 — *Curved feature defined by coordinate dimensions*

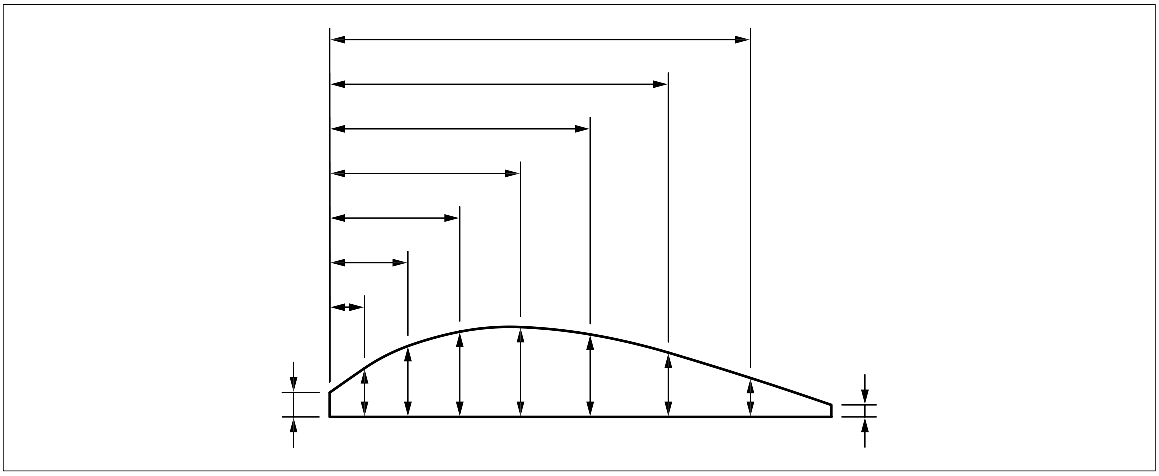
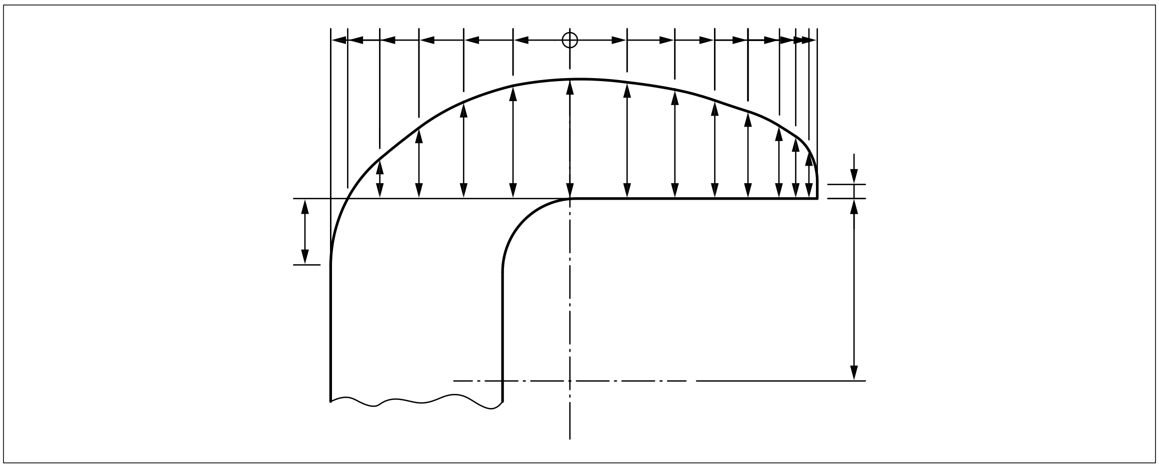


Figure 64 — *Curved features defined by coordinate dimensions*



7.2.14 Arrangements of dimensions

7.2.14.1 General

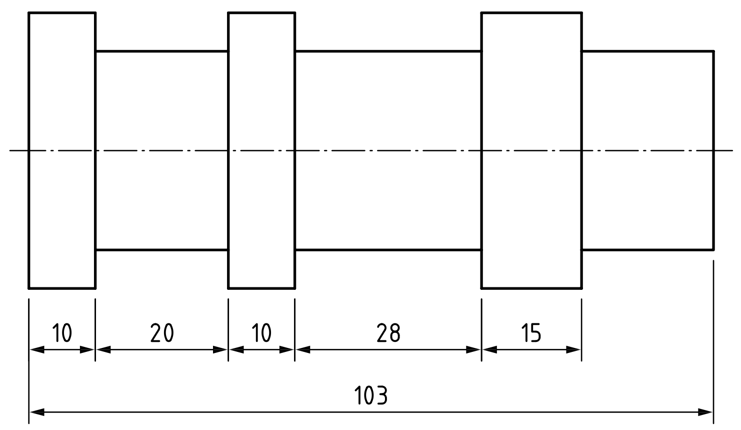
Dimensional values indicated in decimal notation shall use a comma as the decimal marker.

7.2.14.2 Chain dimensioning

Using chain dimensioning, chains of single dimensions shall be arranged in a row (see Figure 65).

Chains of single dimensions shall be used only where the possible accumulation of tolerances does not impinge on the functional requirements of the workpiece.

Figure 65 — *Chain dimensioning*



7.2.14.3 Parallel dimensioning

The dimension lines shall be drawn parallel in one, two or three orthogonal directions, or concentrically (see Figure 66 and Figure 67).

Figure 66 — *Parallel dimensioning – 1*

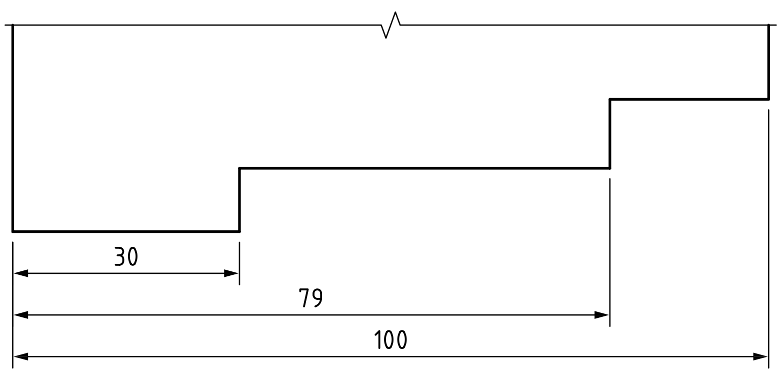
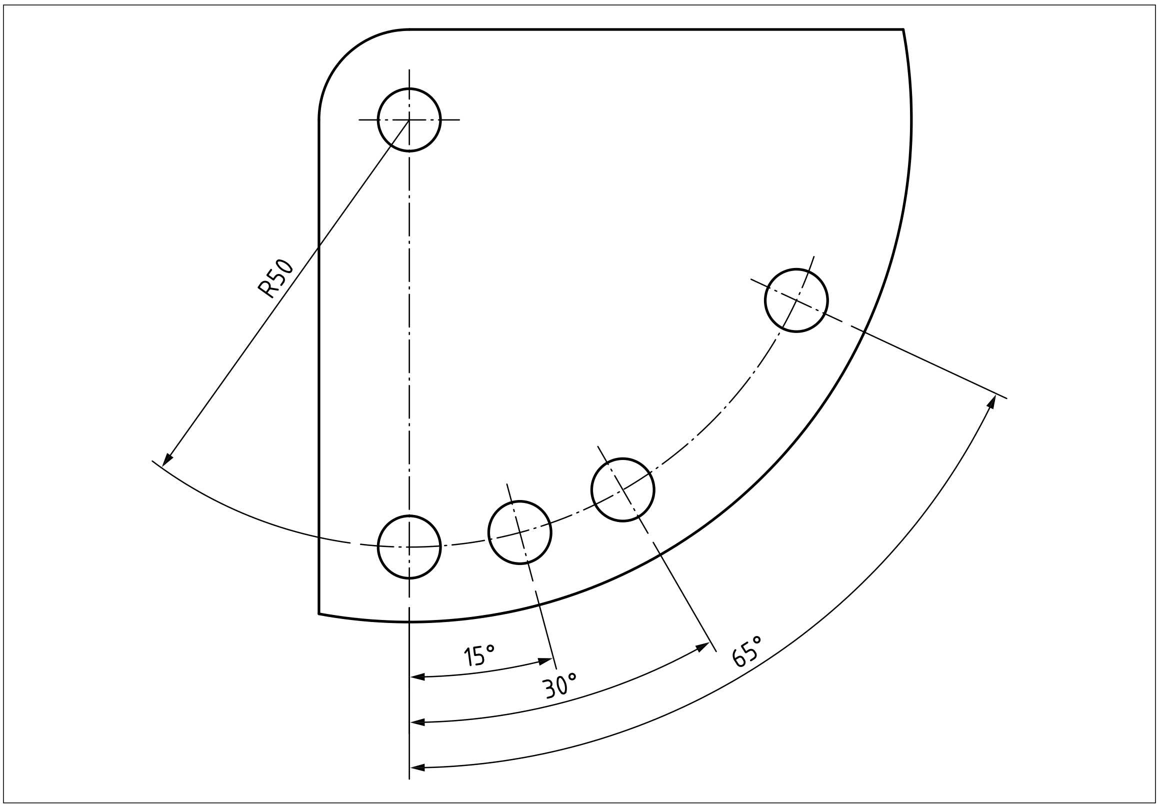


Figure 67 — *Parallel dimensioning – 2*



7.2.14.4 Running dimensioning

7.2.14.4.1 General

NOTE 1 Running dimensioning is simplified parallel dimensioning.

The origin(s) of the dimension line(s) shall be indicated in accordance with **7.2.5.1** (see Figure 29).

Dimensional values shall be placed near the terminator, remote from the origin, either:

a) in line with the corresponding extension line (see Figure 68); or

b) above and clear of the dimension line (see **7.2.14.4.3**, Figure 69).

NOTE 2 An alternative representation of running dimensions may be used where:

a) the origin of the dimension(s) is shown in an appropriate location to indicate where the dimensions are measured from; or

*b) the dimension values are shown on abbreviated dimension lines, where only one arrow is used, directed to the feature to which the dimension value applies (see **7.2.14.4.2**, Figure 68).*

7.2.14.4.2 Running dimensioning in two directions

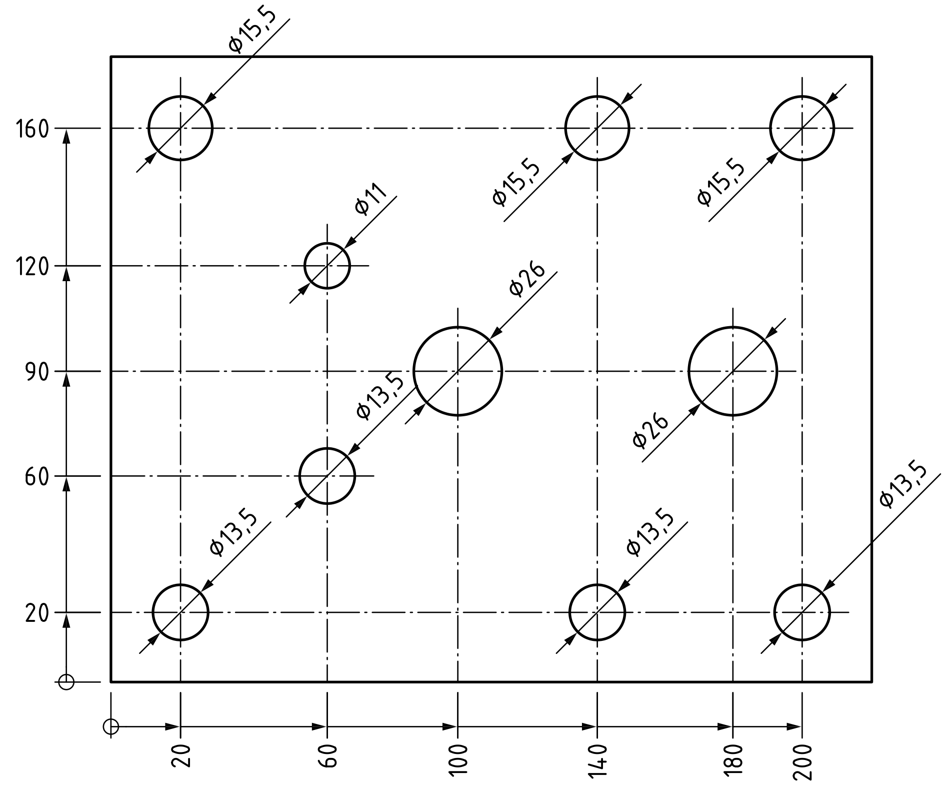
Running dimensioning in two directions shall utilize either:

- one continuous dimension line and one origin; or
- two dimension lines, inclined to each other, and two origins (see Figure 68).

In the case of two directions that are rectangular to each other, the running dimension shall be indicated by two origin symbols.

NOTE Figure 68 shows the running dimension values on the same line as the corresponding origin symbol.

Figure 68 — *Two dimension lines, inclined to each other, and two origins*

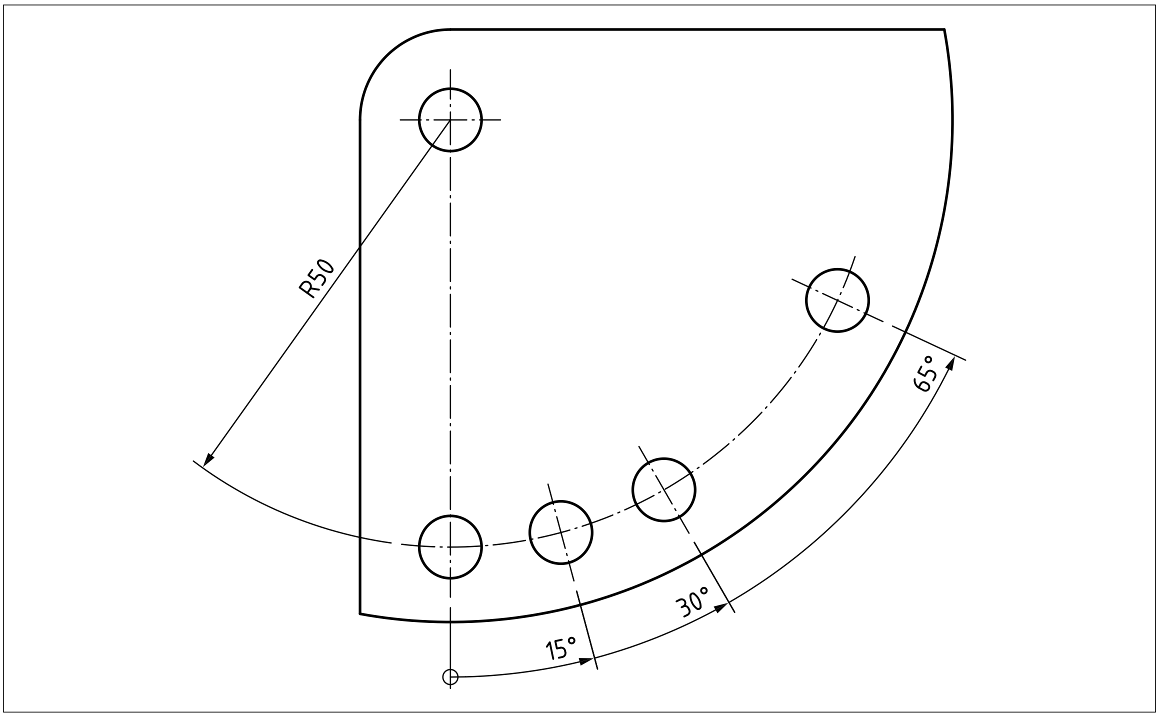


NOTE It is permissible to indicate only one dimension in the case of running dimensions in two directions.

7.2.14.4.3 Running dimensioning of angles

Running dimensioning of angles shall be indicated by one origin symbol and arrowheads to the extension lines of the feature (see Figure 69).

Figure 69 — *Running dimensioning of angles*



7.2.15 Cartesian coordinate dimensioning

*COMMENTARY ON **7.2.15***

Cartesian coordinates are defined starting from the origin by linear dimensions in orthogonal directions (see Figure 70 and Figure 71). Neither dimension lines nor extension lines are drawn.

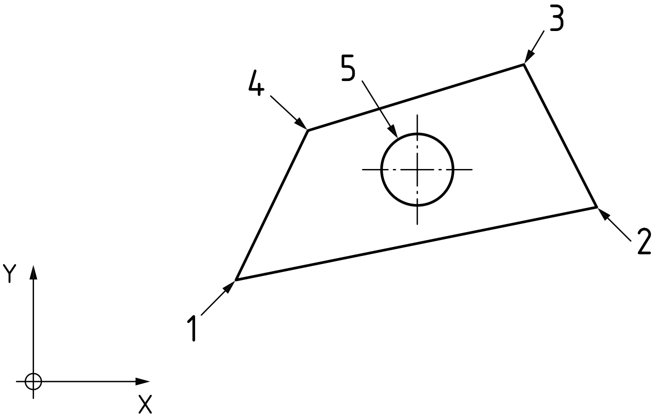
The dimensional values indicated in the negative directions shall have negative signs.

The origin used for coordinate dimensioning shall be indicated (see Figure 70).

NOTE 1 The coordinates may be indicated by a reference letter(s) or number which appears in a table together with the value of the coordinate, or by the direct indication of the coordinates.

NOTE 2 The reference letter(s) or number or the coordinate value may be placed adjacent to the coordinate location or indicated using a leader line.

Figure 70 — *Cartesian coordinate dimensioning – 1*

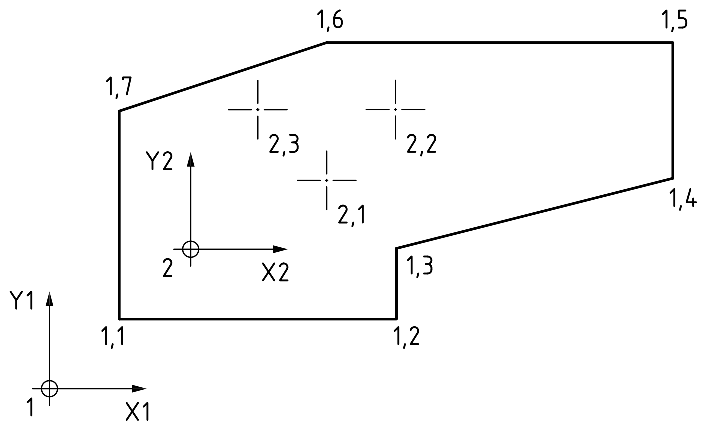


Position	X	Y	*d*
1	10	10	—
2	60	20	—
3	50	40	—
4	20	30	—
5	35	25	Ø10

NOTE Designations for position may be numerical or alphanumerical.

Where the main coordinate system has subsystems, the origin of the coordinate systems and the specific positions within the coordinate systems shall be numbered continuously by Arabic numerals. A comma shall be used as a separation symbol (see Figure 71).

Figure 71 — *Cartesian coordinate dimensioning – 2*



Coordinate origin	**Position**	X1, X2	Y1, Y2	*d*
1	1	0	0	—
1	1,1	10	10	—
1	1,2	50	10	—
1	1,3	50	20	—
1	1,4	80	30	—
1	1,5	80	50	—
1	1,6	30	50	—
1	1,7	10	40	—
1	2	20	20	Ø10
2	2,1	20	10	Ø5
2	2,2	30	20	Ø10
2	2,3	10	20	Ø5

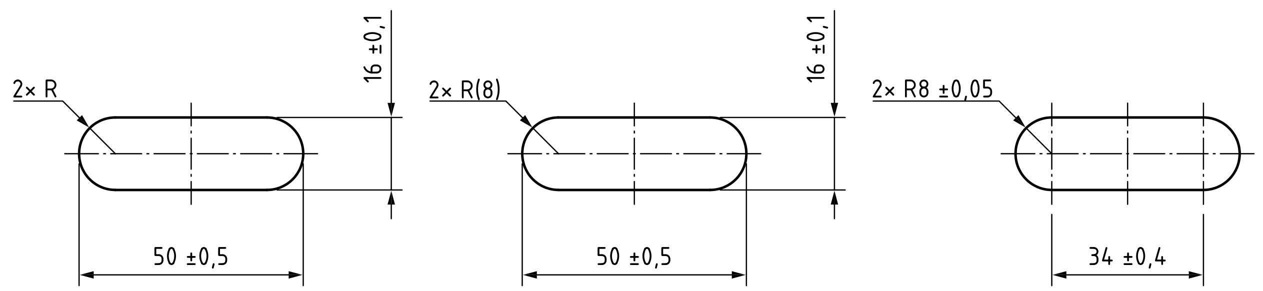
7.3 Dimensioning common features

7.3.1 Dimensioning of slots

Slots shall be specified in accordance with BS EN ISO 129-1 (see Figure 72).

NOTE For further information, see Annex H.

Figure 72 — *Dimensioning slots with rounded ends*



a) Dimensioning slots with rounded ends

b) Dimensioning slots with rounded ends

c) Dimensioning slots with rounded ends

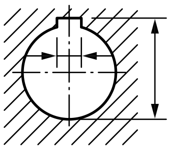
7.3.2 Keyways

Keyways in hubs or shafts shall be dimensioned by one of the methods shown in Figure 73.

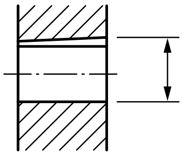
NOTE 1 Limit tolerances and/or geometrical specifications can also be required.

NOTE 2 Further information on keys and keyways is given in BS 4235-1 and BS 4235-2.

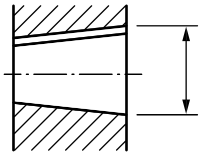
Figure 73 — *Dimensioning of keyways*



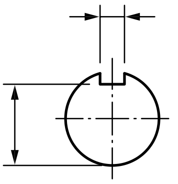
a) Parallel keyway on cylindrical bore



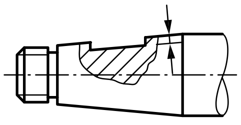
b) Tapered keyway on cylindrical bore



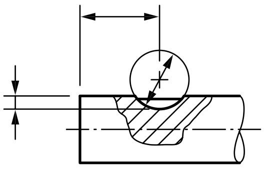
c) Parallel keyway on tapered bore



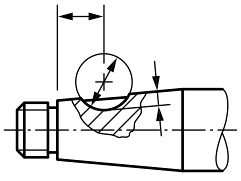
d) Parallel keyway on cylindrical shaft



e) Parallel keyway on tapered shaft



f) Cylindrical keyway on cylindrical shaft



g) Cylindrical keyway on tapered shaft

7.3.3 Screw threads

The definition for metric ISO screw threads shall be in accordance with the following standards, as applicable.

- BS 3643-1, *ISO metric screw threads – Part 1: Principles and basic data.*
- BS 3643-2, *ISO metric screw threads – Part 2: Specification for selected limits of size.*
- BS 4827, *Specification for ISO miniature screw threads – Metric series.*
- BS ISO 261, *ISO general purpose metric screw threads – General plan.*
- BS ISO 262, *ISO general purpose metric screw threads – Selected sizes for bolts, screws, studs and nuts.*
- BS ISO 965-1, *ISO general purpose metric screw threads – Tolerances – Part 1: Principles and basic data.*

Screw threads shall be specified according to functional requirement.

7.4 Dimensional tolerancing

7.4.1 General

Requirements in a TPS which are expressed as dimensions shall be subject to a dimensional tolerance except when they are:

a) TEDs; or

b) auxiliary dimensions.

7.4.2 Methods of specifying dimensional tolerances

The dimensional tolerances shall be specified in one or more of the following ways:

a) as tolerances applied directly to dimensions; or

b) as tolerances applied indirectly to a dimension by way of:

  1) a note;

  2) a table;

  3) a system of general tolerances; or

  4) a reference to another document.

7.4.3 Indication of dimensional tolerances

7.4.3.1 General

Dimensional tolerancing for mechanical engineering shall be in accordance with BS EN ISO 129-1, BS EN ISO 286-1, BS EN ISO 286-2 and the BS EN ISO 14405 series.

NOTE 1 These requirements can also be applied to fields other than mechanical engineering.

Dimensional tolerances applied directly to dimensions shall be shown as:

a) deviation limits;

b) dimension limits; or

c) codes (e.g. from BS EN ISO 286-1 and BS EN ISO 286-2).

All tolerances shall apply to the represented state of the feature in the TPD.

When tolerance or dimensional limits are stacked one above the other, the decimal marker of the upper and lower figures shall be aligned. When a tolerance limit is not shown with a decimal marker, the remaining digits shall be aligned as if the decimal marker had been displayed (see Figure 74, Figure 75, Figure 77, Figure 78 and Figure 80).

NOTE 2 If two limits relating to the same dimension are to be shown, trailing zeros may optionally be used so that both are be expressed to the same number of decimal places (see Figure 74 and Figure 75), except if one of the deviations is zero.

Figure 74 — *Expression of two deviations to the same number of decimal places*

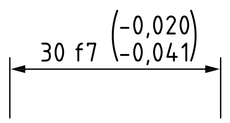
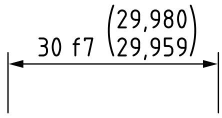
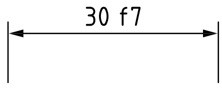


Figure 75 — *Expression of two limits of size to the same number of decimal places*



NOTE For dimensions displayed in accordance with BS EN ISO 286-1, the values of the deviations can be expressed when needed (see Figure 76).

Figure 76 — *Expression of deviations from dimensions displayed in accordance with BS EN ISO 286-1*



7.4.3.2 Deviation limits

The components of a toleranced dimension shall be indicated in the following order (see Figure 77 to Figure 81):

a) the dimensional value; and

b) the deviations limits.

A space shall separate the dimensional values and the tolerance indication, e.g.:

1) 55 ±0,2;

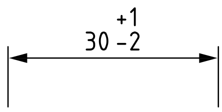
2) 20 min.; and

3) ø10 h7.

Deviation limits shall be expressed in the same unit as the dimensional value.

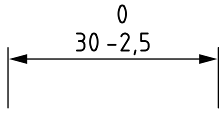
Deviation limits shall be indicated by placing the upper deviation above the lower deviation (see Figure 77 and Figure 79).

Figure 77 — *Components of a toleranced dimension – 1*



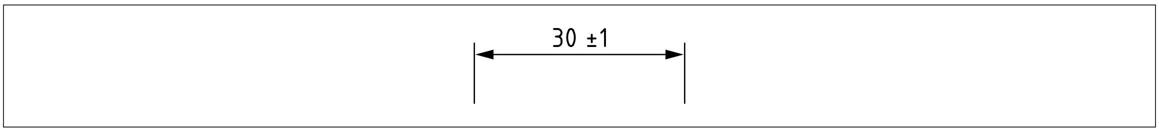
If one of the two limit deviations is zero, this shall be expressed explicitly by the digit zero shown without sign (see Figure 78).

Figure 78 — *Components of a toleranced dimension – 2*



If the tolerance is symmetrical in relation to the dimensional value, the deviation limits shall be indicated only once, preceded by the plus-minus sign (±) (see Figure 79).

Figure 79 — *Components of a toleranced dimension – 3*



7.4.3.3 Display of tolerance values and limit and fit codes

A tolerance code in accordance with BS EN ISO 286-1 shall be sufficient to express deviation limits for a dimension.

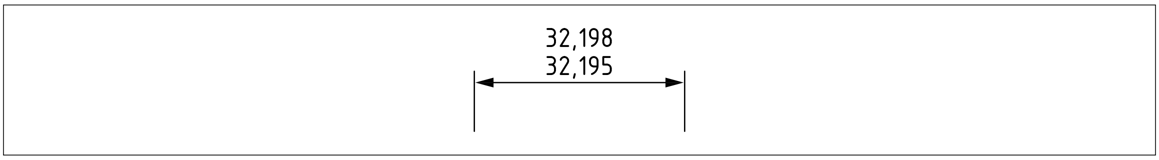
NOTE The tolerance values or dimensional limits that correspond to the code can optionally also be indicated in parentheses (see Figure 74 and Figure 75). Alternatively, the limits can be expressed directly, with the corresponding dimension and code additionally indicated in parentheses.

7.4.3.4 Limits of dimension

7.4.3.4.1 Maximum and minimum limit dimensions

The limits of dimensions shall be indicated by a maximum and a minimum dimension (see Figure 80). The larger dimension shall be placed above the smaller dimension.

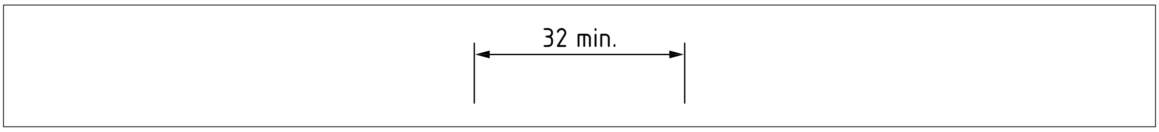
Figure 80 — *Limits of dimensions*



7.4.3.4.2 Single limit dimensions

To limit the dimension in one direction only, the word min. or max. shall be added after the dimensional value (see Figure 81).

Figure 81 — *Single limit dimension*



7.4.4 Functional dimensions

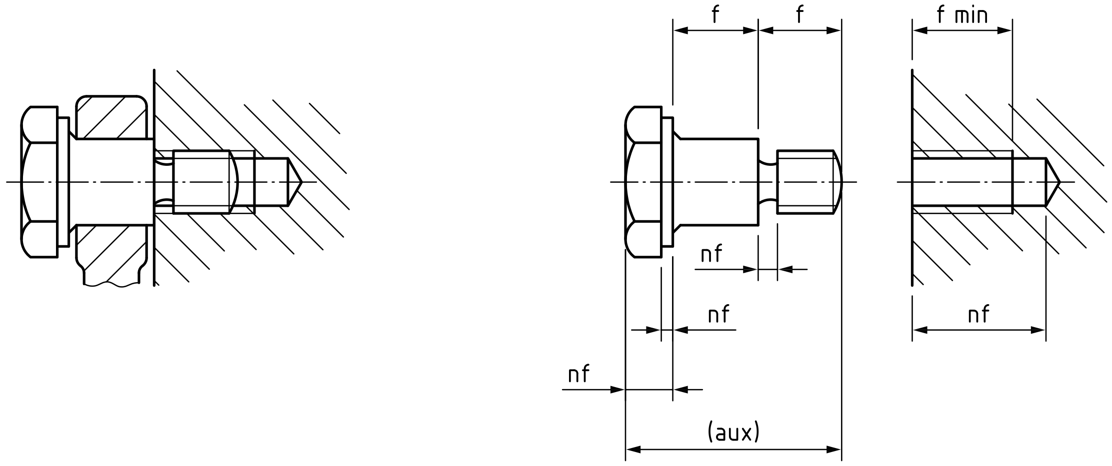
In a functional specification (FUN-SPEC, see Clause **5**), functional dimensions shall be expressed directly (see Figure 82).

NOTE The application of this principle requires the selection of origin or datum features on the basis of the function of the product and the method of locating it in any assembly of which it can be a part.

If any datum feature other than one based on the function of the product is used, tighter tolerances might be necessary to capture functional requirements, which can increase costs (Figure 83).

Expressing functional dimensions does not preclude the use of any dimensioning arrangement providing functional requirements are satisfied.

Figure 82 — *Application of dimensions to suit functional requirements*



Key

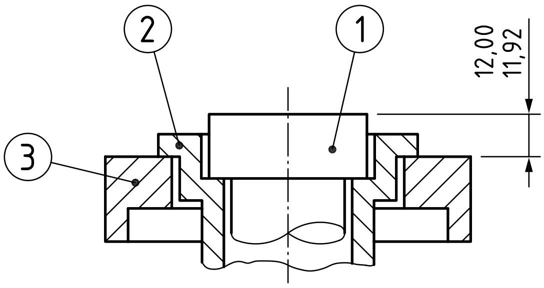
f Functional dimension

nf Non-functional dimension

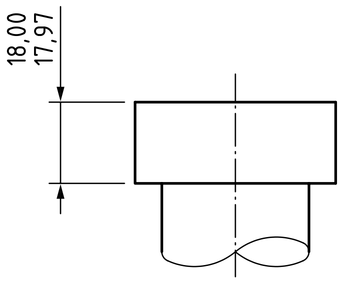
aux Auxiliary dimension without tolerance (for information only)

NOTE Geometric requirements have been omitted for simplicity.

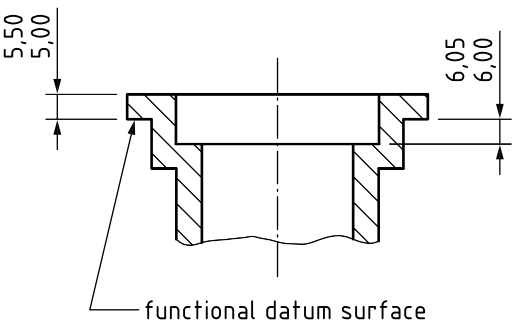
Figure 83 — *Effect of changing datum surfaces from those determined by functional requirements*



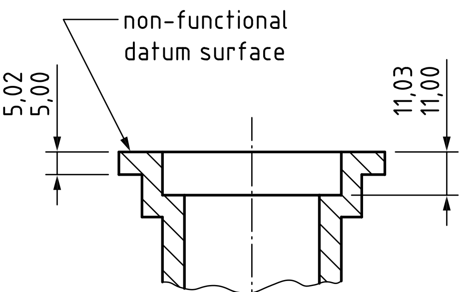
a) Assembly drawing showing a given functional requirement, namely the limits of height of the top face of item 1 above the top face of item 3, with a tolerance of 0,08 mm



b) Detail of head of item 1 showing given limits of size, with a tolerance of 0,03 mm



c) Item 2 dimensioned from a functional datum surface



d) Item 2 dimensioned from a non-functional datum surface

NOTE 1 One direct dimension with a tolerance of 0,05 mm is needed to satisfy the condition shown in a). A nominal flange thickness of 5 mm has been assumed. This value is non-functional in this case.

NOTE 2 Tolerances have had to be reduced; two dimensions with tolerances of, for example, 0,02 mm for the flange and 0,03 mm are now needed to satisfy the condition shown in a).

7.5 Implied dimensions

*COMMENTARY ON **7.5***

When working with 2D drawing, some dimensions used with geometrical specifications can be omitted. The omitted dimensions are referred to as implied dimensions, as their existence is implicit from the context. The concept of implied dimensions is not relevant to 3D CAD data, where all dimensions are explicitly defined.

Stated dimensions and implied dimensions shall be used when working with product definition classification codes 1, 2, 3 and 4 (i.e. 2D drawings used on their own, or in conjunction with a 3D CAD model). Terms such as “equispaced” and “equally spaced” shall not be used.

Dimensions shall be implied by any of the following methods, where there are no contradictory indications:

a) the appearance of perpendicularity;

b) the appearance of parallelism;

c) the appearance of alignment;

d) the appearance of tangency;

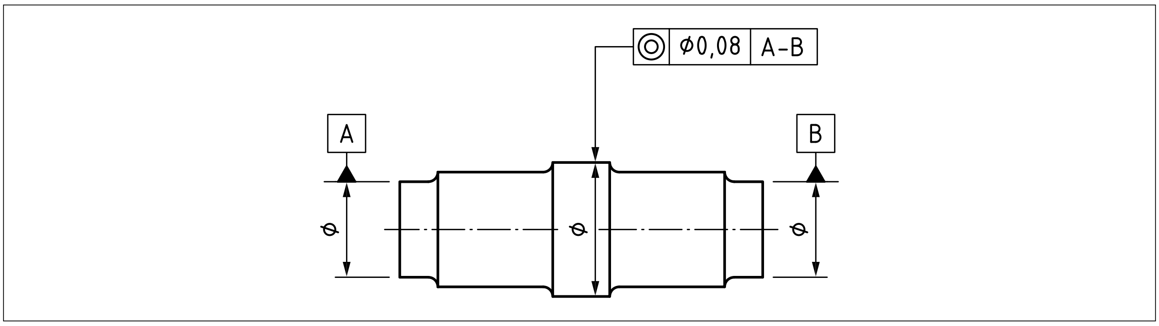
e) the appearance of being equally spaced on a pitch diameter; and

f) a single centreline or centre plane shown passing through two or more features.

NOTE 1 In the example in Figure 84, there is a theoretically exact distance of zero between the nominal datum features and the nominal toleranced feature. It is implied by the following and therefore does not have to be indicated on the drawing:

- *the appearance of alignment;*
- *the single centre line running through all three features; and*
- *the absence of any contradictory indication.*

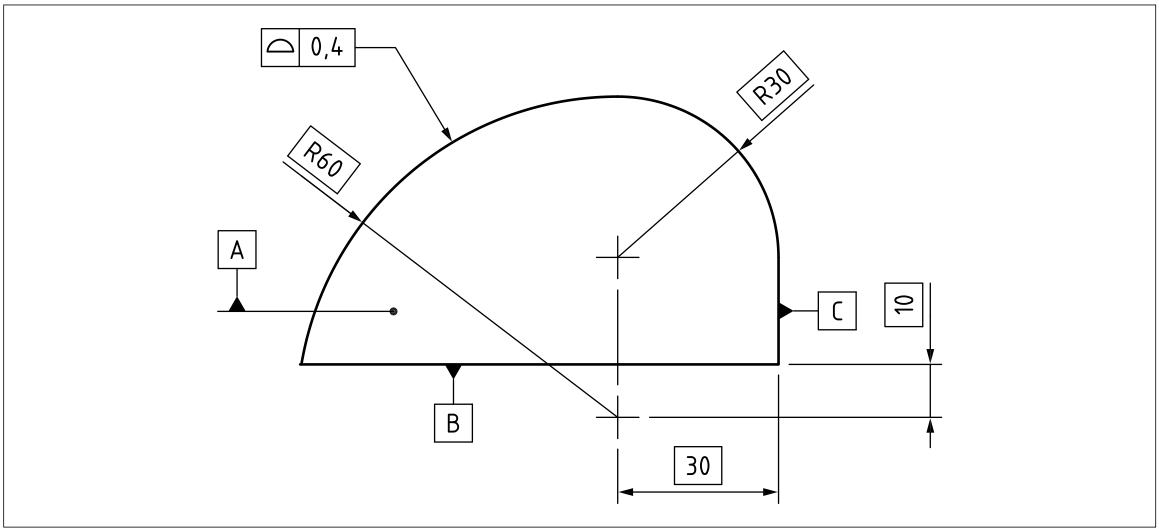
Figure 84 — *Example of implied distance of zero*



NOTE 2 In the example in Figure 85, there is a theoretically exact distance of zero where the R30 feature meets the R60 feature tangentially. It is implied by the following and therefore does not have to be indicated on the drawing:

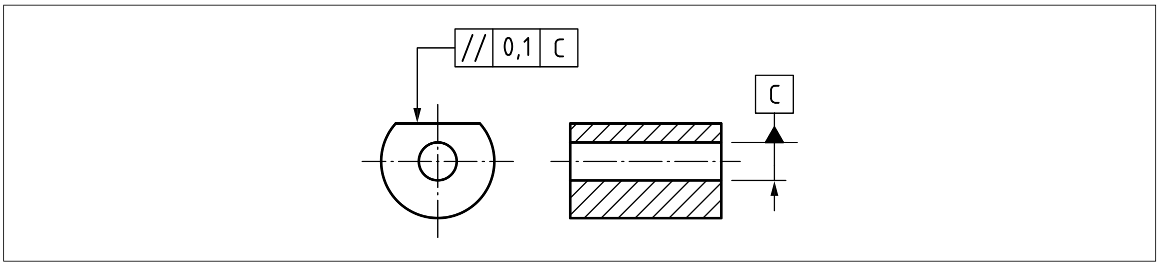
- *the appearance of tangency; and*
- *the absence of any contradictory indication.*

Figure 85 — *Example of implied tangency*



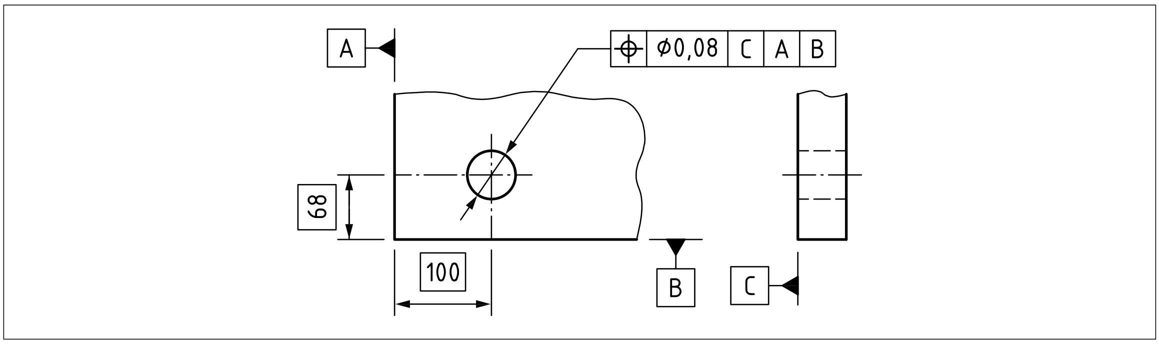
NOTE 3 In the example in Figure 86, there is a theoretically exact angle of 0° or 180° between the nominal datum feature and the nominal toleranced feature. It is implied by the appearance of parallelism and the absence of any contradictory indication, and therefore does not have to be indicated on the drawing.

Figure 86 — *Example of implied 0° or 180°*



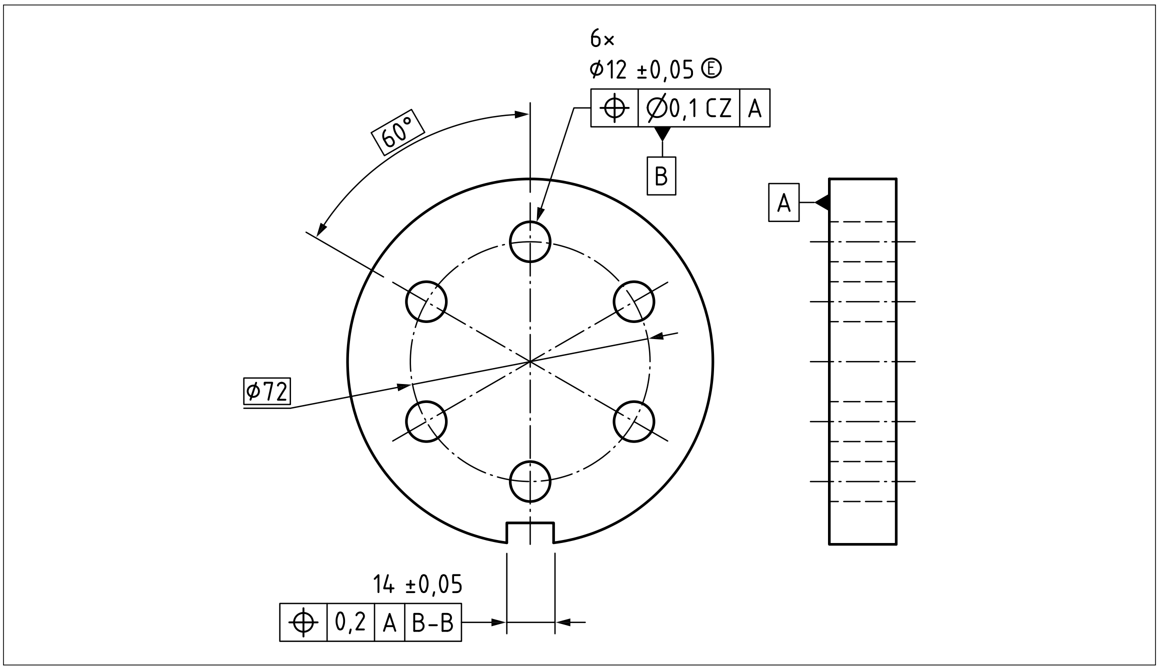
NOTE 4 In the example in Figure 87, there are theoretically exact angles of 90° between each of the nominal datum features. They are implied by the appearance of perpendicularity and the absence of any contradictory indication, and do not have to be indicated on the drawing. There is also a theoretically exact angle of 90° between the nominal hole feature and the nominal datum feature for datum C. It is implied by the appearance of perpendicularity and the absence of any contradictory indication, and does not have to be indicated on the drawing.

Figure 87 — *Example of implied 90°*



NOTE 5 In the example in Figure 88, there are theoretically exact angles of 60° between each of the nominal hole positions on the pitch diameter of Ø72. They are implied by the appearance of being equispaced and the absence of any contradictory indication, and do not have to be indicated on the drawing.

Figure 88 — *Example of implied 60°*



NOTE 6 Where implied dimensions can be easily misinterpreted or overlooked, stated dimensions should be used instead.

NOTE 7 In some cases, implied dimensions, such as those shown in the examples in Figure 84 to Figure 88, are used in conjunction with general dimensional tolerances. This results in specification requirements which are often ambiguous and sometimes contradictory, and should not be relied upon for the effective definition of workpiece geometry.

8 Datums and datum systems

COMMENTARY ON CLAUSE ***8***

This clause sets out some of the main principles and rules of working with datums and datum features (for descriptions see Annex A). A more extensive description of datums and datum systems is given in in BS EN ISO 5459. This clause is not intended to replace BS EN ISO 5459; it is only intended to make some of the key content more accessible.

Some additional options for working with datums are also presented here which are not yet published in ISO standards (e.g. the use of non-default association methods). As some of these options are already in use in industry, they are included in this clause with the caveat that whenever they are used, a clear explanation also appears in the specification, and the additional caveat that, when these options do become available through published ISO standards, their symbology and format might differ from those set out here.

8.1 General

Datums and datum systems shall be used where there is a requirement to define the location or orientation of:

a) a tolerance zone;

b) a virtual condition boundary; or

c) a geometrical feature used in the definition of a specification requirement.

A datum system shall be defined in the datum section of a tolerance indicator (see **9.5.4**) or other specification entity such as an intersection plane indicator (see **10.6.1**).

Where the datum system is composed of more than one datum, the datum system shall define the sequence in which the datums are used.

By default, each datum in a datum section shall lock all the degrees of freedom that it is able to lock.

Additional indications shall be used to limit the degrees of freedom locked by a datum, if necessary.

8.2 Datum systems

*COMMENTARY ON **8.2***

A datum is a set of one or more situation features obtained from the associated features for one or more datum features.

A datum can be:

a) a point;

b) a straight line;

c) a flat plane; or

d) a combination of a), b) and c).

A datum system is an ordered set of one or more datums.

The datum system shall be defined in the datum section of a tolerance indicator (see **9.5.4**) or other specification. The datum system shall be used to constrain some or all of the six degrees of freedom (see **A.9**) for:

a) a tolerance zone; and

b) an ideal feature used in other specification entities such as intersection plane indicators (see **10.6.1**).

Each datum in turn, reading from left to right, shall lock all the degrees of freedom that it is able to lock, and which are required by the geometrical characteristic, and which have not already been locked by a preceding datum, unless otherwise indicated.

8.3 Identifying datum features: Placement of the datum feature indicator

8.3.1 Datums based on single feature

Where a datum is based on a single datum feature, the datum feature shall be identified with a datum feature indicator.

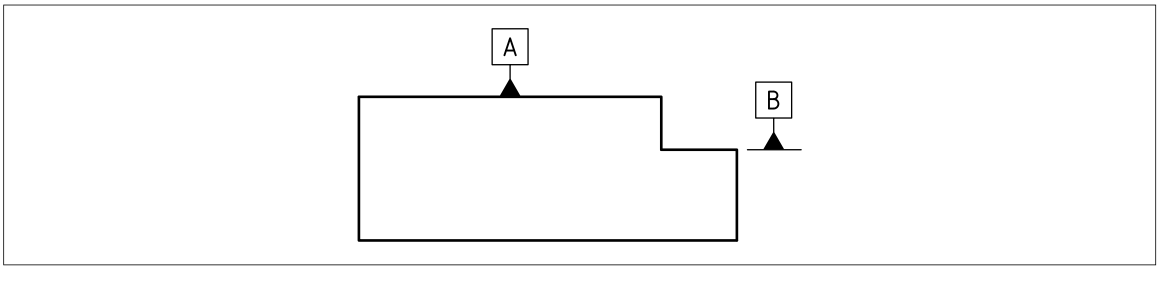
The datum shall be established from the entire surface of the datum feature unless otherwise indicated.

If the datum is based on a limited portion of the datum feature, the datum shall be established using datum targets (see **8.8.1**).

If the feature is not a feature of size, the datum feature indicator shall be placed in one of the following positions:

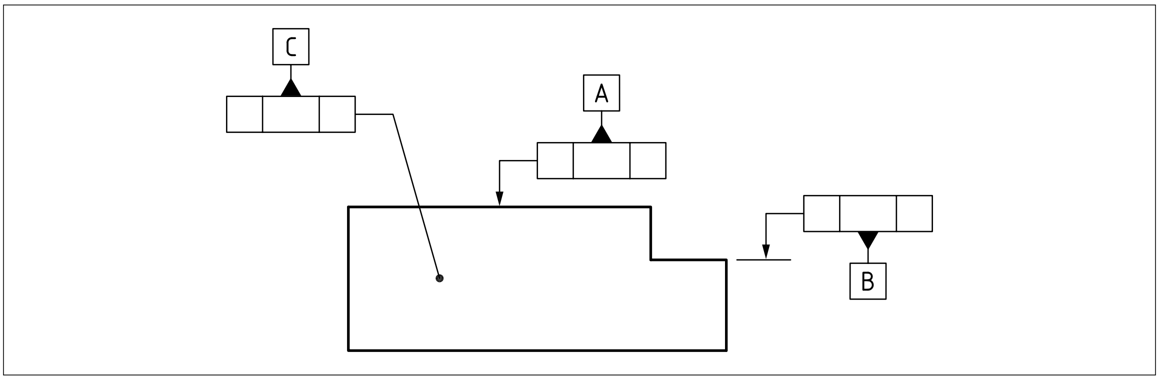
a) attached to an outline or an extension line from an outline, representing the feature in profile (as shown in Figure 89);

Figure 89 — *Datum feature indicators attached to an outline and an extension line from an outline*



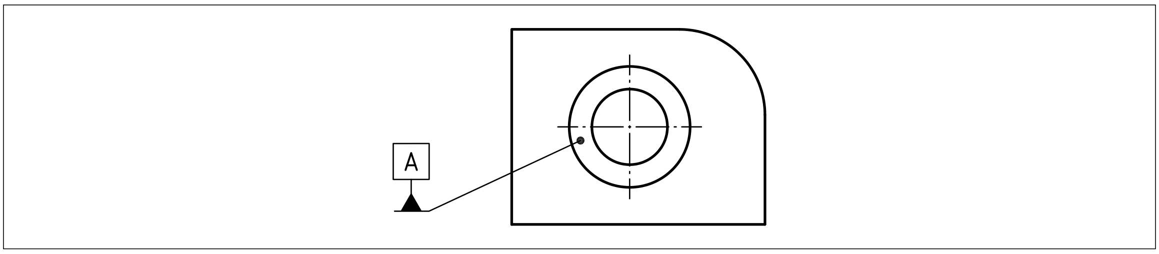
b) attached to a tolerance frame which is attached to the feature (as shown in Figure 90);

Figure 90 — *Datum feature indicators attached to a tolerance frame attached to the feature*



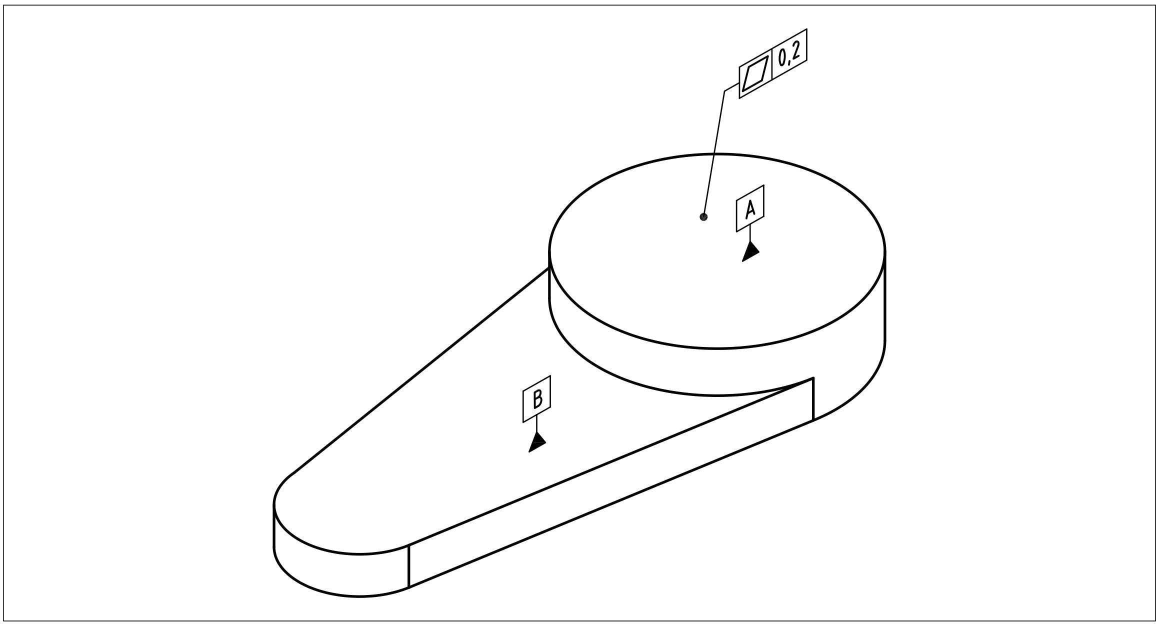
c) attached to a visible surface with a leader line ending in a filled dot on the surface (as shown in Figure 91); or

Figure 91 — *Datum feature indicator attached with leader line ending in a filled dot on surface*



d) attached to a visible surface on a 3D TPS (as shown in Figure 92).

Figure 92 — *Datum feature indicators attached to surface on 3D TPS*

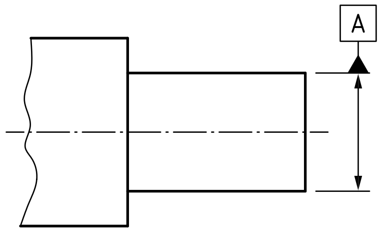


If the datum feature is a feature of size, the datum feature indicator shall be placed in one of the following positions:

1) in line with the dimension line for the size dimension of the feature (as shown in Figure 93); or

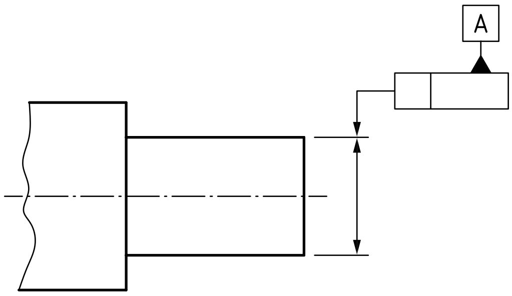
NOTE 1 The datum feature indicator is placed in line with the size dimension to unambiguously identify the feature from which the datum is to be derived.

Figure 93 — *Datum feature indicator placed in line with dimension line for size dimension*



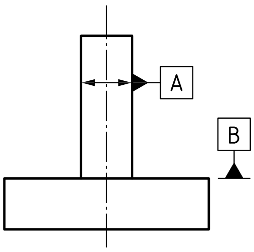
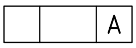
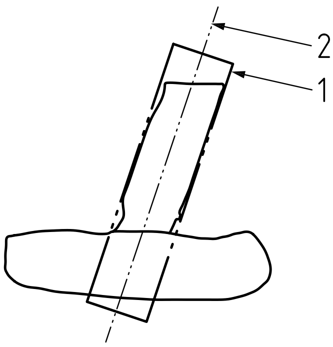
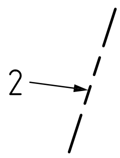
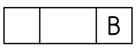
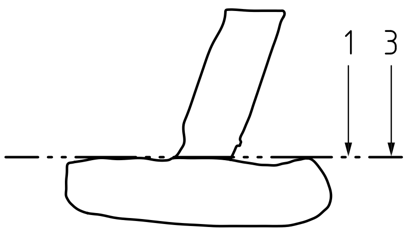
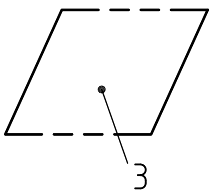
2) attached to a tolerance frame which has its leader line in line with the dimension line for the size dimension of the feature (as shown in Figure 94).

Figure 94 — *Datum feature indicators attached to tolerance frame*



NOTE 2 An example of a datum based on a single datum feature is given in Table 10.

Table 10 — *Example of single datum*

Indication of datum feature	**Indication of datum in tolerance frame**	**Illustration of the meaning**	**Invariance class and situation feature**	**Datum**
			Cylindrical Axis of associated cylinder	
			Planar Associated plane	

Key

1 Associated feature (without orientation constraint)

2 Straight line which is the situation feature of the associated cylinder (its axis)

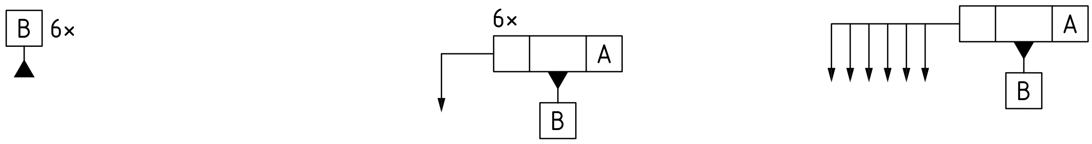
3 Plane which is the situation feature of the associated plane (the associated plane itself)

NOTE Association for single datums is described in BS EN ISO 5459:2024, Annex A.

8.3.2 Datums based on multiple datum features

Where multiple datum features are to be used to define a single datum, this shall be shown in accordance with Figure 95.

Figure 95 — *Datum indications based on multiple datum features*



a) Multiple individual datum feature indication

b) Multiple individual datum feature indication

c) Multiple individual datum feature indication

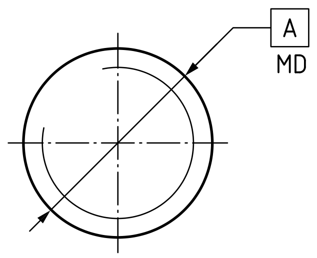
8.3.3 Datums based on threaded features, gears or splines

If the datum feature is a threaded feature, it shall be treated like a cylindrical feature of size. The datum feature indicator shall be placed in line with the thread dimension or attached to a tolerance frame that is attached in line with the thread dimension. By default, the datum shall be a datum axis based on the pitch cylinder of the thread.

If the datum axis is to be derived from the major diameter of the thread, the modifier MD shall be used as an adjacent indication for the datum feature indicator (see Figure 96).

If the datum axis is to be derived from the minor diameter of the thread, the modifier LD shall be used as an adjacent indication for the datum feature indicator.

Figure 96 — *Datum axis based on major diameter of external threaded feature*



If the datum feature is a cylindrical gear form or a splined cylindrical feature, it shall be treated like a cylindrical feature of size. The datum feature indicator shall be placed in line with the size dimension or attached to a tolerance frame that is attached in line with the size dimension for the feature.

The datum definition shall be completed by placing the pitch diameter, MD or LD modifier adjacent to the datum feature indicator (see BS EN ISO 5459).

NOTE There is no default definition of the datum axis based on a gear form or spline.

8.4 Deriving datums from datum features

The datum feature shall be identified in a TPD with a datum feature identifier (see **8.3**).

The datum section of a specification element shall determine whether the datum is primary, secondary or tertiary.

An associated feature shall be constructed for the datum feature using an association criterion.

NOTE 1 The associated feature can be constructed either for the real datum feature or for the extracted datum feature. Constructing the associated feature for the real datum feature involves using physical devices such as surface tables, angle plates, v-blocks and collets. Constructing the associated feature for the extracted datum feature involves a mathematical operation to build the associated feature based on the data which have been sampled from the datum feature.

For a single primary datum feature, the associated feature shall be based on the datum feature alone, with no additional location or orientation constraints.

For a single secondary datum feature, the associated feature shall be based on the datum feature, with additional orientation constraints relating it to the primary datum, unless otherwise indicated.

For a single tertiary datum feature, the associated feature shall be based on the datum feature, with additional orientation constraints relating it to the primary and secondary datums, unless otherwise indicated.

NOTE 2 The datum consists of the situation feature(s) of the associated feature.

Where the setup of a functionally correct datum system requires the associated feature for the secondary and/or tertiary datum to have its location, as well as its orientation, fixed relative to preceding datums, an additional indication shall be used to indicate this (see in BS EN ISO 5459:2024, Annex I).

NOTE 3 See Annex C for examples.

8.5 Indications of situation features in a technical product specification (TPS)

*COMMENTARY ON **8.5***

When a datum feature has a single situation feature, the situation feature has a unique location.

For example, if the datum feature is a cylinder, the situation feature is a straight line, and it can only be centred in the associated cylinder; if the datum feature is a flat surface, the situation feature can only be a tangent plane.

When the datum feature has more than one situation feature, at least one of the situation features does not have a unique location.

For example, if the datum feature is a cone, the situation features are a line and a point. The line (the axis of the cone) has only one location, but the point can be taken to be anywhere along that line.

The location of the point is determined by the specifier; it can be taken as the point where a "gauge diameter" for the cone is defined or as the point at the apex of the cone, or at any other defined location on the axis of the cone.

Where necessary, the situation features shall be identified on a drawing or in a CAD model.

The situation features shall have a theoretically exact relationship with the nominal datum feature.

When these are shown on the specification, the situation features shall be represented as follows:

a) point: by a cross;

b) line: by a long-dashed double-dotted narrow line (line type 05.1 from BS EN ISO 128-2:2022) when shown end on, the line is represented by a point [see a)]; and

c) plane: by a rectangular plane outlined with a long, dashed double-dotted narrow line; when shown edge on, the plane is represented by a line [see b)].

The situation features shall be labelled with an identifier as follows:

1) point: by an identifier consisting of the name of the datum, followed by PT in square brackets;

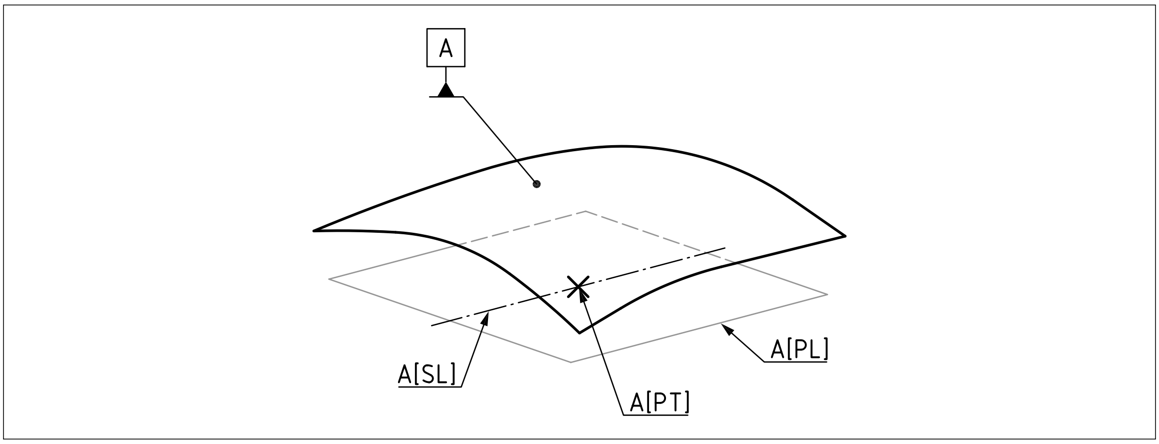
2) line: by an identifier consisting of the name of the datum, followed by SL in square brackets (see Figure 97); and

3) plane: by an identifier consisting of the name of the datum, followed by PL in square brackets (see Figure 97).

Where necessary, these identifiers shall be shown in more than one drawing view.

NOTE See Annex F for further information regarding implementation in CAD.

Figure 97 — *Identifiers for situation features*



8.6 Common datums

8.6.1 General

A common datum (see **A.4**) shall be identified by listing the individual datum feature identifiers in a single datum compartment of the datum section. The individual datum feature identifiers shall be separated by a hyphen, e.g.:

◎	⌀0,2	A–B

Where a common datum is based on a group of identical features, which are identified with a single datum feature indicator, as in Figure 98, the datum feature identifier shall be shown twice, separated by a hyphen, e.g.:

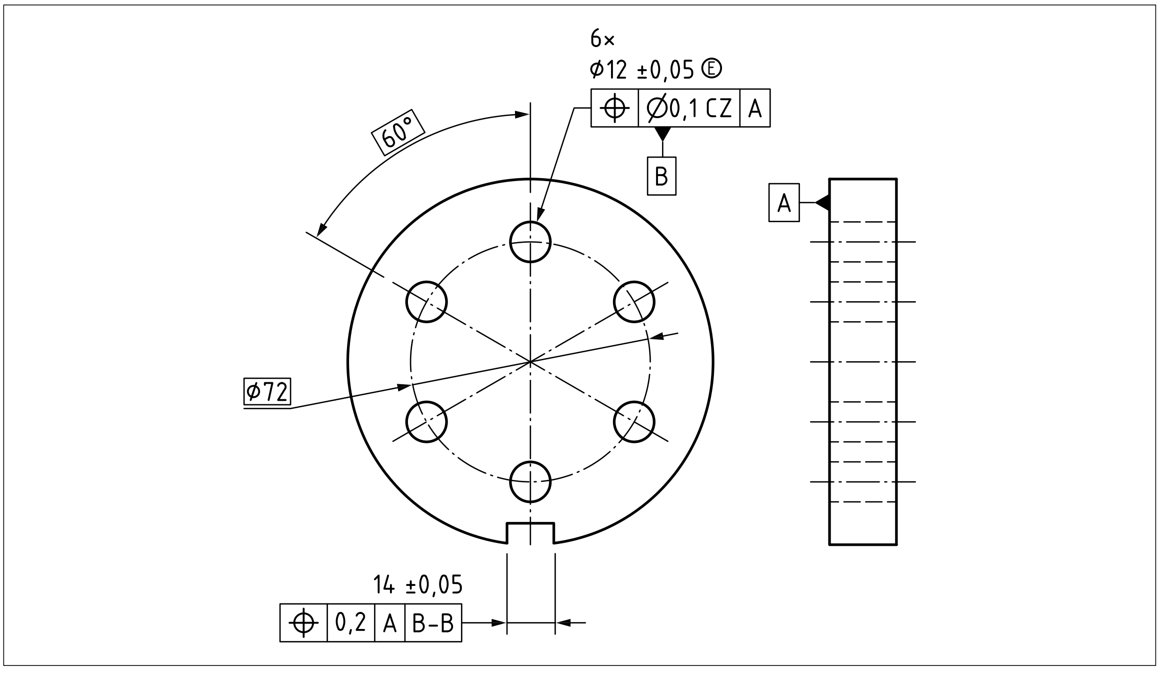
⌖	⌀0,1	A	B–B

When a common datum is used, the relationship between the individual datum features shall be toleranced.

When a common datum is constructed, the theoretically exact relationships between the individual datum features shall also apply to the associated features which are used to construct the datum.

NOTE Where a common datum is used as a secondary or tertiary datum, the associated features are also subject to orientation (but not location) constraints to those preceding datums.

Figure 98 — *Common datum established by a group of identical features*



8.6.2 Common datums based on two coaxial cylinders

A single primary datum based on two nominally coaxial cylindrical features shall consist of a single datum axis (see Figure 99), which is the situation feature for the two associated features:

a) the minimum circumscribing cylinder which can be constructed around datum feature A; and

b) the minimum circumscribing cylinder which can be simultaneously constructed around datum feature B,

where the two minimum circumscribing cylinders are additionally constrained to be coaxial with each other (see Figure 100).

Figure 99 — *Common datum established from two coaxial cylinders*

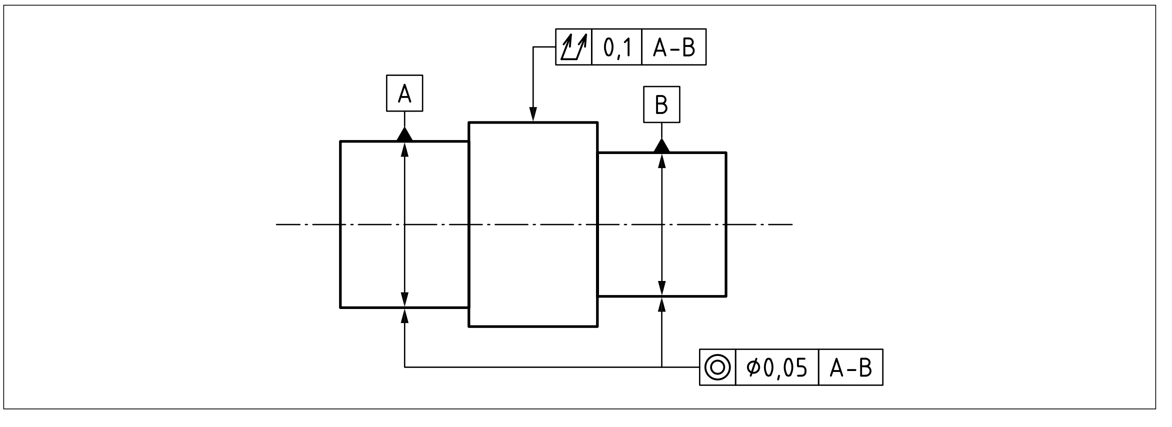
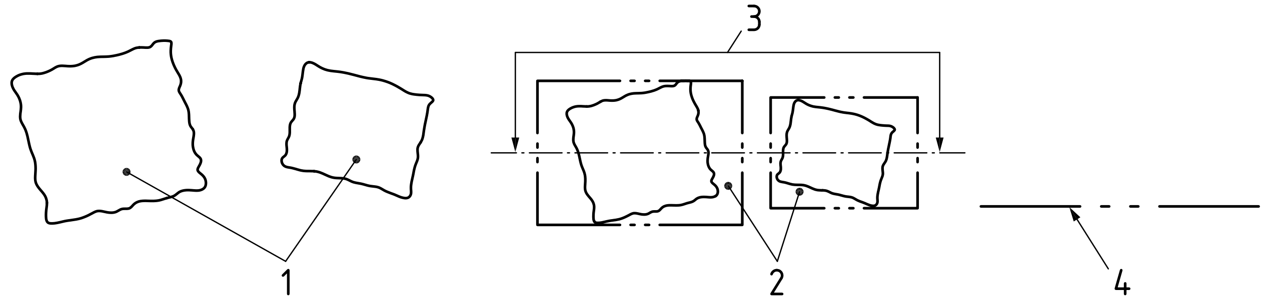


Figure 100 — *Establishing a common datum from two coaxial cylinders*



a) Common datum from two coaxial cylinders

b) Common datum from two coaxial cylinders

c) Common datum from two coaxial cylinders

Key

1 Datum feature: real integral feature

2 Associated features with variable diameter

3 Orientation constraint (parallelism) and location constraint (coaxiality)

4 Single datum – situation feature of the associated feature: straight line (common axis)

8.6.3 Common datums based on two parallel cylinders

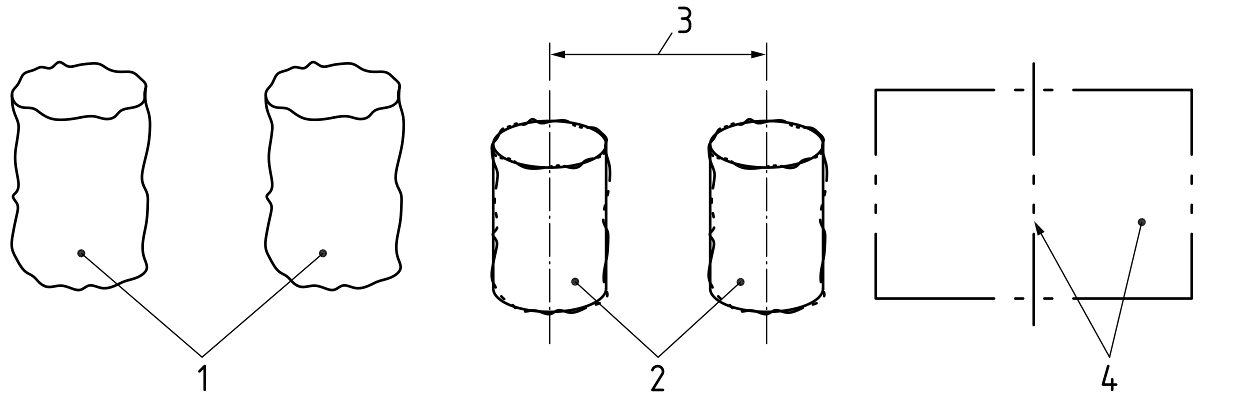
A single primary datum based on two nominally parallel cylindrical holes shall consist of a plane and an axis.

A maximum inscribing cylinder shall be constructed for each of the two holes simultaneously, with the additional constraint that the two inscribing cylinders shall remain parallel with each other, and a fixed distance (*l*) apart.

NOTE The situation features for the two inscribing cylinders are a plane and an axis.

The axis shall be a mean between the axes of the two inscribing cylinders, and the plane shall be a plane through that axis, and the axis of either one of the inscribing cylinders (see Figure 101).

Figure 101 — *Establishing a common datum from two parallel cylinders*



a) Common datum from two parallel cylinders

b) Common datum from two parallel cylinders

c) Common datum from two parallel cylinders

Key

1 Datum features: real integral features

2 Associated features with variable diameters

3 Orientation constraint (parallelism) and location constraint (*l* distance)

4 Common datum – situation feature of the collection surface: plain and straight line

8.6.4 Common datums based on a group of cylinders

A single primary datum based on a set of five nominally identical and parallel cylindrical holes shall consist of a plane and an axis (see Figure 102).

A maximum inscribing cylinder shall be constructed for each of the five holes simultaneously, with the additional constraint that the five inscribing cylinders shall remain parallel with each other and centred on a pitch circle of diameter (Ød) at 72° increments.

NOTE 1 The situation features for the five inscribing cylinders are a plane and an axis.

The axis shall be a mean between the axes of the five inscribing cylinders, and the plane shall be a plane through that axis, and the axis of any one of the inscribing cylinders (see Figure 103).

NOTE 2 These principles can be equally applied to non-parallel cylinders.

Figure 102 — *Pattern of five cylinders: input for reading (drawing indication) and for writing (design intent)*

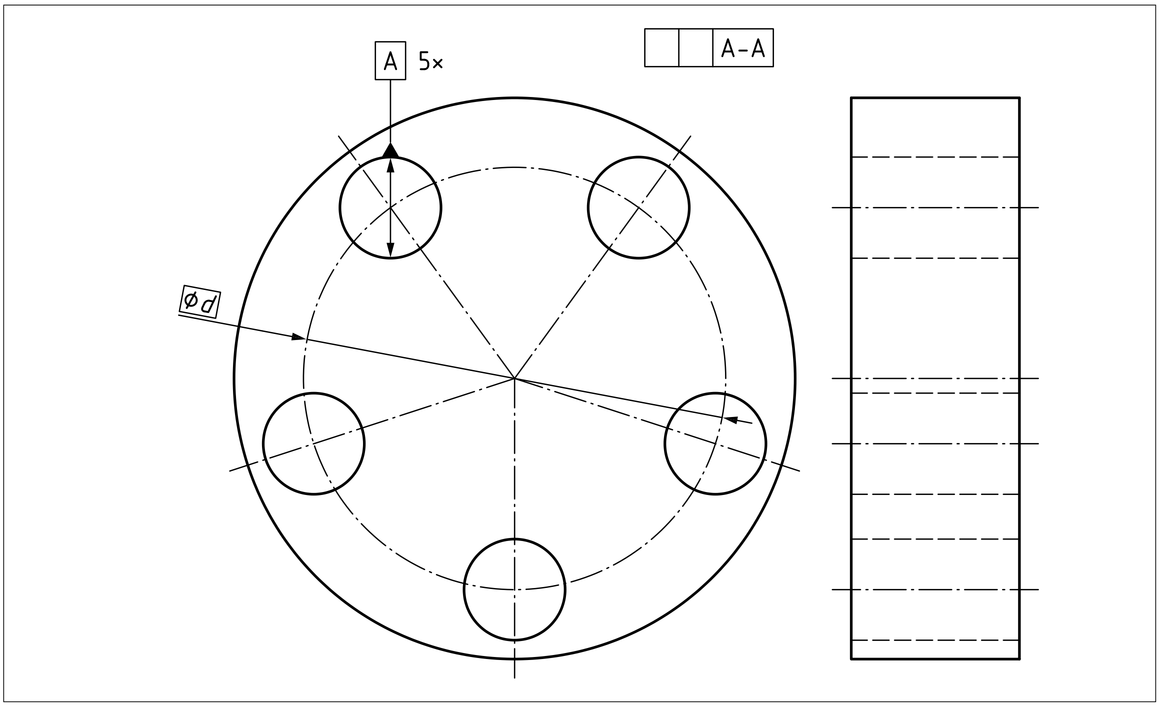
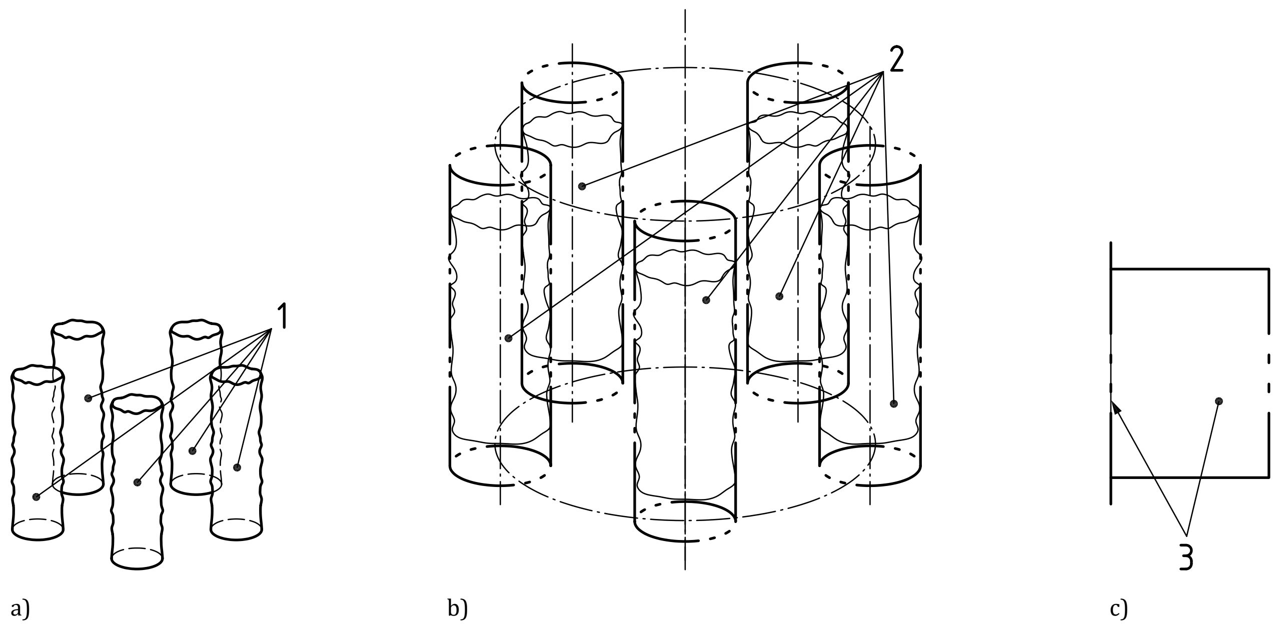


Figure 103 — *Establishing a common datum from a pattern of five cylinders*



Key

1 Datum features: real integral features

2 Associated features with variable diameters, orientation constraint (parallelism) and location constraint [given by the theoretical exact dimensions (TEDs) and angles of 72°]

3 Common datum – situation features of the collection surface: plain and straight line

8.6.5 Association for common datums

To establish a common datum, a collection of ideal single surfaces shall be simultaneously associated with a corresponding set of non-ideal surfaces.

The constraints between the different associated features shall be fixed and shall include:

a) any new intrinsic characteristics defined by the collection of the features (e.g. the pitch diameter for a group of parallel holes in a circular array);

b) any explicit location or orientation constraints between the nominal features defined by TEDs; and

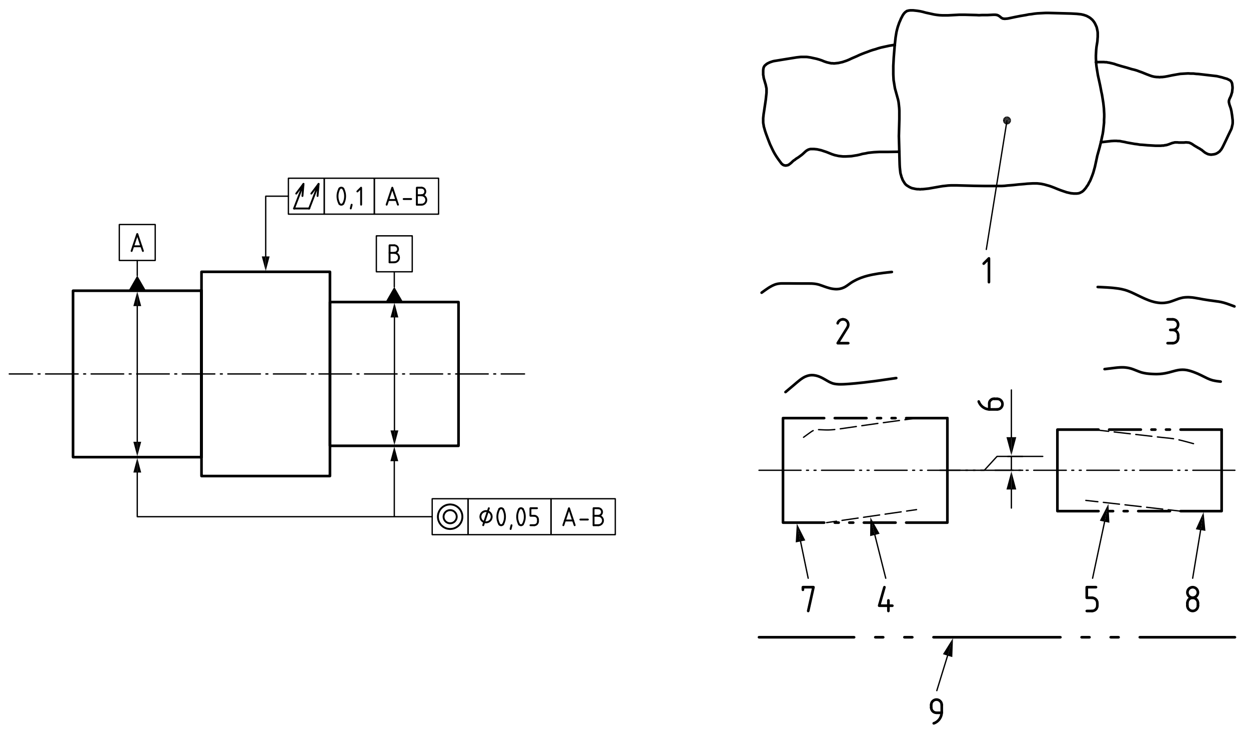
c) any implied location or orientation constraints between the nominal features (implied orientation constraint: 0°, 90°, 180°, 270° and implied location constraint: 0 mm).

The associated features shall also be constructed, by default, to be in contact with the corresponding filtered feature but outside the material.

NOTE 1 The objective function is to simultaneously minimize the maximum distance normal to the associated feature between each associated feature and its filtered feature.

NOTE 2 Figure 104 illustrates the process used to establish a common datum from two surfaces nominally cylindrical and coaxial.

Figure 104 — *Association for a common datum based on two coaxial cylinders*



a) Indication

b) Meaning of common datum A-B

Key

1 Real workpiece

2 Datum feature numbered No. 1 of the collection surface for the common datum

3 Datum feature numbered No. 2 of the collection surface for the common datum

4 Filtered feature of datum feature No. 1

5 Filtered feature of datum feature No. 2

6 Orientation constraint and location constraint between the associated features of the collection features (coaxiality)

7 Associated feature of datum feature No. 1 (see BS EN ISO 5459)

8 Associated feature of datum feature No. 2 (see BS EN ISO 5459)

9 Common datum (in this case, the axis of the collection surface – the two coaxial associated cylinders)

8.7 Datum systems and the six degrees of freedom

A datum system (see **8.2**) shall be used to lock some, or all, of the six degrees of freedom available to a tolerance zone (i.e. in a geometrical specification) or other geometrical entity (e.g. an intersection plane).

The primary datum in a datum system shall, by default, lock all degrees of freedom that it is able to lock.

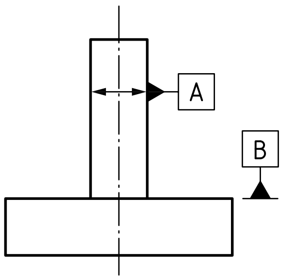
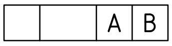
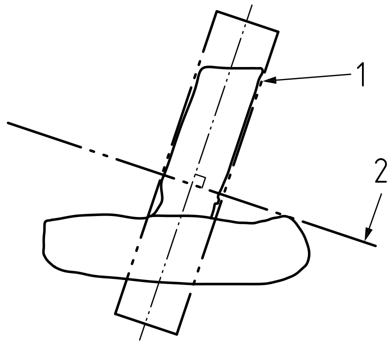
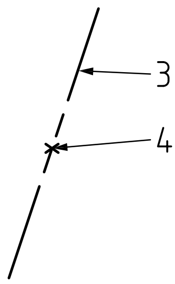
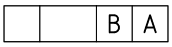
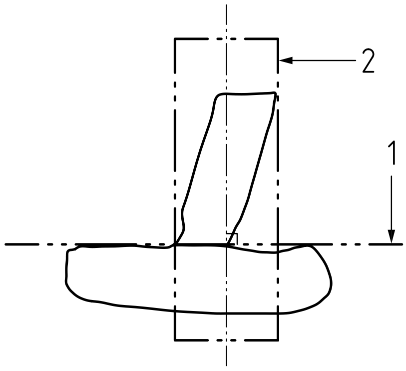
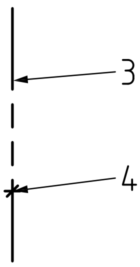
A secondary datum in a datum system shall, by default, lock all degrees of freedom that it is able to lock, and which have not already been locked by the primary datum.

A tertiary datum in a datum system shall, by default, lock all degrees of freedom that it is able to lock, and which have not already been locked by the primary and secondary datum.

*NOTE 1 The effect of datums can be modified with the "orientation-only" symbol (see **8.9.1**), by identifying specific situation features or controlling individual degrees of freedom.*

NOTE 2 Table 11 and Table 12 give examples of datum systems.

Table 11 — *Examples of datum systems 1*

Indication of datum feature	Indication of datum in tolerance frame	Illustration of the meaning on workpiece	Resulting common datum or datum system	Degrees of freedom controlled
				Five degrees of freedom locked. Rotation about the axis (item 3) not locked.
				

Key

1 First associated feature without orientation constraint

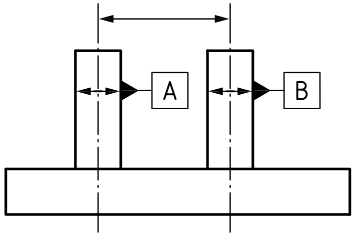
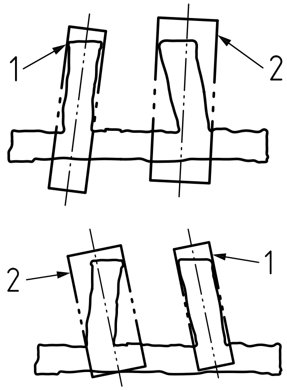
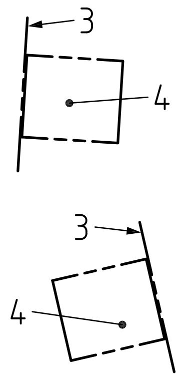
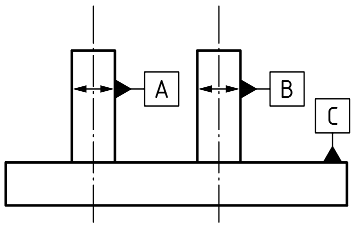
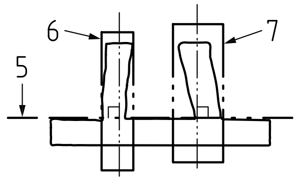
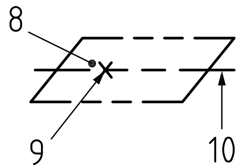
2 Second associated feature with orientation constraint from the first associated feature

3 Straight line which is the situation feature of the associated cylinder (its axis)

4 Point of intersection between the straight line and the plane

NOTE Association for single datums is described in BS EN ISO 5459:2024, Annex A.

Table 12 — *Examples of datum systems 2*

Indication of datum feature	Indication of datum in tolerance frame	Illustration of the meaning on workpiece	Resulting common datum or datum system	Degrees of freedom that can be locked
	A B B A			Five degrees of freedom locked. Translation along the axis (item 3) not locked.
	C A B			All six degrees of freedom locked.

Key

1 First associated cylinder without constraint

2 Second associated cylinder with parallelism constraint from the first associated feature

3 Straight line which is the axis of the first associated cylinder

4 Plane including the axes of the two associated cylinders

5 First associated feature without a constraint

6 Second associated feature with a perpendicularity constraint from the first associated feature

7 Third associated feature with a perpendicularity constraint from the first associated feature (and parallelism constraint from the second feature)

8 Plane which is the first associated feature

9 Point of intersection between the plane and the axis of the second associated feature

10 Straight line which is the intersection between the associated plane and the plane containing the two axes

NOTE 1 The orientation and location of datums are different depending on the datum indication in the tolerance frame. Not all possibilities for establishing the datums are covered.

NOTE 2 Association for single datums is described in BS EN ISO 5459:2024, Annex A.

8.8 Datum targets

COMMENTARY ON ***8.8***

For more information on datums, see Annex A.

8.8.1 General

A datum target shall be used in the following instances:

a) when a datum is based on part of a datum feature, to identify the portion of the datum feature to be used; and

b) to represent a contacting feature with the same nominal geometry.

Where applicable, a datum target shall take the form of:

1) a point;

2) a line; or

3) an area.

A datum target shall be indicated by a datum target indicator constructed from:

i) a datum target frame;

ii) a datum target symbol; and

iii) a leader line linking the two symbols (directly or through a reference line).

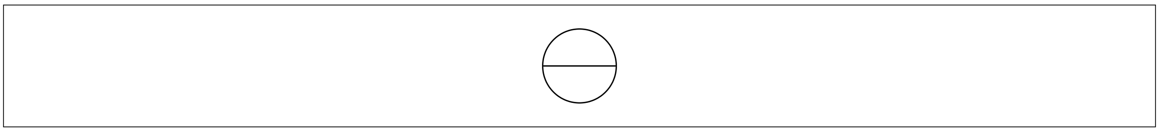
Where necessary, the same datum target shall be indicated on several appropriate views in order to provide an unambiguous definition (see Figure 115 to Figure 118).

NOTE A geometrical specification with a datum reference which is defined by datum targets may be applied to the surface or feature on which those datum targets are identified without creating a circular reference.

8.8.2 Datum target frame

The datum target frame shall be a circle, divided into two compartments by a horizontal line (see Figure 105). The lower compartment shall be reserved for the datum feature identifier, followed by a digit (from 1 to *n*) corresponding to the datum target number. The upper compartment shall be reserved for additional information, such as dimensions of the target area.

Figure 105 — *Single datum target frame*

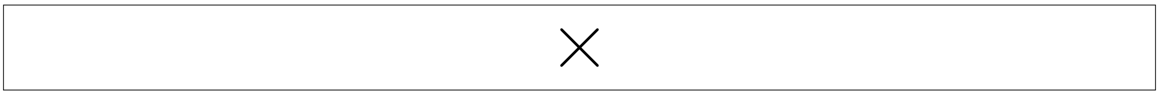


8.8.3 Datum target symbol

NOTE 1 The datum target symbol indicates the type of datum target.

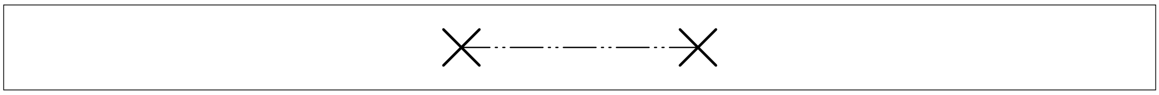
A datum target point shall be identified with a cross (see Figure 106).

Figure 106 — *Datum target point*



A datum target line shall be indicated by a long-dashed double-dotted narrow line (line type 05.1 as given in BS EN ISO 128-2:2022). If the line is open, each end shall be terminated by a cross (see Figure 107). Where applicable, this line shall be straight, circular or a line of any shape.

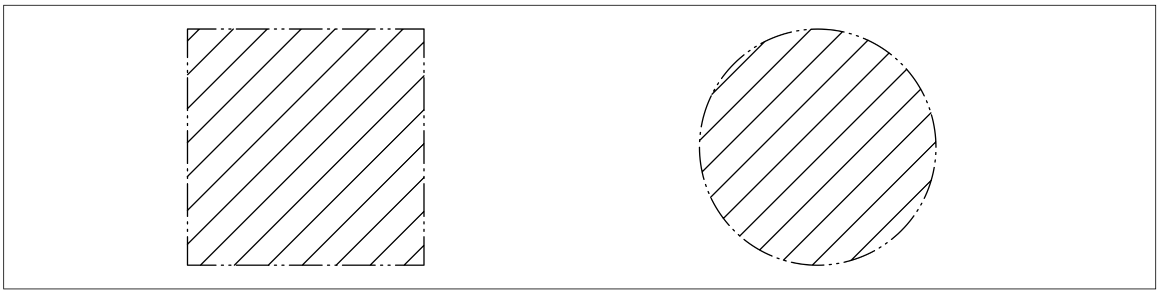
Figure 107 — *Non-closed datum target line*



A datum target area shall be indicated with an outline using a long-dashed double-dotted narrow line (line type 05.1 as given in BS EN ISO 128-2:2022), and cross-hatched (see Figure 108).

*NOTE 2 For 3D TPS a shaded surface may be used (see BS ISO 16792:2015, **10.3.4**).*

Figure 108 — *Datum target area*



8.8.4 Leader line

The datum target frame shall be connected directly, or through a reference line, to the datum target symbol by a leader line terminated with or without an arrow or a dot (see Figure 109 to Figure 111).

Figure 109 — *Indicator for single datum target point*

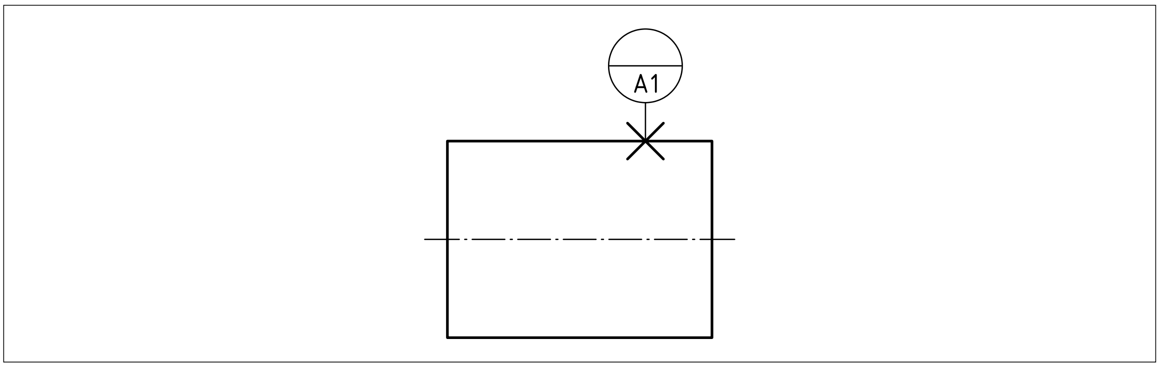


Figure 110 — *Indicator for single datum target line*

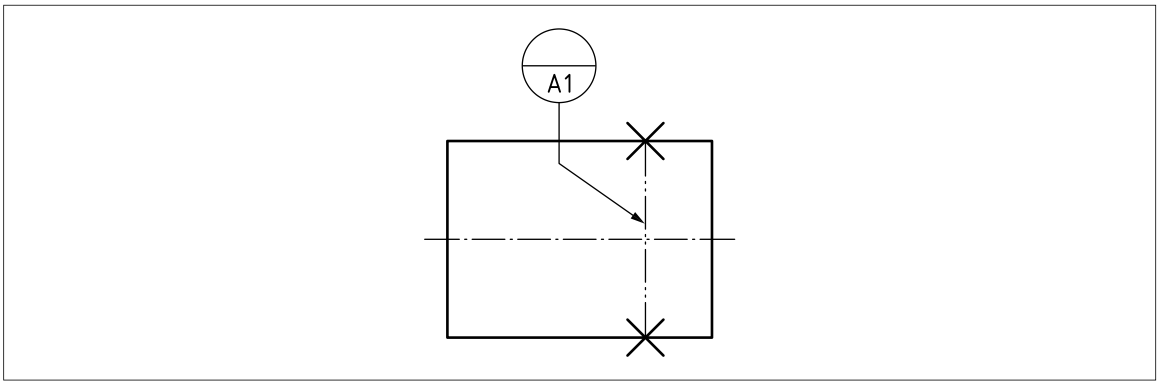
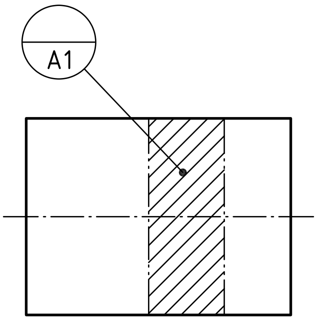


Figure 111 — *Indicator for single datum target area – 1*

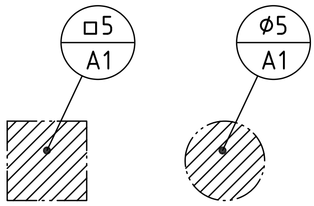


NOTE The orientation of the leader line connecting the frame with the datum target symbol is unimportant.

8.8.5 Datum target areas

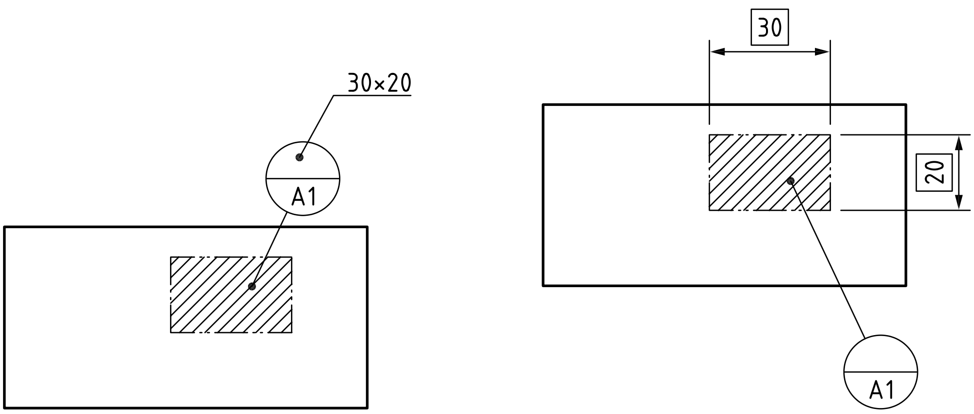
The upper compartment of the datum target frame shall only be used when defining a datum target area (see Figure 112).

Figure 112 — *Indicator for single datum target area – 2*



Where applicable, the size of the area shall, as appropriate, be indicated in the upper compartment or using dimensions elsewhere on the specification (in which case the upper compartment remains blank) (see Figure 113).

Figure 113 — *Direct and indirect indication for single datum target area*



a) Direct indication

b) Indirect indication

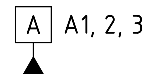
NOTE The dimensions defining the size of the datum target area are theoretically exact.

8.8.6 Datum feature indicators used with datum targets

If a single datum feature is established from one or more datum targets belonging to only one surface, the datum feature identifier identifying the surface shall be repeated close to the datum indicator, followed by the list of numbers (separated by commas) identifying the targets (see Figure 114). Each individual datum target shall be identified by a datum target indicator indicating the datum feature identifier, the number of the datum target and, if applicable, the dimensions of the datum target.

NOTE 1 The list of datum targets can be placed anywhere adjacent to the datum feature identifier.

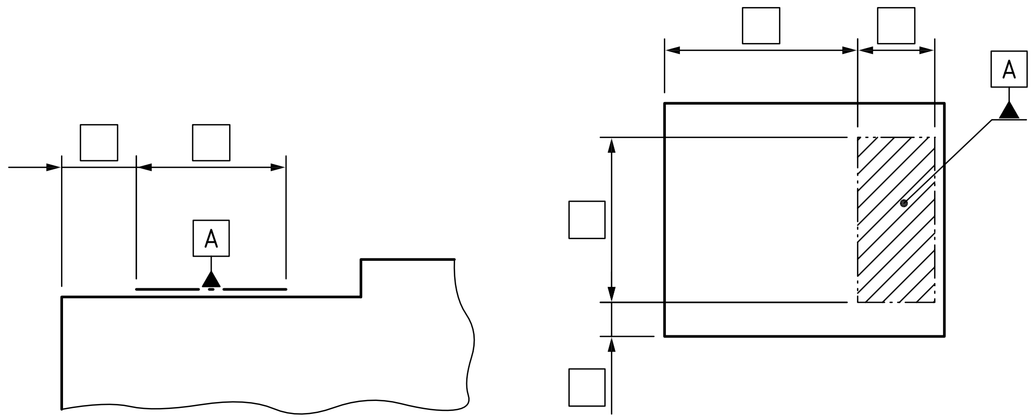
Figure 114 — *Indication of datums established from datum targets*



NOTE 2 In the case of a large number of datum targets (e.g. when defining a datum for a large, flexible panel), an abbreviated indication, such as "A1-20", is acceptable.

NOTE 3 If there is only one datum target, the drawing indication may be simplified by placing the datum indicator as shown in Figure 115 and Figure 116.

Figure 115 — *Simplification of drawing indication when there is only one datum target area*



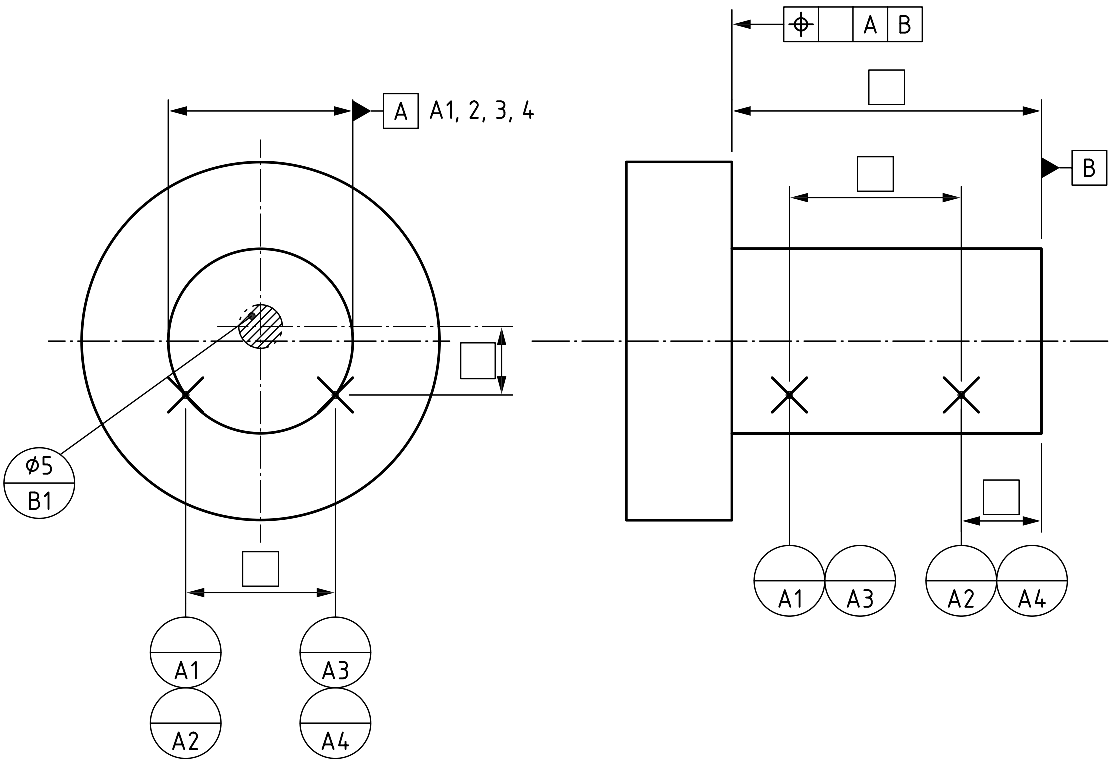
a) Restricted datum target area

b) Restricted datum target area

8.8.7 Datums based on more than one datum target

If a single datum is based on more than one datum target, the relationships between the datum targets shall be defined with theoretically exact dimensions as necessary (see Figure 116).

Figure 116 — *Datums based on more than one datum target*



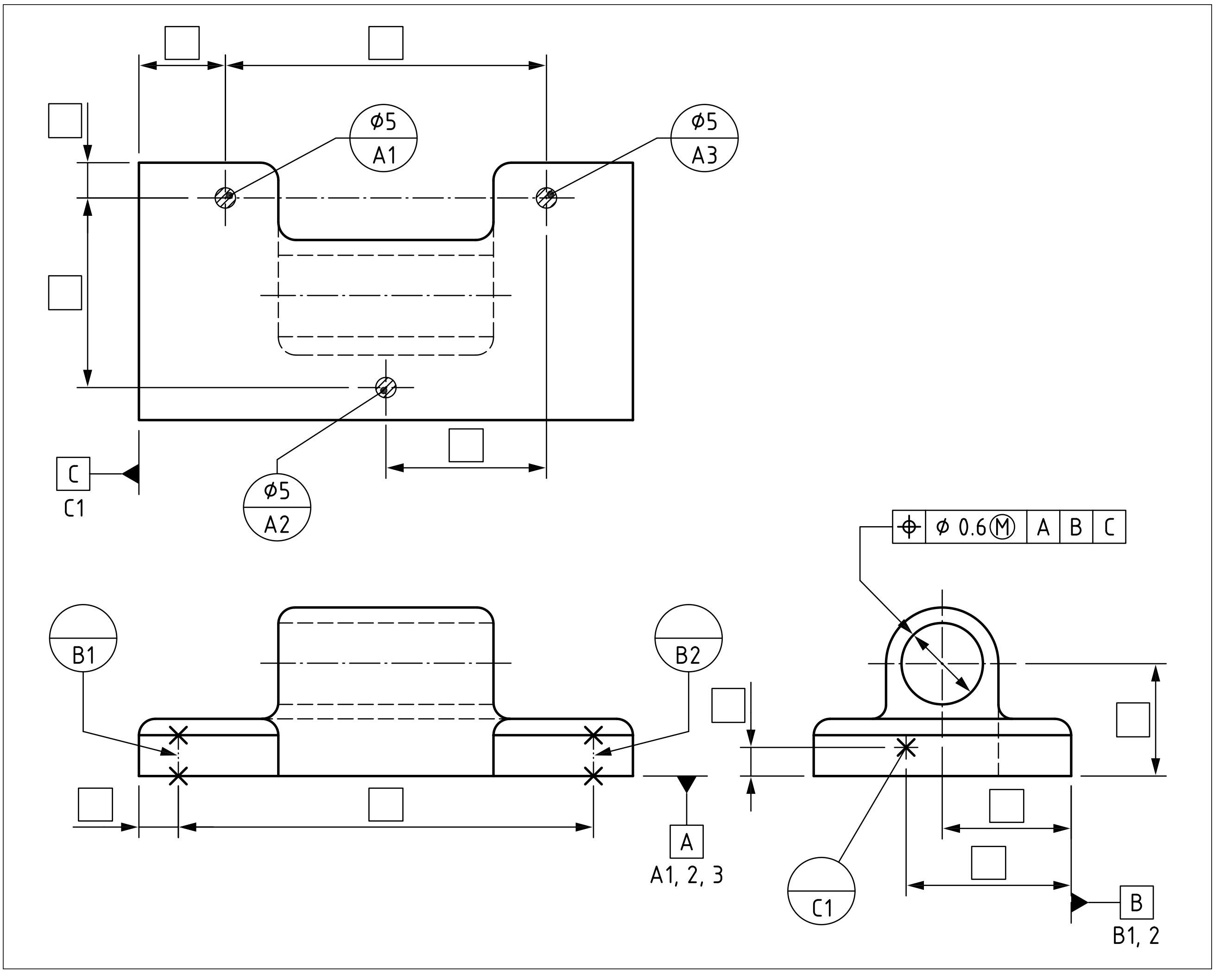
NOTE In this case the TEDs can be used without any corresponding geometrical specification. The tolerance on the location of the datum targets is not part of the definition of the workpiece geometry, so does not need to be defined on the design specification.

When datum targets are used to define datums on a rigid workpiece (see Figure 117), the primary datum shall normally be defined with three datum targets, the secondary datum with two datum targets, and the tertiary datum with one datum target, except where the geometry of the datum feature necessitates the use of more datum targets in order to provide a unique solution (see Figure 116).

If a primary datum plane is defined with more than three datum targets, a statement shall be included in the TPS to explain how the datum is to be constructed, e.g.:

DATUM A IS A LEAST-SQUARES PLANE CONSTRUCTED FROM DATUM TARGETS A1-8.

Figure 117 — *Application of datum targets*



8.9 Modifying the effects of datums

*COMMENTARY ON **8.9***

The effect of a datum can be modified in several ways.

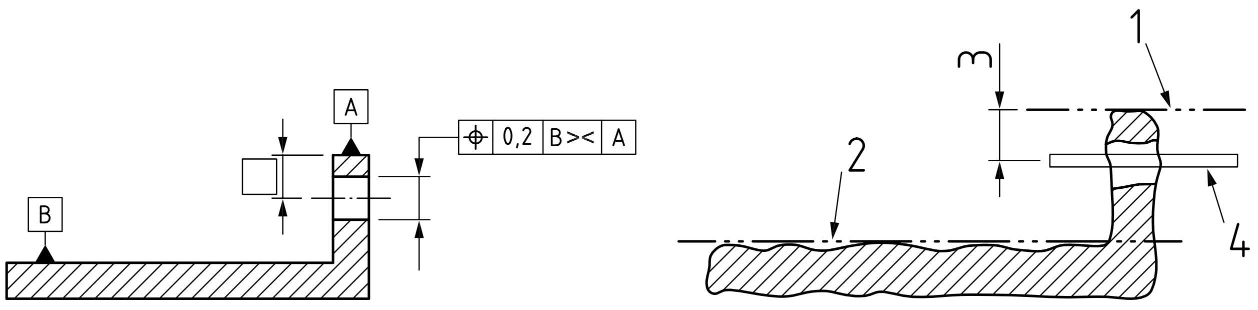
8.9.1 "Orientation-only" modifier

Where an individual datum controls some translational degrees of freedom, but the requirement is for only rotational degrees of freedom to be controlled, the "orientation-only" modifier shall be used in the datum section of an indication (see Figure 118).

The "orientation-only" modifier shall consist of two angle brackets used in the following arrangement, "><".

If the datum reference only controls the orientation of the tolerance zone in any case, the "><" modifier is redundant and shall be omitted.

Figure 118 — *Application of "orientation-only" modifier*



a) Indication

b) Meaning

Key

1 Associated plane with outside material constraint and with orientation constraint from datum B

2 Associated plane with outside material constraint (datum B)

3 Distance relative to the location

4 Tolerance zone with orientation constraint from datum B and location constraint from datum A

NOTE The modifier >< allows only the orientation degrees of freedom of the tolerance zone, and not its translation degree of freedom (managed by the datum A in this case), to be locked from the datum B.

8.9.2 Situation feature modifiers

NOTE 1 A datum consists of a set of one, two or three situation features.

Situation feature modifiers shall be used when a datum section references a subset of these situation features.

Individual situation features shall be identified using Table 13.

Table 13 — *Individual situation features*

Situation feature	**Symbol**
Plane	[PL]
Straight line	[SL]
Point	[PT]

The situation feature modifier shall be placed immediately after the datum feature identifier with no space between them (see Figure 119).

NOTE 2 Where situation feature modifiers are used, more than three datums might be required to restrain all six degrees of freedom.

Figure 119 — *Indication of which modifier is needed in the set of situation features*

| B[PL] |

a) The plane

| B[SL] |

b) The straight line

| B[PT] |

c) The point

| B[PL][SL] |

d) The plane and the straight line

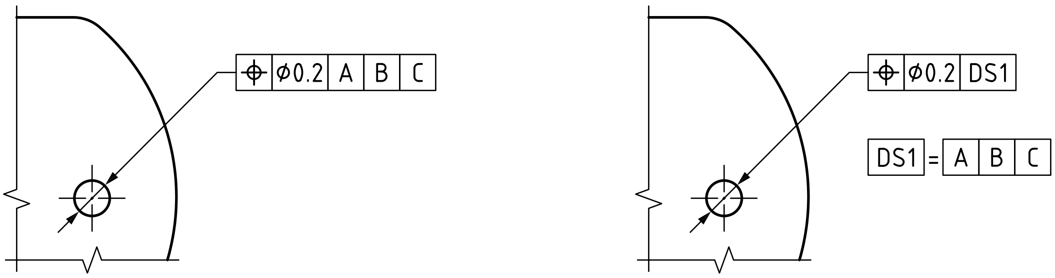
NOTE 1 More than one of these modifiers can be used with a single datum reference.

*NOTE 2 The same symbols are used when labelling individual situation features in a TPS (see Clause **5**).*

"NOTE 3 A datum section indicating one or more datums (single datum or common datum) may be indirectly indicated by using a datum system identifier in only one datum indicator of a datum section. The datum system identifier consists of the symbol DS, optionally followed, without space, by a positive integer n.

The datum system, DS*n*, shall be defined in or near the title block as follows: a datum section, consisting of one datum indicator (one compartment), containing the datum system identifier, followed by an equal sign, followed by a datum section (see examples in Figure 120)."

Figure 120 — *Examples of direct and indirect indications of a datum section*



a) Direct indication of a datum section

b) Indirect indication of a datum section by using a datum system identifier

[SOURCE: BS EN ISO 5459:2024, **7.4.2.11**]

9 Geometrical specifications

9.1 General

Geometrical tolerancing shall conform to the following standards, as appropriate.

- BS EN ISO 1101:2017, *Geometrical product specifications (GPS) – Geometrical tolerancing – Tolerances of form, orientation, location and run out.*
- BS EN ISO 1660, *Geometrical product specifications (GPS) – Geometrical tolerancing – Profile tolerancing.*
- BS EN ISO 2692, *Geometrical product specifications (GPS) – Geometrical tolerancing – Maximum material requirement (MMR), least material requirement (LMR) and reciprocity requirement (RPR).*
- BS EN ISO 5458, *Geometrical product specifications (GPS) – Geometrical tolerancing – Pattern and combined geometrical specification.*
- BS EN ISO 5459, *Geometrical product specifications (GPS) – Geometrical tolerancing – Datums and datum systems.*
- BS EN ISO 7083, *Technical product documentation – Symbols used in technical product documentation – Proportions and dimensions.*
- BS EN ISO 8015, *Geometrical product specifications (GPS) – Fundamentals – Concepts, principles and rules.*
- BS EN ISO 10579, *Geometrical product specifications (GPS) – Dimensioning and tolerancing – Non-rigid parts.*
- BS EN ISO 12180-1, *Geometrical product specifications (GPS) – Cylindricity – Part 1: Vocabulary and parameters of cylindrical form.*
- BS EN ISO 12180-2, *Geometrical product specifications (GPS) – Cylindricity – Part 2: Specification operators.*

- BS EN ISO 12181-1, *Geometrical product specifications (GPS) – Roundness – Part 1: Vocabulary and parameters of roundness*.
- BS EN ISO 12181-2, *Geometrical product specifications (GPS) – Roundness – Part 2: Specification operators*.
- BS EN ISO 12780-1, *Geometrical product specifications (GPS) – Straightness – Part 1: Vocabulary and parameters of straightness*.
- BS EN ISO 12780-2, *Geometrical product specifications (GPS) – Straightness – Part 2: Specification operators*.
- BS EN ISO 12781-1, *Geometrical product specifications (GPS) – Flatness – Part 1: Vocabulary and parameters of flatness*.
- BS EN ISO 12781-2, *Geometrical product specifications (GPS) – Flatness – Part 2: Specification operators*.
- BS EN ISO 16610 (all parts), *Geometrical product specifications (GPS) – Filtration*.
- BS EN ISO 17450-1, *Geometrical product specifications (GPS) – Basic concepts – Part 1: Model for geometrical specification and verification*.
- BS EN ISO 17450-2, *Geometrical product specifications (GPS) – Basic concepts – Part 2: Basic tenets, specifications, operators, uncertainties and ambiguities*.
- BS EN ISO 17450-3, *Geometrical product specifications (GPS) – Basic concepts – Part 3: Toleranced features*.
- BS EN ISO 17450-4, *Geometrical product specifications (GPS) – Basic concepts – Part 4: Geometrical characteristics for quantifying GPS deviations*.
- BS EN ISO 22432, *Geometrical product specifications (GPS) – Features utilized in specification and verification*.
- BS EN ISO 25378, *Geometrical product specifications (GPS) – Characteristics and conditions – Definitions*.

9.2 Basic concepts

"A geometrical specification shall be applied to a feature to define the tolerance zone within which that feature is to be contained (see **9.6**).

The tolerance shall apply to the whole extent of the feature, unless otherwise specified as in **9.9** and **9.14**.

Unless a more restrictive indication is specified, for example by an additional tolerance or explanatory note, the toleranced feature shall be of any form within this tolerance zone."

[SOURCE: BS EN ISO 1101:2017, **4.5**, modified – "required" replaced with "specified"; "additional tolerance or" added, "may" became "shall", "orientation and/or location" removed]

"NOTE 1 A feature is a specific portion of the workpiece, such as a point, a line or a surface; these features can be integral features (e.g. the external surface of a cylinder) or derived features (e.g. a median line or median surface) (see BS EN ISO 14660-1)."

[SOURCE: BS EN ISO 1101:2017, **4.3**, modified – reference to ISO 17450-1 replaced with reference to BS EN ISO 14660-1]

NOTE 2 Indicating geometrical specifications on a drawing does not imply the use of any particular method of production, measurement or gauging.

9.3 Symbols

Symbols for geometrical characteristics shall be as indicated in Table 14 and Table 15 and in BS EN ISO 1101:2017.

Table 14 — *Symbols for geometrical characteristics*

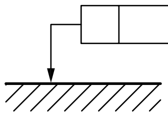
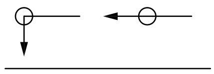
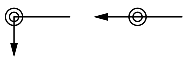
Tolerances	Characteristics	Symbol	Datum system needed
Form	Straightness	⏤	No[A)]
	Flatness	⏥	No
	Roundness	○	No[A)]
	Cylindricity	⌭	No
	Line profile	⌒	No[A)]
	Surface profile	⌓	No
Orientation	Parallelism	∥	Yes
	Perpendicularity	⊥	Yes
	Angularity	∠	Yes
	Line profile	⌒	Yes
	Surface profile	⌓	Yes
Location	Position	⌖	Yes or No
	Coaxiality/concentricity	◎	Yes
	Symmetry	⌯	Yes
	Line profile	⌒	Yes
	Surface profile	⌓	Yes
Run-out	Circular run-out	↗	Yes
	Total run-out	⌰	Yes

NOTE 1 SOURCE: BS EN ISO 1101:2017, Table 2, modified – notes removed.

NOTE 2 For the symbols location, see Figure 130.

A) In some cases form tolerances applied to line elements on a surface might require additional indications, such as intersection plane indicators, which do require datum references.

Table 15 — *Additional symbols*

Description	Symbol	Relevant clauses
Geometrical specification		Clause **9**
Theoretically exact dimensions	$\boxed{50}$	**9.9**
Projected tolerance zone	Ⓟ	**9.11.1**
Maximum material requirement (MMR)	Ⓜ	**9.10**
Least material requirement (LMR)	Ⓛ	**9.10**
Reciprocity requirement	Ⓡ	**9.10.7**
Free state condition (non-rigid parts)	Ⓕ	**9.11.2**
All-around		**10.2**
All-over		**10.3**
Between	←→	**10.1**
Combined zone	CZ	**10.5**
Separate zone	SZ	**10.5**
Unequal zone	UZ	**9.12**
United feature	UF	**10.4**
Minor diameter	LD	**9.11.4**
Major diameter	MD	**9.11.3**
Pitch diameter	PD	**9.11.4**
Any cross-section	ACS	See BS EN ISO 1101:2017
Simultaneous requirement	SIM	**10.5**
Offset zone	OZ	See BS EN ISO 1101:2017

NOTE 1 Further symbols and modifiers are standardized in BS EN ISO 1101:2017.

NOTE 2 SOURCE: BS EN ISO 1101:2017, Table 3 and Table 4.

NOTE Several specification modifiers, e.g. Ⓟ, Ⓜ, Ⓛ, Ⓕ, and CZ can be used simultaneously in the same tolerance frame (see Figure 121).

Figure 121 — *Use of several specification modifiers*

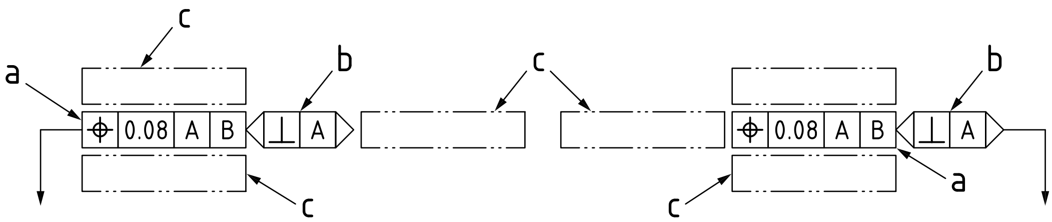
⌖	⌀0,1CZⒻ	AⓂ

9.4 Geometrical specification indication

"A geometrical specification indication consists of a tolerance indicator, optional plane and feature indications and optional adjacent indications (see Figure 122)."

[SOURCE: BS EN ISO 1101:2017, **8.1**, modified – figure number changed]

Figure 122 — *Elements of a geometrical specification indication*



Key

a Tolerance indicator

b Plane and feature indicator

c Adjacent indications

NOTE SOURCE: BS EN ISO 1101:2017, Figure 11.

The geometrical specification indication shall be attached to the relevant feature(s) with a leader line attached to either end of the indication.

9.5 Tolerance indicator

9.5.1 General

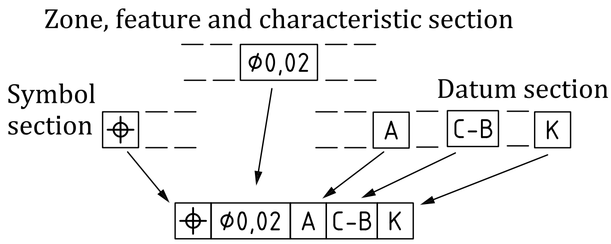
The requirements shall be shown in a rectangular frame, divided into two or more compartments, in accordance with BS EN ISO 1101:2017 (see Figure 123). The frame shall contain the following sections in the following order:

a) the symbol section identifying the geometrical characteristic;

b) the tolerance zone section defining the shape and size of the tolerance zone; and

c) if applicable, a datum section containing the letter or letters identifying the datum system.

Figure 123 — *Three sections of the tolerance indicator*



NOTE 1 The datum section can have one or more compartments.

NOTE 2 SOURCE: BS EN ISO 1101:2017, Figure 12.

9.5.2 Symbol section

The symbol section shall contain one of the symbols given in Table 14. This section shall always be present in a tolerance indicator. No additional modifiers shall be used in this section.

9.5.3 Tolerance zone section

The tolerance zone section shall contain the tolerance value in the unit used for linear dimensions. This section shall always be present in a tolerance indicator.

The tolerance value shall be preceded by the symbol "Ø" if the tolerance zone is circular or cylindrical or by "SØ" if the tolerance zone is spherical, and potentially followed by other modifiers such as CZ, SZ and Ⓜ.

Any modifier following the tolerance value shall be separated from the tolerance value and any other modifiers by a single space, except for modifiers consisting of letters contained in a circle, in which case no space shall be used.

9.5.4 Datum section

The datum section shall contain one or more datum feature identifiers. This section shall be present in the tolerance indicator when location, orientation and run-out requirements are defined.

9.6 Tolerance zones

The width of the tolerance zone shall apply to the specified geometry (see Figure 124), unless otherwise indicated with a direction feature, with the exception of roundness (see **10.6.4** for details on the direction feature and on roundness).

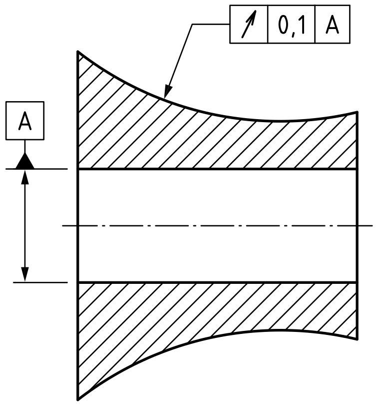
NOTE The orientation of the leader line alone does not influence the definition of the tolerance.

"According to the characteristic to be toleranced and the manner in which it is dimensioned, the tolerance zone shall be one of the following:

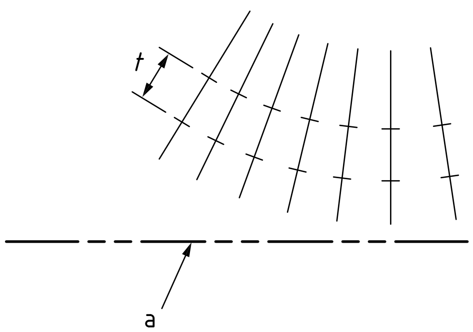
- the space within a circle;
- the space between two coaxial circles;
- the space between additional complex zones as defined in BS EN ISO 1101:2017;
- the space between two equidistant lines or two parallel straight lines;
- the space within a cylinder;
- the space between two coaxial cylinders;
- the space between two equidistant surfaces or two parallel planes; or
- the space within a sphere."

[SOURCE: BS EN ISO 1101:2017, **4.4**, modified – "concentric" changed to "coaxial", third bullet refers to BS EN ISO 1101:2017, some bullets have been omitted]

Figure 124 — *Width of tolerance zone applying to the specified geometry*



a) Drawing indication



b) Interpretation

9.7 Toleranced features

*COMMENTARY ON **9.7***

Full explanation of toleranced features, including the definitions of extracted integral features and extracted derived features, can be found in BS EN ISO 17450-3.

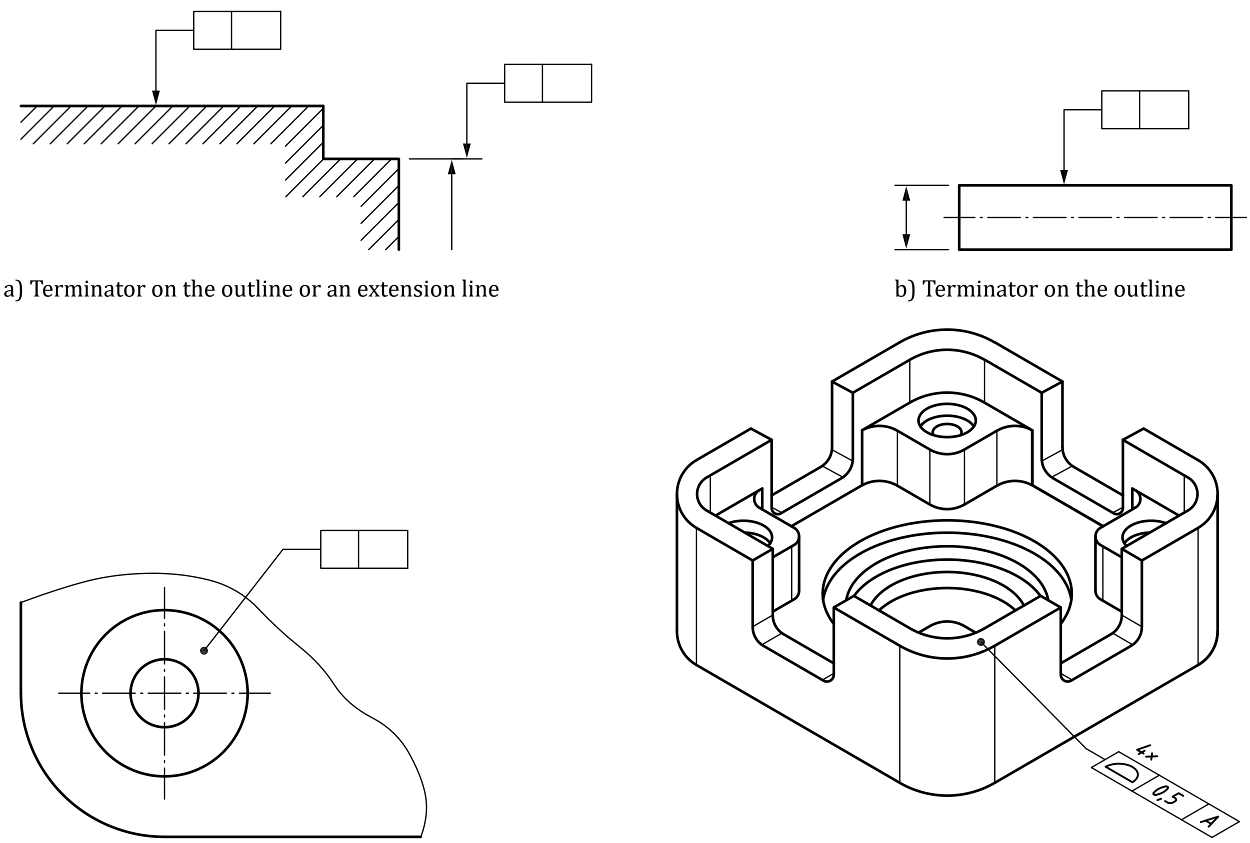
The tolerance frame shall be connected to the toleranced feature by a leader line starting from either side of the frame and ending with a terminator (arrowhead or filled dot) in one of the following ways:

a) on the outline of the feature or an extension of the outline (but clearly separated from the dimension line), with a leader line terminating in an arrowhead, when the tolerance refers to the line or surface itself [see Figure 125a) and Figure 125b)];

b) on the surface of a feature, with a leader line ending in a filled dot on the surface, when the tolerance refers to that surface [see Figure 125c) and Figure 125d)]; or

c) as an extension of the dimension line when the tolerance refers to the median line or median surface, or use of the symbol Ⓐ to identify that the toleranced feature is a derived median line of a cylindrical feature, or the median point if the feature is a sphere or circle (see Figure 126).

Figure 125 — *Terminator on the outline or the surface of a feature*



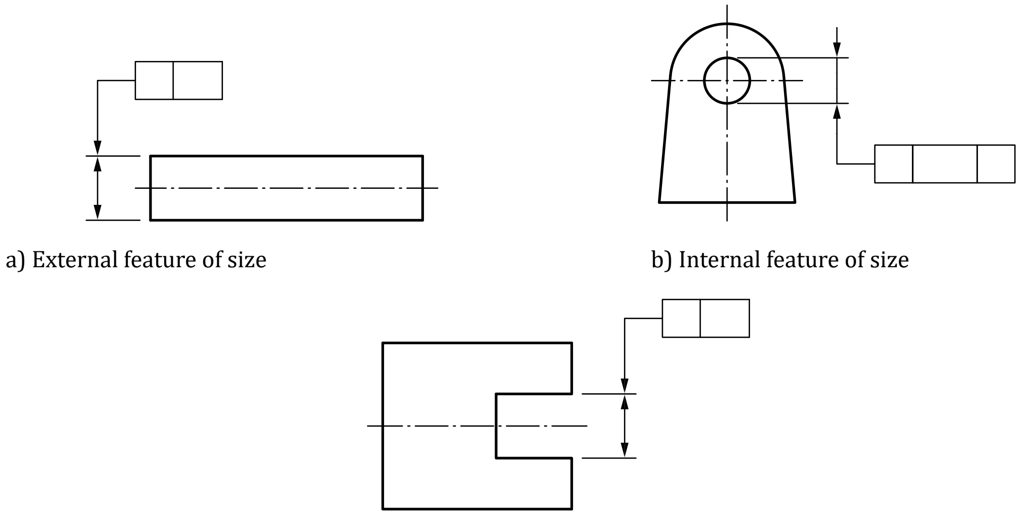
a) Terminator on the outline or an extension line

b) Terminator on the outline

c) Terminator on a leader line connected to the surface

d) Terminator on a leader line connected to the surface

Figure 126 — *Terminator as an extension of the dimension line*



a) External feature of size

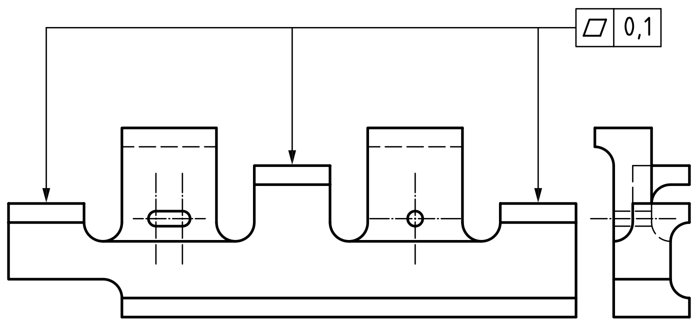
b) Internal feature of size

c) Internal feature of size

When a tolerance applies to more than one feature this shall be indicated by one of the following methods:

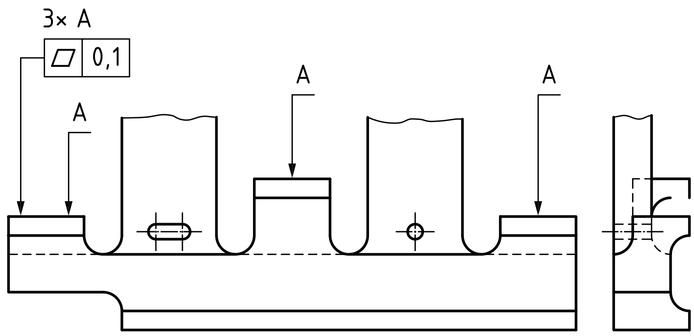
- separate leader lines going to each of the features (see Figure 127);

Figure 127 — *Specification applying to several features individually – 1*



- indicating *n*x above the tolerance indicator, where *n* is the number of features;
- in cases where the *n*x indication would not be sufficiently clear, by placing the indication *n*x A above the tolerance indicator, where *n* is the number of features and A is a letter used to identify each instance (see Figure 128);

Figure 128 — *Specification applying to several features individually – 2*



- use of the between symbol (see **10.1**);
- use of the all-around symbol (see **10.2**); or
- use of the all-over symbol (see **10.3**).

NOTE If it is necessary to specify more than one geometrical specification for a feature, the tolerance indicators may be stacked, one under the other, and attached to the feature with a single leader line (see example of Figure 129 and Figure 130).

Figure 129 — *Tolerance applying to more than one feature*

a) Flatness tolerance applied to six features

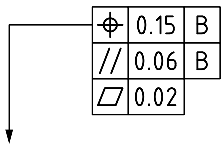
b) Position tolerance applied to six features

"If it is necessary to specify more than one geometrical characteristic for a feature, the requirements may be given in tolerance indicators one under the other for convenience (see the example in Figure 130).

In this case, it is recommended that the tolerance indicators are arranged such that the tolerance values are shown in descending order from top to bottom, as shown in Figure 130.

In this case, the reference line shall be attached to the mid-point of the left-hand end or the right-hand end of one of the tolerance indicators, depending on space, not as an extension of the line between the tolerance indicators. This applies both in 2D and 3D."

Figure 130 — *Stacked tolerance indication*



NOTE 1 When multiple tolerance indicators are applied to the same feature(s), they can be stacked up one above the other. Each tolerance indicator is read separately. Due to the principle of independency, the order of the tolerance indicators does not imply precedence.

NOTE 2 SOURCE: BS EN ISO 1101:2017, Figure 50, modified – note added.

[SOURCE: BS EN ISO 1101:2017, **8.5**, modified – figure number changed]

9.8 Application of geometrical specifications

Geometrical specifications shall be interpreted in accordance with Figure 131.

Figure 131 — *Interpretation of geometrical specifications*

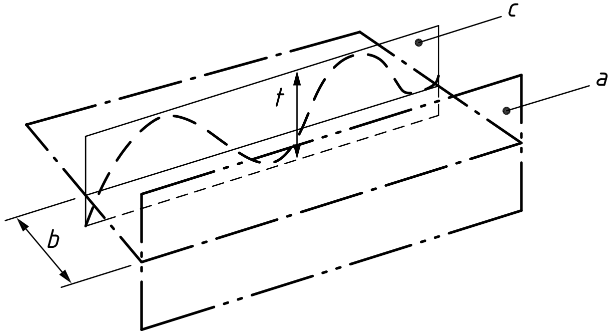
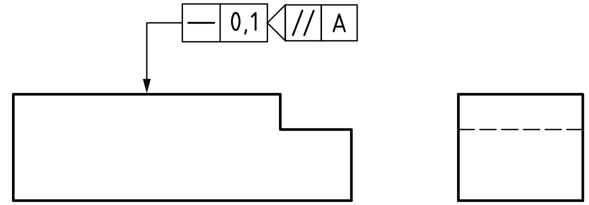
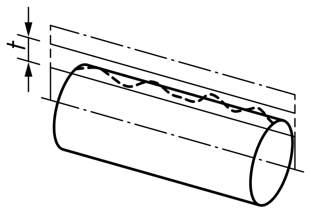
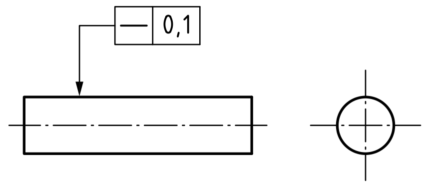
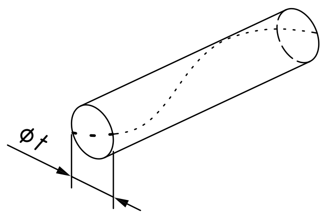
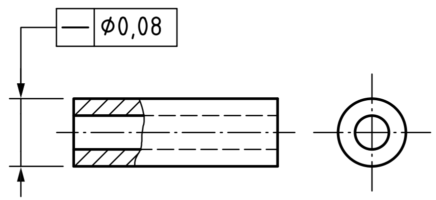
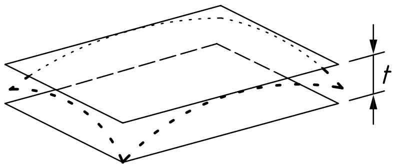
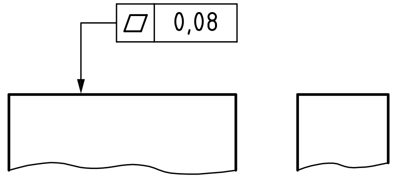
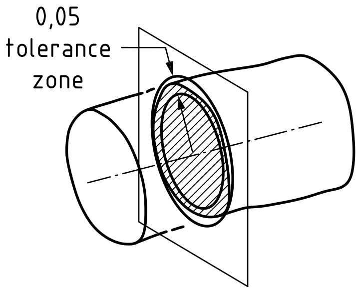
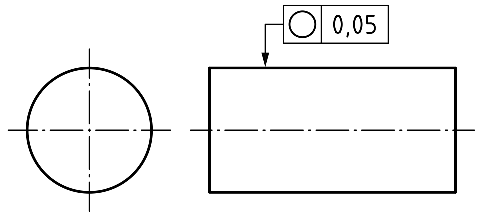
Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Straightness tolerance	
–	The tolerance zone, in the considered plane, is limited by two parallel lines a distance *t* apart, where *t* is the tolerance value, in each plane defined by the intersection plane indicator.  *NOTE a = datum A; b = any distance; c = intersection plane parallel to datum A.*	Any extracted line on the upper surface, within a plane parallel to datum A, shall be contained between two parallel straight lines 0,1 apart. 
–	The tolerance zone is limited by two parallel lines a distance *t* apart, where *t* is the tolerance value, in each plane that includes the axis of the associated cylinder. 	Any extracted line on the cylindrical surface, contained within a plane through the axis of the associated cylinder, shall be contained between two parallel lines 0,1 apart. 
–	The tolerance zone is limited by a cylinder of diameter *t*, where *t* is the tolerance value preceded by the symbol Ø. 	The extracted median line of the cylinder to which the tolerance applies shall be contained within a cylindrical zone of diameter 0,08. 

Figure 131 — *Interpretation of geometrical specifications (continued)*

<table>
<tr><th>Symbol</th><th>Definition of the tolerance zone</th><th>Indication and explanation in 2D</th></tr>
<tr><td></td><td colspan="2">Flatness tolerance</td></tr>
<tr><td>⏥</td><td>The tolerance zone is limited by two parallel planes a distance t apart.

</td><td>The extracted surface shall be contained between two parallel planes 0,08 apart.

</td></tr>
<tr><td></td><td colspan="2">Roundness tolerance</td></tr>
<tr><td>○</td><td>The tolerance zone, in any cross-section perpendicular to the axis of the associated cylinder, is limited by two concentric circles with a difference in radii of t.

</td><td>The extracted circumferential line in any cross-section of a cylindrical surface shall be contained between two coplanar concentric circles with a difference in radii of 0,05.


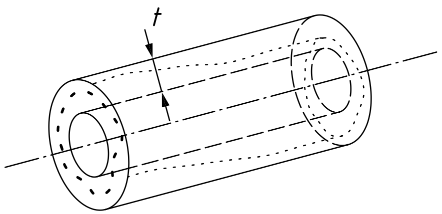
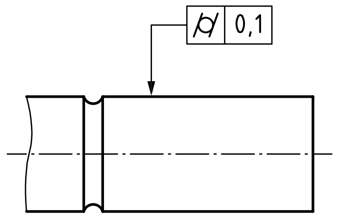
The direction in which the tolerance is applied shall be defined with a direction feature indicator, except in the case of cylindrical and spherical surfaces.</td></tr>
<tr><td></td><td colspan="2">Cylindricity tolerance</td></tr>
<tr><td>⌭</td><td>The tolerance zone is limited by two coaxial cylinders with a difference in radii of t.

</td><td>The extracted cylindrical surface shall be contained between two coaxial cylinders with a difference in radii of 0,1.

</td></tr>
</table>

Figure 131 — *Interpretation of geometrical specifications (continued)*

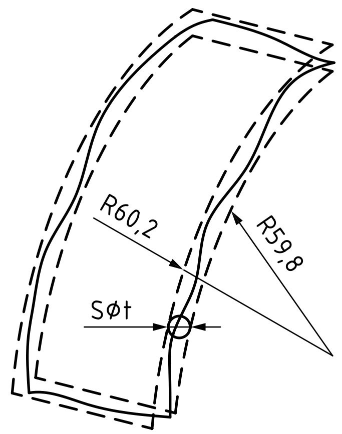
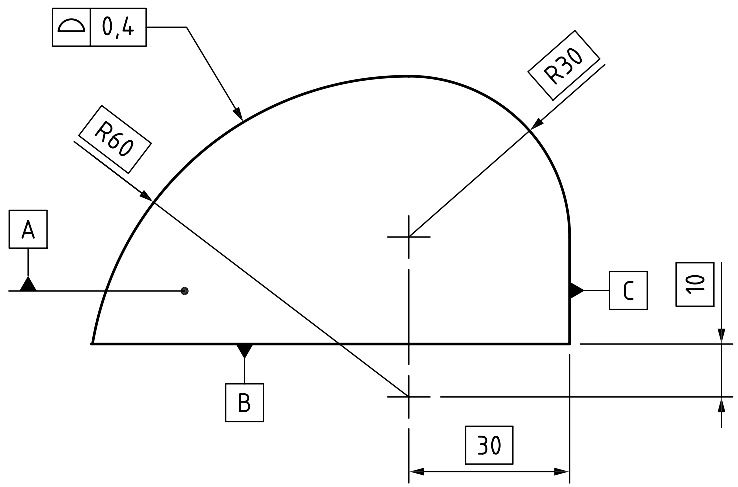
Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Surface profile with no datum system	
⌓	The tolerance zone is limited by two surfaces enveloping spheres of diameter *t*, the centres of which are situated on a surface having the theoretically exact geometrical form. 	The extracted surface shall be contained between two equidistant surfaces enveloping spheres of diameter 0,4, the centres of which are situated on a surface having the theoretically exact geometrical form. 

Figure 131 — *Interpretation of geometrical specifications (continued)*

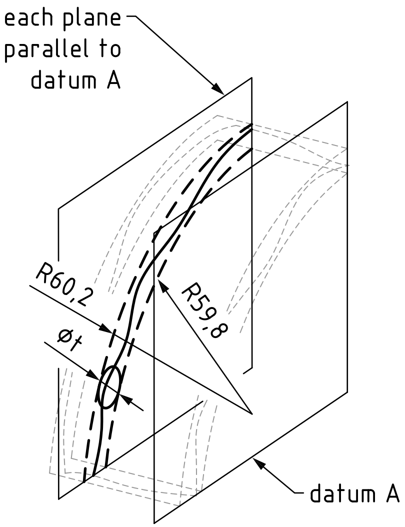
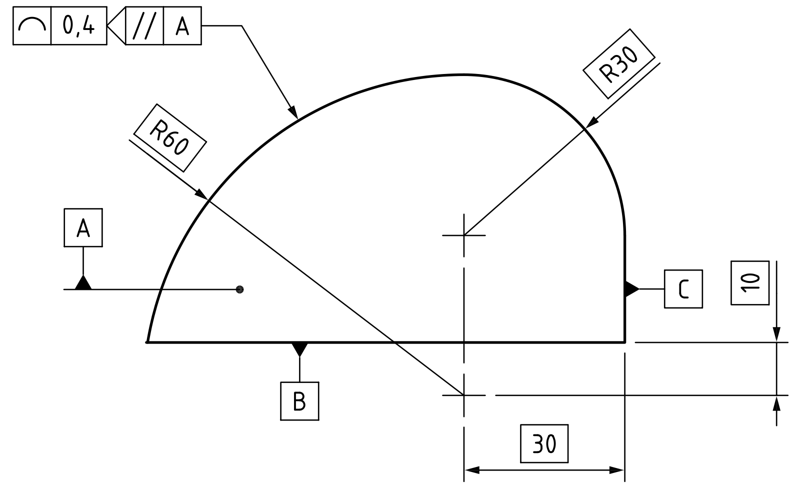
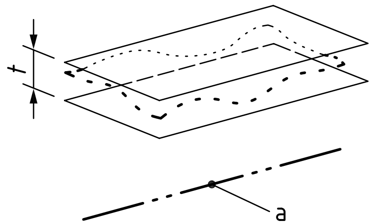
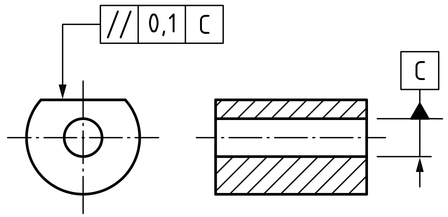
Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Line profile with no datum system	
	The tolerance zone is limited by two lines enveloping circles of diameter *t*, the centres of which are situated on a line having the theoretically exact geometrical form, within each plane defined by the intersection plane indicator. 	An extracted line on the surface within each plane parallel to datum A shall be contained between two equidistant lines enveloping circles of diameter 0,4, the centres of which are situated on a line having the theoretically exact geometrical form. 
	Parallelism tolerance of a surface related to a datum	
//	The tolerance zone is limited by two parallel planes a distance *t* apart and parallel to the datum (a = datum C). 	The extracted surface shall be contained between two parallel planes 0,1 apart, which are parallel to the datum axis C. 

Figure 131 — *Interpretation of geometrical specifications (continued)*

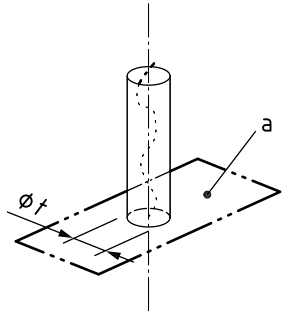
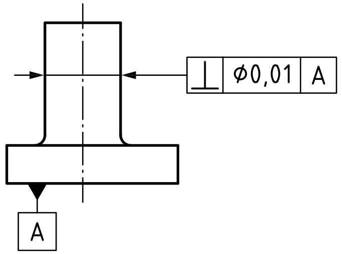
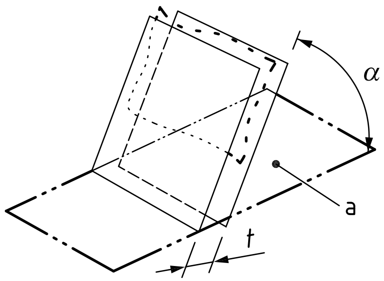
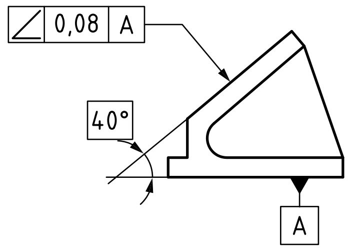
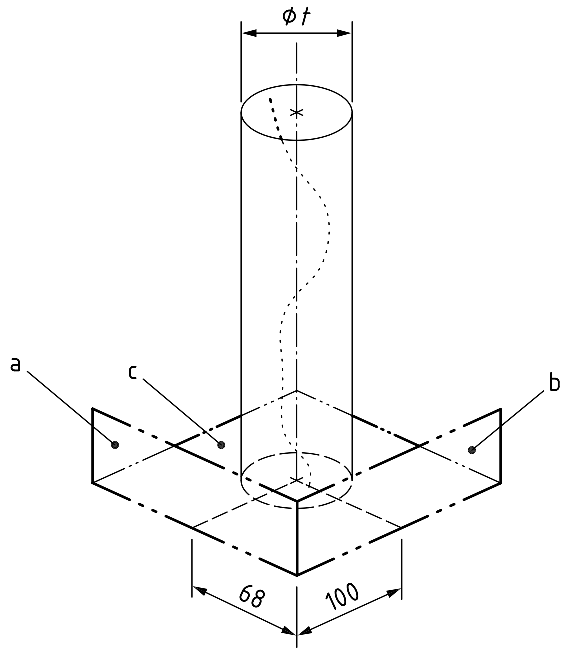
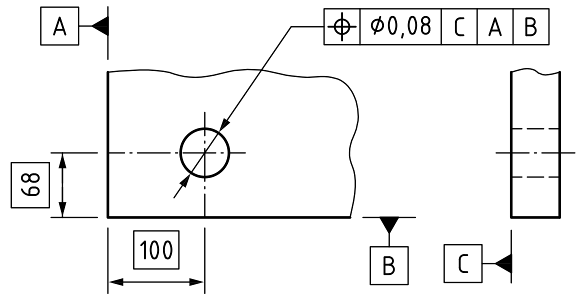
Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Perpendicularity tolerance of a feature of size related to a datum	
⊥	The tolerance zone is limited by a cylinder of diameter *t* perpendicular to the datum if the tolerance value is preceded by the symbol Ø (a = datum A). 	The extracted median line of the cylinder shall be within a cylindrical zone of diameter 0,01 perpendicular to datum plane A. 
	Angularity tolerance of a surface related to a datum	
∠	The tolerance zone is limited by two parallel planes a distance *t* apart and inclined at the specified angle to the datum (a = datum A). 	The extracted surface shall be contained between two parallel planes 0,08 apart that are inclined at a theoretically exact angle of 40° to datum plane A. 

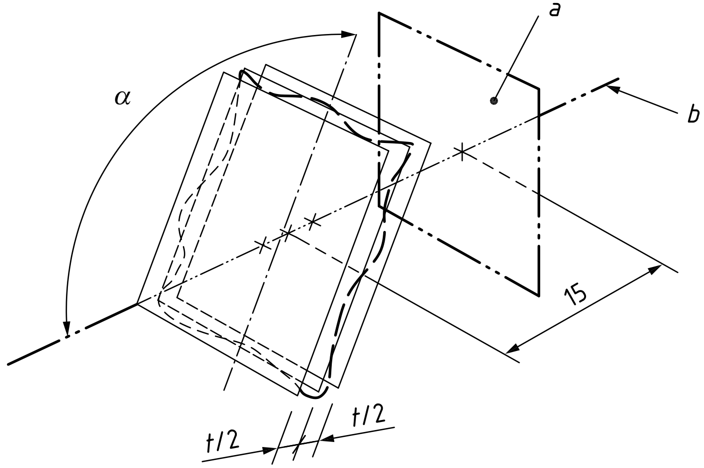
Figure 131 — *Interpretation of geometrical specifications (continued)*

<table>
<tr><th>Symbol</th><th>Definition of the tolerance zone</th><th>Indication and explanation in 2D</th></tr>
<tr><td></td><td colspan="2">Positional tolerance applied to a feature of size</td></tr>
<tr><td>⌖</td><td>The tolerance zone is limited by a cylinder of diameter t if the tolerance value is preceded by the symbol Ø. The axis of the tolerance cylinder shall be fixed by explicit and implicit TEDs (see 9.9) with respect to the datum system.


NOTE a = datum A; b = datum B; c = datum C.</td><td>The extracted median line shall be within a cylindrical zone of diameter 0,08, the axis of which coincides with the theoretically exact location and orientation of the considered hole, with respect to datum planes C, A and B.


NOTE The orientation is defined with an implied 90° TED with respect to datum C.</td></tr>
</table>

Figure 131 — *Interpretation of geometrical specifications (continued)*

<table>
<tr><th>Symbol</th><th>Definition of the tolerance zone</th><th>Indication and explanation in 2D</th></tr>
<tr><td></td><td colspan="2">Positional tolerance applied to a planar surface</td></tr>
<tr><td>⌖</td><td>The tolerance zone consists of two parallel planes separated by the tolerance value t. The centre plane of the tolerance zone is fixed by explicit and implicit (see 9.9) TEDs with respect to the datum system.


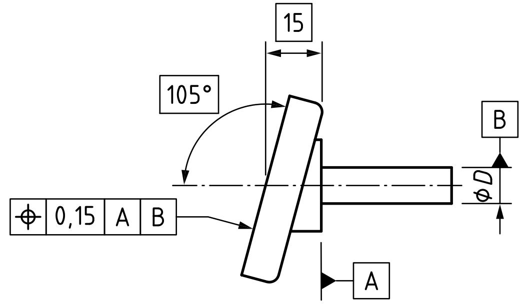
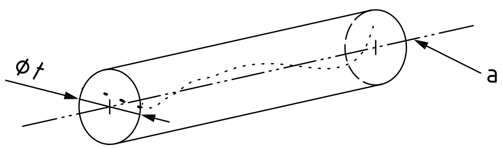
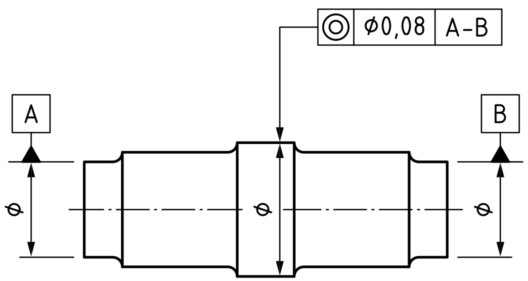
NOTE a = datum A; b = datum B.</td><td>The extracted median surface shall be contained within a tolerance zone of width 0,05, the centre plane of which coincides with the theoretically exact location and orientation of the considered feature, with respect to datum planes A and B.

</td></tr>
<tr><td></td><td colspan="2">Coaxiality tolerance</td></tr>
<tr><td>◎</td><td>The tolerance zone is limited by a cylinder of diameter t; the tolerance value shall be preceded by the symbol Ø. The axis of the cylindrical tolerance zone coincides with the datum (a = datum A – B).

</td><td>The extracted median line of the toleranced cylinder shall be within a cylindrical zone of diameter 0,08, the axis of which is the common datum straight line A – B.

</td></tr>
</table>

Figure 131 — *Interpretation of geometrical specifications (continued)*

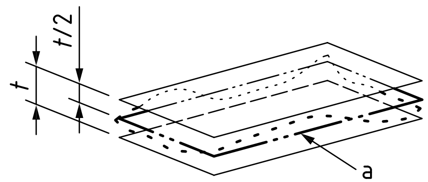
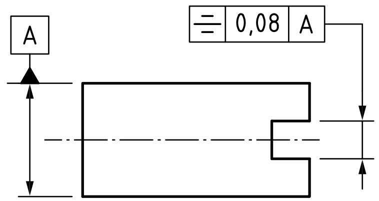
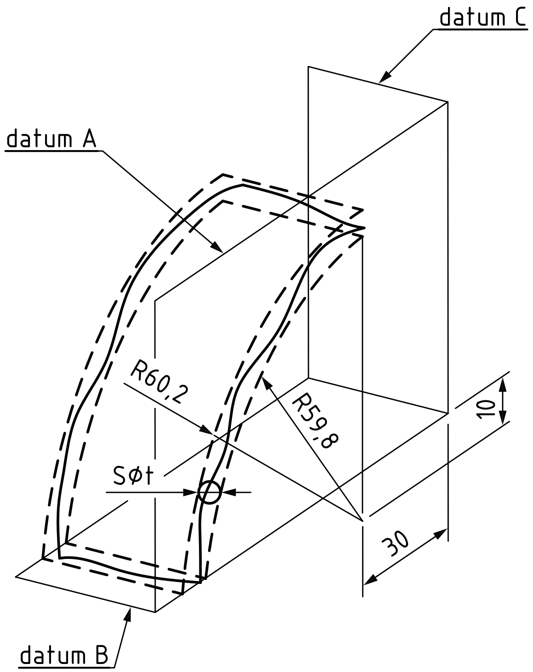
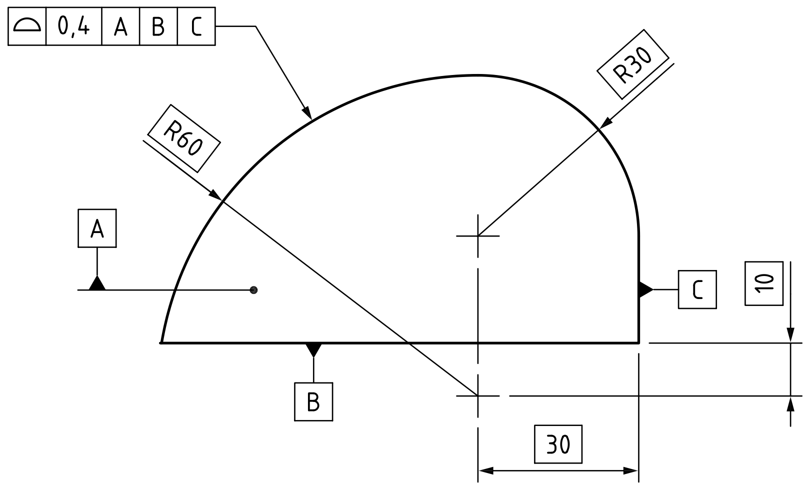
<table>
<tr><th>Symbol</th><th>Definition of the tolerance zone</th><th>Indication and explanation in 2D</th></tr>
<tr><td>⌯</td><td>Symmetry tolerance of a feature of size
The tolerance zone is limited by two parallel planes a distance t apart, symmetrically disposed about the median plane, with respect to the datum (a = datum).
</td><td>The extracted median surface shall be contained between two parallel planes 0,08 apart, which are symmetrically disposed about the datum median plane A.
</td></tr>
<tr><td>⌓</td><td>Surface profile with a datum system
The tolerance zone is limited by two surfaces enveloping spheres of diameter t, the centres of which are situated on a surface having the theoretically exact geometrical form with respect to the datum system.
</td><td>The extracted surface shall be contained between two equidistant surfaces enveloping spheres of diameter 0,4, the centres of which are situated on a surface having the theoretically exact geometrical form with respect to datum system A|B|C.
</td></tr>
</table>

Figure 131 — *Interpretation of geometrical specifications (continued)*

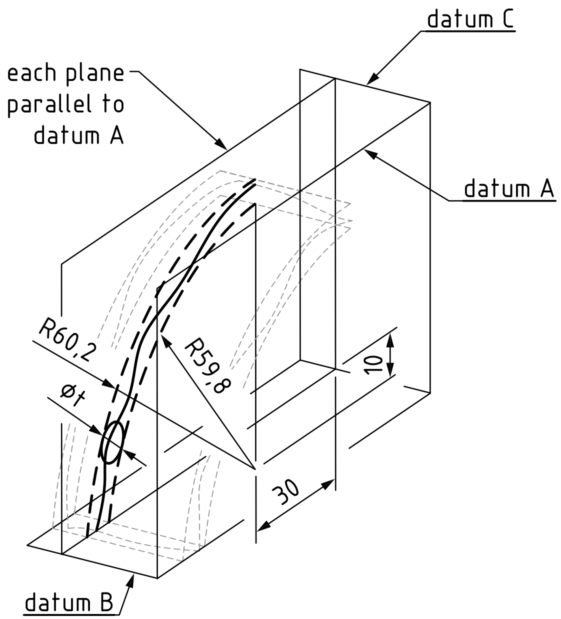
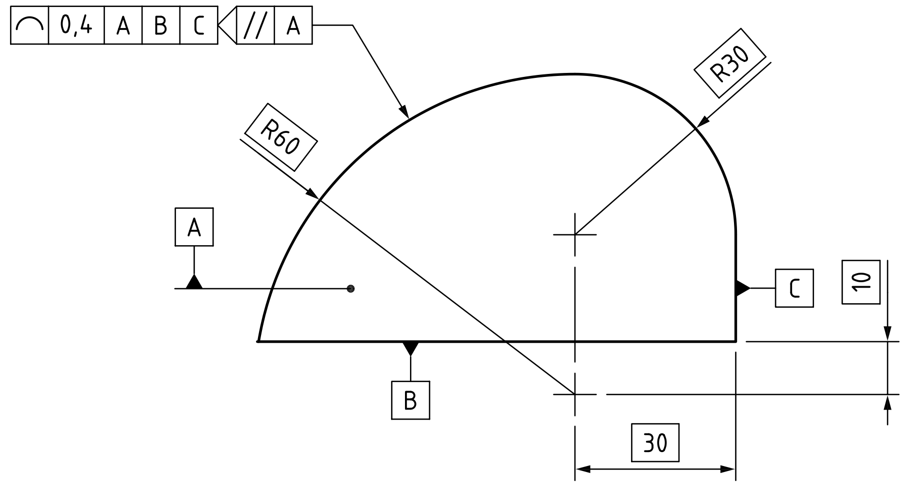
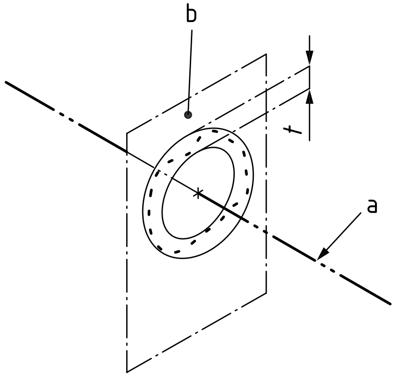
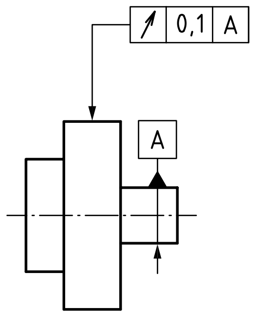
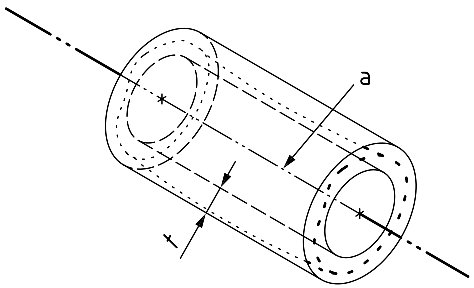
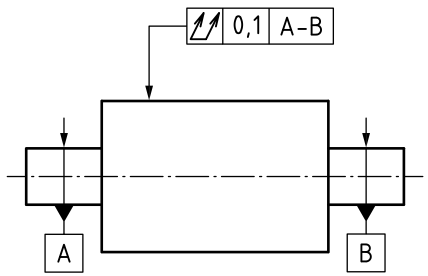
Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Line profile with a datum system	
⌒	The tolerance zone is limited by two lines enveloping circles of diameter *t*, the centres of which are situated on a line having the theoretically exact geometrical form with respect to the datum system, within each plane defined by the intersection plane indicator. 	An extracted line on the surface, in a plane parallel to datum A, shall be contained between two equidistant lines enveloping circles of diameter 0,4, the centres of which are situated on a line having the theoretically exact geometrical form with respect to datum system A|B|C. 

Figure 131 — *Interpretation of geometrical specifications (continued)*

Symbol	Definition of the tolerance zone	Indication and explanation in 2D
	Circular run-out tolerance – Radial	
↗	The tolerance zone is limited within any cross-section perpendicular to the nominal surface by two concentric circles with a difference in radii of *t*, the centres of which coincide with the datum *a*.  *NOTE a = datum; b = cross-section plane.*	The extracted line in any cross-section plane perpendicular to the nominal surface A shall be contained between two coplanar concentric circles with a difference in radii of 0,1 which are centred on datum axis A. 
	Total radial run-out tolerance	
⌰	The tolerance zone is limited by two coaxial cylinders with a difference in radii of *t*, the axes of which coincide with the datum (a = datum A – B). 	The extracted surface shall be contained between two coaxial cylinders with a difference in radii of 0,1 and the axes coincident with the common datum straight line A – B. 

*NOTE 1 For definitions of "extracted features", "associated features", and explanations of median line and median plane, see Clause **3**.*

NOTE 2 BS EN ISO 1660 includes the possibility of applying profile tolerances to derived features.

9.9 Theoretically exact dimensions (TEDs)

*COMMENTARY ON **9.9***

A TED is a linear or angular dimension used to define theoretically exact geometry. A TED can be explicit or implicit. An implicit TED is not indicated. An implicit TED is one of the following:

a) 0 mm;

b) 0°;

c) 90°;

d) 180°;

e) 270°; and

f) the angular distance between equally-spaced features on a complete circle when the location or orientation of the feature or group of features is controlled by a geometrical specification.

When a TED is explicitly indicated on a TPD, the dimension value shall be enclosed in a rectangular frame as shown in Figure 132. Any associated symbols such as ø or R shall also be contained within the frame. A multiplier indication such as *n*× shall be shown outside the frame.

NOTE 1 On a 3D model which is stated to consist of theoretically exact geometry, any dimension obtained through a query is a TED.

No tolerance shall be applied to the value of a TED, including individual or general tolerances.

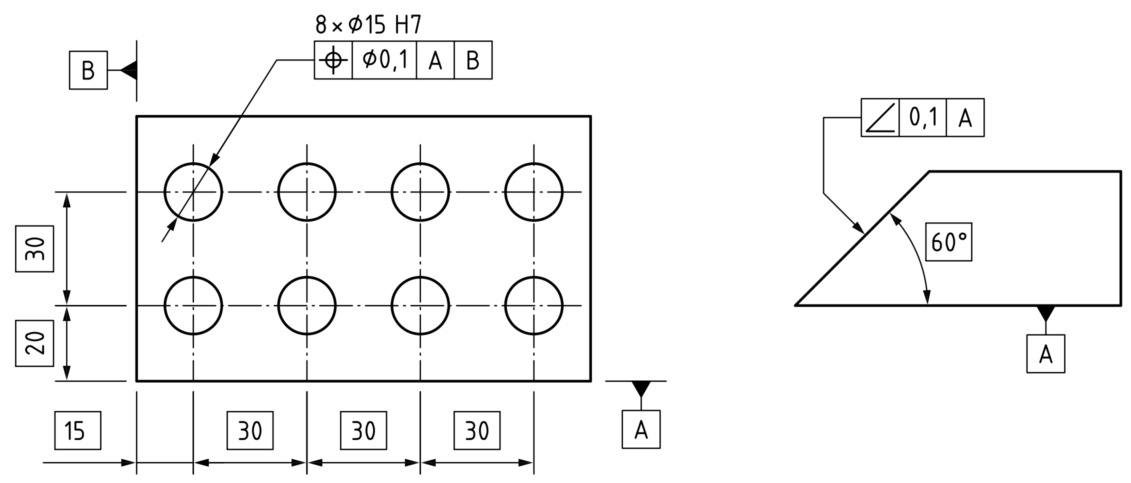
When a geometrical specification is used to tolerance the location of a feature (or group of features), the theoretically exact location of the feature(s) shall be defined with TEDs [see Figure 132a)].

When a geometrical specification is used to tolerance the orientation of a feature (or group of features), the theoretically exact orientation of the feature(s) shall be defined with TEDs [see Figure 132b)].

NOTE 2 Some examples of geometrical tolerancing are given in Figure 129.

When a profile tolerance is applied to a feature (or group of features), the theoretically exact profile or surface shall be defined with TEDs.

Figure 132 — *Use of theoretically exact dimensions (TEDs)*



a) TEDs positioning tolerance zones

b) TED orientating tolerance zone

9.10 Material condition modifiers

NOTE See BS EN ISO 2692 for additional information.

9.10.1 MMR applied to the toleranced feature

*COMMENTARY ON **9.10.1***

The effect of a maximum material virtual condition (MMVC) boundary can be approximated by the concept of "bonus tolerance". The tolerance value given in the tolerance indicator applies to the feature when it is at its maximum material size (MMS). When the feature departs from its MMS, the difference between the actual size and the MMS is considered a "bonus tolerance" which is added to the geometrical specification value as follows:

- *for an internal feature of size (e.g. a bore):*

  Geometrical specification allowance = Geometrical specification value + (Actual Size – MMS);

- *for an external feature of size (e.g. a shaft):*

  Geometrical specification allowance = Geometrical specification value + (MMS – Actual Size).

When the maximum material requirement (MMR) is applied to a feature, it defines a MMVC boundary for the feature, and the following requirements shall be met.

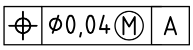
a) The upper size limit shall be satisfied.

b) The lower size limit shall be satisfied.

c) The surface of the feature shall not violate the MMVC boundary (an internal feature shall always remain outside its MMVC boundary, and an external feature shall always remain inside its MMVC boundary).

The MMR for the toleranced feature shall be invoked by placing the Ⓜ modifier immediately after the tolerance value in the tolerance zone section (see Figure 133).

Figure 133 — *Indication of the maximum material requirement – 1*



The MMR shall only be applied to features of linear size.

The size of the MMVC boundary is known as the maximum material virtual size (MMVS) of the feature, and shall be calculated using the following:

1) for an internal feature of size (e.g. a bore): MMVS = MMS – Geometrical specification; and

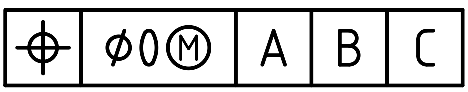
2) for an external feature of size (e.g. a shaft): MMVS = MMS + Geometrical specification.

NOTE The MMVC boundary is equivalent to a functional gauge limit, of fixed size, which can be used to verify the geometrical specification requirement.

9.10.2 MMR applied with a zero tolerance

The MMS of a feature shall be set equal to its MMVS in order to maximize the range of features which can satisfy the tolerance requirement. This shall be indicated using a zero tolerance value with the Ⓜ modifier as shown in Figure 134.

Figure 134 — *Indication of the maximum material requirement – 2*



*NOTE The same outcome can be achieved by using the reciprocity requirement (see **9.10.7**).*

9.10.3 MMR applied to a datum reference in a tolerance indicator

COMMENTARY ON ***9.10.3***

The effect of applying the MMR to a datum feature is that when the datum feature departs from its MMVC limit, the datum can move relative to the datum feature by a corresponding amount. This is sometimes addressed in a similar manner to the "bonus tolerance" concept, with the potential movement of the datum described as a "datum shift" allowance and is calculated in the same way.

When the MMR is applied to a datum reference in a tolerance indicator, the datum shall be derived from the MMVC boundary of the feature instead of the physical surface of the feature. In additional to any other requirements which have been directly defined for the datum feature, the following requirements shall be met.

a) The surface of the datum feature shall not violate the MMVC boundary (an internal feature shall always remain outside its MMVC boundary, and an external feature shall always remain inside its MMVC boundary).

b) The MMR for the datum feature shall be invoked by placing the Ⓜ modifier immediately after the datum name in the datum section of the tolerance indicator (see Figure 135).

Figure 135 — *Indication of the maximum material requirement – 3*

c) The MMR shall only be applied to datum features which are features of linear size.

NOTE 1 Use of the MMR for the datum feature is equivalent to basing the datum on a fixed size gauge limit for the datum feature rather than the actual surface of the datum feature.

NOTE 2 Where MMR is applied to a datum, it should be verified that the datum feature specification either has an MMR or the envelope requirement applied to define a boundary condition. If it does not have a boundary condition defined, the reference to the datum feature with the MMR defines a boundary condition which might be more restrictive than the tolerances applied directly to the datum feature. The result is a specification which appears to contradict itself.

9.10.4 LMR applied to the toleranced feature

COMMENTARY ON ***9.10.4***

The effect of a least material virtual condition (LMVC) boundary can be approximated by the concept of "bonus tolerance". The tolerance value given in the tolerance indicator applies to the feature when it is at its least material size (LMS). When the feature departs from its LMS, the difference between the actual size and the LMS is considered a "bonus tolerance" which is added to the geometrical specification as follows:

- *for an internal feature of size (e.g. a bore):*

  Geometrical specification allowance = Geometrical specification value + (LMS – Actual Size);

- *for an external feature of size (e.g. a shaft):*

  Geometrical specification allowance = Geometrical specification value + (Actual Size – LMS).

When the least material requirement (LMR) is applied to a feature, it defines a LMVC boundary for the feature, and the following requirements shall be met.

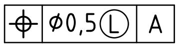
a) The upper size limit shall be satisfied.

b) The lower size limit shall be satisfied.

c) The surface of the feature shall not violate the LMVC boundary; an internal feature shall always remain inside its LMVC boundary, and an external feature shall always remain outside its LMVC boundary.

The LMR for the toleranced feature shall be invoked by placing the Ⓛ modifier immediately after the tolerance value in the tolerance zone section (see Figure 136).

Figure 136 — *Indication of the least material requirement*



The LMR shall only be applied to features of linear size.

The size of the LMVC boundary is known as the least material virtual size (LMVS) of the feature, and shall be calculated using the following:

1) for an internal feature of size (e.g. a bore): LMVS = LMS + Geometrical specification; and

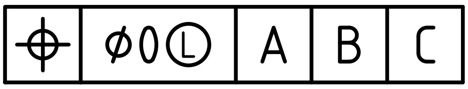
2) for an external feature of size (e.g. a shaft): LMVS = LMS – Geometrical specification.

NOTE The LMVC boundary is inside the material of a feature which satisfies the tolerance requirement, so this requirement cannot be verified with a mechanical gauge.

9.10.5 LMR applied with a zero tolerance

In order to maximize the range of features which satisfies the tolerance requirement, the setting of the LMS of a feature equal to its LMVS shall be indicated by using a zero tolerance value with the Ⓛ modifier as shown in Figure 137.

Figure 137 — *Indication of the LMR applied with a zero tolerance*



*NOTE The same outcome can be achieved by using the reciprocity requirement (see **9.10.7**).*

9.10.6 LMR applied to a datum reference in a tolerance indicator

*COMMENTARY ON **9.10.6***

The effect of applying the LMR to a datum feature is that when the datum feature departs from its LMVC limit, the datum can move relative to the datum feature by a corresponding amount. This is sometimes addressed in a similar manner to the "bonus tolerance" concept, with the potential movement of the datum described as a "datum shift" allowance and is calculated in the same way.

When the LMR is applied to a datum reference in a tolerance indicator, the datum shall be derived from the LMVC boundary of the feature instead of the physical surface of the feature. In additional to any other requirements which have been directly defined for the datum feature, the surface of the datum feature shall not violate the LMVC boundary.

The LMR for the datum feature shall be invoked by placing the Ⓛ modifier immediately after the tolerance value in the tolerance zone section (see Figure 138).

Figure 138 — *Least material requirement applied to a datum reference in a tolerance indicator*

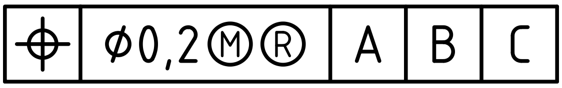
The LMR shall only be applied to datum features which are features of linear size.

NOTE Use of the LMR for the datum feature means that the datum is based on a boundary inside the material of the feature, so the datum cannot be established by a mechanical gauge.

9.10.7 Reciprocity requirement (RPR)

Where the reciprocity requirement (RPR) is used with a geometrical specification applied to a linear feature of size, it shall be indicated as a supplementary requirement to MMC or LMC by placing the Ⓡ symbol after the Ⓜ or Ⓛ symbol in the tolerance zone compartment as shown in Figure 139.

Figure 139 — *Reciprocity requirement*



RPR shall not be indicated in the datum section.

RPR shall not be used when the geometrical specification value is given as zero value (see **9.10.2** and **9.10.5**).

For an external feature, the size of the feature can be larger than the MMS, but the surface shall still be contained within the MMVC boundary. For an internal feature, the size of the feature can be smaller than the MMS, but the surface shall still lie outside the MMVC boundary.

NOTE 1 When the RPR modifier is used with MMR, the requirement to satisfy the MMS limit no longer applies.

NOTE 2 Use of MMR and RPR in a tolerance applied to a feature of size is not compatible with the use of the envelope requirement, as the RPR effectively removes the MMS limit that the envelope requirement depends on.

For an external feature, the size of the feature can be smaller than the LMS, but the surface shall still remain outside the LMVC boundary. For an internal feature, the size of the feature can be larger than the LMS, but the surface shall still be contained within the LMVC boundary.

NOTE 3 When the RPR modifier is used with LMR, the requirement to satisfy the LMS limit no longer applies.

NOTE 4 To achieve the same conditions as reciprocity it is common practice to apply all available tolerance to the linear size dimension and indicate the geometrical specification as 0Ⓜ or 0Ⓛ. In this case reciprocity is not required.

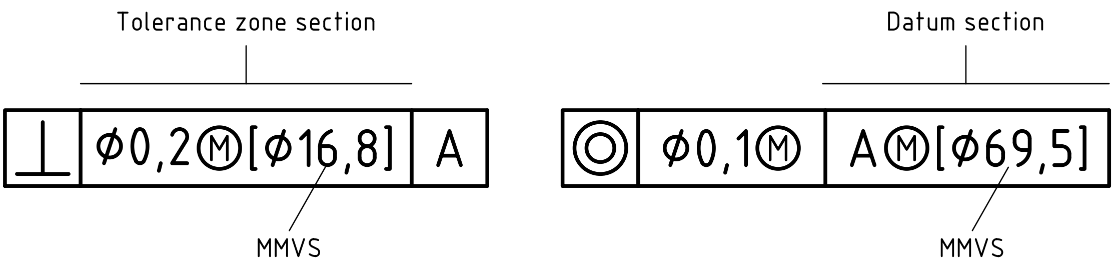
9.10.8 Direct indication of virtual size

NOTE 1 BS EN ISO 2692 introduces the option of indicating the MMVS and LMVS for a feature directly, rather than having to calculate these values.

The MMVS or LMVS value shall be indicated directly, in square brackets, after the material condition modifier.

NOTE 2 This method can be used in the tolerance zone section [see Figure 140a)] and in any appropriate compartment of the datum section of the tolerance indicator [see Figure 140b)].

Figure 140 — *Direct indication of virtual size*



a) Tolerance zone section

b) Datum section

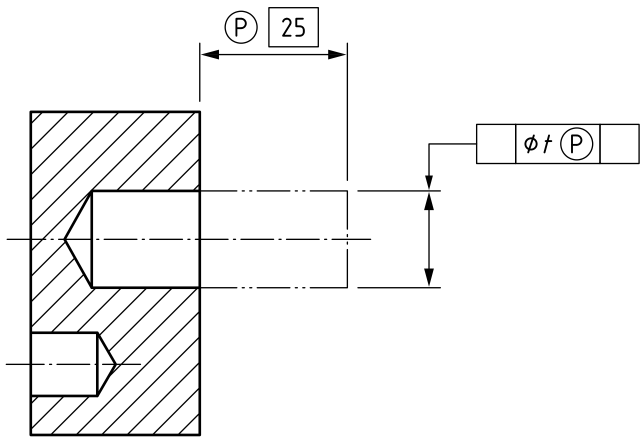
9.11 Additional indications relating to the toleranced feature

9.11.1 Projected tolerance zone

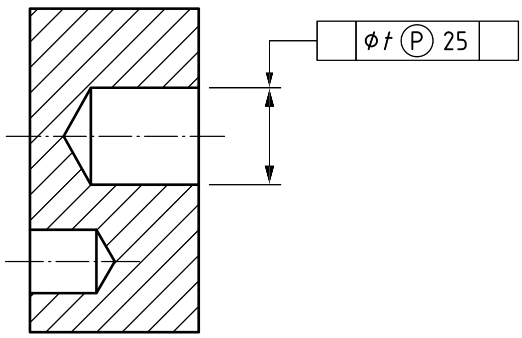
Projected tolerance zones shall be indicated by the specification modifier symbol Ⓟ in the tolerance value compartment (see example in Figure 141).

NOTE See BS EN ISO 1101:2017 for additional information.

Figure 141 — *Two ways of indicating a geometrical specification with projected tolerance modifier*



a) Direct indication of the length of the extension by a TED



b) Indirect indication of the length of the projected toleranced feature in the tolerance frame

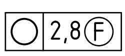
NOTE SOURCE: BS EN ISO 1101:2017, Figure 69 and Figure 70.

9.11.2 Free state condition

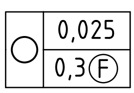
A geometrical specification which applies to a non-rigid part in its free state condition (i.e. with no constraints or clamps) shall be indicated by the specification modifier symbol Ⓕ placed after the specified tolerance value (see Figure 142).

NOTE See BS EN ISO 10579:2013 and BS EN ISO 1101:2017 for additional information.

Figure 142 — *Free state condition*



a) Individual free state tolerance



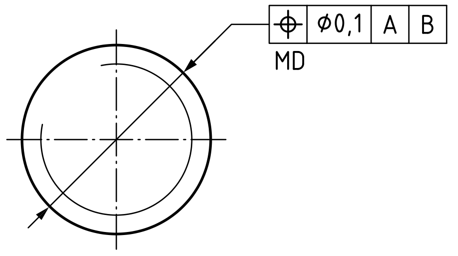
b) Free state tolerance in a tolerance stack

9.11.3 Tolerances applied to threaded features

Geometrical specifications controlling the location or orientation of screw threads shall apply to the axis derived from the pitch cylinder of the thread, unless otherwise specified.

If the tolerance is to be applied to the axis of the major diameter of the thread, the modifier MD shall be used as an adjacent indication for the tolerance indicator (see example of Figure 143).

Figure 143 — *Example of "MD"*



If the tolerance is to be applied to the axis of the minor diameter of the thread, the modifier LD shall be used as an adjacent indication for the tolerance indicator.

9.11.4 Tolerances applied to splines and gears

When a geometrical specification is used to control the location or orientation of a gear profile applied to a bore or shaft, or a spline applied to a bore or shaft, an adjacent indication for the tolerance indicator shall be used to indicate whether the tolerance applies to the axis of the major diameter, minor diameter or pitch diameter, because there is no default.

If the tolerance is to be applied to the axis of the pitch diameter of the gear or spline, the modifier PD shall be used as an adjacent indication for the tolerance indicator.

If the tolerance is to be applied to the axis of the major diameter of the gear or spline, the modifier MD shall be used as an adjacent indication for the tolerance indicator.

If the tolerance is to be applied to the axis of the minor diameter of the thread, the modifier LD shall be used as an adjacent indication for the tolerance indicator.

9.12 Unequal tolerance zone (UZ)

When a profile tolerance is applied to a surface, or when a position tolerance is applied to a plane, the tolerance zone shall be symmetrical around the theoretically exact feature (TEF), unless otherwise indicated.

NOTE This is known as a bilateral tolerance zone.

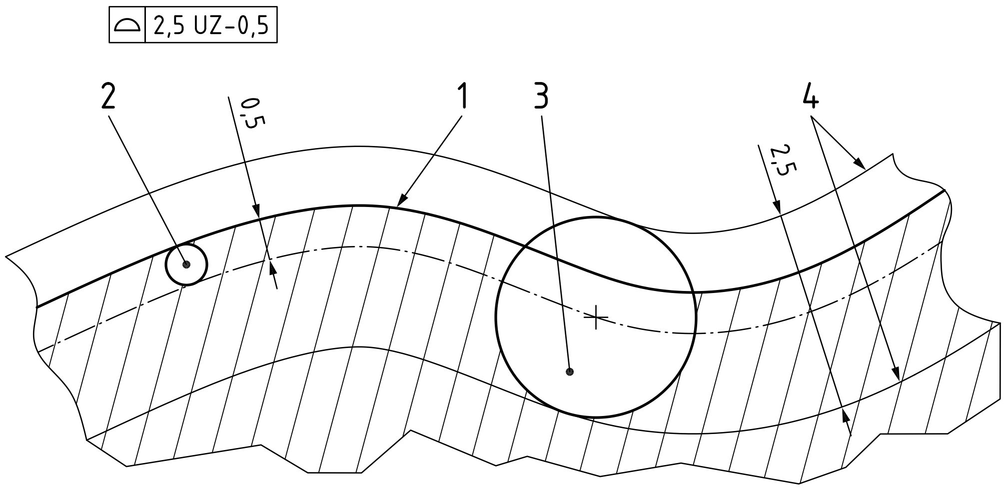
If an unequal (offset) tolerance zone is required, the UZ modifier shall be placed after the tolerance value in the tolerance indicator. The value following the UZ shall indicate the magnitude of the offset of the tolerance zone. A positive value (+) shall indicate that the zone is offset in a direction away from the material of the TEF. A negative value (-) shall indicate that the zone is offset in a direction towards the material of the TEF.

The "+" symbol or "–" symbol shall always be included with the offset value.

EXAMPLE

In Figure 144, the 2,5 tolerance zone is offset into the material (–ve) by 0,5 mm. The size of the zone outside of the material is (2,5/2) – 0,5 = 0,75 mm and the size of zone inside of the material is –(2,5/2) – 0,5 = –1,75 mm.

Figure 144 — *Example of a UZ*



Key

1 Single complex TEF; in this example, the material is below the feature

2 One of the infinite number of spheres defining the offset theoretical feaure, i.e. the reference feature

3 One of the infinite number of spheres defining the tolerance zone along the reference feature

4 Limits of the tolerance zone

NOTE 1 For profile specifications of complex lines or surfaces, the UZ specification element can be used with or without datums.

NOTE 2 SOURCE: BS EN ISO 1101:2017, Figure 21.

To specify a tolerance zone offset entirely into the material of the TEF, or entirely outside the material of the TEF, the value following the UF modifier shall be half the tolerance zone size value.

9.13 Restrictive specifications

9.13.1 Tolerance requirements applied to a specific restricted portion of a feature

To specify a tolerance requirement which is applied to a specific restricted portion of a feature, the restricted portion shall be defined using one of the following methods:

a) in a 2D view, by outlining the surface portions involved using a long-dashed-dotted wide line (in accordance with BS EN ISO 128-2:2022, type 04.2), with the location and dimensions defined by TEDs [see Figure 145c)];

b) by a hatched area with its border indicated as a long-dashed-dotted wide line (in accordance with BS EN ISO 128-2:2022, type 04.2), with the location and dimensions defined by TEDs [see Figure 145a), Figure 145d) and Figure 145e)];

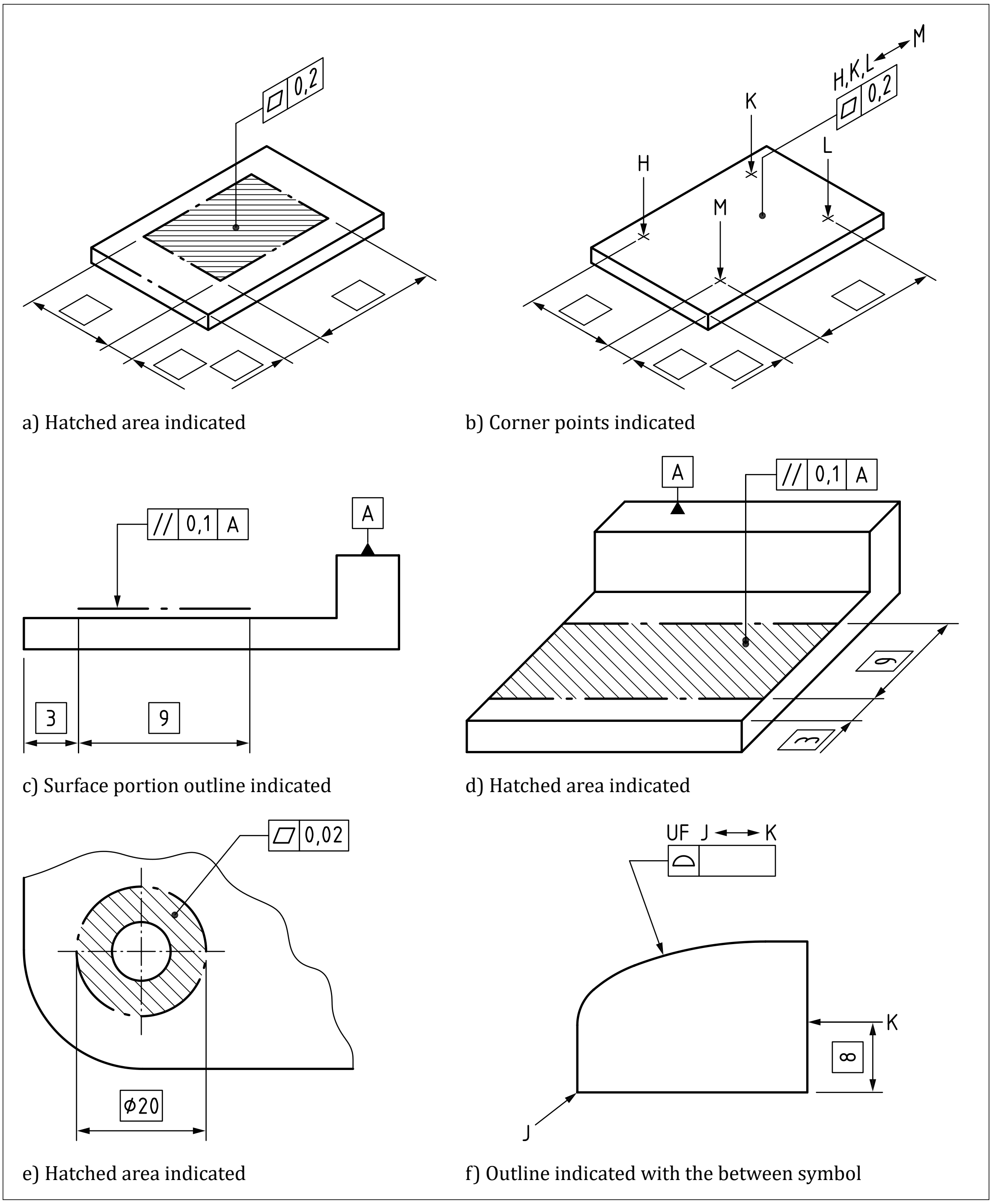
c) by its corner points, indicated as crosses on the integral feature (the location of the points being defined by TEDs), identified by capital letters and leader lines terminated by an arrow, and the letters indicated above the tolerance indicator with a “between” symbol between the last two [see Figure 145b)];

NOTE The border is formed by connecting the corner points with straight segments.

d) by two straight border lines identified by capital letters and leader lines terminated by an arrow (the location of the border lines being defined by TEDs) combined with an indication using the “between” symbol [see Figure 145f)].

The leader line from the geometrical specification indication shall terminate on the restricted area.

Figure 145 — *Tolerance applied to a restricted part of a feature*



a) Hatched area indicated

b) Corner points indicated

c) Surface portion outline indicated

d) Hatched area indicated

e) Hatched area indicated

f) Outline indicated with the between symbol

9.13.2 Tolerance requirements applied to any given restricted portion of a feature

If a tolerance of the same characteristic is applied to a restricted length, lying anywhere within the total extent of the feature, the value of the restricted length shall be added after the tolerance value and separated from it by an oblique stroke [see Figure 146a)].

NOTE 1 If two or more tolerances of the same characteristic are to be indicated, they can be combined as shown in Figure 146b). The use of separate tolerance characteristic symbols or a single combined tolerance characteristic symbol are both permitted and have an identical meaning. Where a single combined tolerance symbol is used this is not equivalent to the composite tolerancing concept in ASME Y14.5 [1].

Figure 146 — *Examples of tolerances of the same characteristic*

a) Single tolerance b) Two or more tolerances

If a tolerance of the same characteristic is applied to a restricted area, lying anywhere within the total extent of the feature, the value of the restricted area shall be added after the tolerance value and separated from it by an oblique stroke (see Figure 147).

The restricted area shall be:

a) circular: indicated by a diameter symbol followed by the diameter value;

b) square or rectangular: indicated by the length and width values separated by an “×”;

c) cylindrical segment: indicated by the axial length and angular circumferential values separated by an “×”;

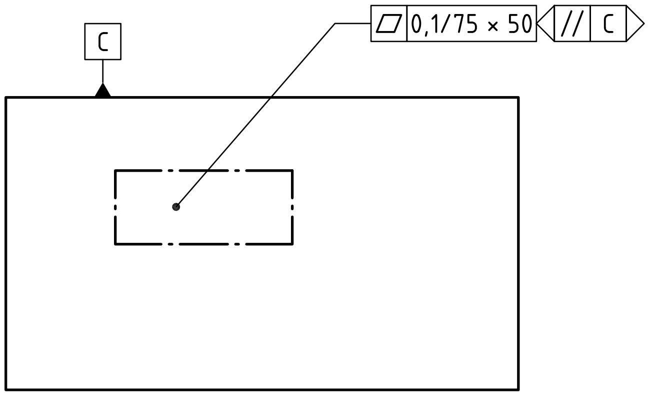
d) spherical segment: indicated by two perpendicular angular circumferential values separated by an “×”; and

e) square area: indicated by the □ symbol followed by a value.

Where it is necessary to define the orientation or alignment of the area, this shall be defined by an orientation plane indicator.

NOTE 2 Where two values are given to define the area, the orientation plane indicator applies to the first of the two values.

Figure 147 — *Tolerance applied to an area*



9.14 Interrelationship of geometrical specifications

*COMMENTARY ON **9.14***

For functional reasons, one or more characteristics can be toleranced to define the geometrical deviations of a feature. Certain types of tolerances which limit the geometrical deviations of a feature can also limit other types of deviations for the same feature.

Location tolerances of a feature shall control location deviation, orientation deviation and form deviation of this feature.

Orientation tolerances of a feature shall control orientation and form deviations of the feature.

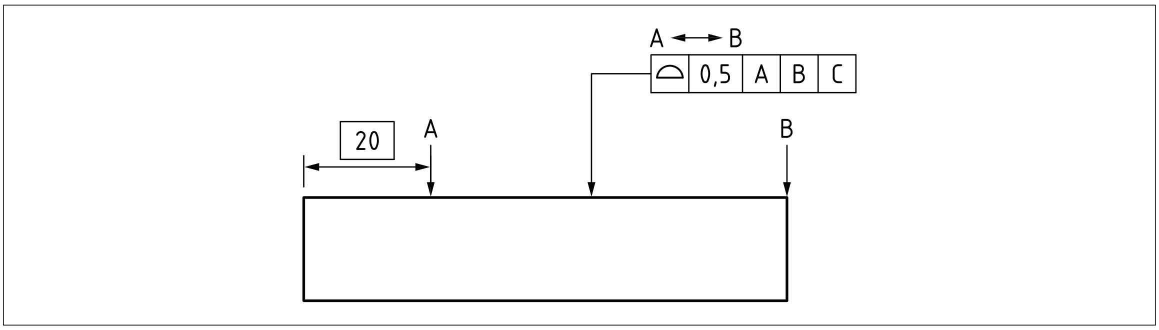
Form tolerances of a feature shall only control form deviations of this feature.

10 Additional indications for geometrical specifications

10.1 Between

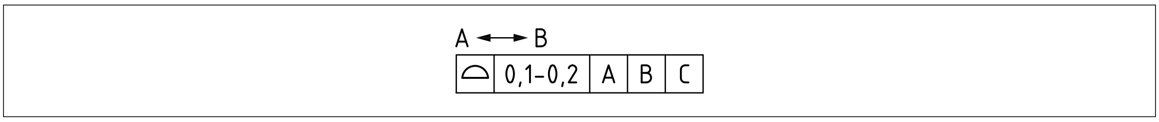
If a geometrical specification is to be applied across multiple contiguous features, or to a restricted portion of a feature(s), the between symbol (↔) shall be used. The specification shall then apply between start and end points which are identified on the TPS using capital letters. The between symbol shall be used in combination with the letters and placed over the tolerance indicator (see Figure 148).

Figure 148 — *Use of between symbol – 1*



Where the tolerance zone is to vary uniformly over the extent of its application, this shall be achieved by identifying the start and finish values for the tolerance in the tolerance zone section with the two values separated by a hyphen. This shall be used in conjunction with the between symbol, with the first tolerance value corresponding to the first capital letter, and the second tolerance values corresponding to the second capital letter (see Figure 149).

Figure 149 — *Use of between symbol – 2*

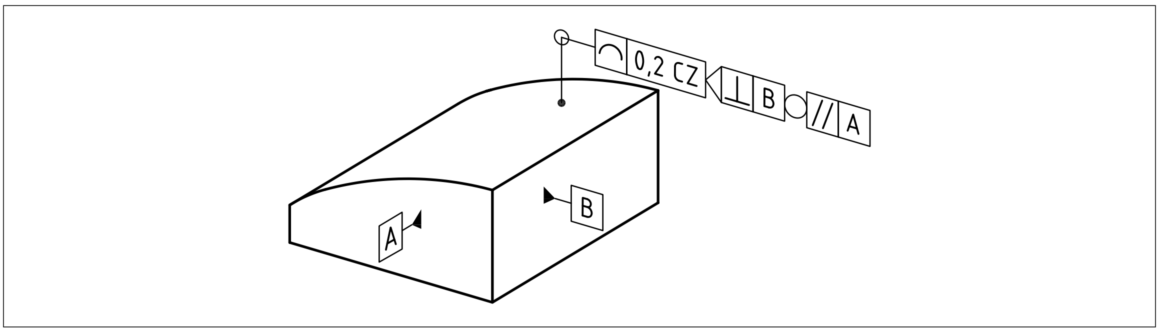


10.2 All-around

If a specification is to be applied to features represented by a closed outline or 2D sections of an outline, the all-around symbol ( ) shall be used, placed at the intersection of the leader line and reference line of the tolerance indicator (see Figure 150).

The all-around symbol shall be accompanied by UF, SZ or CZ unless all degrees of freedom are locked and be accompanied by a collection plane indicator.

Figure 150 — *Use of all-round symbol*



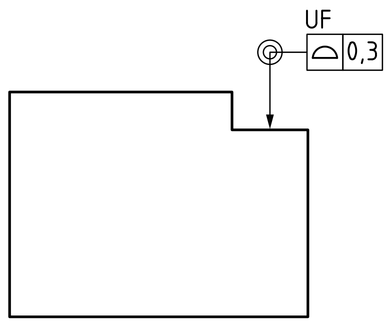
10.3 All-over

If a specification is to apply to an entire workpiece, the all-over symbol ( ) shall be used (see Figure 151).

The all-over symbol shall always be accompanied by UF, SZ or CZ unless all degrees of freedom are locked.

NOTE Where features of different types are included in a workpiece, the use of all-over can lead to ambiguity. In such cases additional definition or notes should be used to clarify the features included in the specification using the all-over symbol.

Figure 151 — *Use of all-over symbol*



10.4 United features

By default, a toleranced feature shall be a complete single feature. Where multiple features are to be treated as a single feature, the features to be united shall be identified using multiple leader lines or *n*× and preceded by UF. The features shall then be treated and toleranced as a single feature.

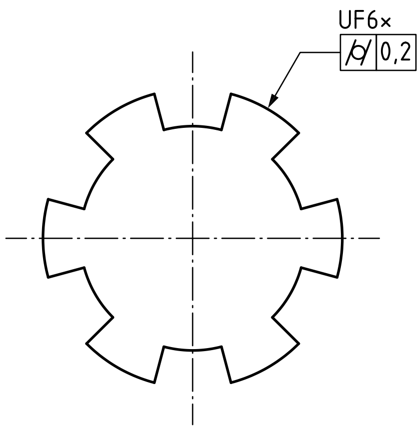
The features being united shall combine to create a single compound integral feature.

NOTE 1 For example, a collection of curved surfaces may be united using UF and toleranced as a single cylindrical feature, e.g. using cylindricity. This would not be possible if treating them as several individual features (see Figure 152).

NOTE 2 Coaxial cylinders with identical diameters may be treated as a single cylinder using UF. Non-coaxial cylinders or cylinders with different diameters cannot. Multiple planar features on a common plane may be treated as a united feature. Planar features that do not reside on a common plane cannot.

NOTE 3 See BS EN ISO 1101:2017 for further details of restrictions on use of the UF modifier.

Figure 152 — *Example united feature*



10.5 Feature patterns (CZ, CZR, SZ and SIM)

10.5.1 Internal and external constraints

NOTE 1 Requirements for pattern specifications are specified in BS EN ISO 5458.

A pattern specification shall be defined when a geometrical specification is applied to a set of identical features, and the tolerance zones shall have their locations and/or orientations locked together.

The relationships between the tolerance zones in the pattern are known as “internal constraints”, each of which shall consist of one of the following:

a) location and orientation constraints; or

b) only orientation constraints.

NOTE 2 If there are no internal constraints, there is no pattern.

The relationships between the pattern of tolerance zones and the datum system are known as “external constraints”, each of which shall consist of one of the following:

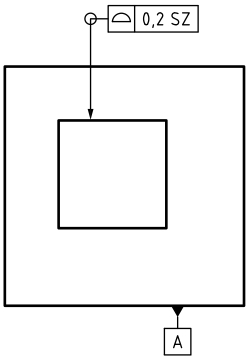
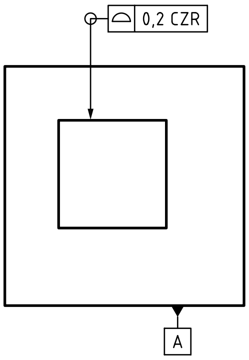
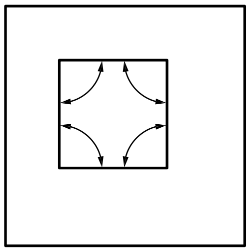
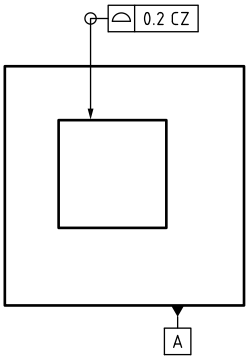
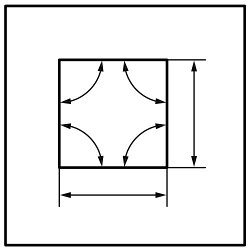
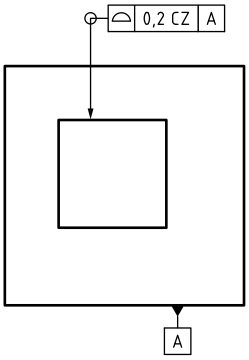
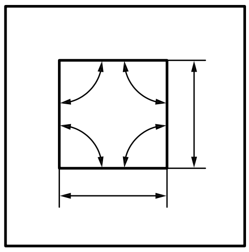
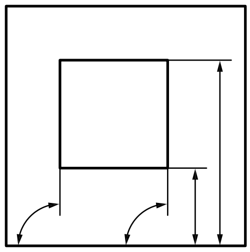
1) location and orientation constraints; or

2) only orientation constraints.

*NOTE 3 If a requirement arises to group dissimilar features together, this can be achieved by using the SIM indicator (see **10.5.4**).*

NOTE 4 Examples of internal and external constraints are given in Table 16.

Table 16 — *Example showing internal and external constraints*

Specification	Internal constraints	External constraints
 TEDs defined in CAD model 123 rev. C.	None. No pattern is specified.	None. No datum or datum system(s) is referenced by the geometrical specification.
 TEDs defined in CAD model 123 rev. C.	CZR introduces internal orientation constraints. 	None. No datum or datum system(s) is referenced by the geometrical specification.
 TEDs defined in CAD model 123 rev. C.	CZ introduces internal location and orientation constraints 	None. No datum or datum system(s) is referenced by the geometrical specification.
 TEDs defined in CAD model 123 rev. C.	CZ introduces internal location and orientation constraints. 	The reference to datum A introduces external constraints to datum A. The type of constraints depends on the tolerance characteristic. 

10.5.2 Specifying a pattern requirement

The geometrical specification shall be applied to a set of identical features by one of the following methods:

a) separate leader lines to each feature;

b) the indication *n*× where *n* is the number of features (see **10.4**);

c) the "all-around" symbol (see **10.2**); and

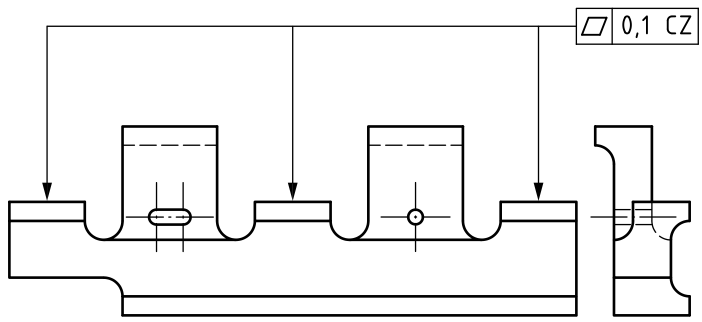
d) the "between" symbol (see **9.13.1**).

The internal constraints shall be defined using either the CZ or the CZR modifier immediately after the tolerance value, separated by a space:

1) the CZ modifier indicates that both location and orientation constraints apply (see Figure 153 and Figure 154); and

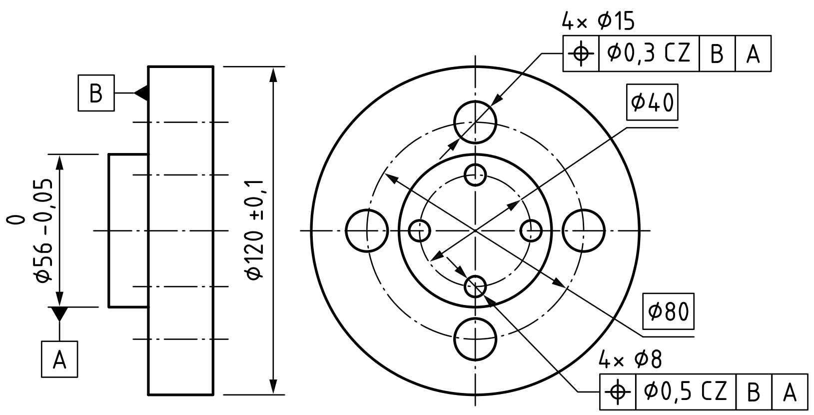
2) the CZR modifier indicates that only orientation constraints apply.

Figure 153 — *Pattern of three flat faces subject to a combined zone flatness tolerance*



NOTE Location constraints (implied distance of 0) and orientation constraints (implied parallelism) apply to the three tolerance zones.

Figure 154 — *Two independent patterns, one of four Ø15 holes, and one of four Ø8 holes*



NOTE Within each group location, constraints (pitch diameter and implied 90° spacing) and orientation constraints (implied parallelism) apply. No constraints exist between the two patterns; the group of Ø8 holes does not have to be rotationally aligned with the group of Ø15 holes.

10.5.3 Nested pattern requirements

NOTE 1 A nested pattern specification is defined when a geometrical specification is applied to a set of identical patterns and the sets of tolerance zones from each pattern have their locations and/or orientations locked together.

If there are *k* groups, each composed of *n* individual features, these shall be identified using one of the following methods:

a) *k*× indicated in front of *n*× in the adjacent indication area with a slash as separator and a space on both sides of the slash. The *k*× and *n*× shall be followed by a space and the identifier letter or symbol to avoid ambiguities (e.g. 4× / 2× or 4× A / 2× B);

NOTE 2 The identification letter can be used to establish a link with individual integral features, or with a group of integral features. When used to identify a group of features, the group may be indicated on a drawing by surrounding the features with a long-dashed double-dotted narrow line (line type 05.1 according to BS EN ISO 128-2:2022) (see Figure 155).

b) *n* leader lines connecting the tolerance indicator to the *n* geometrical features and *k*× indicated in the adjacent indication area;

c) an “all-around” symbol (covering *n* features) defining one group and *k*× indicated in the adjacent indication area; or

d) if the integral feature related to the toleranced feature is a feature of size, the number of groups shall be indicated followed by a space and the group identifier letter if applicable, followed by a space, a slash and a space, followed by the number of features and a space and the nominal size and its specification (general or individual), followed by a space and the feature identifier letter if applicable (e.g. 3× B / 2× 10±0,05 A or 3× / 2× 10±0,05 or 3× / 2× 10).

The nested pattern requirement shall be defined in the geometrical specification using a sequence composed of the CZ, SZ and CZR modifiers (see Table 17 and Table 18).

NOTE 3 Some combinations do not result in meaningful requirements.

NOTE 4 If all the elements of the sequence are SZ, then:

a) this specification does not define a pattern specification; and/or

b) the specification consists of a set of kxn independent tolerance zones, each applying to one geometrical feature, and defining kxn geometrical characteristics.

Figure 155 — *Eight independent specifications*

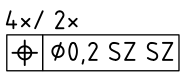


Table 17 — *Sequences of modifiers and their meaning*

Sequence	Meaning
SZ SZ	Indicates independency of all features
SZ CZR	Indicates independency between tolerance zone patterns (first level), each tolerance zone pattern being composed of several tolerance zones with orientation constraints only (no location constraint)
SZ CZ	Indicates independency of tolerance zone patterns (first level)
CZR SZ	Meaningless
CZR CZR	Meaningless
CZR CZ	Indicates dependency between tolerance zone patterns (first level) rotationally only, each tolerance zone pattern being composed of several tolerance zones with orientation and location constraints
CZ SZ	Meaningless
CZ CZR	Meaningless
CZ CZ	Indicates dependency between tolerance zone patterns with orientation and location constraints
SZ SZ CZ	Indicates three levels of repetitions where only the first level creates tolerance zone patterns
SZ CZR CZ	Indicates three levels of repetitions where the tolerance zone pattens of the first level are only constrained rotationally between them
SZ CZ CZ	Indicates three levels of repetitions where the first, with the second level, creates tolerance zone patterns
CZR SZ CZ	Meaningless
CZR CZR CZ	Meaningless
CZR CZ CZ	Indicates three levels of repetitions where the first, with the second level, creates tolerance zone patterns which are only constrained rotationally between them
CZ CZ CZ	Indicates three levels of repetitions where the first, with the second level and the third level, creates a tolerance zone pattern

NOTE Subsequent CZ/CZR/SZ indications apply to the next tolerance zone pattern levels following the same logic.

Table 18 — *Use of CZ, SZ and CZR modifiers*

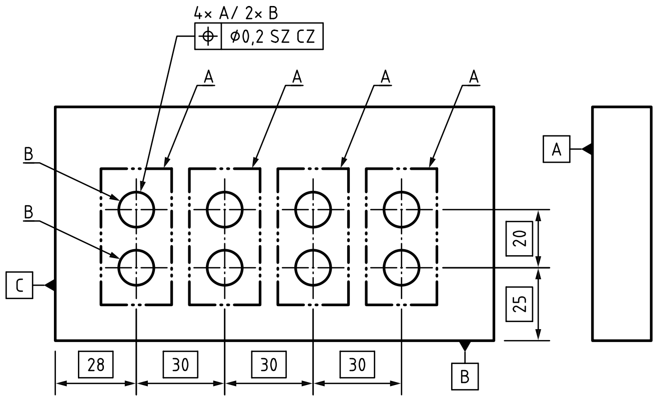
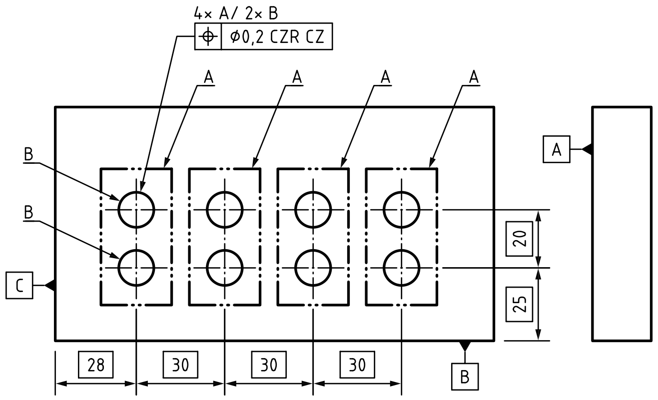
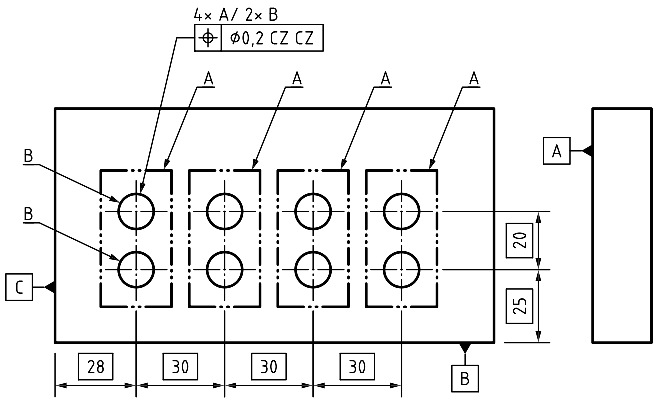
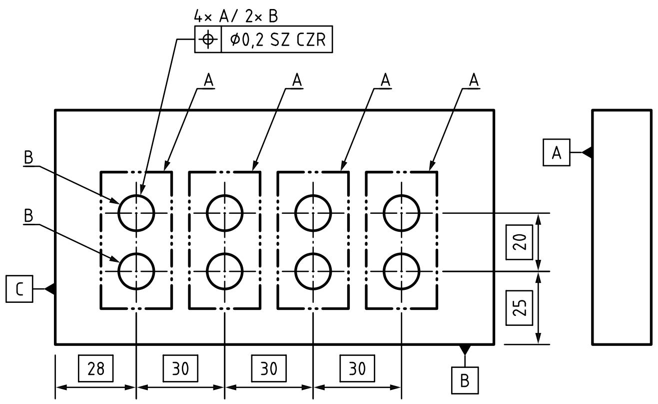
Indication	Meaning
	**SZ CZ with no datum references** Four independent patterns: for each pattern: • toleranced feature = the group of two median lines; and • tolerance zone pattern = two Ø0,2 tolerance zones with an internal location constraint (20 mm) and internal orientation constraint (parallelism).
	**CZR CZ with no datum references** Four dependent patterns: for each pattern: • toleranced feature = the group of two median lines; • tolerance zone pattern = two Ø0,2 tolerance zones with internal and location constraint (20 mm) and internal orientation constraint (parallelism); and • there are also orientation constraints between the four patterns, but no location constraints.
	**CZ CZ with no datum references** Four fully dependent patterns result in a single pattern specification: • toleranced feature = the group of eight median lines; and • tolerance zone pattern = eight Ø0,2; tolerance zones with internal location constraints (30 mm horizontally and 20 mm vertically) and orientation constraints (parallelism).
	**SZ CZR with no datum references** Four independent patterns: for each pattern: • toleranced feature = the group of two median lines; and • tolerance zone pattern = two Ø0,2 tolerance zones with an internal orientation constraint (parallelism) only. The 20 mm distance between the median lines does not apply.

Table 18 — *Use of CZ, SZ and CZR modifiers (continued)*

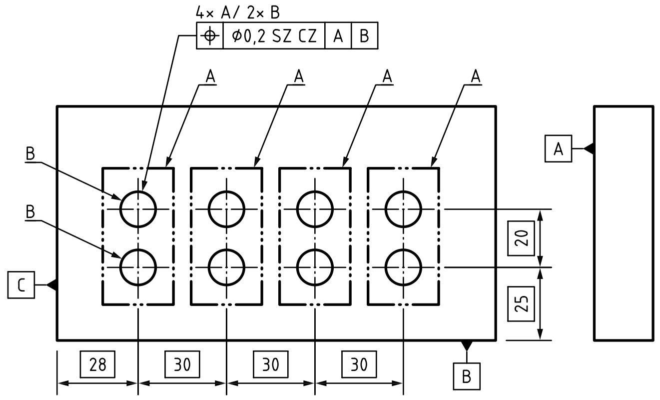
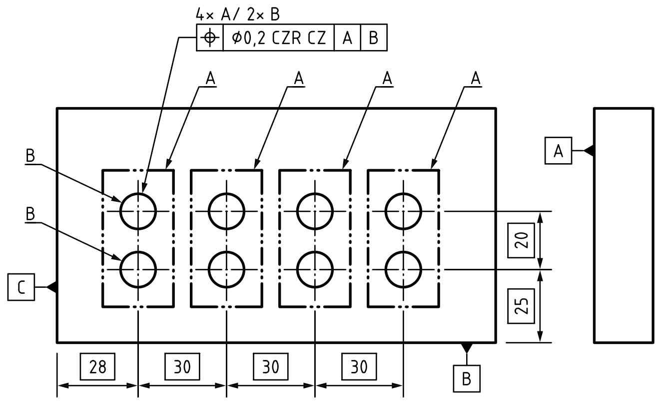
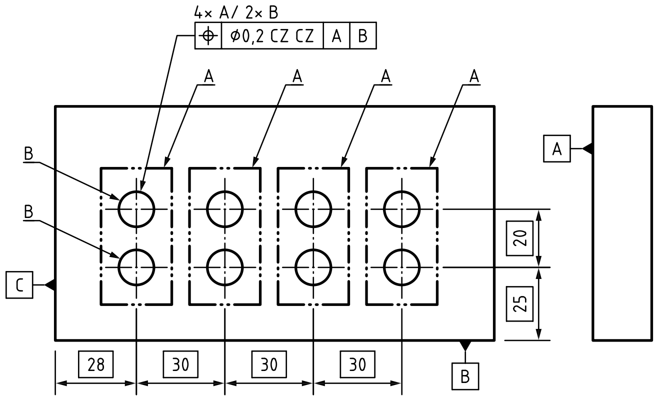
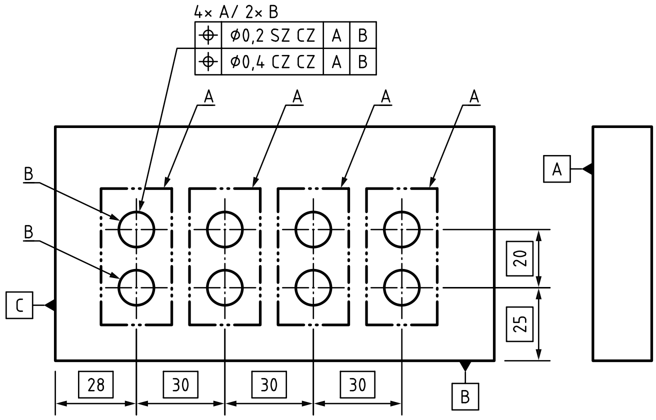
Indication	Meaning
	**SZ CZ with datum references** Four independent patterns constrained in orientation to datum A and location and orientation to datum B (see Note 1). For each pattern: • toleranced feature = the group of two median lines; • tolerance zone pattern = two Ø0,2 tolerance zones with internal location constraint (20 mm) and internal orientation constraint (parallelism); and • there are also external orientation constraints to datum A and datum B for each pattern: the constraint to datum A is orientation (perpendicularity of the pattern of two tolerance zones) and the constraints to datum B are location (distance of the pattern of tolerance zones from datum B is fixed) and orientation (a plane through the centrelines of the two tolerance zones remain perpendicular to datum B). No relationship between the pattern is defined. *NOTE 1 The reference to datum B does not control parallelism of the median lines to datum B because perpendicularity to datum A takes precedence.*
	**CZR CZ with datum references** Four dependent patterns constrained in orientation to datum A and location and orientation to datum B (see Note 2). For each pattern: • toleranced feature = the group of two median lines; • tolerance zone pattern = two Ø0,2 tolerance zones with internal location constraints (20 mm) and internal orientation constraints (parallelism); • there are orientation constraints between the four patterns, but no location constraints: a plane through the centrelines of the two tolerance zones in any pattern remain parallel to the corresponding plane in any other pattern and the distance between patterns can vary; and • there are also external orientation constraints to datum A and datum B for each pattern: the constraint to datum A is orientation (perpendicularity of the pattern of two tolerance zones) and the constraints to datum B are location (distance of the pattern of tolerance zones from datum B is fixed) and orientation (a plane through the centrelines of the two tolerance zones remain perpendicular to datum B). because perpendicularity to datum A takes precedence. *NOTE 2 The reference to datum B does not control parallelism of the median lines to datum B because perpendicularity to datum A takes precedence.*

Table 18 — *Use of CZ, SZ and CZR modifiers (continued)*

Indication	Meaning
	**CZ CZ with datum references** Four dependent patterns constrained in orientation to datum A and location and orientation to datum B (see Note 3) result in one pattern specification. • tolerancer feature = the group of eight median lines; • tolerance zone pattern = eight Ø0,2 tolerance zones with internal location constraint (30 mm × 20 mm) and internal orientation constraint (parallelism); and • there are also external orientation constraints to datum A and datum B for the pattern: the constraint to datum A is orientation (perpendicularity of the pattern of eight tolerance zones) and the constraints to datum B are location (distance of the pattern of tolerance zones from datum B is fixed) and orientation (a plane through the centrelines of any two or more of the eight tolerance zones remain perpendicular to datum B). No relationship between the pattern is defined. *NOTE 3 The reference to datum B does not control parallelism of the median lines to datum B because perpendicularity to datum A takes precedence.*
	These requirements can be stacked. In this example, each pair of holes is located to each other, and to datums A and B, by the Ø0,2 tolerance. In addition, the four groups A are located to each other by the Ø0,4 tolerance.

NOTE 4 The datum system A|B does not constrain all six degrees of freedom. A datum system such as A|B|C which did constrain all six degrees of freedom does make no sense for nested patterns, as the individual features would be fully constrained to the datum system.

NOTE 5 Not all cases illustrated would result in a complete definition for the workpiece. Additional indications could be required.

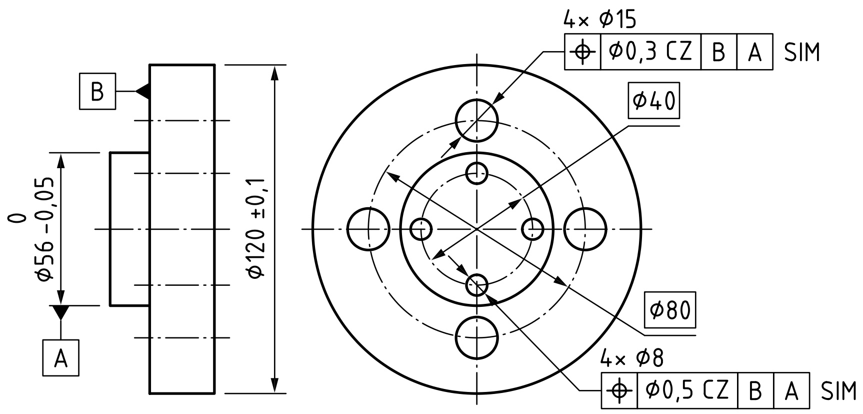
10.5.4 Combined geometrical specifications

NOTE 1 Individual geometrical specifications and pattern specifications can be combined using the simultaneous requirement invoked by the SIM modifier. This makes it possible to create groups of non-identical features.

The SIM modifier shall be placed adjacent to two or more tolerance indicators when two or more tolerances are to be combined into a single requirement.

If there is only one simultaneous requirement on the TPD, the modifier shall be used without an index number (see Figure 156). If there are multiple simultaneous requirements on the TPD, the modifier shall be used with an index number.

Figure 156 — *Two patterns locked together by the SIM indicator*



NOTE Two independent patterns, one of four Ø15 holes, and one of four Ø8 holes, which are then locked together by the SIM indication. Within each group location constraints (pitch diameter and implied 90° spacing) and orientation constraints (implied parallelism) apply. Location and orientation constraints also exist between the two patterns, so the group of Ø8 holes is rotationally aligned with the group of Ø15 holes.

NOTE 2 In Figure 157a), the SIM modifier adjacent to the two tolerance indicators means that the two tolerance zone patterns are combined into a single requirement. All five tolerance zones are locked together by location and orientation constraints.

In Figure 157b), the SIM1 modifier creates one simultaneous requirement, and the SIM2 modifier creates a separate simultaneous requirement. The SIM1 and SIM2 requirements are unrelated to each other.

Figure 157 — *Examples of indication of simultaneous requirements from two separate specifications*

a) Example 1 b) Example 2

10.6 Plane and feature indicators

NOTE A limited set of examples are shown in this subclause but further examples can be found in BS EN ISO 1101:2017.

10.6.1 Intersection plane indicator

10.6.1.1 Purpose of intersection plane indicators

The intersection plane indicator shall be used when necessary to identify the orientation of line elements on a surface to which a geometrical specification is applied.

NOTE The intersection plane indicator replaces the former practice of using the drawing view to determine the plane in which a geometrical specification applies, and also enables a geometrical specification to be applied to line elements in 3D specifications where there is no drawing view.

An intersection plane indicator shall only be used in conjunction with the following types of geometrical specification:

a) straightness;

b) roundness;

c) line profile; and

d) circular run-out,

and shall not be required for the following:

1) a straightness tolerance applied to a cylinder or cone;

2) a roundness tolerance applied to cylindrical, conical or spherical surfaces; and

3) a geometrical specification that is applied to a complete integral feature.

10.6.1.2 Intersection plane indicator symbol

The intersection plane indicator symbol shall consist of a rectangular frame of two or more compartments, with a triangle attached to the left-hand end as shown in Figure 158 and:

a) the first compartment containing one of the geometrical specification symbols set out in Table 19; and

b) the remaining compartments containing one or more datum references (i.e. a datum system).

Figure 158 — *Intersection plane indicator symbol*

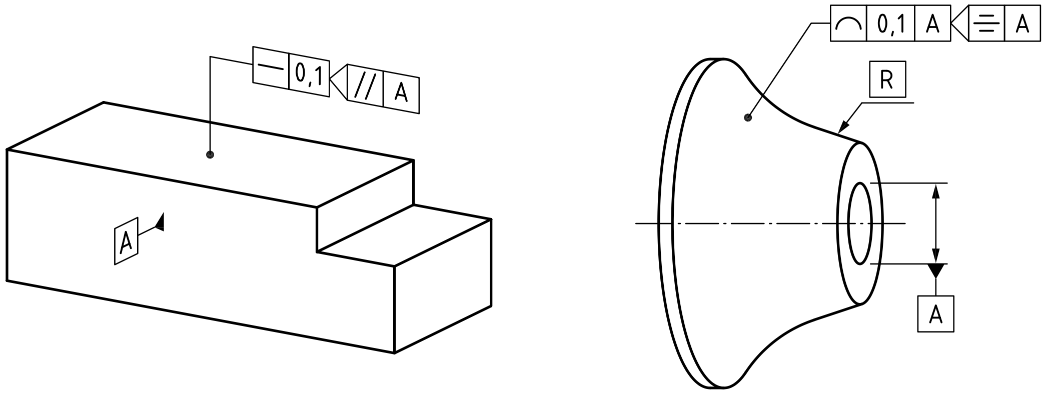
Table 19 — *Geometrical specification symbols used with an intersection plane indicator*

Symbol	Meaning
//	The tolerance applies in each plane parallel to the datum system.
⊥	The tolerance applies in each plane perpendicular to the datum system.
∠	The tolerance applies in each plane at an angle (defined with a TED) to the datum system.
⌯	The tolerance applies in each plane passing through a datum axis.

NOTE When perpendicularity or angularity symbols are used, more than one datum reference might be necessary in order to define a unique set of parallel planes.

NOTE Example use of intersection plane indicators are shown in Figure 159.

Figure 159 — *Example use of intersection plane indicators*



a) The straightness specification applies in each plane parallel to datum A

b) The line profile specification applies in each plane through (including) datum axis A

10.6.2 Orientation plane indicator

10.6.2.1 Purpose of orientation plane indicators

The orientation plane indicator shall be used when necessary to identify the orientation of a tolerance zone.

NOTE In most cases, the orientation of a tolerance zone is fully defined without additional indications.

The orientation plane indicator shall also be used for orientating a restricted area application of a geometrical specification.

10.6.2.2 Orientation plane indicator symbol

The orientation plane indicator symbol shall consist of a rectangular frame of two or more compartments, with a triangle attached to each end as shown in Figure 160 and:

a) the first compartment containing one of the geometrical specification symbols set out in Table 20; and

b) the remaining compartments containing one or more datum references (i.e. a datum system).

Figure 160 — *Orientation plane indicator symbol*

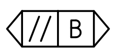
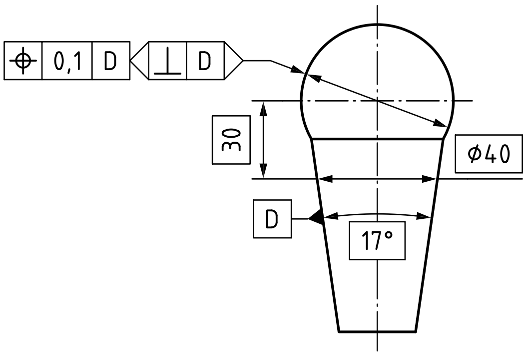


Table 20 — *Geometrical specification symbols used with an orientation plane indicator*

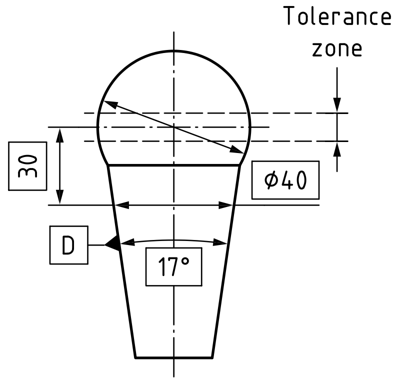
Symbol	Meaning
//	For a tolerance zone composed of two parallel planes, the two parallel planes are parallel to the datum system. For a cylindrical tolerance zone, the axis of the tolerance zone is parallel to the datum system.
⊥	For a tolerance zone composed of two parallel planes, the two parallel planes are perpendicular to the datum system. For a cylindrical tolerance zone, the axis of the tolerance zone is perpendicular to the datum system.
∠	For a tolerance zone composed of two parallel planes, the two parallel planes are at an angle (defined with TEDs) to the datum system. For a cylindrical tolerance zone, the axis of the tolerance zone is at an angle (defined with TEDs) to the datum system.

NOTE The orientation plane indicator is not required when the orientation of the tolerance zone is unambiguously defined by the datum system in the tolerance indicator. Example uses of orientation plane indicators are shown in Figure 161.

Figure 161 — *Example uses of orientation plane indicators*



a) Indication



The centre of the sphere lies between two parallel planes, 0.1 mm apart, centred 30 mm from the Φ40 mm section of the conical datum feature, and perpendicular to datum D.

b) Interpretation

10.6.3 Collection plane

10.6.3.1 Purpose of collection planes

The collection planes shall be used when the “all-around” symbol is applied to a geometrical specification. The collection plane identifies a family of parallel planes, which identifies the features covered by the “all-around” indication.

NOTE 1 The geometrical specification applies to the set of features which create a continuous closed line of intersection with the plane that also passes through the point at which the leader line for the specification is applied.

NOTE 2 The collection plane indicator replaces the former practice of using the drawing view to define the plane in which the “all-around” symbol is applied. This allows the “all-around” symbol to be used in 3D indications, and to be applied to a wider range of geometries.

NOTE 3 While the former practice of relying on the drawing view plane is now “deprecated”, it is still permitted for 2D drawings.

NOTE 4 Explanatory notes might sometimes be required to explain intent in addition to the rules and explanation for this symbol given in BS EN ISO 1101:2017.

10.6.3.2 Collection plane symbol

The collection plane symbol shall consist of a rectangular frame of two or more compartments, with a circle attached to the left-hand end as shown in Figure 162 and:

a) the first compartment containing one of the geometrical specification symbols set out in Table 21; and

b) the remaining compartments containing one or more datum references (i.e. a datum system).

Figure 162 — *Collection plane indicator symbol*

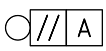
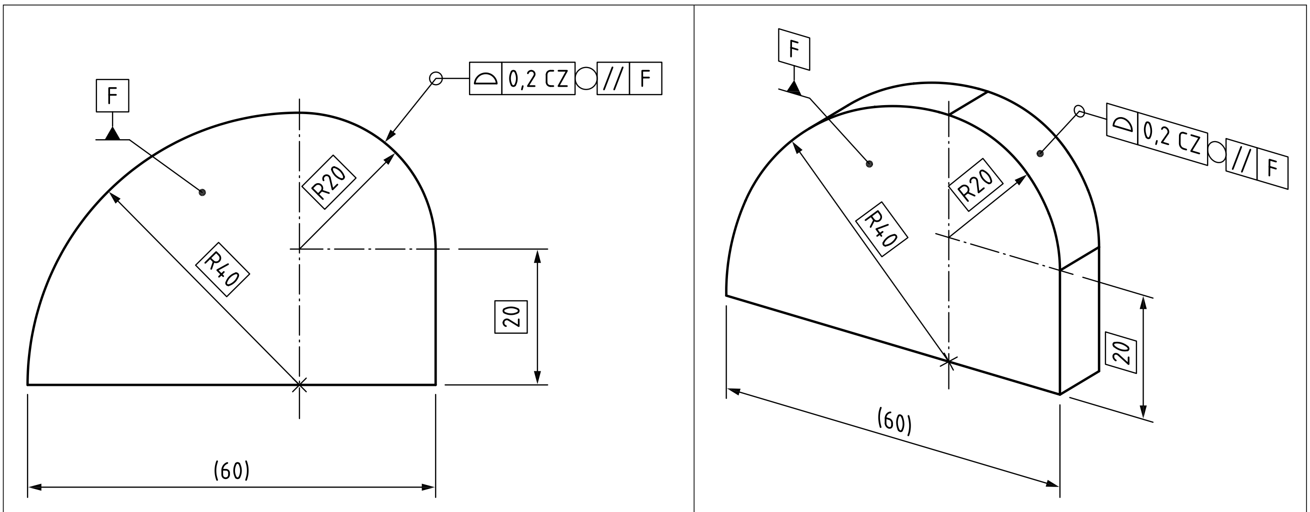


Table 21 — *Geometrical specification symbols used with a collection plane indicator*

Symbol	Meaning
//	The "all-around" requirement applies in a plane parallel to the datum system.
⊥	The "all-around" requirement applies in a plane perpendicular to the datum system.
∠	The "all-around" requirement applies in a plane at an angle (defined with a TED) to the datum system.

NOTE Example uses of intersection plane indicators are shown in Figure 163. The surface profile tolerance applies to all the features which make up the continuous profile shown in the 2D indication in Figure 163a). Figure 163b) shows the same requirement identified in a 3D CAD specification.

Figure 163 — *Example uses of collection plane indicators*



a) 2D drawing indication

b) 3D drawing indication

10.6.4 Direction feature

10.6.4.1 Purpose of direction features

The direction feature shall be used when it is necessary to identify the direction in which a geometrical specification is applied. In most cases, the default shall be that a geometrical specification applies in a direction perpendicular to the surface of the feature, and no direction feature symbol is required in these cases. A direction feature shall be used if the geometrical specification requirement is to be applied in a direction which is not perpendicular to the surface.

The direction feature shall be used when a roundness tolerance is applied to a revolute surface other than a cylinder or a sphere.

*NOTE When a roundness tolerance is applied to a cylinder or sphere, there is a default direction for the application of the tolerance (see Clause **15**). When roundness is applied to any other revolute surface, there is no default direction.*

10.6.4.2 Direction feature symbol

The direction feature symbol shall consist of a rectangular frame of two or more compartments, with a short leader terminating in a filled arrowhead attached to the left-hand end as shown in Figure 164 and:

a) the first compartment containing one of the geometrical specification symbols set out in Table 22; and

b) the remaining compartments containing one or more datum references (i.e. a datum system).

Figure 164 — *Direction feature symbol*

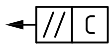
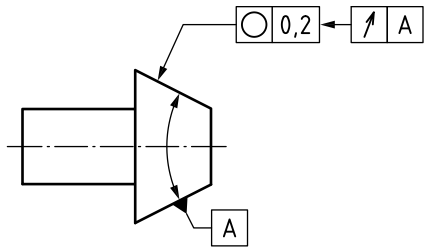


Table 22 — *Geometrical specification symbols used with a direction feature indicator*

Symbol	Meaning
//	The width of the tolerance zone is parallel to the datum system.
⊥	The width of the tolerance zone is perpendicular to the datum system.
∠	The width of the tolerance zone is at an angle (defined with a TED) to the datum system.
↗	The width of the tolerance zone is normal to the surface as for a run-out tolerance.

NOTE An example use of the direction feature is shown in Figure 165. The roundness tolerance applies perpendicular to the surface of the conical feature, instead of perpendicular to the axis of the conical feature.

Figure 165 — *An example use of the direction feature*



11 General tolerances

11.1 General tolerance specifications

*COMMENTARY ON **11.1***

The term "general tolerance" is used to describe a tolerance, or range of tolerances, which apply to dimensions or features which do not have their own individual tolerances.

General tolerances can be defined as dimensional tolerances, geometrical specifications, surface texture requirements or edge condition requirements.

11.1.1 General

The following rules in **11.1.2** to **11.1.4** shall apply to the use of general tolerances, unless alternative rules or extended rules are indicated with a statement in the TPS or a related document.

NOTE Rules for general tolerance can be indicated with a reference to a standard for general tolerances such as BS EN ISO 22081 or BS ISO 8062-4. In most cases, an indication of a tolerance category, grade or value can also be required.

11.1.2 Application of general dimensional tolerances

General dimensional tolerances shall apply to any dimension displayed in a TPS which is shown with a nominal value, and is:

a) not subject to an individual dimensional tolerance;

b) not a TED; and

c) not an auxiliary dimension.

NOTE 1 See BS EN ISO 14405-2 for an explanation of the ambiguities inherent in using dimensional tolerances for any properties other than size.

NOTE 2 A general dimensional tolerance can only be applied to those sizes, distances, angles, etc., which are identified with a nominal dimension.

11.1.3 Application of general geometrical specifications

The general geometrical specification shall apply to any surface which is:

a) not subject to an individual geometrical specification;

b) not subject to an individual dimensional tolerance;

c) not subject to a general dimensional tolerance; and/or

d) not used as a datum feature which is referenced by the general geometrical specification.

NOTE If any geometrical specification other than a surface profile tolerance is used, then additional rules might be required.

11.1.4 Application of general tolerances for surface texture and edge conditions

General surface texture requirements shall apply to any feature which is not subject to an individual surface texture specification.

General edge condition requirements shall apply to any feature which is not subject to an individual edge condition specification.

11.2 General dimensional tolerances

*COMMENTARY ON **11.2***

This subclause describes methods for specifying general tolerance requirements. Any numerical values used are for illustrative purposes only, and do not imply or recommend any specific tolerance values.

General dimensional tolerances might consist of a single value, e.g. a drawing note stating:

"All dimensions ±0,2 unless otherwise stated".

11.2.1 General dimensional tolerance with a single value

General dimensional tolerances consisting of a single value shall be indicated on the TPD with a note, such as: "All dimensions ±0,2 unless otherwise stated" or an equivalent indication.

11.2.2 General dimensional tolerances with range of values

General dimensional tolerances consisting of a range of values shall be indicated on the TPD along with the system used for allocating the tolerance values, such as:

a) a table of values (see Table 23);

b) the use of flag notes; and

c) labelling or classification of dimensions.

Table 23 — *Tolerances allocated according to dimension value*

Dimension value		Tolerance
Over	**Up to**	
0[A)]	10	±0,1
10	50	±0,2
50	200	±0,4
200	—	±0,8

A) This can result in requirements where the tolerance is larger than the dimension value, e.g. 0,05 ±0,1 resulting in possible negative values. Alternatively, a positive minimum value can be indicated, such as "over 3 / up to 10" in which case the system does not cover dimensions with a value below 3.

NOTE A system of general dimensional tolerances which links tolerance values to the number of decimal places in the dimension is widely used, e.g.:

DIMENSION X ± 0,5

DIMENSION X.X ± 0,2

DIMENSION X.XX ± 0,05.

This method should be avoided because:

a) it is contrary to the decimal principle given in BS EN ISO 8015;

b) it depends on the number of decimal places being correct for every nominal dimension without an individual tolerance, so there is a significant risk of error;

c) the value of the tolerance is not related to the value of the dimension; and

d) when changes are made to the 3D CAD and the drawing is regenerated, dimension values change and sometimes the number of decimal places change, which can result in inadvertent changes to tolerance values.

11.3 General tolerances and 3D CAD data

General dimensional tolerances shall not be applied to 3D CAD data, unless their application is limited, e.g. to hole diameters.

NOTE 1 If general dimensional tolerances are applied to 3D CAD data without limitations, there are an infinite number of dimensions to which they can apply, which might also lead to tolerancing conflicts.

NOTE 2 When general tolerances are to be applied to 3D CAD data, a surface profile tolerance (⌓) is usually used. This should be set up to avoid circular references to the datum features, which can normally be achieved by tolerancing the datum features separately. BS EN ISO 22081 provides a method and rules for doing this.

11.4 General geometrical and size specifications (BS EN ISO 22081)

COMMENTARY ON ***11.4***

BS EN ISO 22081 defines a system for applying general tolerances to size dimensions and/or to surfaces but does not provide tolerance values.

The general tolerance applied to surfaces is a surface profile tolerance with a datum system. The general tolerances applied to size dimensions can be linear and/or angular. Some of the key rules are listed below.

The following rules shall be applied for datum systems.

a) If general tolerances are to be applied to surfaces, a datum system shall be defined.

b) The datum system shall control all necessary degrees of freedom.

c) Datum features used for the general tolerance datum system shall be toleranced individually, and not be subject to general tolerances.

The following rules shall be applied for general tolerances applied to size dimensions.

1) The general dimensional tolerances shall only be applied to size dimensions for features of size which are identified in the specification.
2) The features of size shall be identified on a drawing with a dimension which is not a TED, or an auxiliary dimension, or subject to a specific size tolerance, and displays only a nominal value.
3) The feature of size shall be identified in a CAD model with a dimension as above, or as a CAD attribute.

The following rules shall be applied for general tolerances applied to surfaces.

i) The general tolerance shall be surface profile with a datum system.
ii) The general tolerance shall not apply to surfaces which:
   - are subject to an individual or general size tolerance;
   - have an individual geometrical specification applied either to the surface or to a median line or median plane derived from the surface;
   - are used as datum features for the datum system used in the general tolerance; or
   - are indicated with simplified representations (e.g. threaded holes) or omitted from the 3D CAD model (such as modified edges or draft angles in certain cases).

The general tolerances shall be indicated in or near the title block of the specification, or the appropriate view state of a 3D specification (with appropriate values and datum systems substituted for those shown in Figure 166).

Figure 166 — *General tolerances – ISO 22081*

General tolerance ISO22081

⌓	0,5	A	B	C

Linear sizes: ±0,2Ⓜ

Angular sizes: ±2°

NOTE There are also options for replacing single general tolerance values with a range of values defined in BS EN ISO 22081:2021, Annex A.

11.5 General dimensional values (BS EN 22786-1)

COMMENTARY ON ***11.5***

This British Standard provides general dimensional tolerances which correspond to customary workshop accuracy for parts formed by metal machining operations or from sheet metalwork. It is based on data collected prior to its publication in 1989. BS EN 22768-1 does not distinguish between dimensions which are defining size or location properties. It lists four tolerance classes, from fine (f) to very coarse (v). For each class, tolerances are given for different dimension value ranges. BS EN 22768-1:1993, Table 1 and Table 2 are provided for linear dimensions, for radii and chamfer on external edges, and for angular dimensions. Table 24 shows the tolerances on linear dimensions.

When general dimensional tolerance values from BS EN 22768-1 are to be used in a TPD, the indication "GENERAL TOLERANCES ISO 2768-x" shall appear on the TPD, where "x" is replaced by the letter, "f", "m", "c" or "v" in accordance with Table 24.

NOTE 1 BS EN 22768-1 is currently under revision. The new revision is expected in 2026 or 2027. The new revision is expected to restrict the use of these general tolerance values to size dimensions for features of linear or angular size.

NOTE 2 The tolerance values from BS EN ISO 2768 (under preparation) can be used in conjunction with BS EN ISO 22081.

Table 24 — *Tolerance limits for general specifications of linear sizes*

Tolerance class		Nominal linear size							
Designation	Description	0,5 up to 3	over 3 up to 6	over 6 up to 30	over 30 up to 120	over 120 up to 400	over 400 up to 1 000	over 1 000 up to 2 000	over 2 000 up to 4 000
f	Fine	± 0,05	± 0,05	± 0,1	± 0,15	± 0,2	± 0,3	± 0,5	—
m	Medium	± 0,1	± 0,1	± 0,2	± 0,3	± 0,5	± 0,8	± 1,2	± 2
c	Coarse	± 0,2	± 0,3	± 0,5	± 0,8	± 1,2	± 2	± 3	± 4
v	Very coarse	—	± 0,5	± 1	± 1,5	± 2,5	± 4	± 6	± 8

NOTE SOURCE: BS EN 22768-1:1993, Table 1.

11.6 General tolerances for specific manufacturing processes

General tolerances for specific manufacturing processes shall conform to the following standards, as appropriate.

- ISO 8062 series: This series of standards cover general tolerances for cast components.
- BS EN ISO 8062-1, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 1: Vocabulary*.
- PD CEN ISO/TS 8062-2, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 2: Rules*.
- BS EN ISO 8062-3, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 3: General dimensional and geometrical tolerances and machining allowances for castings using ± tolerances for indicated dimensions*.
- BS ISO 8062-4, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 4: Rules and general tolerances for castings using profile tolerancing in a general datum system*.
- BS EN ISO 13920, *Welding – General tolerances for welded constructions – Dimensions for lengths and angles, shape and position*.
- BS ISO 20457, *Plastics moulded parts – Tolerances and acceptance conditions*.
- BS EN ISO 17295:2023, *Additive manufacturing – General principles – Part positioning, coordinates and orientation*.
- BS EN 2066, *Aerospace series – Extruded section in aluminium alloys – General tolerances*.
- BS ISO 13583-1, *Centrifugally cast steel and alloy products – Part 1: General testing and tolerances*.
- BS EN 10058, *Hot rolled flat steel bars and steel wide flats for general purposes – Dimensions and tolerances on shape and dimensions*.
- BS EN ISO 9013, *Thermal cutting – Classification of thermal cuts – Geometrical product specification and quality tolerances*.
- BS ISO 3302-1, *Rubber – Tolerances for products – Part 1: Dimensional tolerances*.
- BS ISO 3302-2, *Rubber – Tolerances for products – Part 2: Geometrical tolerances*.

12 Statistical tolerancing

*COMMENTARY ON CLAUSE **12***

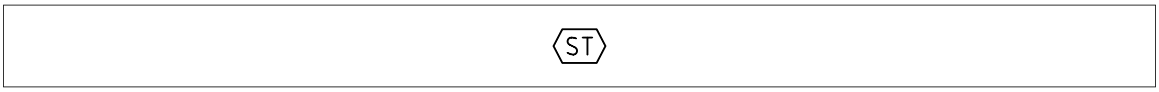
Tolerances can be applied to features of a TPS to facilitate the prediction of assembly conditions in the final product, e.g. clashes, gaps. The process of assigning tolerances to individual features involves identifying the final functional limits and then arithmetically subdividing and sharing this tolerance to the sub-assemblies and component parts.

Statistical methods of assigning tolerances, such as square root of the sum of the squares (RSS), can be used to subdivide the allowable assembly tolerance amongst the sub-assemblies and components. Differentiating statistically derived tolerances via symbology and/or a note identifies that the tolerance has been used in statistical analysis and, as such, when added to other contributory tolerances, could exceed the overall product requirement specification limits.

A statistical control of the manufactured population, as well as the individual feature, might also therefore be required to guarantee the desired output for the final product. The success of statistical tolerancing is dependent on the manufacturing processes producing the features being capable and stable within the parameters defined by the statistical control. By default, a GPS specification is an individual specification.

Where a tolerance is to be controlled using statistical methods, the population specification shall be identified using the statistical tolerance symbol (see Figure 167).

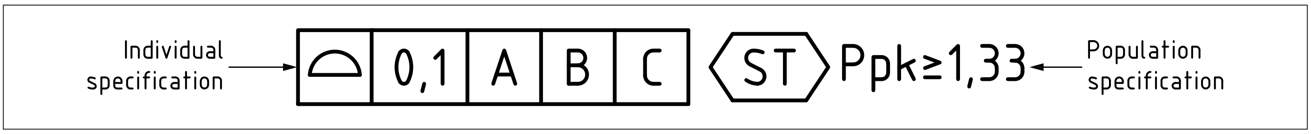
Figure 167 — *Statistical tolerance symbol*



The symbol shall be placed after the tolerance value, tolerance indicator or any linear, size or angle dimension for an individual feature specification, and before the population specification (statistical requirement) (see Figure 168). Both the individual and population specifications shall be satisfied independently to demonstrate conformity of the workpiece.

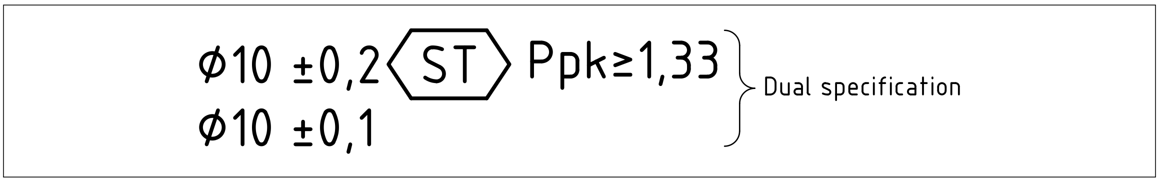
NOTE 1 The statistical control criteria apply to a set of values which consist of the same individual characteristic taken from a sample of individual workpieces in a population.

Figure 168 — *Population specification*



NOTE 2 Where two individual specifications are given for the same feature with different tolerance values and where one also has a complementary statistical population specification (dual specification), either the individual specification with an associated population specification OR the more restrictive individual specification can be used to prove conformity (see Figure 169).

Figure 169 — *Dual specification*



The following shall be agreed between the supplier and customer:

a) the statistical control criteria e.g. Ppk and Cpk, for demonstrating conformity to the population specification; and

b) the population sample size to be tested, method of assessing conformity and any physical validation required.

During the application of statistical tolerances as knowledge in the field of statistical analysis and control, the appropriate population controls shall be applied in any specific application. Statistical tolerances shall only be applied to product features where the variation and distribution can be assessed and monitored on the shop floor to check the physical output aligns with the statistical assumptions made in any tolerance analysis.

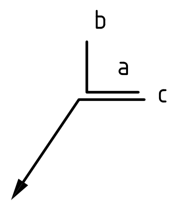
13 Edges

Where it is necessary to indicate how “blunt” or “sharp” an edge is permitted to be, edges shall be specified in accordance with BS EN ISO 13715. These indications shall only be used for unmodified edges, where, for example, no chamfer or radius is applied. The indications shall be used to indicate how much additional material can be present at an edge, which is known as a “passing”, or how much material can be missing from an edge, known as an “undercut”.

NOTE 1 The form of the passing or undercut is not defined.

The requirement shall be defined using the symbol in Figure 170 with indications in any of the locations marked a, b or c.

Figure 170 — *Edge condition indication*



Key

a Indication applies to both surfaces meeting at the edge

b Indication applies to the vertical surface leading to the edge

c Indication applies to the horizontal surface leading to the edge

The indications in locations a, b or c shall consist of one of the symbols “+”, “–” or “±” optionally followed by a value: the value determines the extent of the undercut or passing.

NOTE 2 Table 25 explains the meanings of the indications.

Table 25 — *Meaning of indications*

Indication	Meaning
+	a passing is permitted, size undefined
$+t$	a passing of between zero and t is permitted
$+t_1$ $+t_2$	a passing of between t_1 and t_2 is permitted
–	an undercut is permitted, size undefined
$-t$	an undercut of between zero and t is permitted
$-t_1$ $-t_2$	an undercut of between t_1 and t_2 is permitted
$+t_1$ $-t_2$	a passing of up to t_1 or undercut of up to t_2 is permitted
$\pm t$	a passing of up to t or undercut of up to t is permitted
where t, t_1 and t_2 are values in millimetres	

NOTE 3 Some examples of how edges can be specified are given in Table 26, while additional examples can be found in BS EN ISO 13715.

Table 26 — *Examples of indication of edges*

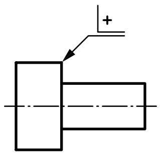
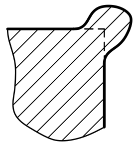
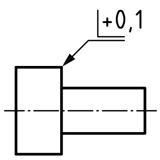
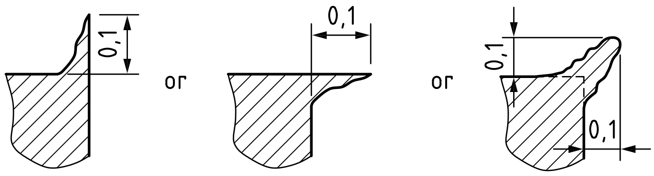
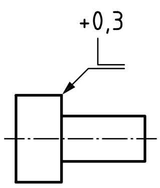
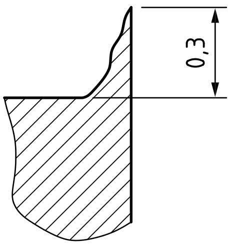
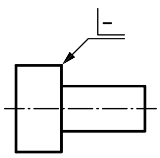
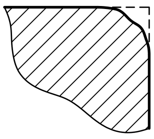
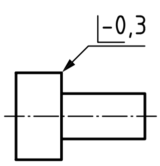
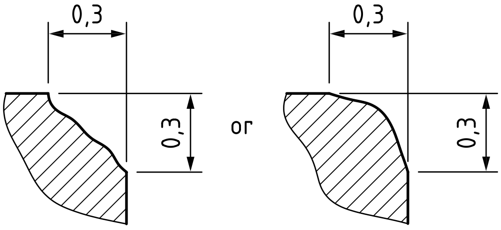
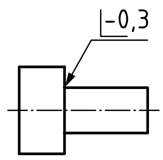
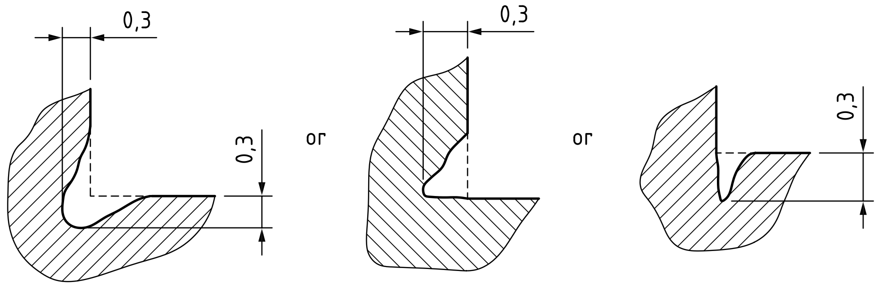
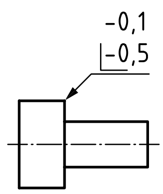
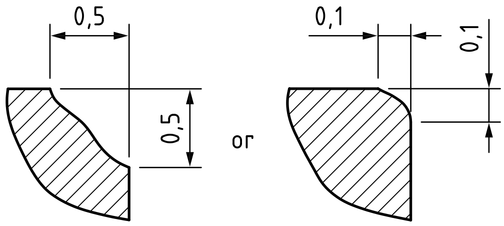
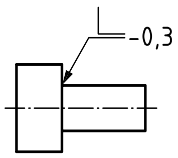
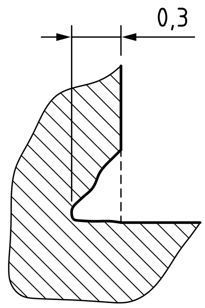
Indication	Meaning	Explanation
		External edge with passing permitted; size undefined
		External edge with passing up to 0,1 mm permitted in either direction
		External edge with passing acceptable up to 0,3 mm; direction defined
		External edge with undercut permitted, size undefined
		External edge with undercut up to 0,3 mm permitted in either direction

Table 26 — *Examples of indication of edges (continued)*

Indication	Meaning	Explanation
		Internal edge with undercut up to 0,3 mm permitted in either direction
		External edge with undercut between 0,1 mm and 0,5 mm in either direction
		Internal edge with undercut up to 0,3 mm permitted; direction defined

NOTE SOURCE: BS EN ISO 13715:2019, Table B.1, modified – updated title, selection of examples, reworded explanations

14 Welding symbols

14.1 The use of welding symbols on a TPS

The use of welding symbols and how they are displayed in a TPS shall conform to BS EN ISO 2553.

For welding of thermoplastic components, only the basic reference line indications, with the tail information (see **14.2** and **14.5**) shall be used.

NOTE Values are nominal with no system for applying tolerances to them.

14.2 Welding symbol basic requirements

"A reference line and arrow line are required elements. Additional elements may be included to convey specific information, such as the optional tail.

It is preferable to show the welding symbol on the same side of the joint that the weld is to be made, i.e. the arrow side (see **6.10**).

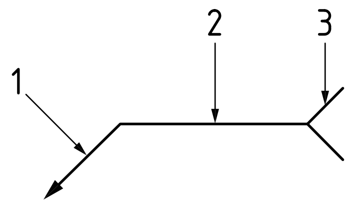
The thickness of the arrow lines, reference line, elementary symbols and lettering shall be in accordance with BS EN ISO 128 (all parts) and BS EN ISO 3098-2.

In order not to overburden drawings, reference should be made to notes in the drawing or other design-related documents."

[SOURCE: BS EN ISO 2553:2019, **4.1**, modified – "such as the optional tail" added]

NOTE The symbol shown in Figure 171 is often used to indicate the location of tack welds.

Figure 171 — *Basic welding symbol (joint details and type not specified)*



Key

1 Arrow line

2 Reference (continuation leader) line

3 Tail

NOTE SOURCE: BS EN ISO 2553:2019, Figure 1.

14.3 Arrow line

*COMMENTARY ON **14.3***

Two or more arrow lines can be combined with a single reference line to indicate the locations of identical welds (see Figure 172).

"An arrow line shall be used to indicate the joint to be welded.

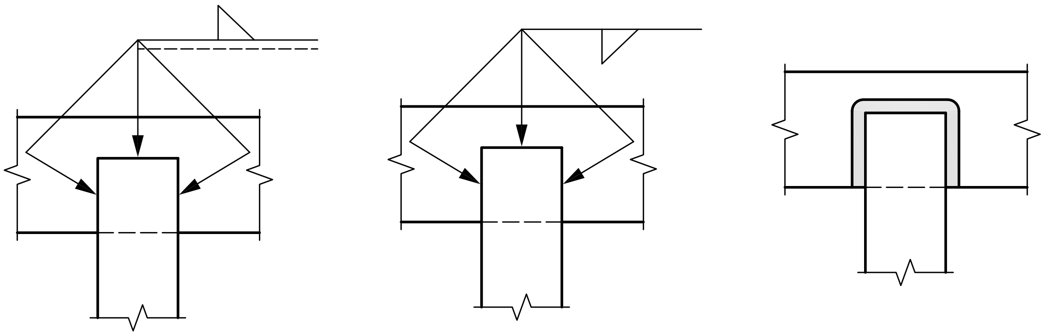
The arrow line shall:

- point to and be in contact with a solid line comprising part of the joint on the drawing (visible line);
- be drawn at an angle to and joined to a reference line and completed with a closed filled arrowhead.

NOTE The arrow line may be joined to either end of the reference line."

[SOURCE: BS EN ISO 2553:2019, **4.6**]

Figure 172 — *Examples of use of multiple arrow lines*



a) System A welding symbol

b) System B welding symbol

c) Weld produced using either system

NOTE SOURCE: BS EN ISO 2553:2019, Figure 4.

14.4 Reference line

"The reference line, when combined with elementary symbols, is used to indicate the side of the joint on which the weld is to be made.

NOTE 1 The reference line can be drawn parallel to the side edge of the drawing (whole welding symbol rotated by 90°), but this should only be done when space does not permit drawing parallel to the bottom edge."

[SOURCE: BS EN ISO 2553:2019, **4.7.1**]

"Of the two different approaches to designating the arrow side and other side on drawings (System A and System B), System A shall be the default.

*NOTE 2 These are as detailed in BS EN ISO 2553, **4.3**.*

Systems A and B shall not be mixed and drawings shall clearly indicate which system is used, including units of measurement in accordance with BS EN ISO 129-1."

[SOURCE: BS EN ISO 2553:2019, **4.3**]

"**Reference line – System A:** The reference line consists of two parallel lines of equal length: a continuous line and a dashed line (see Figure 173).

The dashed line may be drawn above or below the continuous line but shall preferably be drawn below.

The dashed line should be omitted for symmetrical welds, and for spot and seam welds made at the interface between two components.

Reference line – System B: The reference line shall be drawn as a continuous line (see Figure 173).

[SOURCE: BS EN ISO 2553:2019, **4.7.1**]"

"**Weld location arrow side/other side**: The arrow side is the side of the joint to which the arrowhead is pointing.

The other side is the opposite side of the joint to which the arrowhead is pointing. The arrow side and other side always form part of the same joint.

The other side of a joint shall not be confused with a hidden weld forming part of a different joint.

Arrow side/other side – System A: Elementary symbols shall be located on the continuous line when the weld is to be made on the arrow side of the joint (see Figure 173).

Elementary symbols shall be located on the dashed (identification) line when the weld is to be made on the other side of the joint.

Arrow side/other side – System B: Elementary symbols shall be located below the reference line when the weld is to be made on the arrow side of the joint (see Figure 173).

Elementary symbols shall be located above the reference line when the weld is to be made on the other side of the joint.

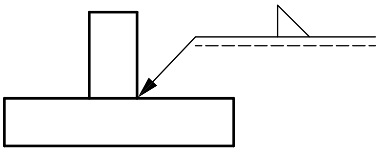
NOTE 3 In System A, the component of the reference line on which the elementary symbol is placed determines the side of the joint which is to be welded – the dashed line can be drawn above or below the solid line.

NOTE 4 In System B, the position of the elementary symbol above or below the reference line determines the side of the joint on which the weld is made."

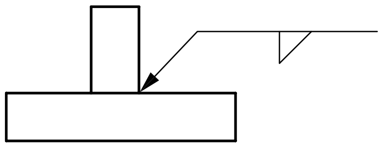
[SOURCE: BS EN ISO 2553:2019, **4.7.2.1**]

NOTE 5 The other side option is typically used where space is restricted in the TPS and it is not possible to position the welding symbol adjacent to the joint.

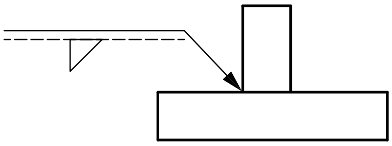
Figure 173 — *Reference lines illustrating arrow side and other side for System A and System B*



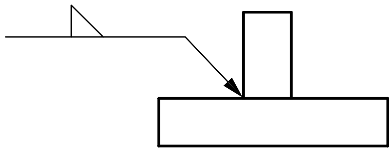
a) System A: Arrow side (symbol on solid component of reference line)



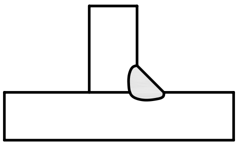
b) System B: Arrow side (symbol below reference line)



c) System A: Other side (symbol on dashed component of reference line)



d) System B: Other side (symbol above reference line)



e) Same weld produced using all four options

NOTE SOURCE: BS EN ISO 2553:2019, Table 5.

14.5 Tail

"The tail is an optional element which might be added to the end of the continuous reference line (see Figure 174) where additional complementary information is included as part of the welding symbol, for example:

a) quality level, for example, in accordance with BS EN ISO 5817, BS EN ISO 10042, BS EN ISO 13919 (all parts) or BS ISO 5623;

b) the welding process, for example, reference number in accordance with BS EN ISO 4063, or abbreviation;

c) filler material, for example, in accordance with BS EN ISO 14171 or BS EN ISO 14341;

d) welding position, for example in accordance with BS EN ISO 6947; and

e) supplementary information to be considered when making the joint.

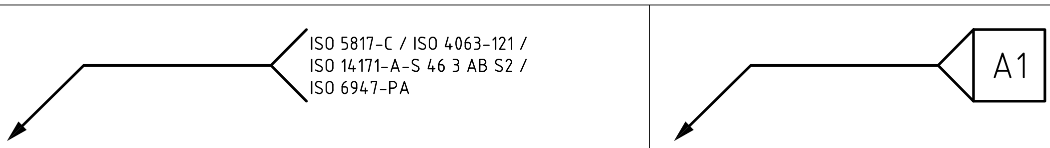
The information shall be listed and separated by a forward slash (/) [see Figure 174a)].

A closed tail shall only be used to indicate reference to a specific instruction, e.g. reference to a welding procedure specification (WPS), welding procedure qualification record (WPQR) or other document [see Figure 174b)]."

[SOURCE: BS EN ISO 2553:2019, **4.8**, modified – BS ISO 5623 added]

Repetition of additional information on symbols on a drawing shall be avoided. A single general note on the drawing shall be used instead.

Figure 174 — *Examples of the use of a tail on welding symbols*



a) Open tail

b) Closed tail

NOTE SOURCE: BS EN ISO 2553:2019, Figure 6.

14.6 Elementary symbols

"For metal welds, elementary symbols, in accordance with Table 27, may be added to the reference line in both systems A and B to indicate the type of weld to be made.

Elementary symbols form part of the welding symbol and shall be drawn attached to the reference line generally at the mid-point.

Elementary symbols can be complemented by:

a) supplementary symbols;

b) dimensions; and

c) complementary information.

The orientation of the elementary symbols shall not be changed to that shown."

[SOURCE: BS EN ISO 2553:2019, **4.4**, modified – references removed]

Table 27 — *Elementary symbols for metal welds*

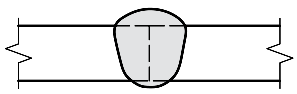
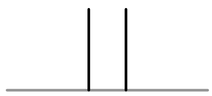
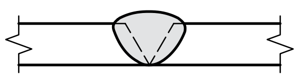
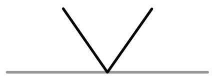
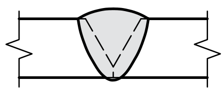
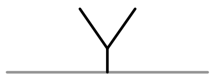
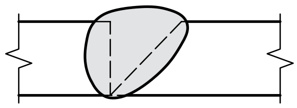
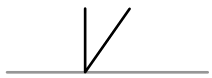
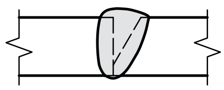
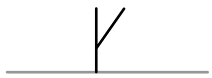
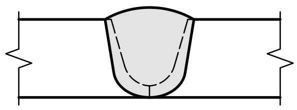
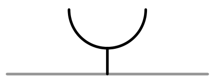
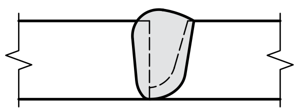
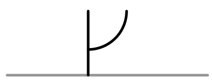
Designation	Illustration of weld type[A)]	Symbol[B)]
Square butt[C)]		
Single-V butt[C)]		
Single-V butt with broad root face[C)]		
Single-bevel butt[C)]		
Single-bevel butt with broad root face[C)]		
Single-U butt[C)]		
Single-J butt[C)]		

Table 27 — *Elementary symbols for metal welds (continued)*

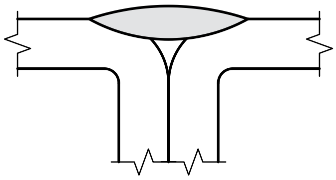
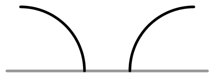
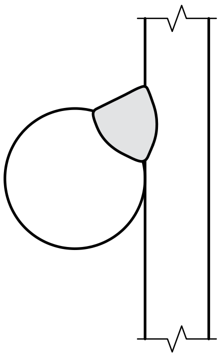
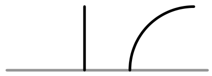
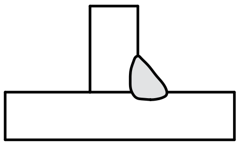
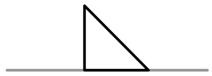
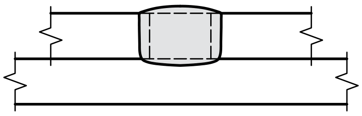
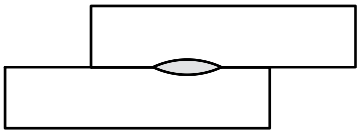
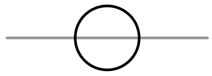
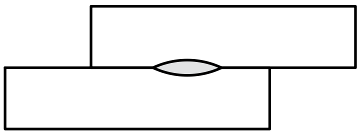
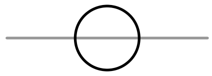
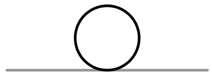
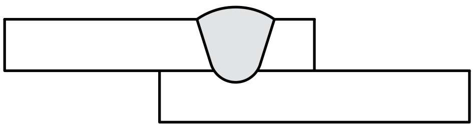
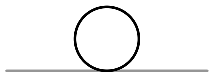
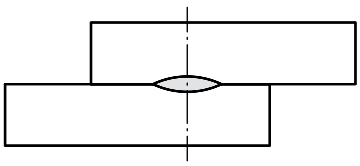
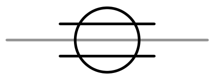
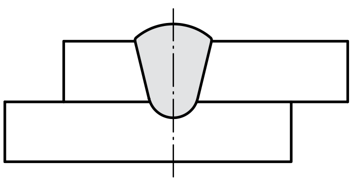
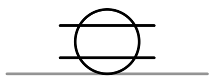
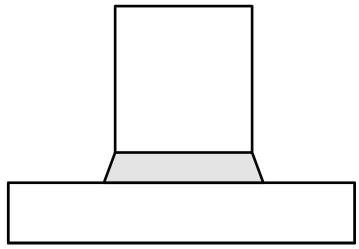
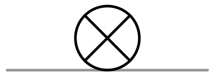
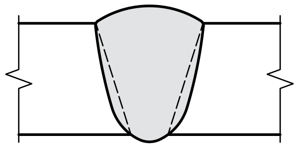
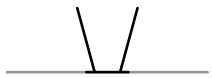
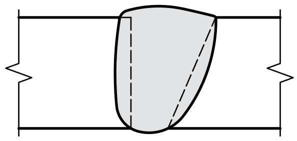
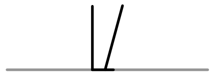
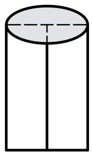
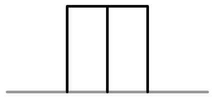
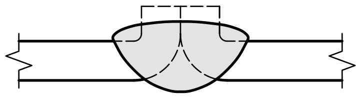
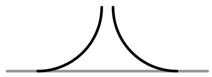
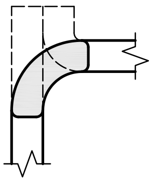
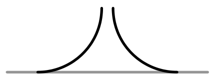
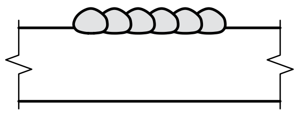
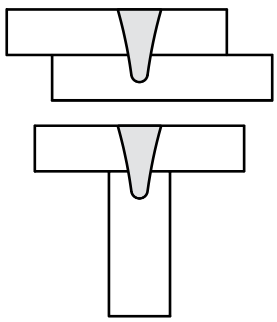
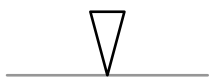
Designation	Illustration of weld type[A]	Symbol[B]
Flare V		
Flare bevel		
Fillet		
Plug		
Resistance spot[D]		
Projection		System A  System B 
Fusion spot		
Resistance seam[D]		
Fusion seam		

Table 27 — *Elementary symbols for metal welds (continued)*

Designation	**Illustration of weld type**[A)]	**Symbol**[B)]
Stud		
Steep-flanked single-V butt[C)]		
Steep-flanked single-bevel butt[C)]		
Edge[D)]		
Flanged butt		
Flanged corner		
Overlay		
Stake[D)]		

NOTE SOURCE: BS EN ISO 2553:2019, Table 1, modified – number column and weld type removed, for metal welds added on the title.

A) Dashed lines show joint preparation prior to welding.

B) The grey line is not part of the symbol. It indicates the position of the reference line.

C) Butt welds are full penetration unless otherwise indicated by dimensions on the welding symbol or by reference to other information, for example the WPS.

D) Symbol can also be used for joints with more than two members.

14.7 Broken arrow lines

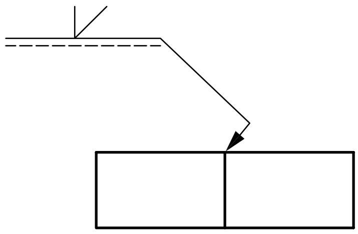
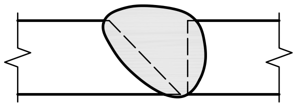
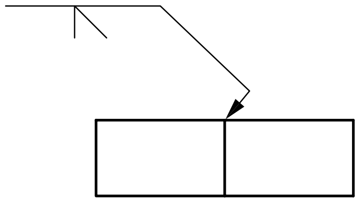
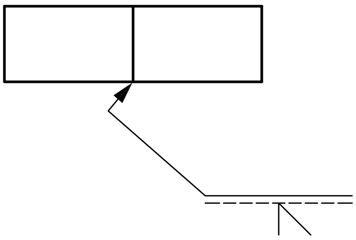
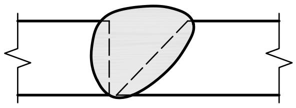
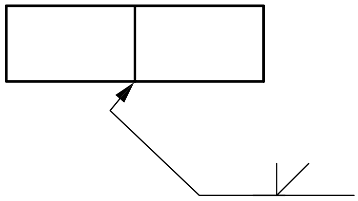
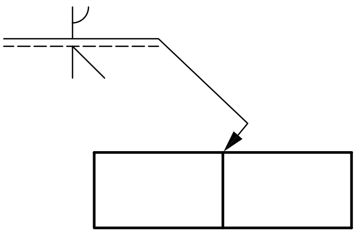
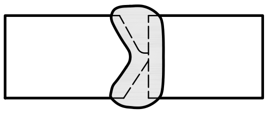
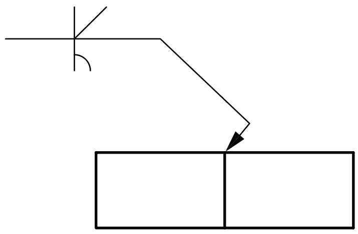
"For butt welds in metal plates (excluding T-butt welds) when a specific joint member is required to be prepared (e.g. single-bevel or single-J-butt welds), the arrow line shall have a break and point toward that member.

The arrow line need not be broken if it is obvious or if there is no preference as to which member is to be prepared.

Examples of the use of broken arrow lines are given in Table 28."

[SOURCE: BS EN ISO 2553:2019, **4.6.3**, modified – "joints in plates" replaced with "welds in metal plates"]

Table 28 — *Example uses of broken arrow line for metal welds*

System A welding symbol	Illustration of weld type	System B welding symbol
		
		
		

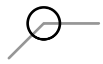
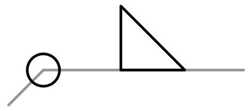
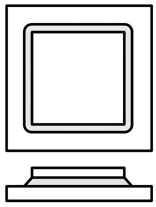
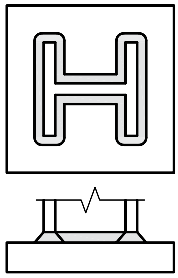
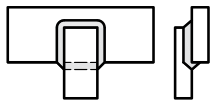
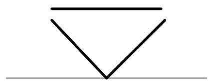
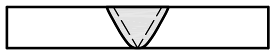
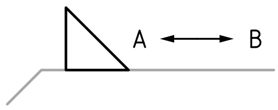
NOTE SOURCE: BS EN ISO 2553:2019, Table A.1, modified – number column removed, for metal welds added on the title.

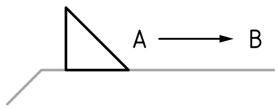
14.8 Supplementary symbols

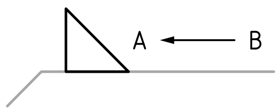
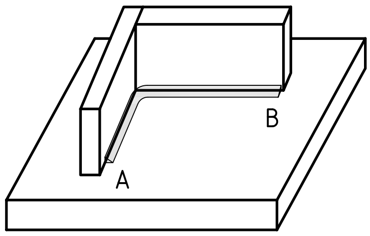
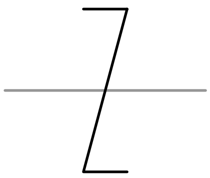
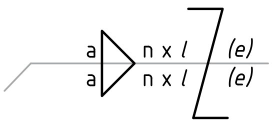
Additional information for the required joint shall be provided using supplementary symbols, where required, such as information about the shape of the weld or how the welded joint is made.

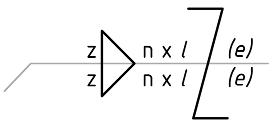
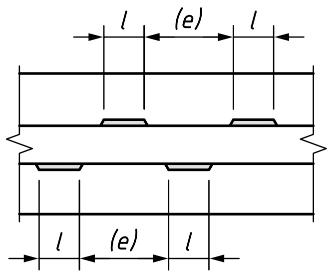
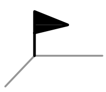
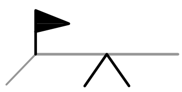
NOTE The most commonly used supplementary symbols for metal weld processes are given in Table 29. For the full list of supplementary symbols, descriptions and diagrams available, see BS EN ISO 2553:2019, Table 3.

Table 29 — *Common supplementary symbols for metal welds*

<table>
<tr><th>Designation</th><th>Symbol[A)]</th><th>Application[A)]</th><th>Illustration of weld</th></tr>
<tr><td>Weld all-round</td><td></td><td></td><td>
Example A

Example B

Example C</td></tr>
<tr><td>Flush[B)]
(flat-finished)</td><td></td><td></td><td></td></tr>
<tr><td>Weld between two points</td><td></td><td>

or


or

</td><td>
</td></tr>
<tr><td>Staggered intermittent weld</td><td></td><td>

or

</td><td>
</td></tr>
<tr><td>Field weld</td><td></td><td></td><td>No example</td></tr>
</table>

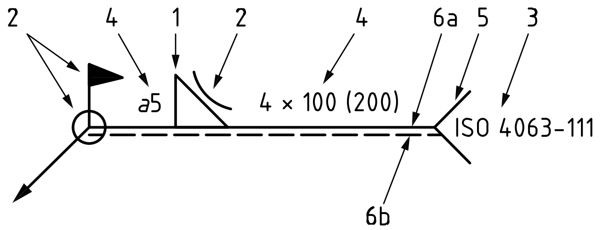
NOTE SOURCE: BS EN ISO 2553:2019, Table 3, modified – selected examples, for metal welds added on the title.

A) The grey line is not part of the symbol and is included to show the position of symbol on reference line and/or the arrow line only.

B) Welds that require approximately flush, convex or concave faces without post-weld finishing are specified by use of the flush, convex or concave contour symbol.

Figure 175 demonstrates the correct application of drawing principles (system A). It is illustrative only and is not intended to represent good design practices or to replace code or specification requirements. For more examples of the use of welding symbols see BS EN ISO 2553:2019.

Figure 175 — *Comprehensive welding symbols showing the location of weld elements*



Key

1 Elementary symbol (fillet weld)

2 Supplementary symbol (concave fillet weld contour, field weld, weld all-around)

3 Complementary information [shielded metal arc welding (SMAW)/process 111 in accordance with BS EN ISO 4063]

4 Dimensions (5 mm nominal throat thickness intermittent fillet weld, composing four weld elements 100 mm in length with 200 mm spacing between elements)

5 Tail

6a Reference line (continuous)

6b Dashed line (identification line) – system A only

NOTE SOURCE: BS EN ISO 2553:2019, Figure A.1a), modified – "identification line" added.

15 Surface texture specification details

*COMMENTARY ON CLAUSE **15***

Although the surface of a workpiece might appear smooth, there is a complex surface structure or texture comprising peaks and valleys, with its form and orientation resulting from manufacturing and surface treatment processes. The texture of surfaces is critical to the durability and performance of interacting systems, particularly in the case of lubricated sliding and rolling contacts. It is therefore useful to be able to specify the surface texture and orientation required for the desired functionality.

Many of the surface variations repeat at regular intervals as they are consequences of machining feed rates, particular harmonics of vibration or factors such as the grain structure of the material, so surface texture specifications deal with different frequencies and amplitudes of these variations.

Surface texture is specified and evaluated either as profile surface texture or as areal surface texture. The full range of parameters and settings involved in these specifications is immense, and cannot be adequately summarized here, so this clause should be treated simply as an introduction to some of the terminology and symbology used, and reference should be made to the standards listed below for a full understanding of the application and interpretation of these specifications.

15.1 General

Surface texture specifications shall conform to:

a) BS EN ISO 21920 series for profile surface texture;

b) BS EN ISO 25178 series for areal surface texture; and

c) BS EN ISO 16610 series for filtration.

NOTE 1 Some examples of surface texture specifications and a summary of the changes from BS EN ISO 1302 (withdrawn) can be found in Annex M.

Specifications based on the BS EN ISO 21920 series and BS EN ISO 25178 series shall use the new versions of the surface texture symbol that are defined in those standards.

NOTE 2 The new symbols are used to differentiate the specifications from indications based on BS EN ISO 1302 (withdrawn).

NOTE 3 In cases where a TPS makes use of the BS EN ISO 1302 (withdrawn) version of the surface texture symbol, surface texture specifications should be interpreted in accordance with BS EN ISO 1302 (withdrawn).

15.2 Profile surface texture

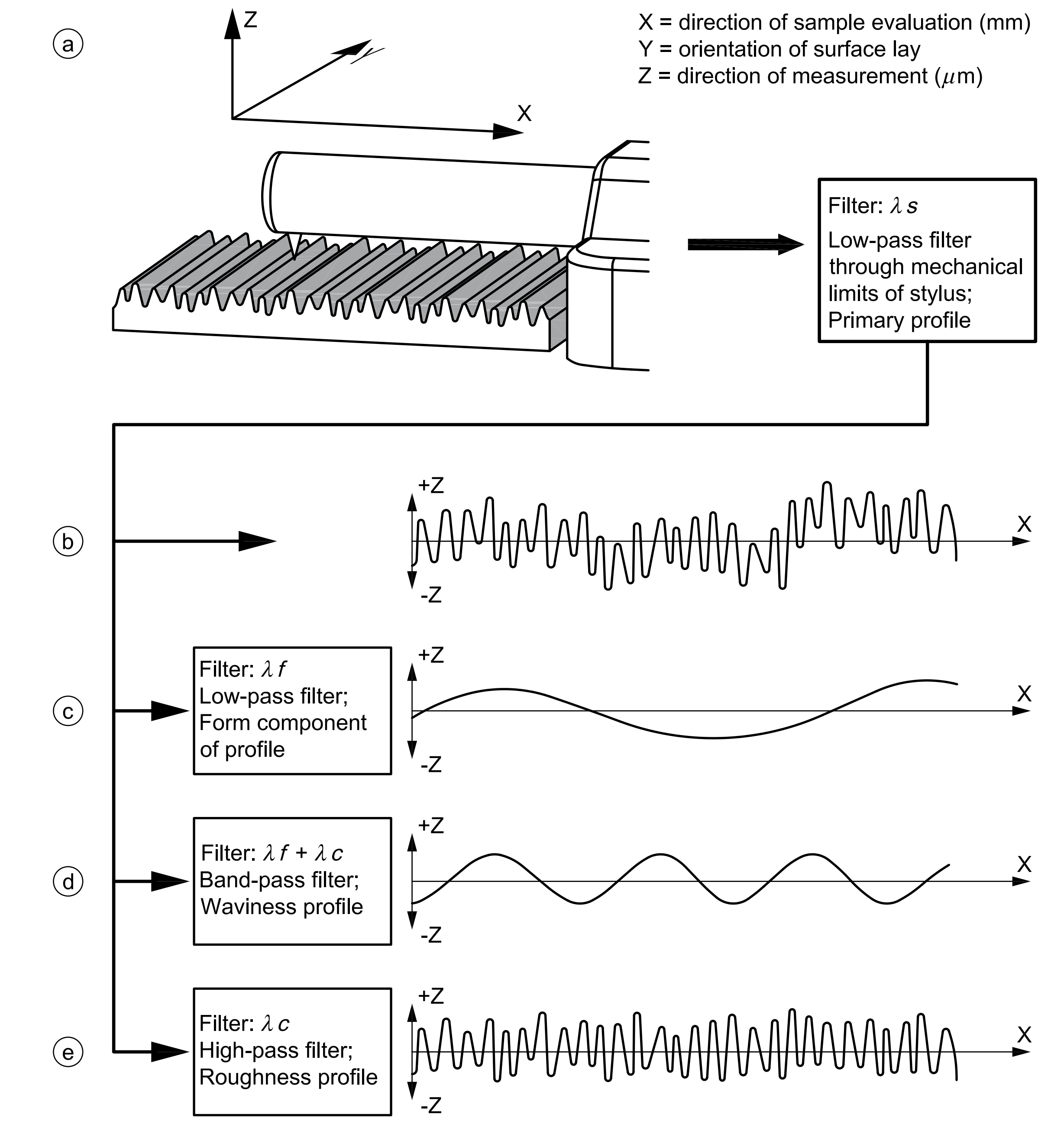
15.2.1 General

When verifying surface profile, it shall be measured to establish a range of geometrical parameters.

NOTE 1 The measurement should be taken perpendicular to the direction of the lay unless otherwise indicated.

NOTE 2 Typically, a stylus device is drawn across the surface in the X direction, where Y is nominally parallel to the lay of the surface texture and Z is normal to the surface as shown in Figure 176.

Figure 176 — *Use of stylus device*



NOTE The measuring device generates the mechanical profile prior to the application of filters for the separation of roughness from waviness, form and signal noise.

15.2.2 Profiles

COMMENTARY ON ***15.2.2***

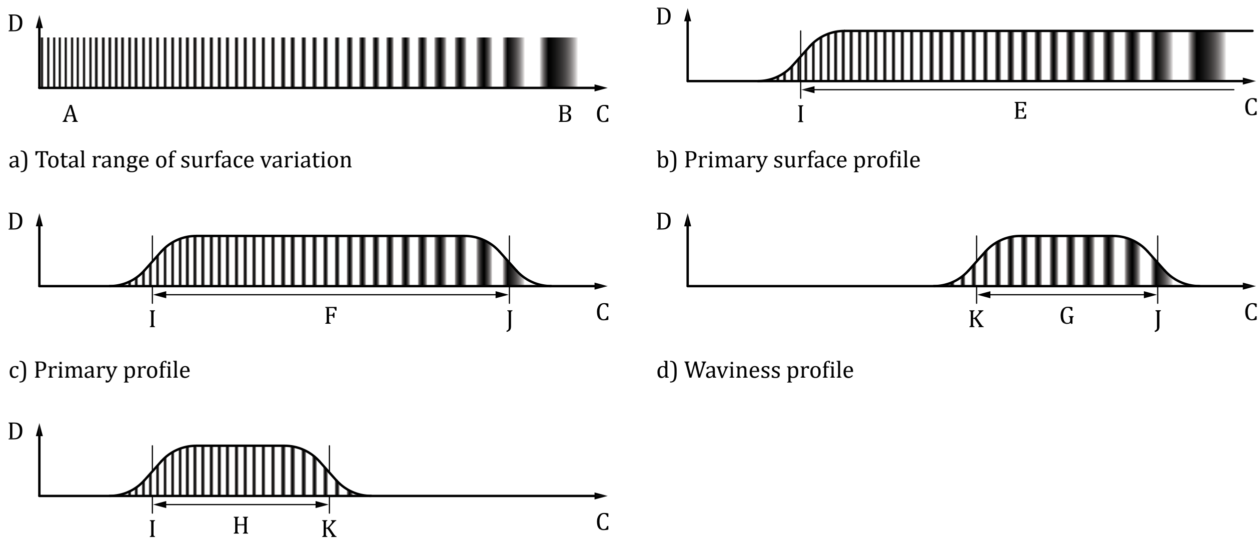
The primary surface profile is the total profile after application of the low-pass S-filter with N_{is} *nesting index, removing what is sometimes known as "microroughness" from the profile. In practice, this filter is sometimes the mechanical limit of the measurement stylus. The remaining profile includes waviness, roughness and longer wavelength components of form.*

The form elements of the profile, such as curvature of a surface, are removed with a profile F-operation, leaving the primary profile. The primary profile can be characterized with a P-parameter. The primary profile may be subject to an S-filter with Nic nesting index, removing the roughness and leaving the waviness profile which can be characterized with a W-parameter. The primary profile may be subject to an L-filter with Nic nesting index, removing the waviness and leaving the roughness profile which can be characterized with an R-parameter (see Figure 177).

The Nic nesting index marks the boundary between roughness and waviness.

The measurement result is dependent on the evaluation length and the filter settings. For many parameters default settings are defined, but these can be changed if required.

Figure 177 — *Relationship between the primary surface profile, P-profile, W-profile and R-profile*



a) Total range of surface variation

b) Primary surface profile

c) Primary profile

d) Waviness profile

e) Roughness profile

Key

A Small lateral scale

B Large lateral scale

C Scale axis

D Amplitude axis

E Lateral scale component of primary surface profile

F lateral scale component of P-profile

G Lateral scale component of W-profile

H Lateral scale component of R-profile

I Profile S-filter nesting index N_{is}

J Profile F-operation nesting index N_{if}

K Profile L-filter nesting index N_{ic}

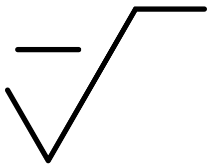
NOTE SOURCE: BS EN ISO 21920-2:2022, Figure 9.

15.2.3 Profile surface texture indications

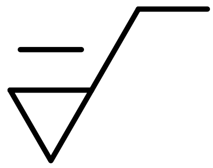
15.2.3.1 Graphical symbols

The graphical symbols shall conform to Figure 178.

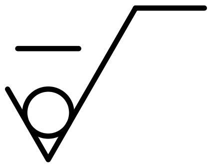
Figure 178 — *Graphical symbols indication profile surface texture*



a) Any manufacturing process permitted



b) Material shall be removed



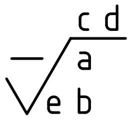
c) Material removal is not permitted

NOTE 1 The bar above the symbol is to distinguish this symbol from the ones used in BS EN ISO 1302 (withdrawn) and those used for indication of areal surface texture specifications.

NOTE 2 SOURCE: BS EN ISO 21920-1:2022, Figure 4.

Surface texture requirements shall be placed in the following locations relative to the graphical symbol as given in Figure 179.

Figure 179 — *Structure of profile surface texture symbol*



Key

a Single profile surface texture requirement

b Additional profile surface texture requirements

NOTE 1 These requirements can be placed on separate rows and the height of the graphical symbol extended to accommodate them.

c Manufacturing process

d Surface lay type and direction

e Profile direction relative to the surface lay

NOTE 2 SOURCE: BS EN ISO 1302:2002 (withdrawn), Figure 6, modified – d moved next to c and e moved next to b.

15.2.3.2 Minimal indications and complete indications for profile surface texture

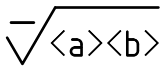
*COMMENTARY ON **15.2.3.2***

Defined defaults exist for parameters Rz, Ra, Rp, Rv, Rq, Rmax, Rt and Pt, which are contained in BS EN ISO 21920-3.

For examples of profile surface texture specification and areal surface texture specifications, and summary of changes from BS EN ISO 1302 (withdrawn), see Annex K.

A minimal indication shall be used, incorporating a parameter and tolerance value alone, when all defaults apply to the specification, as shown in Figure 180.

Figure 180 — *Minimal indications for parameter with defined defaults (all defaults apply)*



Key

a Symbol for parameter

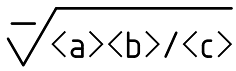
b Tolerance limit value of the profile surface texture parameter

〈 〉 Placeholder has to be specified

NOTE SOURCE: BS EN ISO 21920-1:2022, Figure 5.

For other parameters which do not have fully defined defaults, a minimal indication shall consist of a parameter, a tolerance limit value and a setting class, *Scn*, or nesting index N_{ic}, as shown in Figure 181.

Figure 181 — *Minimal indications for parameter without defined defaults*



Key

a Symbol for parameter

b Tolerance limit value of the profile surface texture parameter

c Symbol for the specified setting class *Scn* or a default value of profile L-filter nesting index N_{ic} for R-parameters or profile S-filter nesting index N_{ic} for W-parameters

〈 〉 Placeholder has to be specified

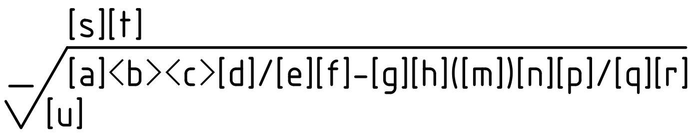
NOTE 1 Oblique strokes, "/", are used to separate the specification sections.

NOTE 2 SOURCE: BS EN ISO 21920-1:2022, Figure 6.

When non-default settings are required, they shall be included in the specification. The specification elements for evaluation length R-parameters shall be arranged in the sequences shown for the complete indications given in Figure 182.

The specification elements for section length R-parameters shall be arranged in the sequences shown for the complete indications given in Figure 183.

Figure 182 — *Indication for evaluation length R-parameters including all optional specification elements*



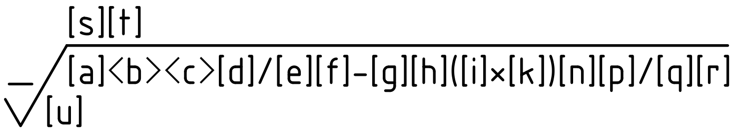
Key

a	Tolerance type
b	Symbol for R-parameter
c	Tolerance limit value of the profile surface texture parameter
d	Tolerance acceptance rule
e	Profile S-filter type
f	Profile S-filter nesting index
g	Profile L-filter type
h	Profile L-filter nesting index
m	Evaluation length
n	Profile F-operator association method and element
p	Profile F-operator nesting index
q	Method of profile extraction
r	Placeholder for the OR(*n*) symbol for other requirements
s	Manufacturing process
t	Surface lay and direction of lay
u	Profile direction relative to the surface lay
〈 〉	Placeholder has to be specified
[ ]	Placeholder can be specified if different from the default or an additional requirement

NOTE 1 Evaluation length R-parameters include Ra, Rq, Rt, Rzx and Rk (see BS EN ISO 21920-2).

NOTE 2 SOURCE: BS EN ISO 21920-1:2022, Figure 7.

Figure 183 — *Indication for section length R-parameters including all optional specification elements*



Key

a	Tolerance type
b	Symbol for R-parameter
c	Tolerance limit value of the profile surface texture parameter
d	Tolerance acceptance rule
e	Profile S-filter type
f	Profile S-filter nesting index
g	Profile L-filter type
h	Profile L-filter nesting index
i	Section length
k	Number of sections
n	Profile F-operator association method and element
p	Profile F-operator nesting index
q	Method of profile extraction
r	Placeholder for the OR(*n*) symbol for other requirements
s	Manufacturing process
t	Surface lay and direction of lay
u	Profile direction relative to the surface lay
〈 〉	Placeholder has to be specified
[ ]	Placeholder can be specified if different from the default or an additional requirement

NOTE 1 Section length R-parameters include Rz, Rp and Rv (see BS EN ISO 21920-2).

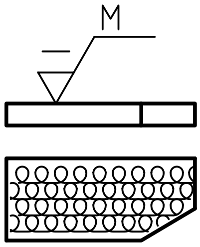
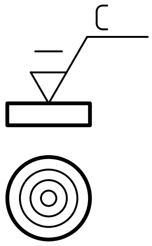
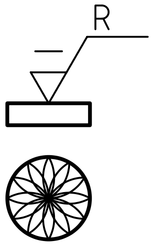
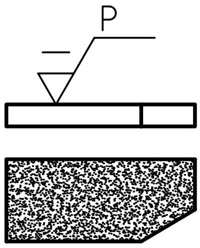
NOTE 2 The parentheses around ([i] x [k]) bring out the relationship between these specification elements as the evaluation length results from this product.

NOTE 3 SOURCE: BS EN ISO 21920-1:2022, Figure 8.

15.2.4 Symbols for surface lay type and direction

A surface lay requirement which does not require a direction reference shall be indicated by the symbols given in Table 30.

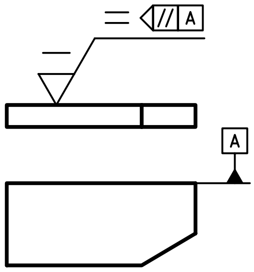
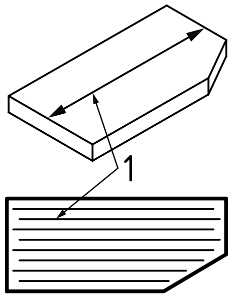
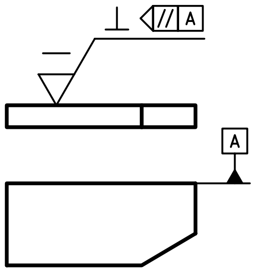
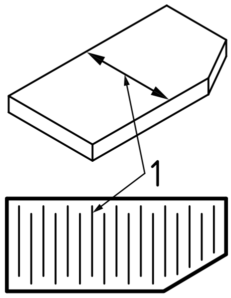
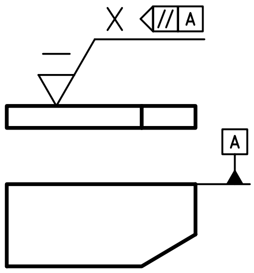
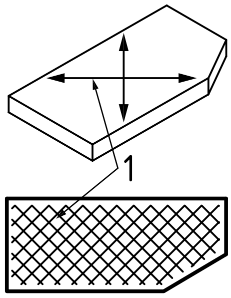
Table 30 — *Symbols, interpretation and examples for the indication of surface lay*

Graphical symbol	Interpretation and example	
M	Multidirectional	
C	Circular relative to centre of surface to which symbol applies	
R	Radial relative to centre of surface to which symbol applies	
P	Lay is particulate, non-directional or protuberant	

NOTE SOURCE: BS EN ISO 21920-1:2022, Table 1.

A surface lay requirement which does require a direction reference shall be indicated by the symbols given in Table 31, making use of the intersection plane indicator.

Table 31 — *Symbols, examples of indication and interpretation of surface lay and direction of lay*

Graphical symbol	Example of indication	Interpretation	Required direction of lay
=		Parallel to the direction, given by the intersection plane indicator and the datum feature indicator	
⊥		Perpendicular to the direction, given by the intersection plane indicator and the datum feature indicator	
X		Crossed in two oblique directions relative to the direction, given by the intersection plane indicator and the datum feature indicator	

Key

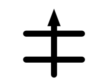
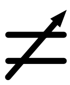
1 Required direction of lay

NOTE SOURCE: BS EN ISO 21920-1:2022, Table 2.

15.2.5 Profile direction

The profile direction shall be indicated when the profile is obtained in a non-default direction, using the symbols in Table 32.

Table 32 — *Graphical symbols for the indication of the profile direction*

Graphical symbol	Interpretation
	Perpendicular to the predominant direction
	Parallel to the predominant direction
	Circular to centre of surface to which symbol applies
	At a defined angle to the predominant direction (0° < n < 90°)

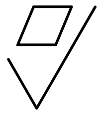
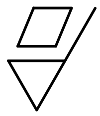
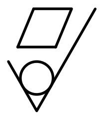
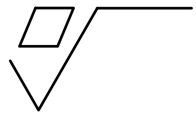
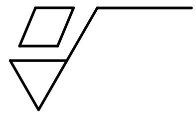
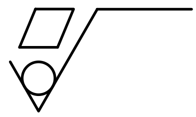
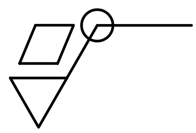
NOTE SOURCE: BS EN ISO 21920-1:2022, Table 3, modified – reference to Figure 19 removed.

15.3 Areal surface texture indications

When the same surface texture is required on all surfaces around a workpiece outline (integral features), represented on the drawing by a closed outline of the workpiece, a circle shall be added to the complete graphical symbol as given in Table 33 and Figure 184.

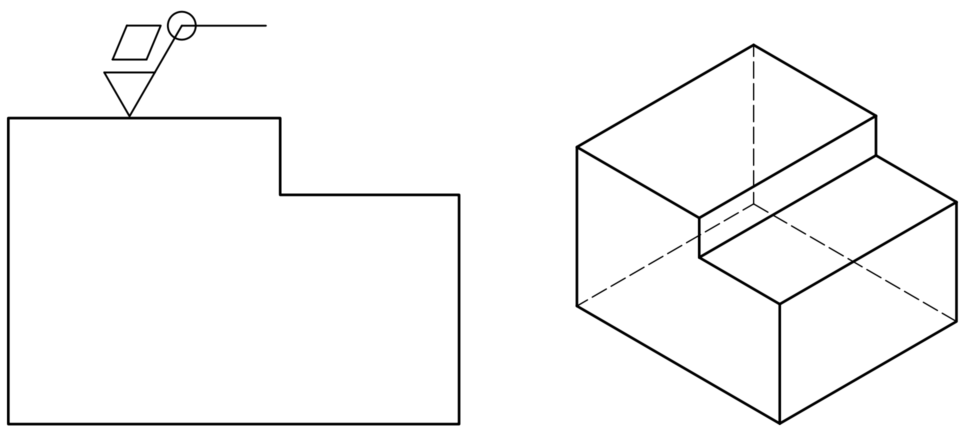
Surfaces shall be indicated independently if any ambiguity may arise from the all-around indication.

Table 33 — *Graphical symbols for the indication of areal surface texture*

No.	Description	Symbol
1	Basic graphical symbol for areal surface texture	
2	Expanded graphical symbol indicating removal of material required	
3	Expanded graphical symbol indicating removal of material not permitted	
4	Complete graphical symbol Any manufacturing process permitted	
5	Complete graphical symbol Material shall be removed	
6	Complete graphical symbol Material shall not be removed	
7	Complete graphical symbol With “all-around” modifier	

NOTE SOURCE: BS EN ISO 25178-1:2016, Table 1.

Figure 184 — *Areal surface requirement for all six surfaces represented by outline on workpiece*



NOTE SOURCE: BS EN ISO 25178-1:2016, Figure 1.

The complementary surface texture requirements in the form of:

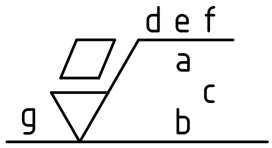
a) surface texture parameters;

b) numerical values; and

c) a transmission band

shall be located at the specific positions in the complete graphical symbol in accordance with Figure 185.

Figure 185 — *Positions of surface texture requirements in the complete graphical symbol*



Key

a Single profile surface texture requirement

NOTE 1 Generally, the different parts of the indication are separated by a single space but to avoid misinterpretation, a double space (double blank) is inserted between the parameter designation and the limit value.

b Additional profile surface texture requirements (these can be placed on separate rows, and the height of the graphical symbol extended to accommodate them)

c Intersection plane for indication of orientation of the evaluation area

NOTE 2 By default, the orientation of the evaluation area is determined by the position of the graphical symbol on the drawing, but an intersection plane indicator may be added here for clarification.
If the orientation of the lay is the same as the orientation of the evaluation area, this intersection plane indicator covers both.

d Manufacturing requirements

e Surface lay type

f Intersection plane for indication of orientation of the surface lay

NOTE 3 By default, the orientation of the lay is determined by the position of the graphical symbol on the drawing and an intersection plane indicator can be used here to indicate a different orientation.

g Machining allowance (in millimetres)

NOTE 4 SOURCE: BS EN ISO 25178-1:2016, Figure 1, modified – key added.

16 Conformity and non-conformity

COMMENTARY ON CLAUSE ***16***

Tolerance limits are used to describe a range of acceptable values for a particular property, such as the size or location of a feature. This range of acceptable values defines a "tolerance zone" or "specification zone" for the property.

Measurements are carried out to determine if the manufactured item conforms to the specification zone. However, no measurement process is perfect, and variability can arise from a number of sources, such as (but not limited to):

a) the resolution of the measurement device;

b) environmental factors such as temperature or vibration;

c) human factors such as the operation or manipulation of the measurement device;

d) physical factors such as the difficulty of taking the measurement; and

e) geometrical factors such as the form of the feature being measured.

Consequently, there is always a degree of "measurement uncertainty" associated with any measurement, and the measurement value corresponds to a range of possibilities for the quantity being measured. This means that, particularly when a measurement is close to tolerance limits, a measurement value that appears to be within the tolerance limits might represent an actual value which is outside the tolerance limits. This can result in an incorrect decision regarding conformity or non-conformity to a specification. The greater the measurement uncertainty, the greater the risk of an incorrect decision.

The BS EN ISO 14253 series provides rules which can be used to take measurement uncertainty into account.

16.1 Measurement uncertainty

Measurement uncertainty shall be calculated according to one of the following methods:

a) statistical analysis of repeated measurements (type A evaluation) in accordance with ISO/IEC Guide 98-3;

b) analysis based on other available data, knowledge or assumptions (type B evaluation) in accordance with ISO/IEC Guide 98-3; and

c) the procedure for uncertainty management method (PUMA); an iterative method described in BS EN ISO 14253-2.

NOTE A normal distribution is often assumed for measurement uncertainty, with standard uncertainty, u, corresponding to one standard deviation.

16.2 Decision rules

[SOURCE: BS EN ISO 14253-1:2017, **4.1**, modified – selected text minor modifications throughout.]

16.2.1 General

Decision rules shall be used to define how measurement uncertainty and shall be handled when determining conformity or non-conformity with a specification.

Decision rules shall define:

a) acceptance, rejection and, if required, transition zones;

b) what actions take place for measurements in the transition zones, e.g. classify part as poor quality rather than outright reject;

c) a policy for repeated measurements if required, e.g. measurements within 10% of specification limits to be taken twice and averaged; and

d) a policy for rejected measurement data, e.g. maintain a record of rejected measurement along with a reason for rejection (such as excess vibration due to traffic, or spurious reading due to operator error).

Guard bands shall be used when defining stringent or relaxed acceptance or rejection rules. A stringent rule shall use the guard bands to reduce the risk of an incorrect decision. A relaxed rule shall use the guard bands to increase the risk of an incorrect decision.

NOTE ISO/TR 14253-6 provides generalized decision rules.

16.2.2 Guard bands

By default, guard bands shall be set to achieve a 95% level of confidence that a decision on conformity or non-conformity is correct.

The guard band value, g, shall be calculated by multiplying the standard uncertainty, u, by a coverage factor, k:

$$g = k \times u$$

To achieve the default 95% probability of a correct decision, for a normal distribution of measurement uncertainty, a coverage factor of k = 1.65 shall be used:

$$g = 1.65 \times u$$

NOTE The coverage factor, k, can be set to a different value if a different level of confidence is required.

16.2.3 Simple acceptance/rejection decision rule

COMMENTARY ON ***16.2.3***

This is a binary decision rule: there are only two possible outcomes. No guard bands are used and there are no transition zones.

The specification zone shall be the acceptance zone.

Any measurement up to and including the specification limits shall be taken to indicate conformity.

Any measurement outside the specification limits shall be taken to indicate non-conformity.

NOTE With the rule, measurement uncertainty is ignored. There is a risk of accepting non-conforming parts and of rejecting conforming parts.

16.2.4 Stringent acceptance/relaxed rejection decision rule

COMMENTARY ON ***16.2.4***

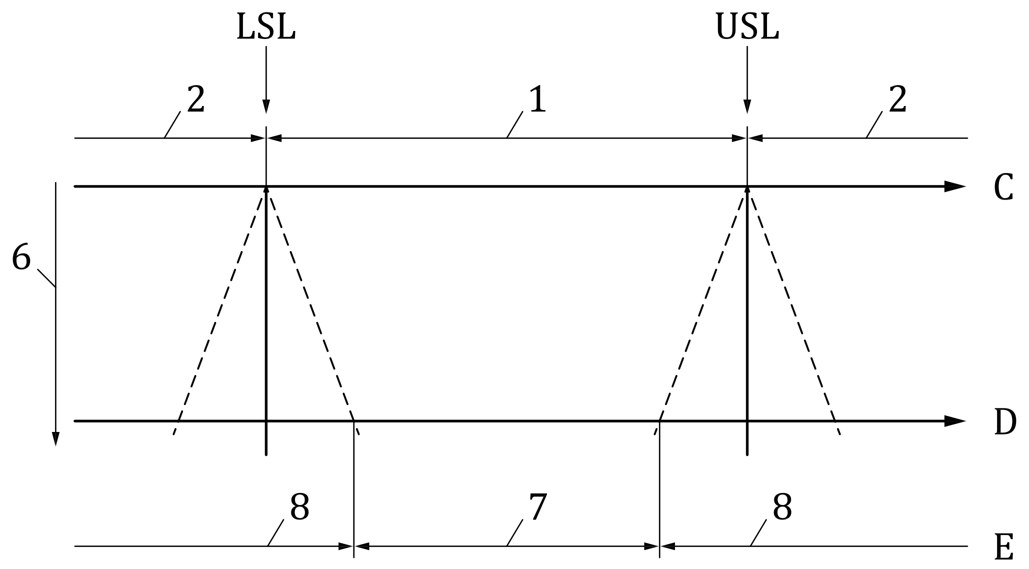
This is a binary decision rule: there are only two possible outcomes. There are no transition zones.

The specification zone shall be reduced by the guard bands to define the acceptance zone (see Figure 186).

The specification zone shall be reduced by the guard bands to define the rejection zone (see Figure 186).

NOTE With the rule, there is a reduced risk of accepting non-conforming parts and a reduced risk of rejecting conforming parts.

Figure 186 — *Stringent acceptance with relaxed rejection*



Key

C	Design/specification phase	2	Characteristic out of specification
D	Verification phase	6	Increasing measurement uncertainty, *U*
E	Verifying conformity	7	Acceptance zone (when verifying conformity)
1	Characteristic within specification	8	Rejection zone (when verifying conformity)

NOTE SOURCE: BS EN ISO 14253-1:2017, Figure 4, modified – numbering changed.

16.2.5 Relaxed acceptance/stringent rejection decision rule

*COMMENTARY ON **16.2.5***

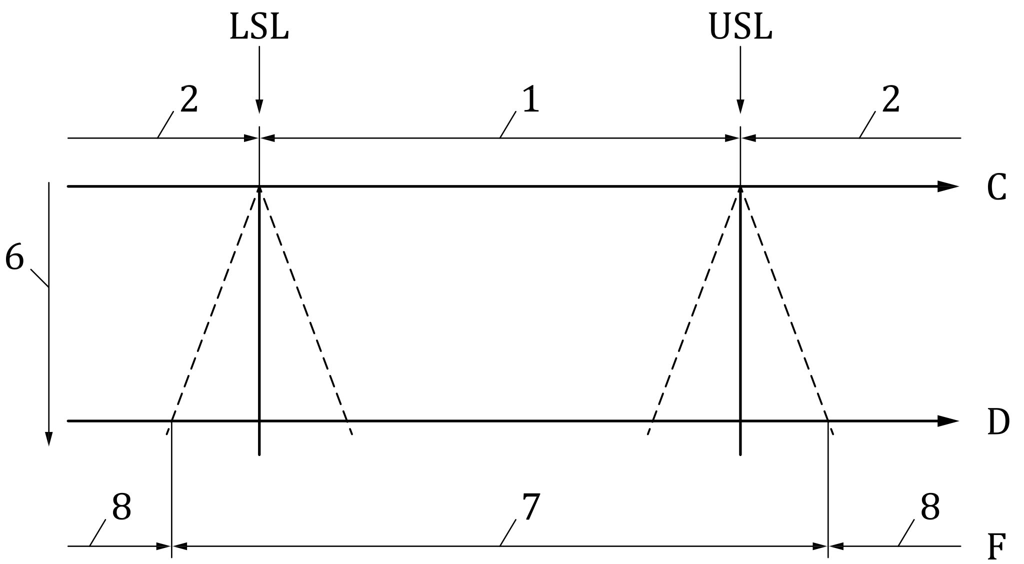
This is a binary decision rule: there are only two possible outcomes. There are no transition zones.

The specification zone shall be extended by the guard bands to define the acceptance zone (see Figure 187).

The specification zone shall be extended by the guard bands to define the rejection zone (see Figure 187).

NOTE With the rule, there is a reduced risk of accepting non-conforming parts and a reduced risk of rejecting conforming parts.

Figure 187 — *Relaxed acceptance with stringent rejection*



Key

C	Design/specification phase	2	Characteristic out of specification
D	Verification phase	6	Increasing measurement uncertainty, *U*
F	Verifying conformity	7	Acceptance zone (when verifying conformity)
1	Characteristic within specification	8	Rejection zone (when verifying conformity)

NOTE SOURCE: BS EN ISO 14253-1:2017, Figure 8, modified – numbering changed.

16.2.6 Stringent acceptance/stringent rejection decision rule

COMMENTARY ON ***16.2.6***

There are three possible outcomes.

The specification zone shall be reduced by the guard bands to define the acceptance zone (see Figure 188).

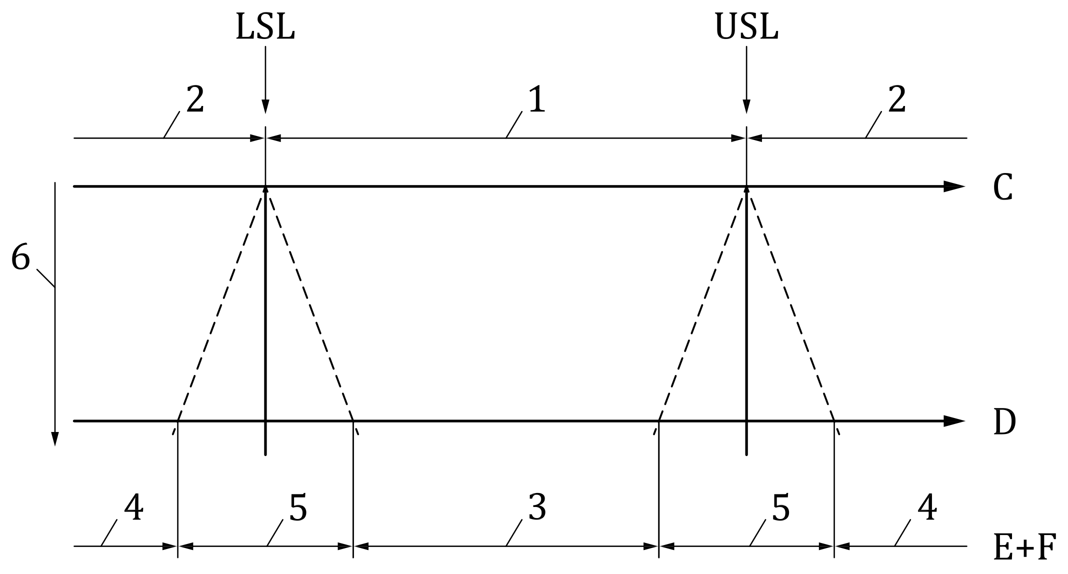
The specification zone shall be extended by the guard bands to define the rejection zone (see Figure 188).

The transition zone shall be the uncertainty zone between the acceptance and rejection zones.

A rule shall be defined for any measurement values in the transition zone (e.g. set aside for rechecking with more accurate instruments).

NOTE With the rule, there is a reduced risk of accepting non-conforming parts and a reduced risk of rejecting conforming parts.

Figure 188 — *Stringent acceptance with stringent rejection*



Key

C	Design/specification phase	2	Out of specification
D	Verification phase	3	Zone where conformity is verified
E	Verifying conformity	4	Zone where non-conformity is verified
F	Verifying non-conformity	5	Uncertainty zone
1	Specification zone (in specification)	6	Increasing measurement uncertainty, *U*

NOTE SOURCE: BS EN ISO 14253-1:2017, Figure 6, modified – numbering changed (9 becomes 6).

16.2.7 Other decision rules

Decision rules using other combinations of simple, stringent, and relaxed acceptance and rejection shall be used according to commercial requirements, such as stringent acceptance with simple rejection.

NOTE In the case of stringent acceptance with simple rejection, the specification zone is reduced by the guard bands to define the acceptance zone, while the specification zone limits define the rejection zone. The transition zone is the zone between the acceptance and specification zones. With this rule, there is a reduced risk of accepting non-conforming parts.

17 Symbols and abbreviations

17.1 General

Abbreviations (text equivalents) used in a TPS shall be the same in the singular and the plural.
A full stop shall not be used except where an abbreviation forms a word (e.g. NO. as an abbreviation for "number").

*NOTE 1 Where possible, abbreviations should be avoided (see **17.2**).*

Symbols used for physical quantities and units of measurement shall conform to the following standards, as appropriate.

- BS EN ISO 80000-1, *Quantities and units – Part 1: General.*
- BS EN ISO 80000-2, *Quantities and units – Part 2: Mathematics.*
- BS EN ISO 80000-3, *Quantities and units – Part 3: Space and time.*

- BS EN ISO 80000-4, *Quantities and units – Part 4: Mechanics*.
- BS EN ISO 80000-5, *Quantities and units – Part 5: Thermodynamics*.
- BS ISO 80000-7, *Quantities and units – Part 7: Light*.
- BS EN ISO 80000-8, *Quantities and units – Part 8: Acoustics*.
- BS EN ISO 80000-9, *Quantities and units – Part 9: Physical chemistry and molecular physics*.
- BS EN ISO 80000-10, *Quantities and units – Part 10: Atomic and nuclear physics*.
- BS EN ISO 80000-11, *Quantities and units – Part 11: Characteristic numbers*.
- BS EN ISO 80000-12, *Quantities and units – Part 12: Condensed matter physics*.
- BS EN 80000-13, *Quantities and units – Part 13: Information science and technology*.
- BS EN 80000-14, *Quantities and units – Part 14: Telebiometrics related to human physiology*.

NOTE 2 BS EN 80000-6 (withdrawn) gives symbols on electromagentism.

17.2 Standard symbols and abbreviations

*COMMENTARY ON **17.2***

In the existing environment of outsourcing across national borders, every effort is being made to make the use of GPS independent of language through the adoption of standard symbology. It is for this reason that the continued use of abbreviations is deprecated.

Where particular specification requirements cannot be expressed using the available GPS system, full text description shall be employed.

NOTE 1 It is suggested that, where such a requirement occurs frequently, this be drawn to the attention of the relevant ISO committee through the appropriate BSI Technical Committee.

For diagrams used in technical applications, a library of harmonized graphical symbols is published in the following series of standards with close cooperation between ISO and IEC and these symbols shall be applied wherever practicable to improve the universal applicability of the TPS.

- BS ISO 14617-1, *Graphical symbols for diagrams – Part 1: General rules*.
- BS ISO 14617-2, *Graphical symbols for diagrams – Part 2: Graphical symbols*.

NOTE 2 BS ISO 14617-3 (withdrawn), BS ISO 14617-4 (withdrawn), BS ISO 14617-5 (withdrawn), BS ISO 14617-6 (withdrawn), BS ISO 14617-7 (withdrawn), BS ISO 14617-8 (withdrawn), BS ISO 14617-9 (withdrawn), BS ISO 14617-10 (withdrawn), BS ISO 14617-11 (withdrawn) and BS ISO 14617-12 (withdrawn) give also additional symbols.

Symbols appropriate to TPS are provided and detailed throughout the suite of documents cross-referenced from this British Standard, and these shall be used where appropriate.

Where, in particular technical fields, certain abbreviations are in common use and generally understood, it is accepted that these can continue to be used, but new abbreviations shall not be introduced.

NOTE 3 Former practice has resulted in certain abbreviations becoming accepted as symbols and these should not be considered to provide precedence for the proliferation of abbreviations.

18 Document handling

18.1 Types of documentation

18.1.1 General

*COMMENTARY ON **18.1.1***

The careful targeting of TPD to known or intended users can greatly assist the accuracy with which the specification is converted into the final product.

While precision and avoidance of ambiguity are always paramount, the means employed to convey this information shall be shown to match the capability, or potential capability, of the available or achievable manufacturing facility.

NOTE Specification beyond this level is unlikely to produce satisfactory results and can prove expensive, both in terms of the cost of the over-specification itself and in terms of inadequate or unacceptable product.

18.1.2 Presentation media

18.1.2.1 General

The presentation of the drawings shall conform to the following standards, as appropriate.

- BS EN ISO 5457, *Technical product documentation – Sizes and layout of drawing sheets*.
- BS EN ISO 7200, *Technical product documentation – Data fields in title blocks and document headers*.
- BS ISO 7573, *Technical product documentation – Parts lists*.

18.1.2.2 Format

Drawing sheets and other documents shall be presented in one of the following formats:

a) landscape: intended to be viewed with the longest side of the sheet horizontal; or

b) portrait: intended to be viewed with the longest side of the sheet vertical.

18.2 Security

18.2.1 Introduction

NOTE Many TPSs have minimal requirements for security, other than that provided by general handling and storage procedures.

Where specific need for a general level of security is identified, the requirements of **18.2.2**, **18.2.3** and **18.2.4** shall be met.

18.2.2 General security

Procedures for managing the security of TPDs and TPSs shall conform to the following standard.

- BS EN ISO 11442, *Technical product documentation – Document management*.

18.2.3 Enhanced security

Where enhanced security is claimed, the requirements of Annex D shall be met, in addition to those in **18.2.4**.

18.2.4 Security level identification

The level of security attributed to any given TPS shall be clearly identified by the relevant marking placed adjacent to the title or title block of every TPD making up that TPS.

18.3 Storage

Methods for storage and retrieval of the document shall conform to the following standards, as appropriate.

- BS EN ISO 6428, *Technical drawings – Requirements for microcopying*.
- BS EN ISO 11442, *Technical product documentation – Document management*.

18.4 Marking

18.4.1 General

TPDs shall be marked in a prominent location to indicate which standards, or system of standards, are to govern their interpretation.[2)]

TPDs prepared in accordance with this British Standard shall be prepared in accordance with the ISO system.

NOTE The marking of a TPD or TPS with the number of this standard constitutes a claim that the appropriate requirements of all relevant cross-referenced standards, in addition to the requirements directly specified in this British Standard, have been met. Attention is drawn to the date of issue principle.

18.4.2 Enhanced security

TPDs conforming to Annex D, shall be marked in a prominent location with the number of this standard, followed by the suffix "/D", i.e.:

CONFORMS TO BS 8888:2025/D

18.5 Protection notices

Where appropriate to place restrictions on the use of TPD, the following standard shall be applied.

- BS ISO 16016, *Technical product documentation – Protection notices for restricting the use of documents and products*.

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2) Marking BS 8888:2025 on or in relation to a product represents a manufacturer's declaration of conformity, i.e. a claim by or on behalf of the manufacturer that the product meets the requirements of the standard. The accuracy of the claim is solely the claimant's responsibility. Such a declaration is not to be confused with third-party certification of conformity.

Annex A (informative)
Datums and datum systems: Definitions and explanations

A.1 Datum feature

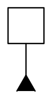
A datum feature is a real integral feature or part of a real integral feature.

NOTE The datum feature is a feature on the workpiece itself or a feature on a tool or fixture to which the workpiece is attached.

A.2 Datum feature indicator

A datum feature indicator is a symbol used in a specification to identify a datum feature (see Figure A.1).

Figure A.1 — *Datum feature indicator*



NOTE The datum feature indicator is used to label a datum feature, not the datum itself.

A.3 Datum feature identifier

A datum feature identifier is one or more capital letters from the Latin alphabet used to name a datum feature.

NOTE Letters I, O, Q and X are not used.

A.4 Common datum

A common datum is a datum based on more than one datum features.

A.5 Datum section

A datum section is the section in a tolerance indicator or other specification indicator, consisting of one or more compartments containing datum feature identifiers (see Figure A.2).

NOTE A datum section can be used as a part of a tolerance indicator, an intersection plane indicator, an orientation plane indicator, a collection plane indicator or a direction feature indicator (see BS EN ISO 1101:2017).

Figure A.2 — *Datum section of a tolerance indicator defining a datum system*

a) Tolerance with a datum system consisting of two datums

b) Tolerance with a datum system consisting of three datums

A.6 Primary datum

The primary datum is a datum that is not influenced by constraints from other datums and listed first in a datum section.

A.7 Secondary datum

The secondary datum is a datum in a datum system which has its orientation constrained to the primary datum and listed second in a datum section.

NOTE In BS EN ISO 5459:2024, the secondary datum is described as only having orientation constraints to the primary datum. This is contrary to how datum systems are set up in some instances. BS EN ISO 5459:2024, Annex I, covers this issue in detail.

A.8 Tertiary datum

Tertiary datum is a datum in a datum system which has its orientation constrained to the primary and secondary datums.

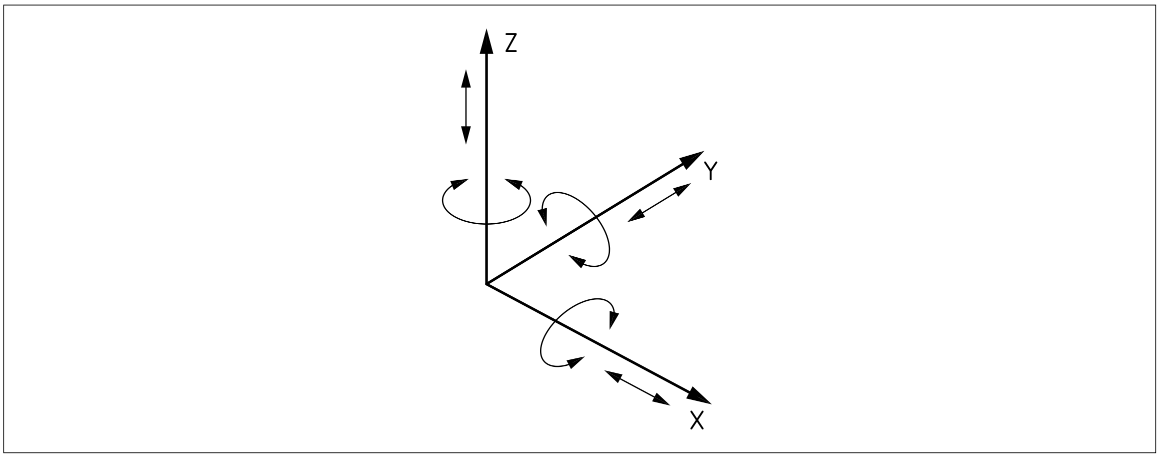
NOTE In BS EN ISO 5459:2024, the tertiary datum is described as only having orientation constraints to the primary datum. This is contrary to how datum systems are set up in some instances. BS EN ISO 5459:2024, Annex I, covers this issue in detail.

A.9 Six degrees of freedom

Six degrees of freedom are the freedom of a rigid body, in three-dimensional space, when not subject to any constraints, to translate along and rotate around, each of the axes of a Cartesian coordinate system (see Figure A.3).

A tolerance zone or geometric entity which is not subject to any constraints has three translational degrees of freedom and three rotational degrees of freedom. To control the location or orientation properties of a feature, it is necessary to constrain some or all of the degrees of freedom for the tolerance zone.

Figure A.3 — *Degrees of freedom*



A.10 Situation feature

A situation feature is a point, line, plane or combination of these elements, from which the location of a geometrical feature can be defined.

A situation feature is a geometrical attribute of an ideal feature. The situation features for any real (non-ideal) manufactured surface are the situation features for its associated (ideal) feature.

The situation feature for a planar surface is a flat plane, the situation feature for a cylinder is a straight line (the axis of the cylinder), and the situation feature for a sphere is a point (the centre point of the sphere) (see Table A.1).

Locking the location or orientation of a situation feature causes some of the degrees of freedom of the geometric entity to be locked.

Some geometric entities have more than one situation feature. For example, a cone has two situation features, a straight line (axis) and a point.

Table A.1 — *Situation features*

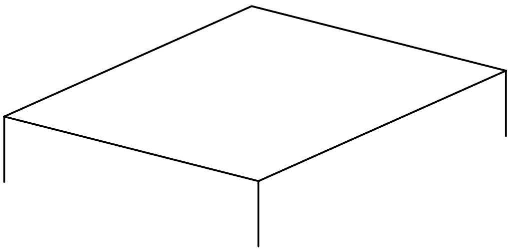
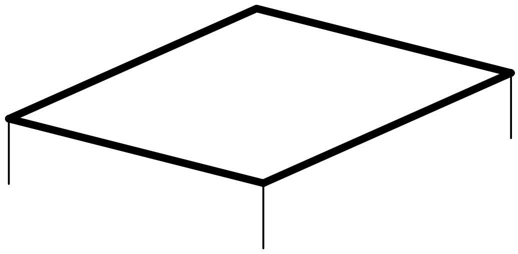
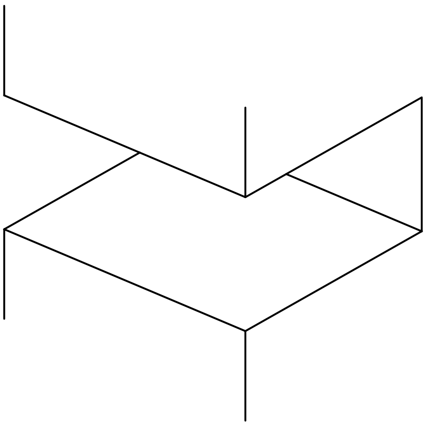
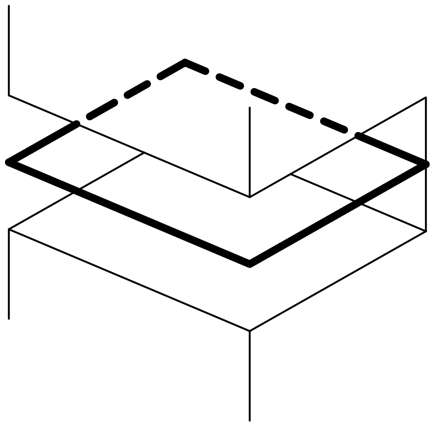
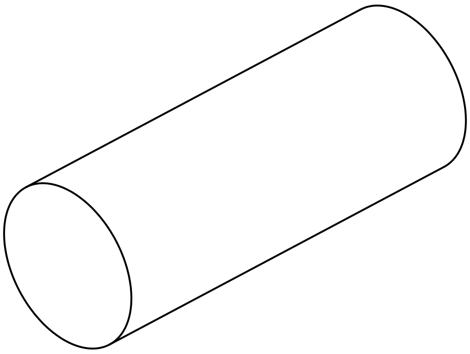
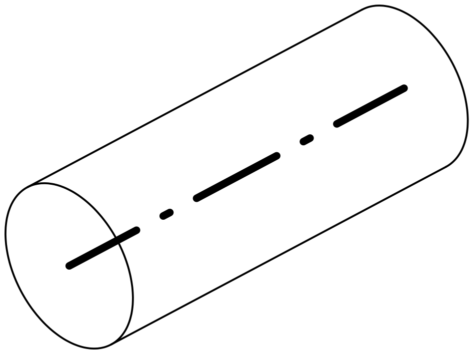
Feature type	**Situation feature(s)**
Plane: 	Plane: 
Two parallel opposed planes: 	Plane: 
Cylinder: 	Straight line: 

Table A.1 — *Situation features (continued)*

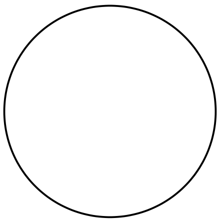
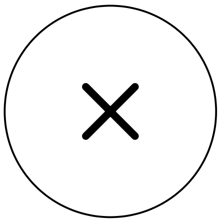
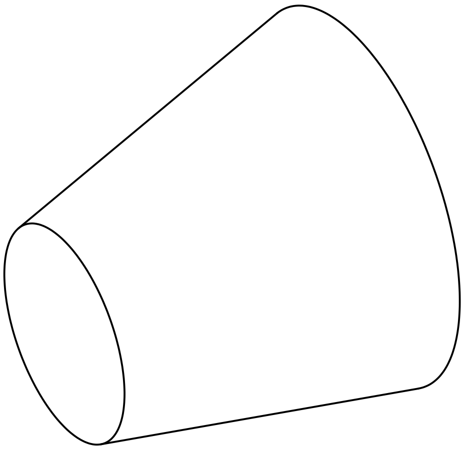
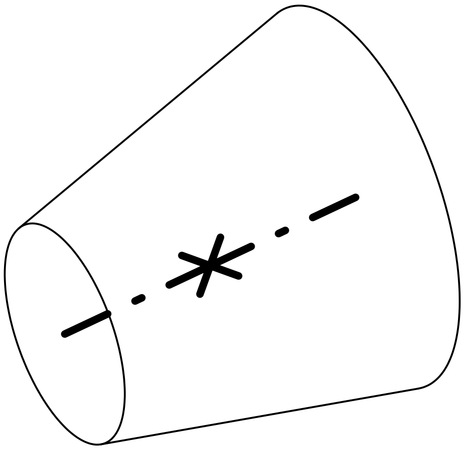
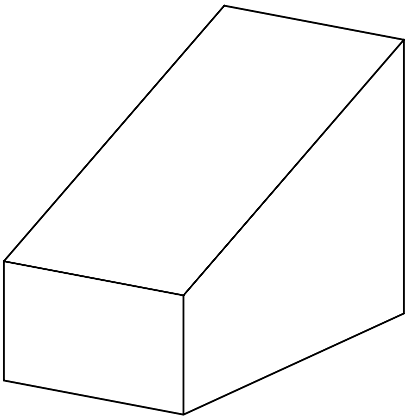
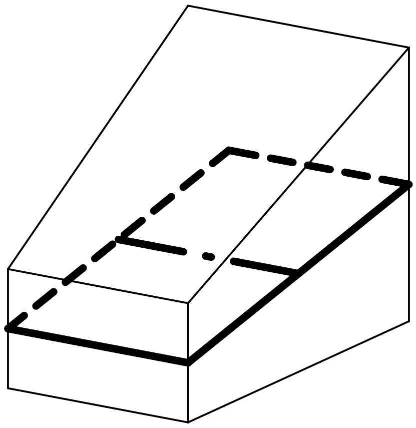
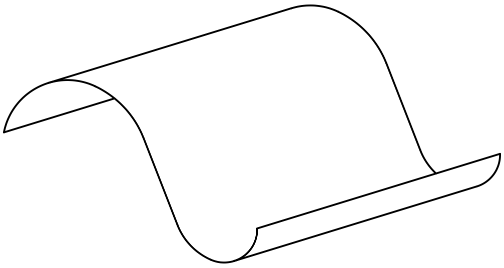
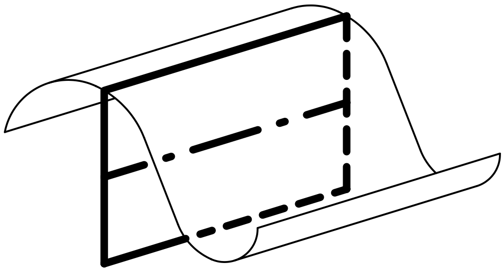
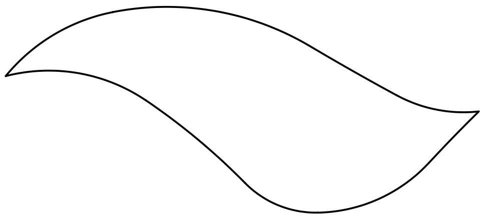
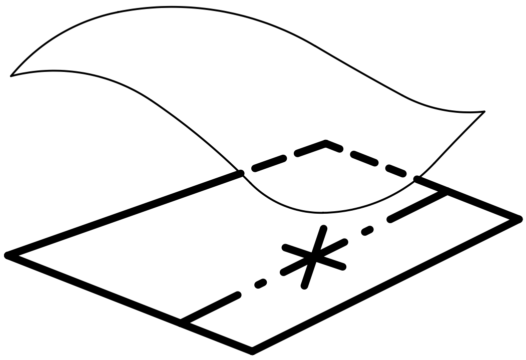
Feature type	Situation feature(s)
Sphere: 	Point: 
Cone: 	Straight line and point: 
Wedge (prismatic): 	Plane and straight line: 
Extruded profile: 	Plane and straight line: 

Table A.1 — *Situation features (continued)*

Feature type	Situation feature(s)
Complex: 	Plane, straight line and point: 

A.11 Associated feature

An associated feature is an ideal feature aligned with a non-ideal real or extracted feature through an association operation in accordance with BS EN ISO 17450-1 and BS EN ISO 5459.

NOTE 1 The associated features used to establish datums are normally intended to simulate the contact or interface with other components.

NOTE 2 See Annex B for further information about association.

A.12 Association

An association is a feature operation used to fit an ideal feature (or features) to a non-ideal feature (or features) according to one or more association criteria.

A.13 Association criterion

An association criterion is an objective function, with or without association constraints, used for an association operation.

EXAMPLE 1 (see Figure B.3):

The default association for a datum plane based on a nominally flat datum feature is for the associated plane to be in contact with the datum feature, but outside the material, using the total least-squares objective function with an outside the material constraint.

NOTE 1 This is different to:

a) a minimax association (the former default in BS EN ISO 5459:2024);

b) a total least-squares association without a material constraint; and

c) a total least-squares association with an offset.

EXAMPLE 2 (see Figure B.5):

The default association for a datum axis based on a cylindrical datum feature is to use the minimum circumscribed feature for an external cylinder, or the maximum inscribed feature for an internal feature.

NOTE 2 See Annex B for a full list of the different association criteria.

A.14 Objective function (for association)

An objective function (for association) is the method or algorithm which defines how the associated feature is fitted to a real feature.

Objective functions include:

a) minimax;

b) Gaussian (least-squares);

c) minimum circumscribed; and

d) maximum inscribed.

NOTE See Annex B for further details.

A.15 Association constraints

Association constraints are additional constraints which can be combined with the objective function when constructing an associated feature.

Association constraints include:

a) location constraints (e.g. when constructing a secondary or tertiary datum);

b) orientation constraints (e.g. when constructing a secondary or tertiary datum);

c) material constraints (e.g. outside the material, inside the material, etc); and

d) size constraints (e.g. size fixed or size variable).

NOTE See Annex B for further details.

Annex B (informative) Associations

B.1 Association operations

The association operation is the method used to “fit” the ideal geometry to the non-ideal feature. Traditionally, when this was achieved using a mechanical device such as a surface table, angle plate or chuck, there was no need to define an association operation, as there was only one possibility.

Advances in metrological equipment have meant that devices such as coordinate measuring machines (CMMs), articulated arms and scanners of various kinds can offer a multiplicity of different association operations. Some of these are used when defining datums, while others are used when evaluating tolerances; some work better with some kinds of geometry, and others with different kinds of geometry. The term “best fit” is frequently used, often very loosely, and can refer to any of a number of association operations, including the following:

a) minimax (also known as “Chebyshev” or $L_{\infty \text{ norm}}$): the associated feature is arranged to minimize the maximum distance between the associated feature and the datum feature;

b) minimum circumscribed: the associated feature is the smallest ideal feature which fully envelops the datum feature;

c) maximum inscribed: the associated feature is the largest ideal feature which can be fully enveloped by the datum feature; and

d) least-squares (also known as “Gaussian” or $L_{2 \text{ norm}}$): the associated feature is an ideal feature fitted to the datum feature using the total least-squares algorithm.

NOTE The name is usually taken from the mathematical algorithm on which the operation is based.

Physical association operations, involving mechanical devices, are applied to the real surface of the datum feature. Digital association operations, using devices like CMMs, are applied to the extracted feature (the measurement data obtained from sampling the real surface).

In addition, the following factors can affect the association.

1) **Location and orientation constraints**

If the association is being performed in the definition of a secondary or tertiary datum, it includes additional orientation or location constraints to maintain the correct relationship with the preceding datums.

2) **Material constraints**

In some cases, the associated feature can be established entirely outside the material of the workpiece, or entirely inside the material of the workpiece.

3) **Intrinsic constraints**

Some datum features can be defined either with a toleranced dimension or with a TED. In some cases, the associated feature could be defined with either a toleranced dimension or a TED.

B.2 Default association operations

When defining datums, default association operations should be used unless otherwise stated (see Table B.1).

Table B.1 — *Default association operations used when defining datums*

Datum feature	Internal or external	Default association operation	Default material constraint	Intrinsic characteristic	Default intrinsic constraint	Situation feature
Plane	—	Least-squares (Gaussian)	Outside the material	—	—	Plane
Two parallel opposed planes	External (flange)	Minimum circumscribed[A]	Outside the material	Width (distance between the two parallel planes)	Variable	Plane
	Internal (slot)	Maximum inscribed[A]				
Cylinder	External (shaft)	Minimum circumscribed[A]	Outside the material	Diameter	Variable	Line (axis)
	Internal (hole)	Maximum inscribed[A]				
Sphere	External (ball)	Minimum circumscribed[A]	Outside the material	Diameter	Variable	Point
	Internal (socket)	Maximum inscribed[A]				
Cone	External	Minimax	Outside the material	Angle	Variable or fixed[B]	Line and point
	Internal	Minimax				
Wedge	External	Minimax	Outside the material	Angle	Variable or fixed[B]	Plane and line
	Internal	Minimax				
Prismatic	—	Minimax	Outside the material	—	—	Plane and line
Complex	—	Minimax	Outside the material	—	—	Plane, line and point

[A] Association with features of linear size, using the minimum circumscribing or maximum inscribing operation, can in some cases result in more than one possible solution. In such cases, the minimax association is used instead.

[B] If the feature angle is defined with a tolerance (e.g. 30° ±1°), the associated feature has a variable angle. If the feature angle is defined with a TED (e.g. 30°), the associated feature has a fixed angle. See BS EN ISO 5459 for additional details.

NOTE SOURCE: BS EN ISO 5459:2024, Table A.1 and Table A.2, modified – headings changed, simplified content, prismatic added, notes of Table A.2 removed.

B.3 Non-default association operations

It is also possible to use alternative association operations. These can be invoked by using the following modifiers next to the datum reference in a tolerance frame. As these are yet to be standardized in a published ISO standard, it is imperative that the use of any of these modifiers is fully explained with a key or a description on the specification.

Non-default association algorithms are invoked with the following symbols in Table B.2.

Table B.2 — *Objective function symbols*

Objective function	
Symbol	**Designation**
G	Least-squares (Gaussian)
C	Minimax (Chebyshev)
K	Minimum volume
X	Maximum inscribed
N	Minimum circumscribed
L_p	Minimizing the p-norm distance (p is the parameter; the value of p shall be a positive integer different from 1, 2 and ∞)

NOTE 1 The objective functions maximum inscribed and minimum circumscribed are only applicable to features of linear size.

NOTE 2 SOURCE: BS EN ISO 5459:2024, Table 12.

The symbol for the non-default association should be accompanied with a symbol defining the material constraint to be used for the association criteria.

Material constraints or offset options can be invoked with the following symbols in Table B.3 and Table B.4.

Table B.3 — *Material constraint*

Material constraint or material offset	
Symbol	**Designation**
E	External material
I	Internal material
M	Without material constraint

NOTE SOURCE: BS EN ISO 5459:2024, Table 13.

Table B.4 — *Material offset*

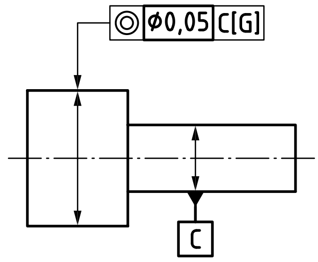
Material constraint or material offset	
Symbol	**Designation**
+	Shifted tangent outside material
–	Shifted tangent inside material
x%	Material ratio percentage, with x the value of the percentage

NOTE SOURCE: BS EN ISO 5459:2024, Table 14.

NOTE GE is not the same as G+. For G+, a first associated feature is established without material constraint (GM) and then shifted or scaled to just contact the material. For GE, the associated feature is established one time with outside material constraint. The GE and G+ association criteria generally yield different associated features.

An example of an individual specification operator for a datum association is given in Figure B.1.

Figure B.1 — *Example of individual specification operator for a datum association*



NOTE SOURCE: BS EN ISO 5459:2024, Figure 46, modified – title changed, C[G] replaced C[GM].

Non-default intrinsic constraints can be invoked with the following symbols:

Intrinsic constraint	Symbol
Size fixed	[SF]
Size variable	[SV]

B.4 Examples of datums obtained from different datum features

B.4.1 Datum based on a planar datum feature

The datum feature is identified on the TPD as a nominally flat plane.

The real datum feature on the manufactured workpiece is a non-ideal surface.

By default, the associated feature is a flat plane in contact with the datum feature, outside the material, constructed using the least-squares (Gaussian) association with the outside the material constraint.

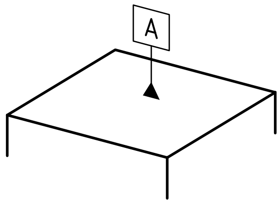
The datum is the situation feature for the associated feature, which is a flat plane.

A primary datum based on the datum feature is a plane, constructed with no additional constraints. This can control three degrees of freedom (see Figure B.2).

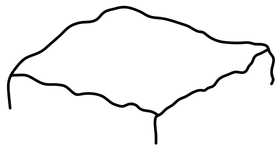
A secondary datum based on the datum feature is a plane, constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is a plane, constructed with orientation constraints to the primary and secondary datums.

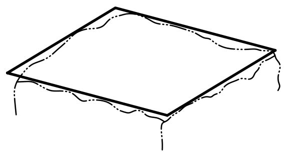
Figure B.2 — *Primary datum based on a planar datum feature*



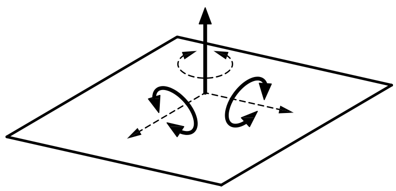
a) Nominal datum feature identified on TPD.



b) Real datum feature: The datum feature is a plane. The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.



c) Associated feature: The associated feature is a flat plane. The situation feature is a flat plane.



d) Datum as the situation feature for the associated feature: This datum can constrain three degrees of freedom:

- translation along an axis perpendicular to the plane; and
- rotation about two perpendicular axes aligned with the plane.

Key

Locked

Unlocked

B.4.2 Datum based on two parallel, opposed planes (external)

The datum feature is identified on the TPD as two parallel opposed planes (e.g. a flange).

The real datum feature on the manufactured workpiece is composed of two non-ideal surfaces.

By default, the associated feature is a minimum circumscribed feature in contact with the datum feature, outside the material (see Figure B.3).

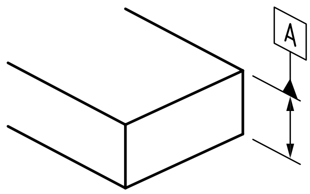
The datum is the situation feature for the associated feature, which is a median plane.

A primary datum based on the datum feature is a median plane, constructed with no additional constraints. This can control three degrees of freedom (see Figure B.3).

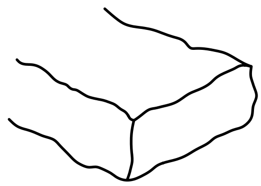
A secondary datum based on the datum feature is a plane, constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is a plane, constructed with orientation constraints to the primary and secondary datums.

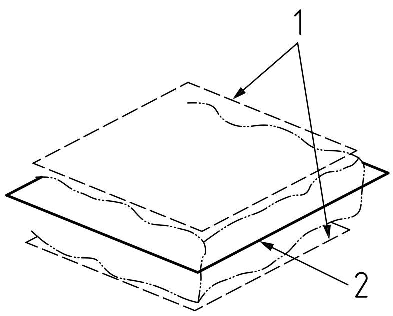
Figure B.3 — *Primary datum based on two parallel opposed planes*



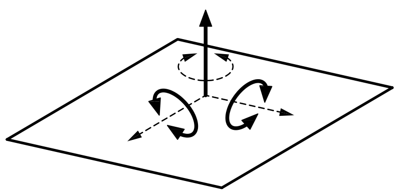
a) The nominal datum feature identified in the TPD is a prismatic surface.



b) Real datum feature: The datum feature is two parallel opposed planes. The real datum feature is the manufactured surfaces. The extracted datum feature is the data set obtained by sampling the surfaces.



c) Associated feature: The associated feature is two parallel opposed planes. The situation feature is a median plane.



d) Degrees of freedom that can be controlled: This datum can constrain three degrees of freedom:

- translation along an axis perpendicular to the median plane; and
- rotation about two perpendicular axes aligned with the median plane.

Key

Locked

Unlocked

1 Associated feature

2 Median plane

B.4.3 Datum based on two parallel, opposed planes (internal)

The datum feature is identified on the TPD as two parallel opposed planes (e.g. a slot).

The real datum feature on the manufactured workpiece is composed of two non-ideal surfaces.

By default, the associated feature is a maximum inscribed feature in contact with the datum feature, outside the material.

The datum is the situation feature for the associated feature, which is a median plane.

A primary datum based on the datum feature is a median plane, derived from an associated feature constructed with no additional constraints. This can control three degrees of freedom.

A secondary datum based on the datum feature is a plane, constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is a plane, constructed with orientation constraints to the primary and secondary datums.

B.4.4 Datum based on an external cylindrical feature

The datum feature is identified on the TPD as a nominally cylindrical external feature.

The real datum feature on the manufactured workpiece is a non-ideal cylinder.

The associated feature is a minimum circumscribing cylinder constructed around the datum feature (unless otherwise indicated).

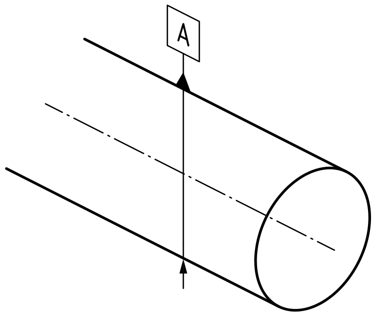
The datum is the situation feature for the minimum circumscribing cylinder, which is its axis.

A primary datum based on the cylindrical datum feature is an axis derived from the minimum circumscribing cylinder with no additional constraints. This can control four degrees of freedom (see Figure B.4).

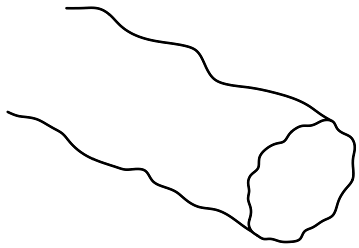
A secondary datum based on the datum feature is an axis, derived from a minimum circumscribing cylinder constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is an axis, derived from a minimum circumscribing cylinder constructed with orientation constraints to the primary and secondary datums.

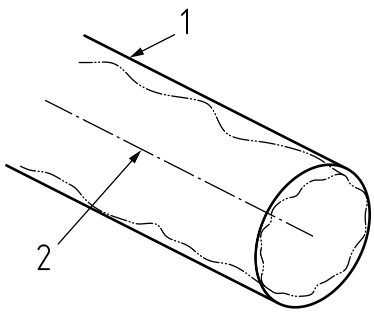
Figure B.4 — *Primary datum based on an external cylindrical feature*



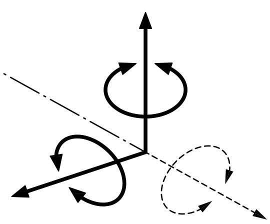
a) Nominal datum feature identified on TPD.



b) Real datum feature: The datum feature is an external cylinder. The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.



c) Associated feature: The associated feature is the minimum circumscribing cylinder. The situation feature for the associated feature is its axis.



d) Degrees of freedom that can be controlled: This datum can constrain four degrees of freedom:

- translation along two perpendicular axes which are also perpendicular to the axis of the associated feature; and
- rotation about two perpendicular axes which are also perpendicular to the axis of the associated feature.

Key

↑	Locked
⇡	Unlocked
1	Associated feature
2	Axis

B.4.5 Datum based on an internal cylindrical feature

The datum feature is identified on the TPD as a nominally cylindrical internal feature.

The real datum feature on the manufactured workpiece is a non-ideal cylinder.

The associated feature is a maximum inscribing cylinder constructed within the datum feature (unless otherwise indicated).

The datum is the situation feature for the maximum inscribing cylinder, which is its axis.

A primary datum based on the cylindrical datum feature is an axis, derived from a maximum inscribing cylinder constructed with no additional constraints. This can control four degrees of freedom.

A secondary datum based on the datum feature is an axis, derived from a minimum circumscribing cylinder constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is an axis, derived from a minimum circumscribing cylinder constructed with orientation constraints to the primary and secondary datums.

B.4.6 Datum based on an external spherical feature

The datum feature is identified on the TPD as a nominally spherical external feature.

The real datum feature on the manufactured workpiece is a non-ideal sphere.

The associated feature is a minimum circumscribing sphere constructed around the datum feature (unless otherwise indicated).

The datum is the situation feature for the minimum circumscribing sphere, which is its centre point.

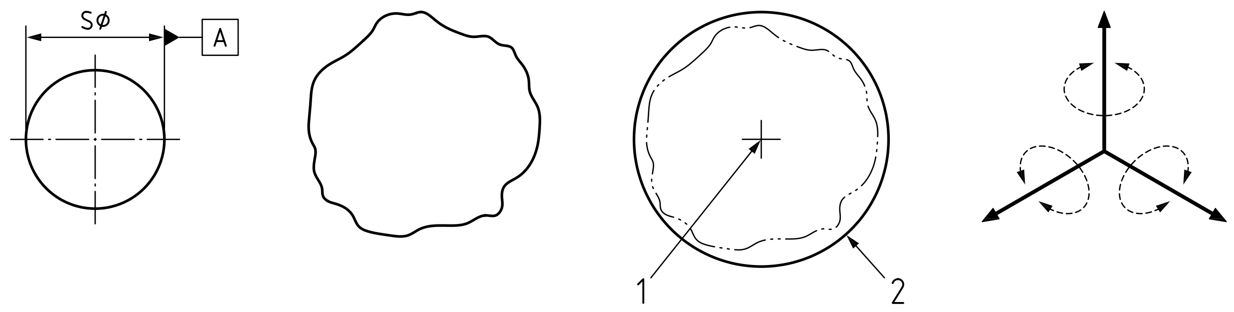
A primary datum based on the spherical datum feature is a point, derived from the minimum circumscribing sphere constructed with no additional constraints. This can control three degrees of freedom (see Figure B.5).

A secondary or tertiary datum based on the datum feature would also be a point derived from the minimum circumscribing sphere.

A tertiary datum based on the datum feature is a point, derived from the minimum circumscribing sphere, constructed with location constraints to the primary and secondary datums.

Figure B.5 — *Primary datum based on an external spherical feature*

Specification | **Real datum feature** | **Associated feature** | **Degrees of freedom that can be controlled**



a) Nominal feature is identified in the TPD.

b) The datum feature is an external sphere. The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.

c) The associated feature is the minimum circumscribing sphere. The situation feature for the associated feature is its centre point.

d) This datum can constrain three degrees of freedom:

- translation along each of three mutually perpendicular axes which have their origin located at the centre point of the associated feature.

Key

1 Centre point

2 Associated feature

Locked

Unlocked

B.4.7 Datum based on an internal spherical feature

The datum feature is identified on the TPD as a nominally spherical internal feature.

The real datum feature on the manufactured workpiece is a non-ideal sphere.

The associated feature is a maximum inscribing sphere constructed within the datum feature (unless otherwise indicated).

The datum is the situation feature for the maximum inscribing sphere, which is its centre point.

A primary datum based on the spherical datum feature is a point derived from the maximum inscribing sphere with no additional constraints. This can control three degrees of freedom.

A secondary or tertiary datum based on the datum feature would also be a point derived from the maximum inscribing sphere.

A tertiary datum based on the datum feature is a point derived from the maximum inscribing sphere constructed with location constraints to the primary and secondary datums.

B.4.8 Datum based on an external or internal cone

The datum feature is identified on the TPD as a nominally conical feature.

The real datum feature on the manufactured workpiece is a non-ideal conical surface.

The associated feature is a minimax cone, in contact with the feature but outside the material.

The datum is the set of situation features for the associated cone, which are its centreline (axis) and a point.

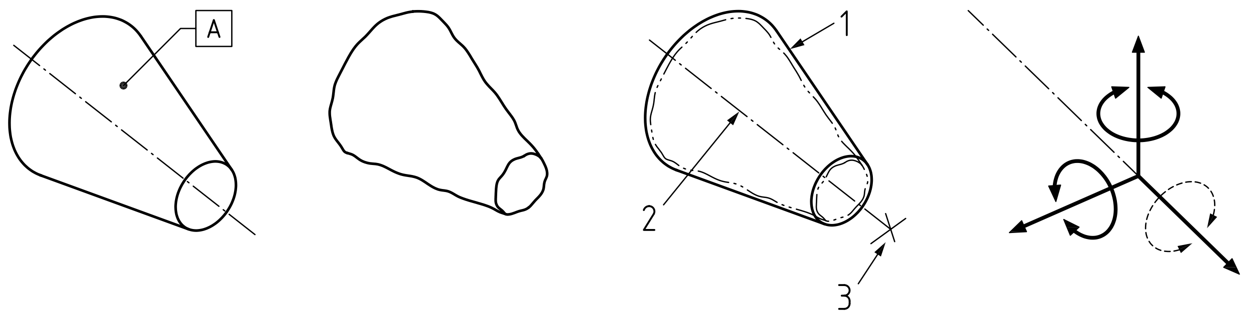
A primary datum based on the conical datum feature is an axis and a point derived from an associated cone constructed with no additional constraints. This datum can control five degrees of freedom (see Figure B.6).

A secondary datum based on the datum feature is an axis and a point derived from an associated cone constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is an axis and a point derived from an associated cone constructed with orientation constraints to the primary and secondary datums.

NOTE If the cone is defined with a theoretically exact angle, the angle of the associated feature is fixed at the same angle. If the cone is defined with a tolerance applied to the cone angle, the angle of the associated feature is variable.

Figure B.6 — *Primary datum feature based on a conical feature*



Specification	Real datum feature	Associated feature	Degrees of freedom that can be controlled
a) Nominal feature is identified in the TPD.	b) The datum feature is a cone. The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.	c) The associated feature is a minimax cone. The situation features for the associated feature are a straight line (axis) and a point. In this case, the axis has a unique location but the point does not.	d) This datum can constrain five degrees of freedom: • translation along all three axes; and • rotation about two perpendicular axes which are also perpendicular to the axis of the associated feature.

Key

1 Associated feature

2 Axis

3 Point

Locked

Unlocked

B.4.9 Datum based on an external or internal wedge

The datum feature is identified on the TPD as two planes which are not parallel.

The real datum feature on the manufactured workpiece is a non-ideal wedge.

The associated feature is a minimax wedge, in contact with the feature but outside the material.

The datum is the set of situation features for the associated wedge, which are its centre plane and a straight line.

A primary datum based on the datum feature is a plane and a straight line derived from an associated wedge constructed with no additional constraints. This datum can control five degrees of freedom (see Figure B.7).

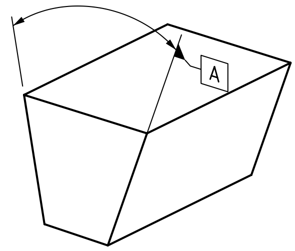
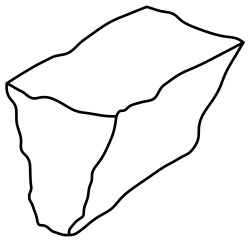
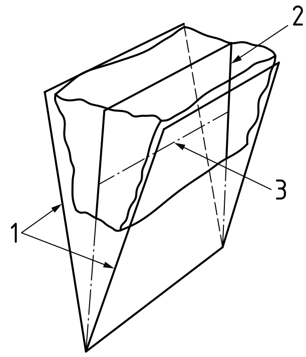
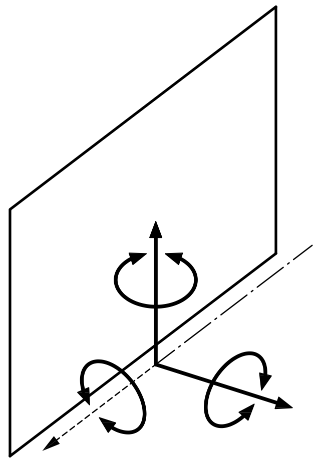
A secondary datum based on the datum feature is a plane and a straight line derived from an associated wedge constructed with orientation constraints to the primary datum.

A tertiary datum based on the datum feature is a plane and a straight line derived from an associated wedge constructed with orientation constraints to the primary and secondary datums.

The straight line situation feature lies within the plane situation feature and is parallel to the apex of the wedge but has no unique location on that plane. The line can be taken to be at the apex of the wedge or at any another specified position within the plane and parallel to the apex.

NOTE If the wedge is defined with a theoretically exact angle, the angle between the planes of the associated feature is fixed at the same angle. If the wedge is defined with a toleranced value for the angle, the angle between the planes of the associated feature is variable.

Figure B.7 — *Primary datum based on a wedge*

Specification	**Real datum feature**	**Associated feature**	**Degrees of freedom that can be controlled**
			
a) Nominal feature is identified in the TPD.	b) The datum feature is a wedge. The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.	c) The associated feature is the minimax wedge. The situation features for the associated feature are its centre plane and a straight line. In this case, the centre plane has a unique location but the straight line does not.	d) This datum can constrain five degrees of freedom: • translation along an axis perpendicular to the median plane, and translation along an axis within the median plane but perpendicular to the situation feature axis; and • rotation about all three axes.

Key

1 Associated feature

2 Centre plane

3 Straight line

Locked

Unlocked

B.4.10 Prismatic feature

The datum feature is identified on the TPD as a prismatic surface (see Figure B.8).

The real datum feature on the manufactured workpiece is a non-ideal surface (see Figure B.8).

The associated feature is an ideal version of the prismatic surface, constructed to be in contact with the real datum feature, outside the material, and arranged such that the maximum distances between the associated feature and the real surface are minimized.

The datum is the situation feature for the associated prismatic surface, which is a plane and a line (see Figure B.8).

A primary datum based on the prismatic datum feature is a plane and a line based on an associated feature constructed with no additional constraints. This can control five degrees of freedom (see Figure B.8).

A secondary datum based on the prismatic datum feature is a plane and a line based on an associated feature constructed with orientation constraints to the primary datum.

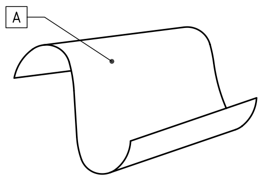
A tertiary datum based on the prismatic datum feature is a plane and a line based on an associated feature constructed with orientation constraints to the primary and secondary datums.

The plane situation feature might have no unique location, although it is aligned with the "direction of extrusion" of the prismatic feature. The line situation feature lies within the plane situation feature and be aligned with the "direction of extrusion" of the prismatic feature but has no unique location on that plane. The line can be taken to be at any specified location within the plane and parallel to the direction of extrusion.

NOTE For a prismatic feature like the one shown in Figure B.8, a Gaussian (least-squares) association with no material constraint might make more sense.

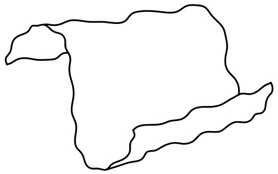
Figure B.8 — *Primary datum based on a prismatic surface*

Identification of datum feature



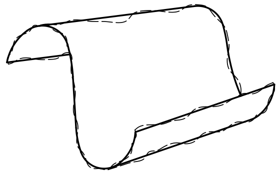
a) The nominal datum feature identified in the TPD is a prismatic surface.

Real datum feature



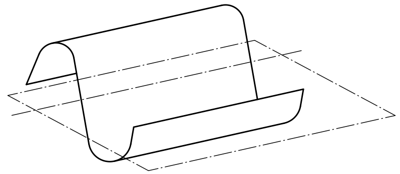
b) The real datum feature is the manufactured surface. The extracted datum feature is the data set obtained by sampling the surface.

Associated feature



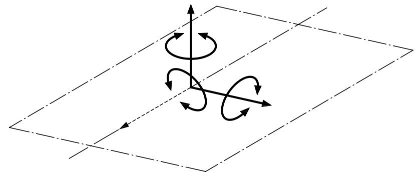
c) By default, the associated feature is a minimax associated surface outside the material, but a least squares associated surface may well make more sense.

Situation features



d) The situation feature for the associated feature are a plane and a straight line. The situation features are oriented in the direction of extrusion but neither has a unique location.

Degrees of freedom that can be controlled



e) This datum can constrain five degrees of freedom:

- translation along an axis perpendicular to the situation feature plane, and translation along an axis within the situation feature plane but perpendicular to the situation feature line; and
- rotation about all three axes.

Key

Locked

Unlocked

B.4.11 Complex surface

The datum feature is identified on the TPD as a complex surface.

The real datum feature on the manufactured workpiece is a non-ideal surface.

The associated feature is an ideal version of the complex surface, constructed to be in contact with the real datum feature, outside the material, and arranged such that the maximum distances between the associated feature and the real surface are minimized.

The datum is the situation feature for the associated complex surface, which is a plane, a line and a point.

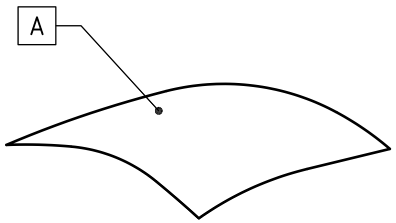
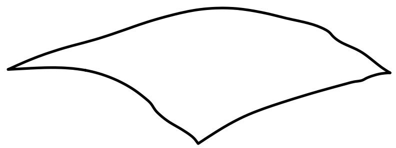
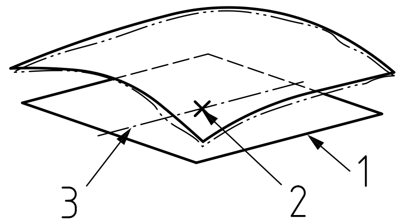
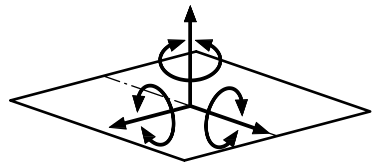
A primary datum based on the complex datum feature is a plane, a line and a point based on an associated feature constructed with no additional constraints. This can control six degrees of freedom (see Figure B.9).

A complex surface is not normally used as a secondary or tertiary datum feature.

The plane situation feature might have no unique location at all, so it can be placed at the discretion of the specifier. The line situation feature needs to lie within the plane situation feature, but has no unique location on that plane, so may be placed at any defined location within that plane. The point situation feature needs to lie on the line situation feature, but has no unique location on that line, so may be placed at any defined point on that line.

NOTE For a complex feature like the one shown, a Gaussian (least-squares) association with no material constraint may make more sense.

Figure B.9 — *Primary datum based on a complex surface*

Specification	Real datum feature	Associated feature and situation features	Degrees of freedom that can be controlled
			
a) Nominal feature is a complex surface identified in the TPS.	b) The real datum feature is the manufactured surface.	c) The associated feature by default is a minimax associated surface outside the material. The datum consists of three situation features, a plane, a straight line and a point, none of which have a unique location or orientation (they can be embedded in the CAD data in any location that suits the application).	d) This datum can constrain six degrees of freedom: • translation along all three axes; and • rotation about all three axes.

Annex C (informative) Implementation in CAD

Most CAD software:

- is limited in its ability to implement the full requirements of ISO standards;
- does not correctly distinguish between datums and datum features; and
- is not currently capable of dealing with datums consisting of more than one situation feature.

To implement these concepts in CAD, it might be necessary to deviate from the rules and methods specified in BS 8888 and ISO standards.

The following subclauses do not replace or override the requirements of this British Standard and ISO standards; they are simply offered to assist with the practical implementation of some of the datum requirements.

In Figure C.1:

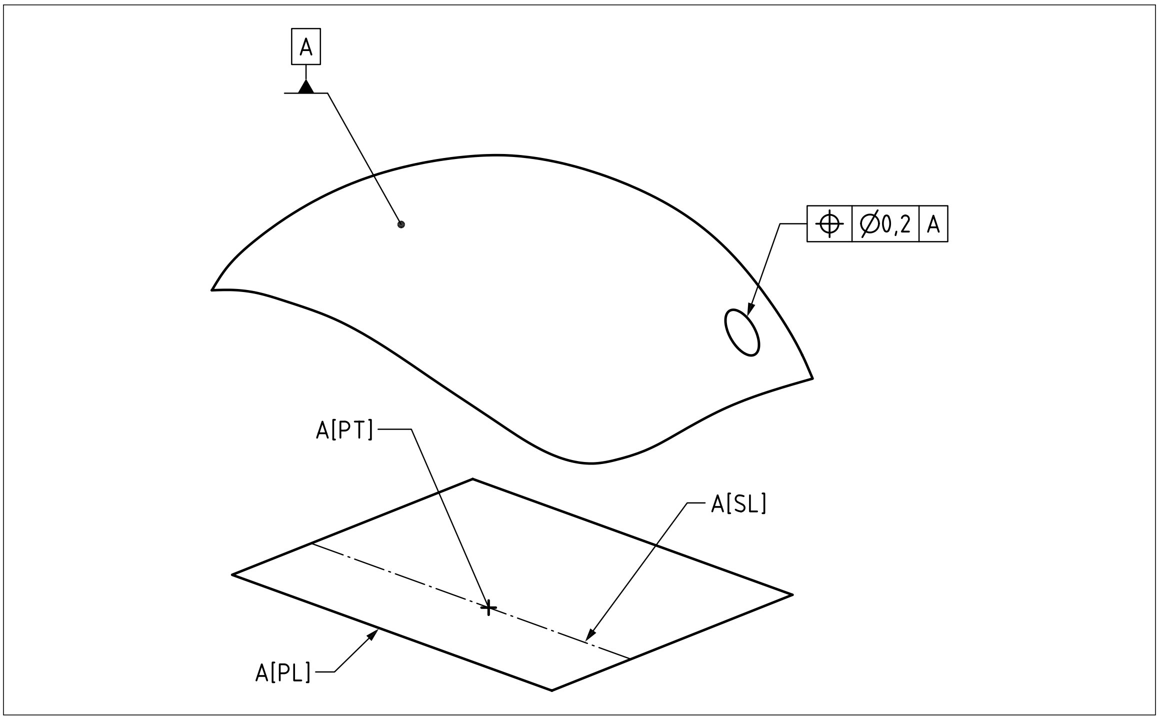
a) datum A is a single datum;

b) the datum feature is a complex surface;

c) datum A consists of a plane, line and point; and

d) if a position tolerance is applied to a hole in the part, a reference to datum A is sufficient to control all six degrees of freedom for the tolerance zone.

Figure C.1 — *Implementation according to BS 8888 and BS EN ISO 5459*



Where necessary, each situation feature can be treated as a separate datum. Thus the datum system could be treated as a datum plane, a datum line and a datum point, each with a separate name.

As the plane, line and point are representing a single datum, changing the sequence makes no difference to the resulting datum system.

Some additional explanatory notes may be required in the TPD.

Annex D (normative)
Document security: Enhanced

D.1 Enhanced security

Where requirements for enhanced security are known to exist, the procedures identified in this annex shall be applied in addition to those specified in **18.2.3**.

D.2 Identification of security classification

Any required security classification and/or caveat shall be inserted in the TPS immediately after classified information is incorporated.

Each sheet shall be classified according to its content.

The security classification shall appear at the top and bottom of A4 sheets and at the top left and bottom right-hand corners of sheets larger than A4.

The security classification shall either be:

a) larger than the largest text used in the TPS; or

b) bolder and the same size as the largest text used in the TPS.

D.3 Marking for enhanced security

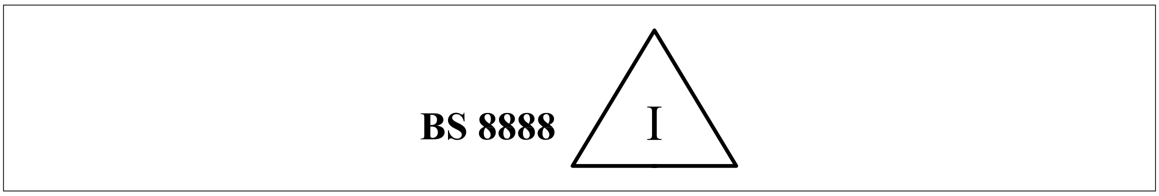
Technical product document sets prepared in accordance with the requirements of **D.1** and **D.2** shall be identified by the addition of the suffix "/D" to the identifier of this standard, i.e. "BS 8888/D", in a prominent location.

Annex E (informative)
Tolerancing system: Former practice

It was former practice to mark a TPS with the indication “BS 8888”, supplemented by the letter “I” contained within an equilateral triangle (see Figure E.1) or the letter “D” contained within an equilateral triangle (see Figure E.2).

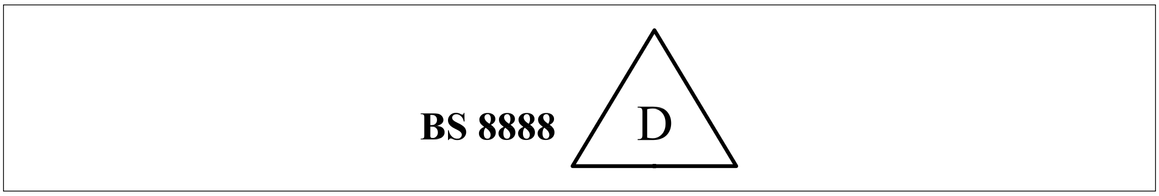
The triangle “I” symbol (see Figure E.1) was taken to indicate that the principle of independency was to be used to govern the interpretation of size and form requirements. While this had the same meaning as “TOLERANCING ISO 8015”, its meaning might not be apparent to an interpreter who was familiar with ISO standards but not BS 8888, so its use is no longer recommended.

Figure E.1 — *BS 8888 independency system symbol*



The triangle “D” symbol (see Figure E.2) was taken to indicate that the principle of dependency was to be used to govern the interpretation of size and form requirements. While this was included to maintain consistency with earlier versions of BS 8888 and BS 308 (withdrawn), the interpretation of the principle of dependency as a general requirement is not fully defined within the ISO system, so there might be ambiguities in its interpretation. In view of such possible ambiguities, and the fact that the symbol might not be understood by interpreters who were familiar with the ISO system but not BS 8888, the use of this symbol is no longer recommended.

Figure E.2 — *BS 8888 dependency system symbol*



Annex F (informative)
Selected ISO fits: Hole and shaft basis

F.1 General

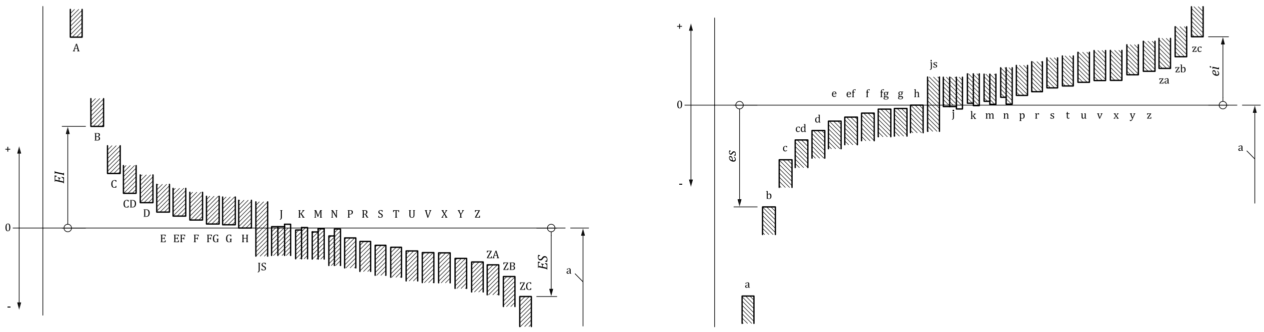
Selected ISO fits for hole and shaft basis are given in Table F.1 and Table F.2. The tolerances are stated for specific size ranges and comprised the fundamental deviation identifier (see Figure F.1) where capital letters are used for holes and lowercase for shafts, and the standard tolerance grade, which is given by Table F.1.

Table F.1 — *Values of standard tolerance grades for nominal sizes up to 3 150 mm*

Nominal size mm		Standard tolerance grades																			
		IT01	**IT0**	**IT1**	**IT2**	**IT3**	**IT4**	**IT5**	**IT6**	**IT7**	**IT8**	**IT9**	**IT10**	**IT11**	**IT12**	**IT13**	**IT14**	**IT15**	**IT16**	**IT17**	**IT18**
Above	**Up to and including**	**Standard tolerance values**																			
		µm											mm								
—	3	0,3	0,5	0,8	1,2	2	3	4	6	10	14	25	40	60	0,1	0,14	0,25	0,4	0,6	1	1,4
3	6	0,4	0,6	1	1,5	2,5	4	5	8	12	18	30	48	75	0,12	0,18	0,3	0,48	0,75	1,2	1,8
6	10	0,4	0,6	1	1,5	2,5	4	6	9	15	22	36	58	90	0,15	0,22	0,36	0,58	0,9	1,5	2,2
10	18	0,5	0,8	1,2	2	3	5	8	11	18	27	43	70	110	0,18	0,27	0,43	0,7	1,1	1,8	2,7
18	30	0,6	1	1,5	2,5	4	6	9	13	21	33	52	84	130	0,21	0,33	0,52	0,84	1,3	2,1	3,3
30	50	0,6	1	1,5	2,5	4	7	11	16	25	39	62	100	160	0,25	0,39	0,62	1	1,6	2,5	3,9
50	80	0,8	1,2	2	3	5	8	13	19	30	46	74	120	190	0,3	0,46	0,74	1,2	1,9	3	4,6
80	120	1	1,5	2,5	4	6	10	15	22	35	54	87	140	220	0,35	0,54	0,87	1,4	2,2	3,5	5,4
120	180	1,2	2	3,5	5	8	12	18	25	40	63	100	160	250	0,4	0,63	1	1,6	2,5	4	6,3
180	250	2	3	4,5	7	10	14	20	29	46	72	115	185	290	0,46	0,72	1,15	1,85	2,9	4,6	7,2
250	315	2,5	4	6	8	12	16	23	32	52	81	130	210	320	0,52	0,81	1,3	2,1	3,2	5,2	8,1
315	400	3	5	7	9	13	18	25	36	57	89	140	230	360	0,57	0,89	1,4	2,3	3,6	5,7	8,9
400	500	4	6	8	10	15	20	27	40	63	97	155	250	400	0,63	0,97	1,55	2,5	4	6,3	9,7
500	630	—	—	9	11	16	22	32	44	70	110	175	280	440	0,7	1,1	1,75	2,8	4,4	7	11
630	800		—	10	13	18	25	36	50	80	125	200	320	500	0,8	1,25	2	3,2	5	8	12,5
800	1 000	—	—	11	15	21	28	40	56	90	140	230	360	560	0,9	1,4	2,3	3,6	5,6	9	14
1 000	1 250	—	—	13	18	24	33	47	66	105	165	260	420	660	1,05	1,65	2,6	4,2	6,6	10,5	16,5
1 250	1 600	—	—	15	21	29	39	55	78	125	195	310	500	780	1,25	1,95	3,1	5	7,8	12,5	19,5
1 600	2 000	—	—	18	25	35	46	65	92	150	230	370	600	920	1,5	2,3	3,7	6	9,2	15	23
2 000	2 500	—	—	22	30	41	55	78	110	175	280	440	700	1 100	1,75	2,8	4,4	7	11	17,5	28
2 500	3 150	—	—	26	36	50	68	96	135	210	330	540	860	1 350	2,1	3,3	5,4	8,6	13,5	21	33

NOTE SOURCE: BS EN ISO 286-1:2010, Table 1.

Figure F.1 — *Fundamental deviations of holes and shafts*



a) Holes (internal features of size) b) Shafts (external features of size)

NOTE SOURCE: BS EN ISO 286-1:2010, Figure 7.

F.2 Determination of limit deviations using Table F.1 and Table F.2

"The tolerance class is decomposed into the fundamental deviation identifier and the standard tolerance grade number.

EXAMPLE

Toleranced size for a hole 90 F7 Ⓔ and for a shaft 90 f7 Ⓔ,

where:

90	is the nominal size in millimetres;
F	is the fundamental deviation identifier for a hole;
f	is the fundamental deviation identifier for a shaft;
7	is the standard tolerance grade number;
Ⓔ	is the envelope requirement according to BS EN ISO 14405-1 (if necessary)."

[SOURCE: BS EN ISO 286-1:2010, **4.3.2.1**]

"For general engineering purposes, preferred fits are given in Figure F.2.

Based on the decision taken, the tolerance grades and the fundamental deviation (placement of tolerance interval) should then be chosen for the hole and the shaft to give the corresponding minimum and maximum clearances or interferences that best meet the required conditions of use.

For normal ordinary engineering purposes, only a small number of the many possible fits is required. Figure F.2 indicates those fits which meet many of the needs of an average engineering organization. For economic reasons, the first choice for a fit should, whenever possible, be made from the tolerance classes shown in the frames (see Figure F.2).

Satisfactory fits are obtained by the combinations of basic holes system [see Figure F.2a)] or for special applications the combinations of basic shafts system [see Figure F.2b)]."

Figure F.2 — *Preferred fits for general engineering*

Basic hole	Tolerance classes for shafts																	
	Clearance fits							Transition fits				Interference fits						
H6						g5	h5	js5	k5	m5		n5	p5					
H7					g5	**g6**	**h6**	**js6**	**k6**	m6	**n6**		**p6**	**r6**	**s6**	t6	u6	x6
H8				e7	**f7**		**h7**	js7	k7	m7					s7		u7	
			d8	**e8**	f8		h8											
H9			d8	**e8**	f8		h8											
H10	b9	c9	**d9**	e9			**h9**											
H11	**b11**	**c11**	d10				h10											

a) Preferred fits of the hole-basis system

Basic shaft	Tolerance classes for holes																	
	Clearance fits							Transition fits				Interference fits						
h5						G6	H6	JS6	K6	M6		N6	P6					
h6					F7	**G7**	**H7**	**JS7**	**K7**	M7	**N7**		**P7**	**R7**	**S7**	T7	U7	X7
h7				E8	**F8**		**H8**											
h8			D9	E9	F9		H9											
h9				E8	**F8**		**H8**											
			D9	**E9**	F9		**H9**											
	B11	C10	**D10**				H10											

b) Preferred fits of the shaft-basis system

NOTE SOURCE: BS EN ISO 286-1:2010, Figure 13.

[SOURCE: BS EN ISO 286-1:2010, **5.3.4**]

NOTE Table F.2 and Table F.3 give selected ISO fits for holes and shafts in accordance with BS EN ISO 286-1:2010, Table 1 to Table 4.

Table F.2 — *Selected ISO fits: Hole basis*

Nominal sizes		Clearance fits												Transition fits				Interference fits				Nominal sizes	
		Tolerances		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**			
Over	To	H11	c11	H9	d10	H9	e9	H8	f7	H7	g6	H7	h6	H7	k6	H7	n6	H7	p6	H7	s6	Over	To
mm	mm	0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		mm	mm
–	3	+60 0	−60 −120	+25 0	−20 −60	+25 0	−14 −39	+14 0	−6 −16	+10 0	−2 −8	+10 0	0 −6	+10 0	+6 0	+10 0	+10 +4	+10 0	+12 +6	+10 0	+20 +14	–	3
3	6	+75 0	−70 −145	+30 0	−30 −78	+30 0	−20 −50	+18 0	−10 −22	+12 0	−4 −12	+12 0	0 −8	+12 0	+9 +1	+12 0	+16 +18	+12 0	+20 +12	+12 0	+27 +19	3	6
6	10	+90 0	−80 −170	+36 0	−40 −98	+36 0	−25 −61	+22 0	−13 −28	+15 0	−5 −14	+15 0	0 –	+15 0	+10 +1	+15 0	+19 +10	+15 0	+24 +15	+15 0	+32 +23	6	10
10	18	+110 0	−95 −205	+43 0	−50 −120	+43 0	−32 −75	+27 0	−16 −34	+18 0	−6 −17	+18 0	0 −11	+18 0	+12 +1	+18 0	+23 +12	+18 0	+29 +18	+18 0	+39 +28	10	18
18	30	+130 0	−110 −240	+52 0	−65 −149	+52 0	−40 −92	+33 0	−20 −41	+21 0	−7 −20	+21 0	0 −13	+21 0	+15 +2	+21 0	+28 +15	+21 0	+35 +22	+21 0	+48 +35	18	30
30	40	+160 0	−120 −280	+62 0	−80 −180	+62 0	−50 -112	+39 0	−25 −50	+25 0	−9 −25	+25 0	0 −16	+25 0	+18 +2	+25 0	+33 +17	+25 0	+42 +26	+25 0	+59 +43	30	40
40	50		−130 −290																			40	50
50	65	+190 0	−140 −330	+74 0	−100 −220	+74 0	−60 −134	+46 0	−30 −60	+30 0	−10 −29	+30 0	0 −19	+30 0	+21 +2	+30 0	+39 +20	+30 0	+51 +32	+30 0	+72 +53	50	65
65	80		−150 −340																		+78 +59	65	80
80	100	+220 0	−170 −390	+87 0	−120 −260	+87 0	−72 −159	+54 0	−36 −71	+35 0	−12 −34	+35 0	0 −22	+35 0	+25 +3	+35 0	+45 +23	+35 0	+59 +37	+35 0	+93 +71	80	100
100	120		−180 −400																		+101 +79	100	120

Table F.3 — *Selected ISO fits: Shaft basis*

		Clearance fits												Transition fits				Interference fits					
Nominal sizes		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Tolerances**		**Nominal sizes**	
Over	To	h11	C11	h9	D10	h9	E9	h7	F8	h6	G7	h6	H7	h6	K7	h6	N7	h6	P7	h6	S7	Over	To
mm	mm	0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		0,001 mm		mm	mm
–	3	0 −60	+120 +60	0 −25	+60 +20	0 −25	+39 +14	0 −10	+20 +6	0 −6	+12 +2	0 −6	+10 0	0 −6	0 −10	0 −6	−4 −14	0 −6	−6 −16	0 −6	−14 −24	–	3
3	6	0 −75	+145 +70	0 −30	+78 +30	0 −30	+50 +20	0 −12	+28 +10	0 −8	+16 +4	0 −8	+12 0	0 −8	+3 −9	0 −8	−4 −16	0 −8	−8 −20	0 −8	−15 −27	3	6
6	10	0 −90	+170 +80	0 −36	+98 +40	0 −36	+61 +25	0 −15	+35 +13	0 −9	+20 +5	0 −9	+15 0	0 −9	+5 −10	0 −9	−4 −19	0 −9	−9 −24	0 −9	−17 −32	6	10
10	18	0 −110	+205 +95	0 −43	+120 +50	0 −43	+75 +32	0 −18	+43 +16	0 −11	+24 +6	0 −11	+18 0	0 −11	+6 −12	0 −11	−5 −23	0 −11	−11 −29	0 −11	−21 −39	10	18
18	30	0 −130	+240 +110	0 −52	+149 +65	0 −52	+92 +40	0 −21	+53 +20	0 −13	+28 +7	0 −13	+21 0	0 −13	+6 −15	0 −13	−7 −28	0 −13	−14 −35	0 −13	−27 −48	18	30
30	40	0 −160	+280 +120	0 −62	+180 +80	0 −62	+112 +50	0 −25	+64 +25	0 −16	+34 +9	0 −16	+25 0	0 −16	+7 −18	0 −16	−8 −33	0 −16	−17 −42	0 −16	−34 −59	30	40
40	50		+290 +130																			40	50
50	65	0 −190	+330 +140	0 −74	+220 +100	0 −74	+134 +60	0 −30	+76 +30	0 −19	+40 +10	0 −19	+30 0	0 −19	+9 −21	0 −19	−9 −39	0 −19	−21 −51	0 −19	−42 −72	50	65
65	80		+340 +150																		−48 −78	65	80
80	100	0 −220	+390 +170	0 −87	+260 +120	0 −87	+159 +72	0 −35	+90 +36	0 −22	+47 +12	0 −22	+35 0	0 −22	+10 −25	0 −22	−10 −45	0 −22	−24 −59	0 −22	−58 −93	80	100
100	120		+400 +180																		−66 −101	100	120

[SOURCE: BS EN ISO 286-1:2010, Table 1, Table 2, Table 3 and Table 4]

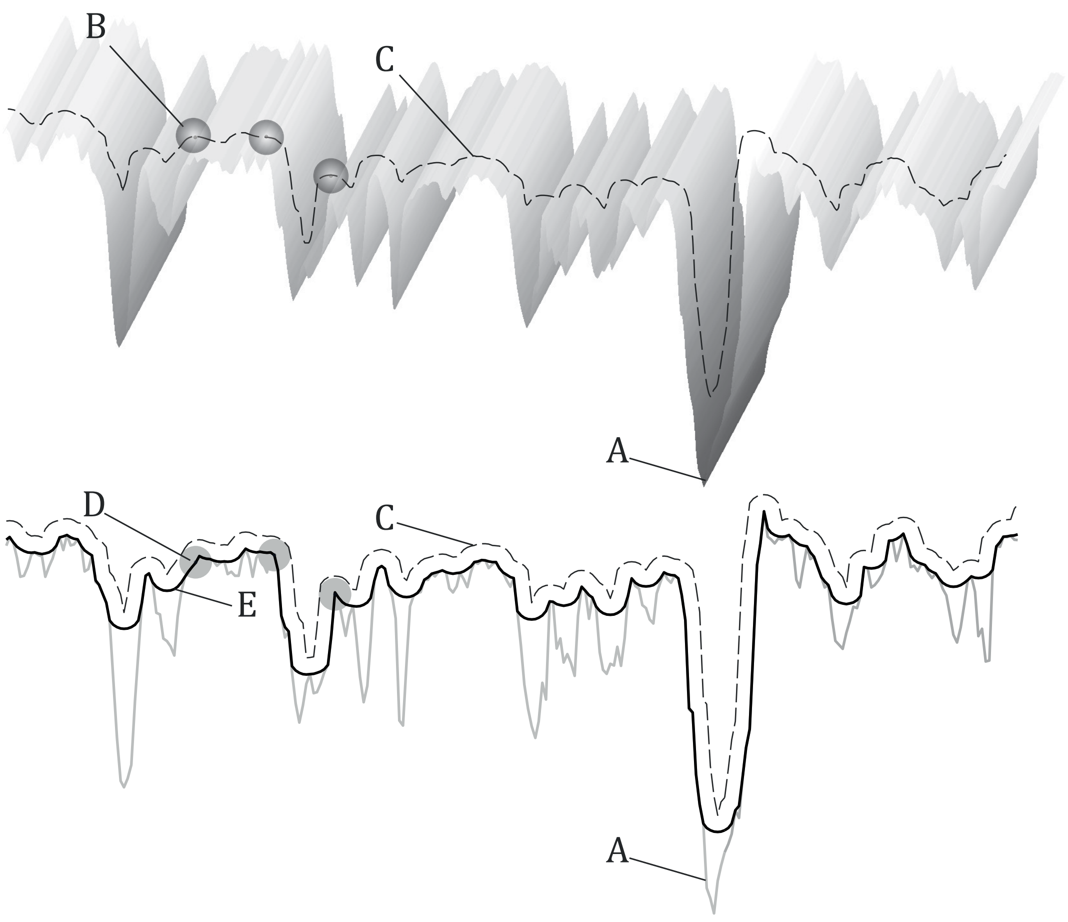
Annex G (informative)
Filtration

G.1 Introduction

The data of a measurement taken for a manufactured surface can be incomplete. For instance, some information regarding the surface is lost due to the resolution of the measurement process.

The data representing the manufactured surface is thus filtered by the measurement process. The manner in which the measurement process filters the measurement data might be controlled, for instance by defining the diameter of a probe tip, the point extraction strategy or the extent over which a trace is taken. If a contacting spherical probe was used to take a trace, this results in mechanical filtering and the filtered profile is known as the mechanical profile as shown in Figure G.1.

Figure G.1 — *Mechanical profile, showing the effect of mechanical filtering*



Key

A Non-ideal surface (e.g. manufactured surface)

B Contacting sphere (e.g. tip of measuring probe)

C Path of B (e.g. travelling in a plane)

D Mathematically offset sphere

E Mechanical profile (path of D) (e.g. mechanically filtered data representing A)

NOTE SOURCE: BS EN ISO 21920-2:2022, Figure 1, modified – key changed, diagram removed.

In addition to the filtration, additional digital filters might be imposed on the measurement data (e.g. on the mechanical profile), in order to isolate a specific range of structures on a feature to which a specification is applied, or simply to reduce the quantity of measurement data to a manageable level. Filters are usually applied digitally, but in some cases, mechanical filtering can be applied and can simulate the effects of digital filters, such as morphological filters. Filters can also be used to reject data that is not relevant to the properties under consideration. The choice of filter is often influenced by the functional requirements of the surface.

Filtration of measurement data is unavoidable, but traditionally geometrical specifications have tended to avoid defining filters on specifications. Defining filters on geometrical specifications reduces ambiguity during the verification process. This British Standards provides the means to define the filtration properties requirements. The practice of defining filters for surface texture definitions is well established and default rules for filters for surface texture are given in the BS EN ISO 21920 and BS EN ISO 25178 series. The possibility of defining filter properties for geometrical specifications of form, such as flatness and roundness, has existed since 2003, when the BS EN ISO 12180 series, BS EN ISO 12181 series, BS EN ISO 12780 series and BS EN ISO 12781 series first appeared. BS EN ISO 1101:2017 now provides the tools for applying filters to any geometrical specification.

Filters are usually applied to the frequency, or wavelength, of the variations in the feature in order to select a specific range but can also be used to eliminate outliers or noise in data.

G.2 Filter types

Many different types of filters are available, and these are documented in the BS EN ISO 16610 series and listed in Table G.1 and Table G.2.

Table G.1 — *Basic semantics in designating filters*

Filter	**Type**	**Category**
F = Filter	A = Areal (3D)	L = Linear M = Morphological R = Robust
	P = Profile (2D)	L = Linear M = Morphological R = Robust

NOTE SOURCE: BS EN ISO 16610-1:2015, Table 1.

Table G.2 — *Filter symbols*

Symbol	Designation(s)	Name	ISO Document(s)
G	FALG, FPLG	Gaussian	BS EN ISO 16610-21, BS EN ISO 16610-61
S	FALS, FPLS	Spline	BS EN ISO 16610-22, BS EN ISO 16610-62
SW	FALPSW, FPLPSW	Spline Wavelet	BS EN ISO 16610-29, BS EN ISO 16610-69
CW	FALPCW, FPLPCW	Complex Wavelet	BS EN ISO 16610-29, BS EN ISO 16610-69
RG	FARG, FRPG	Robust Gaussian	BS EN ISO 16610-31, BS EN ISO 16610-71
RS	FARS, FPRS	Robust Spline	PD ISO/TR 16610-32
OH	FAMOH, FPMOH	Opening Horizontal Segment	BS EN ISO 16610-41
OD	FPMOD	Opening Disc	BS EN ISO 16610-41
CH	FAMCH, FPMCH	Closing Horizontal Segment	BS EN ISO 16610-41
CD	FPMCD	Closing Disc	BS EN ISO 16610-41
AH	FAMAH, FPMAH	Alternating Horizontal Segment	BS EN ISO 16610-49
AD	FPMAD	Alternating Disc	BS EN ISO 16610-49
F	—	Fourier (harmonics)	N/A
H	—	Hull	N/A

NOTE SOURCE: BS EN ISO 1101:2017, Table C.1, modified – only published documents included.

Filters can be classified as:

a) **Long-wave pass filters:** Long-wave pass filters allow longer wavelength properties to pass through and filter out the shorter wavelength properties. This is probably the most common type of filter. Other names used for this filter include “low-pass” and “S-Filter”;

b) **Short-wave pass filters:** Short-wave pass filters allow shorter wavelength properties to pass through and filter out the longer wavelength properties. Other names used for this filter include “high-pass” and “L-Filter”; and

c) **Band pass filters:** Band pass filters have longer and shorter wavelength limits, allowing a specified range of wavelengths through.

Another way of classifying filters is as profile and areal filters, given in Table G.1:

1) **Profile filters:** Profile filters are applied to a two-dimensional profile; and

2) **Areal filters:** Areal filters are applied to three-dimensional areas.

G.3 Filters and nesting indices

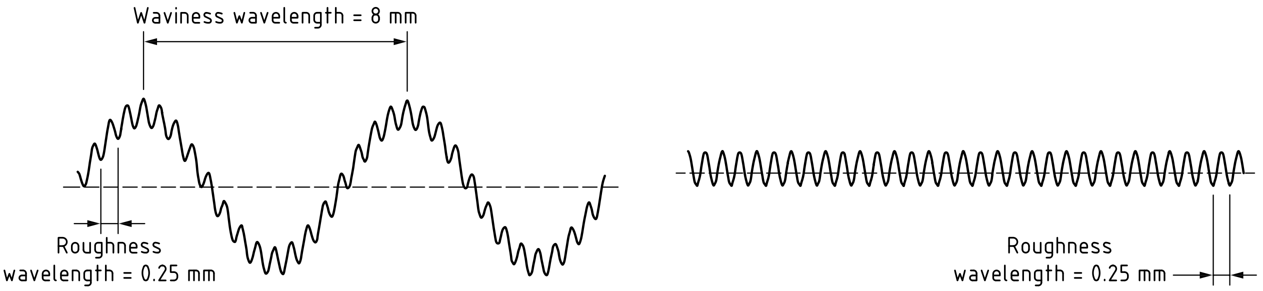
When a filter is defined, there are two components: one is the type of filter, and the other is the parameter (or parameters) required for that type of filter.

An analogy sometimes used is that of a sieve, used to filter out objects larger than the holes in the sieve. The type of filter is the type of sieve; some have round holes, some have square holes and some have slots arranged in various ways. The parameters associated with the filter define the properties of the aperture (such as the size of the hole or the width of the slot).

The parameter associated with a filter (in the analogy, the size of the aperture in the sieve) is referred to as the "nesting index". As the nesting index approaches zero, the size of the aperture approaches zero and nothing can get through, so all data representing the surface are captured, and nothing is filtered out.

Figure G.2 shows an example of the effect of filtering on a profile that contains waviness and roughness variations. If only the roughness variation was of interest, the roughness component of the profile can be isolated by filtering using a nesting index that removes the waviness component of the profile.

Figure G.2 — *Effect of filtering using a filter with a 0.25 mm cut-off*



a) Unfiltered profile   b) Filtered profile (Waviness removed)

NOTE Wavelengths greater than 0.25 mm are filtered out.

When a filter is applied to an "open" feature, a feature which terminates at an edge or a point, such as a line element or plane, the nesting index is defined in millimetres.

When a filter is applied to a "closed" surface, a feature which is continuous, such as a circular line element, or a cylinder in the circumferential direction, the nesting index is defined in "undulations per revolution", or UPR.

In the case of a morphological filter, the nesting index is the size of the structuring element.

Table G.3 shows nesting indices used with each filter type.

Table G.3 — *Nesting indices*

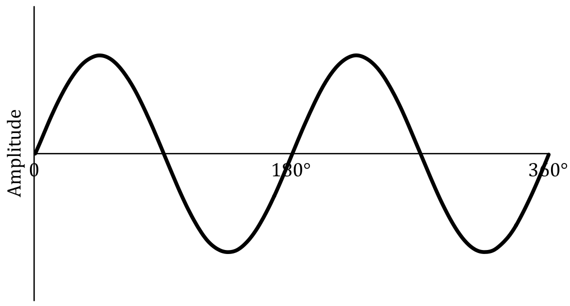
Symbol	Name	Nesting index
G	Gaussian	Cut-off length and/or cut-off UPR
S	Spline	Cut-off length and/or cut-off UPR
SW	Spline Wavelet	Cut-off length and/or cut-off UPR
CW	Complex Wavelet	Cut-off length and/or cut-off UPR
RG	Robust Gaussian	Cut-off length and/or cut-off UPR
RS	Robust Spline	Cut-off length and/or cut-off UPR
OB	Opening Ball	Ball radius
OH	Opening Horizontal Segment	Segment length
OD	Opening Disc	Disc radius
CB	Closing Ball	Ball radius
CH	Closing Horizontal Segment	Segment length
CD	Closing Disc	Disc radius
AB	Alternating Ball	Ball radius
AH	Alternating Horizontal Segment	Segment length
AD	Alternating Disc	Disc radius
F	Fourier (harmonics)	N/A
H	Hull	N/A

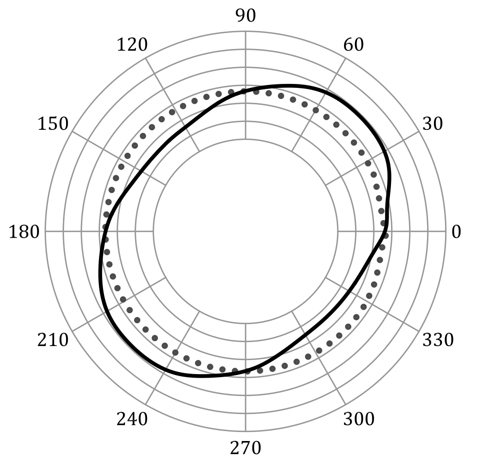
NOTE SOURCE: BS EN ISO 1101:2017, Table C.2, modified – last two index entries replaced with N/A.

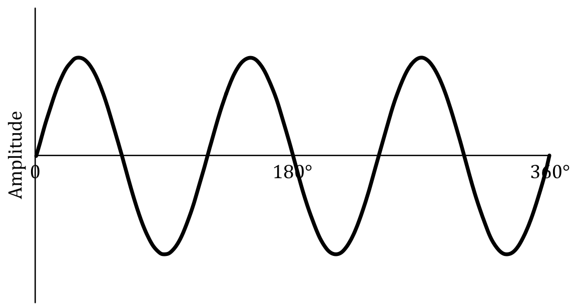
G.4 Harmonic content

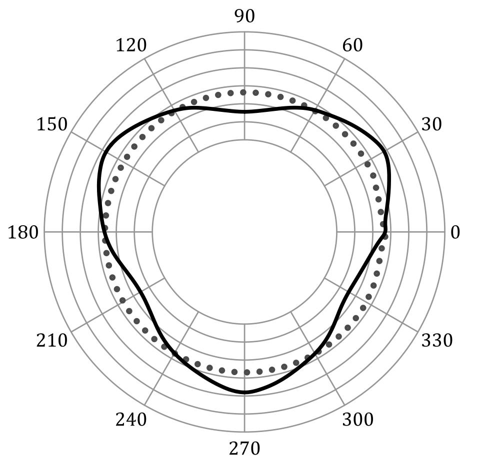
A roundness profile (closed profile) can be decomposed into a number of harmonics (i.e. sinusoidal components) called the Fourier series. The Fourier series consists of a fundamental sinusoid whose wavelength is the circumference of the circle or one UPR. This is the first harmonic of the profile, i.e. 1 UPR. The other harmonics consist of the higher UPR, e.g. the second harmonic has 2 UPR, the third harmonic has 3 UPR and the Nth harmonic has N UPR. All constituent harmonics combined together form the complete roundness profile. Figure G.3 shows some examples of isolated harmonics.

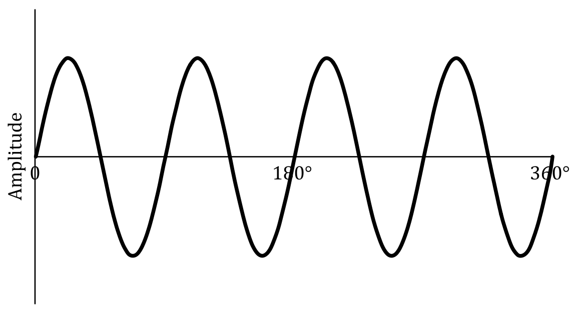
Figure G.3 — *Harmonic groups*

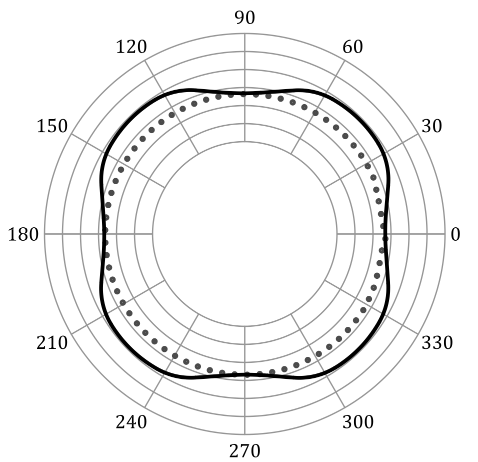
<table>
<tr><th>Waveform</th><th>Harmonic profile</th></tr>
<tr><td colspan="2">second Harmonic (Ovality) – 2 UPR</td></tr>
<tr><td>


</td><td>


</td></tr>
<tr><td colspan="2">third Harmonic (Tri-Lobe) – 3 UPR</td></tr>
<tr><td>


</td><td>


</td></tr>
<tr><td colspan="2">fourth Harmonic – 4 UPR</td></tr>
<tr><td>


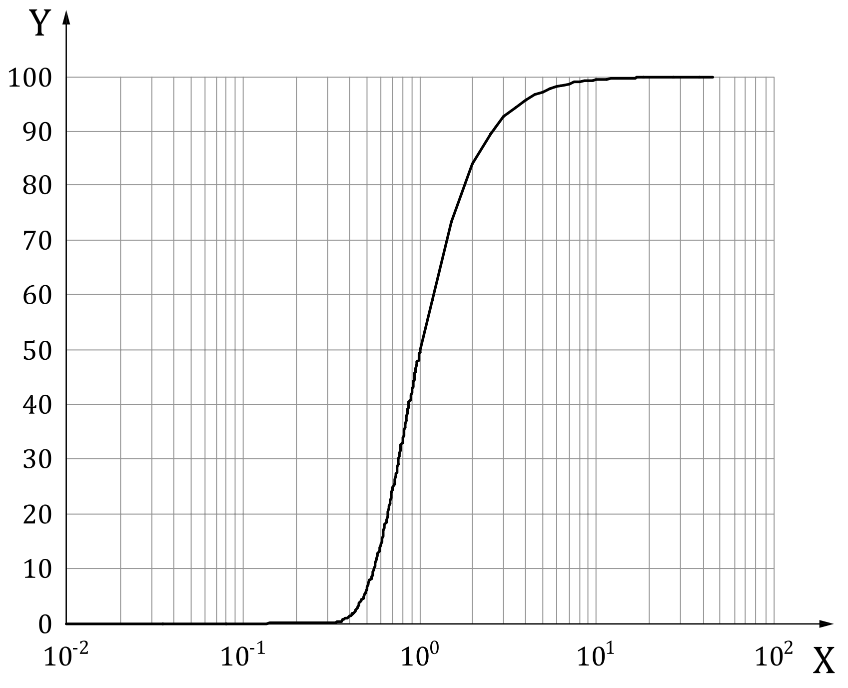
</td><td>


</td></tr>
</table>

G.5 Transmission characteristics of linear filters

Linear filters, e.g. Gaussian filter, have a short-wave and long-wave transmission characteristic that can be viewed in the frequency domain via a Fourier transformation. This characteristic indicates the amount by which the amplitude of a sinusoidal profile is attenuated as a function of its wavelength. Figure G.4 shows the long-wave transmission function of a Gaussian filter for an open profile. It shows how 50% of the amplitude of wavelengths of a sinusoidal profile at the selected cut-off value (λc) are transmitted.

Figure G.4 — *Long-wave transmission function of the Gaussian filter for an open profile*



Key

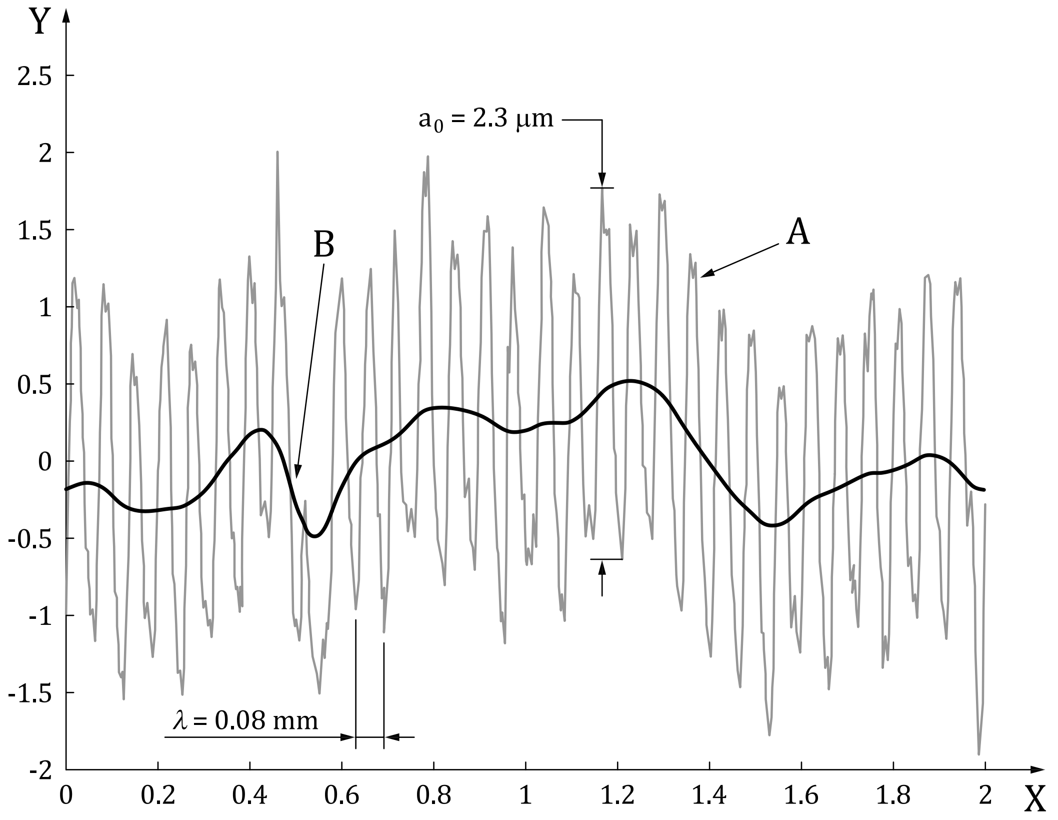
X λ/λ_c

Y Amplitude transmission a_1/a_0, in per cent

NOTE SOURCE: BS EN ISO 16610-61:2015+A1:2020, Figure 5, modified – A and B replaced with Y and X.

Figure G.5 shows how the shorter wavelength (high frequency) component of an unfiltered surface profile (i.e. a line on a surface) is removed via filtering, leaving the longer wavelength (low frequency) component behind. If plotted on the transmission function in Figure G.4, this gives X = λ/λ_C = 0,08/0.8 = 0,1, and Y = $a1/a0$ = 2,5/$a0$ = 0. Therefore, amplitude of profile component at this wavelength is fully attenuated by the filter. The filtered profile appears as an averaging line in the middle of the unfiltered profile, and this is generally typical of all linear filters (e.g. Gaussian, Spline and Spline Wavelet filters). Filters can also be used for areal features, i.e. planes, cylinders, etc., but for ease of illustration, profiles are used here.

Figure G.5 — *Open Gaussian filter with (λc = 0,8 mm), using the moment retainment criterion for end effect correction, applied on a milled surface*



Key

X Distance x, mm

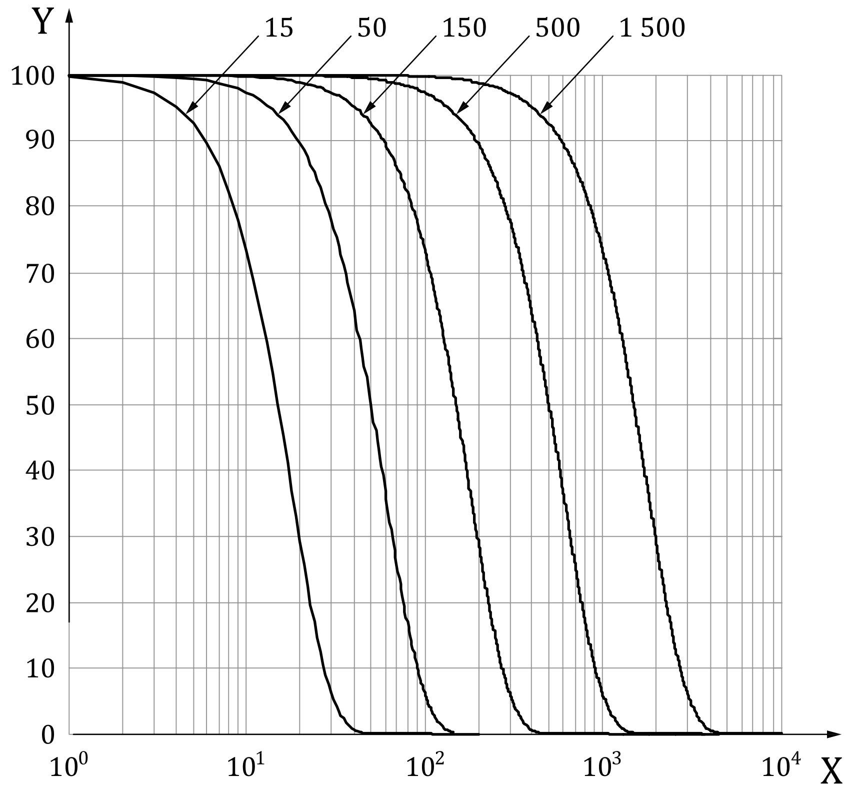
Y Height y, µm

A Unfiltered profile

B Filtered profile

Figure G.6 shows the transmission characteristic of the long-wave profile component for closed profile filters of different UPR values.

Figure G.6 — *Transmission characteristic of the long-wave profile component for closed profile filters of different UPR values*



Key

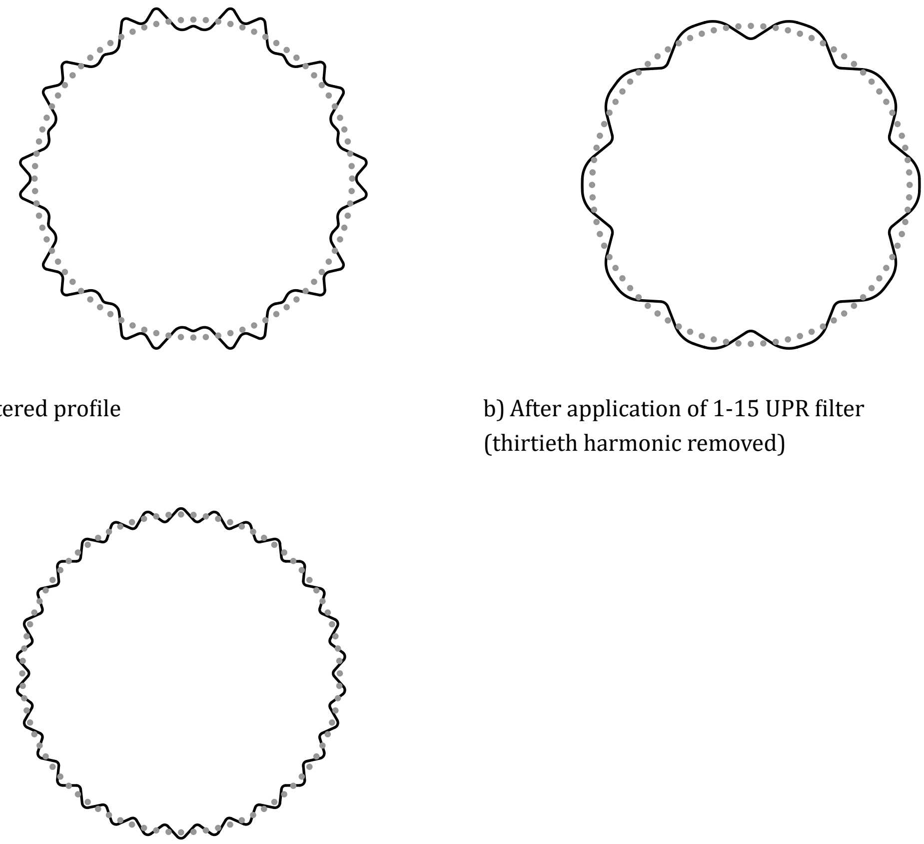
X UPR

Y amplitude transmission a_1/a_0, in per cent

NOTE SOURCE: BS EN ISO 16610-61:2015+A1:2020, Figure 4, modified – A and B replaced with Y and X.

Figure G.7 shows how a roundness profile that has a dominant tenth and thirtieth harmonic, i.e. 10 UPR and 30 UPR respectively, can have its individual harmonics isolated via filtering. Figure G.7b) shows the application of a 1-15 UPR filter on the unfiltered profile, thus all harmonics outside of this range are attenuated in amplitude by increasing amounts the further the harmonic is away from the selected range, i.e. the thirtieth harmonic. Figure G.7c) shows the application of a 15-50 UPR filter on the unfiltered profile, thus all harmonics outside of this range are attenuated, i.e. the tenth harmonic.

Figure G.7 — *Roundness profile with a dominant tenth and thirtieth harmonic and individual harmonics isolated via use of filters on unfiltered profile*



a) Unfiltered profile

b) After application of 1-15 UPR filter (thirtieth harmonic removed)

c) After application of 15-50 UPR filter (tenth harmonic removed)

NOTE This is a simplified illustration that assumes an ideal transmission characteristic of the filters.

G.6 Surface texture

G.6.1 General

A primary surface profile (i.e. ultra short wavelengths, e.g. noise, below cut-off λs removed from unfiltered profile) can have its shorter wavelength components (roughness), medium wavelength components (waviness) and longer wavelength components (form) derived from filters, as shown in Figure G.8 and Figure G.9.

Figure G.8 — *Transmission characteristic of High/Band/Long pass filters*

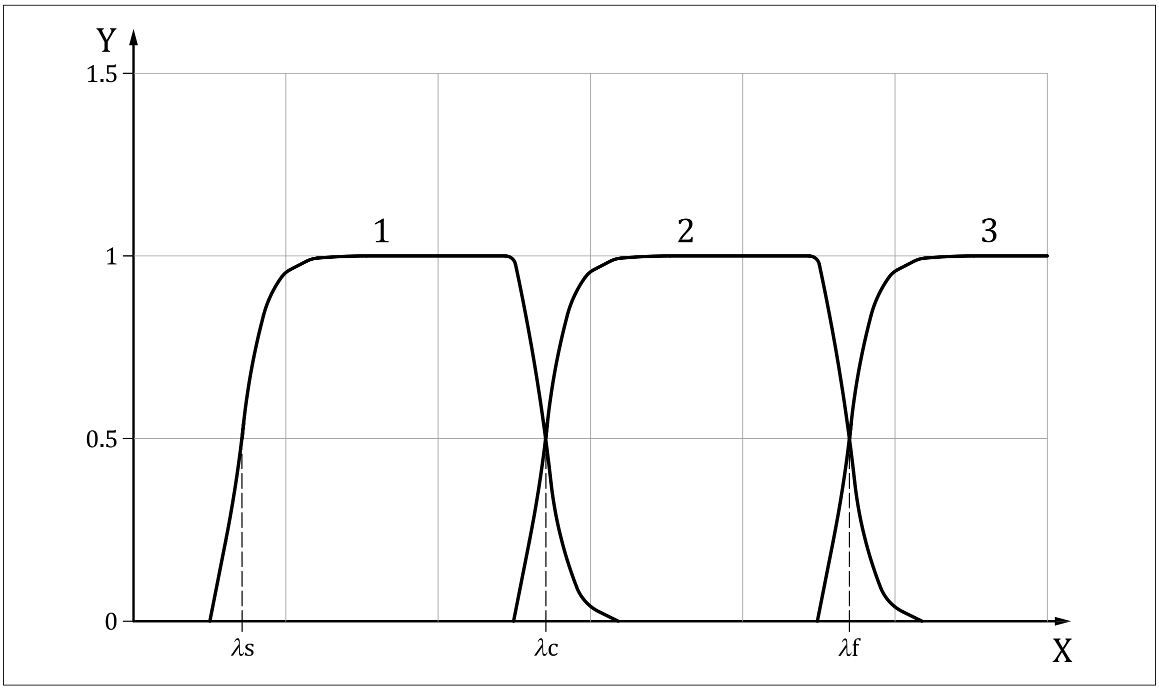
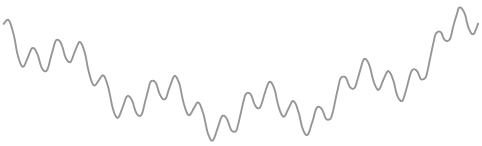
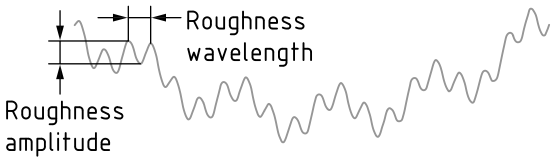


Figure G.9 — *Different components of a profile separated via filters*

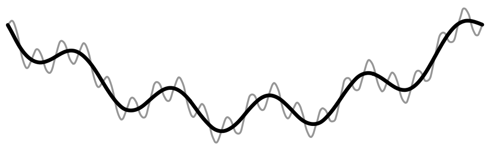


a) Roughness, waviness and form: Simplified Primary Surface profile (ultra short wavelengths (e.g. noise) below cut-off λs removed)

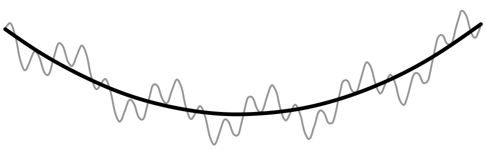


b) Roughness trend (pre-filtered): Simplified Roughness profile (black) derived from use of L-filter with cut-off λc

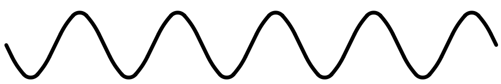
c) Roughness trend (filtered): Simplified Roughness profile (black) derived from use of L-filter with cut-off λc



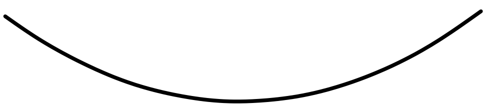
d) Waviness (pre-filtered): Simplified Waviness profile (black) derived from use of S-filter with cut-off λc and L-filter with cut-off λf



e) Form trend (pre-filtered): Simplified Form profile (black) derived from use of S-filter with cut-off λf



f) Waviness (filtered): Simplified Waviness profile (black) derived from use of S-filter with cut-off λc and L-filter with cut-off λf

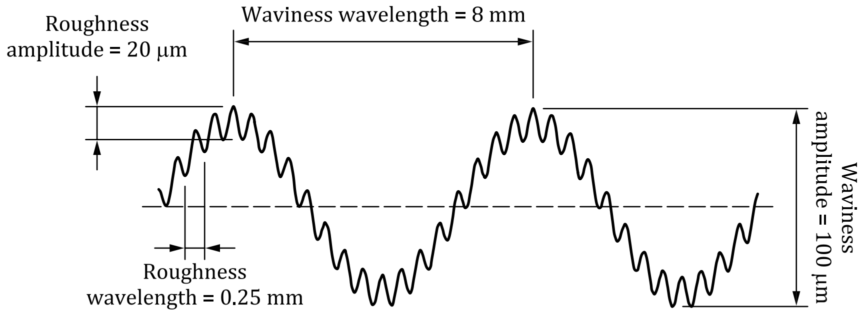


g) Form trend (filtered): Simplified Form profile (black) derived from use of S-filter with cut-off λf

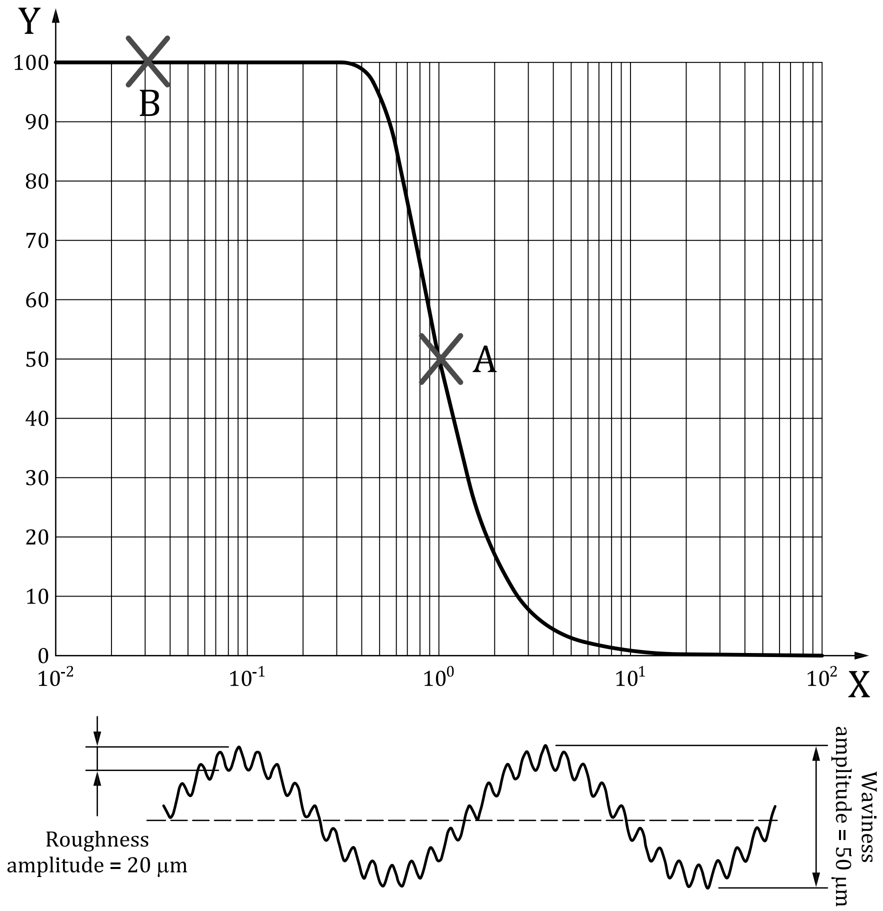
Generally, the surface texture standards focus on controlling the properties of roughness (R-parameters) and waviness (W-parameters), whereas the geometrical tolerancing standards (e.g. BS EN ISO 1101:2017) focus on control of large wavelength properties, i.e. form. However, there can be some overlap when it comes to waviness, as this can be controlled via a surface texture specification or a form tolerance specification, depending on the filter settings specified.

Figure G.10 shows the effects of filtering on an unfiltered profile using different cut-off values. It can be seen that the choice of cut-off value can have a significant effect on the filtered profile, highlighting the importance of choosing a suitable cut-off value.

Figure G.10 — *Effects of filtering on unfiltered profile using different cut-off values*

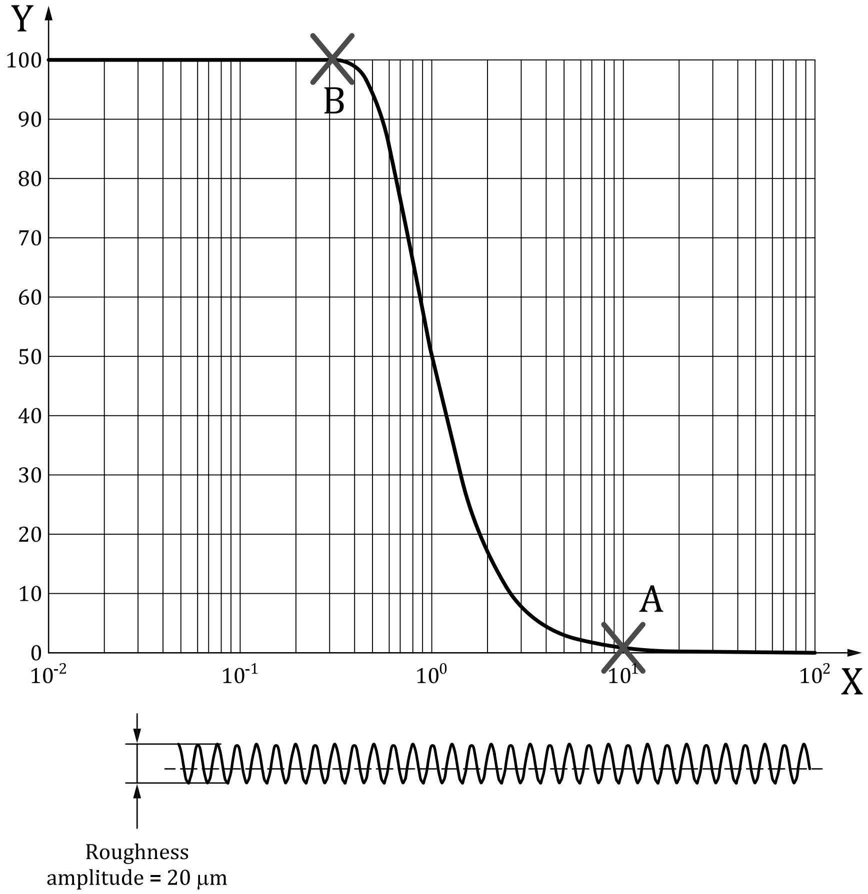


a) Unfiltered profile with no cut-off applied



b) Filtered profile after cut-off λc = 8 mm applied

Figure G.10 — *Effects of filtering on unfiltered profile using different cut-off values (continued)*



c) Filtered profile after cut-off λc = 0,8 mm applied

NOTE The amplitude transmission % is derived from the short-wave transmission function of the Gaussian filter for an open profile.

G.6.2 Defaults filters

BS EN ISO 21920-3 contains default filter settings for determination of roughness (R-parameters). However, the choice of filter settings for waviness (W-parameters) is highly dependent on the functional requirements and no nesting index defaults exist so these are specified.

G.6.3 Examples

The BS EN ISO 21920 and BS EN ISO 25178 series provide several examples showing how the filter settings can be specified. Clause **15** also provides examples. Figure G.11 shows an example of non-default filter settings being used on a roughness specification, with explanation in Table G.4.

Figure G.11 — *Indication for an evaluation length R-parameter including all definable specification elements by using no default*

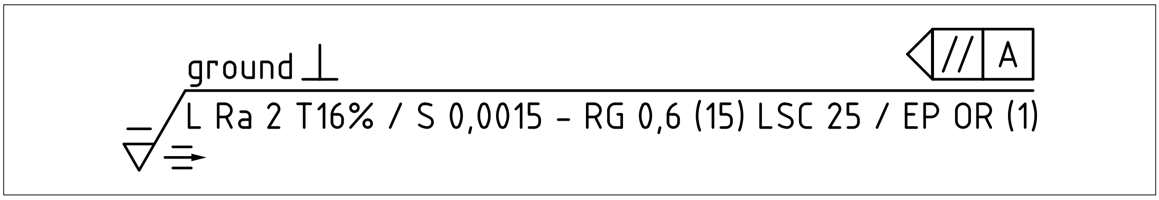


Table G.4 — *Explanation of filter settings for example in Figure G.11*

Specification elements	Explanation
S	Profile S-filter type: Spline
0,001 5	Profile S-filter nesting index Nis: 0,001 5 mm
RG	Profile L-filter type: Robust Gaussian
0,6	Profile L-filter nesting index Nic: 0,6 mm
NOTE For other symbols used in the indication, refer to BS EN ISO 21920 series and BS EN ISO 25178 series.	

G.7 Geometric tolerances

G.7.1 General

In a geometrical specification, the mathematical representation of the feature to which the tolerance applies to can be modified by applying a filter. This enables the tolerance to be "focussed" on particular aspects of the surface. For instance, a tolerance could be applied just to higher frequency variations in a surface, or just to lower frequency variations in a surface.

The type of filter can be specified using one of the symbols shown in Table G.3, followed by a value which defines the nesting index for the filter. There is no space between the symbol for the filter and the number(s) indicating the nesting index.

The nesting index can be defined as a single value, or a range (e.g. upper and lower cut-offs).

Most nesting indices are long-wave or short-wave pass filters:

a) for a long-wave pass filter, the value is followed by a "-" symbol;

b) for a short-wave pass filter, the value is preceded by a "-" symbol; and

c) for a range of values long-wave pass and short-wave pass limits are shown separated by the "-" sign.

Table G.5 illustrates further how a nesting index can be used to specify filters.

NOTE The nesting index values in Table G.5 are not defaults, but used only to illustrate how long-wave, short-wave and band pass filters can be specified.

Table G.5 — *Indication for specifying long-wave, short-wave and band pass filters*

Nesting index	Meaning
0,8–	Long-wave pass filter, removes wavelengths shorter than 0.8 mm
–2,5	Short-wave pass filter, removes wavelengths longer than 2.5 mm
0,8-2,5	Band pass filter: long-wave pass filter removes wavelengths shorter than 0.8 mm, and short-wave pass filter removes wavelengths longer than 2.5 mm
5	Single values, with no "-" symbol before or after, are only used with Fourier filters when a tolerance is being applied to a single harmonic

Filters can be used on geometric specifications for controlling location, orientation and form.

Filters which include a short-wave pass filter (e.g. short-wave pass filters and band pass filters) are only used for tolerances dealing with form or surface texture, as any information relating to location or orientation has been removed by the filter.

Sometimes two different filters can be applied to the feature in different directions. The two filter indications are separated by a "x".

If the feature is open in both directions, such as a plane, then an intersection plane indicator is used to define the direction in which the first filter applies, and the second filter is applied in a direction perpendicular to the first.

If the feature is open in one direction, such as a cylinder, then the filter applied in the open direction is indicated first, and the filter applied in the closed direction is indicated second.

When the toleranced feature is a derived feature or an associated feature, the filtration is applied on the extracted integral feature before the derivation or association operation.

Short-wave and band pass filters can only be used for form specifications, i.e. specifications that do not reference datums, because they remove the location and orientation attributes from the toleranced feature.

G.7.2 Default filters

No default filters are currently defined for geometrical specifications. In the absence of a specific filter being indicated, the specification is ambiguous, and it can be assumed that the tolerance applies to the feature without any filtration (see G.1).

G.7.3 Examples

The use of filters with geometrical specifications is new, and guidance and examples are relatively sparse. However, a range of examples is given in BS EN ISO 1101:2017, Annex D and in Table G.6.

Table G.6 — *Examples of use of filters with geometrical specifications*

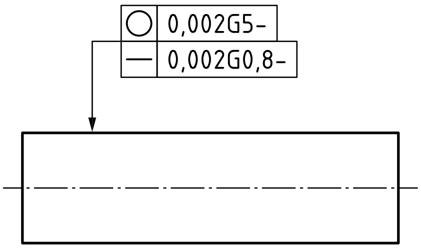
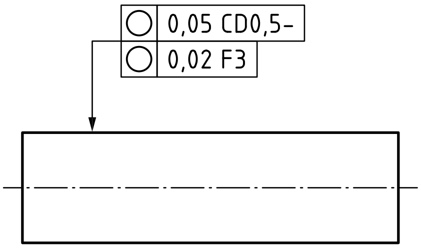
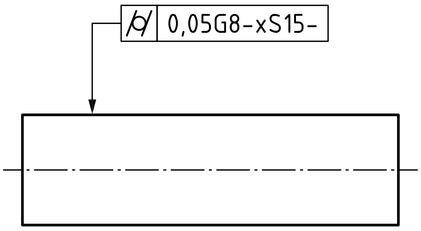
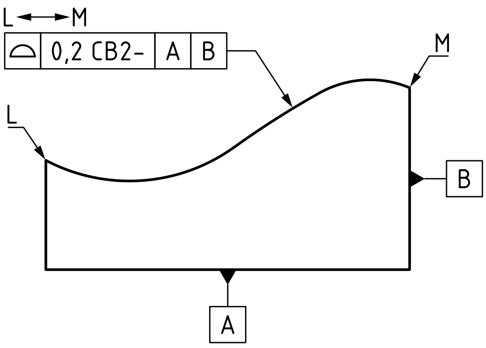
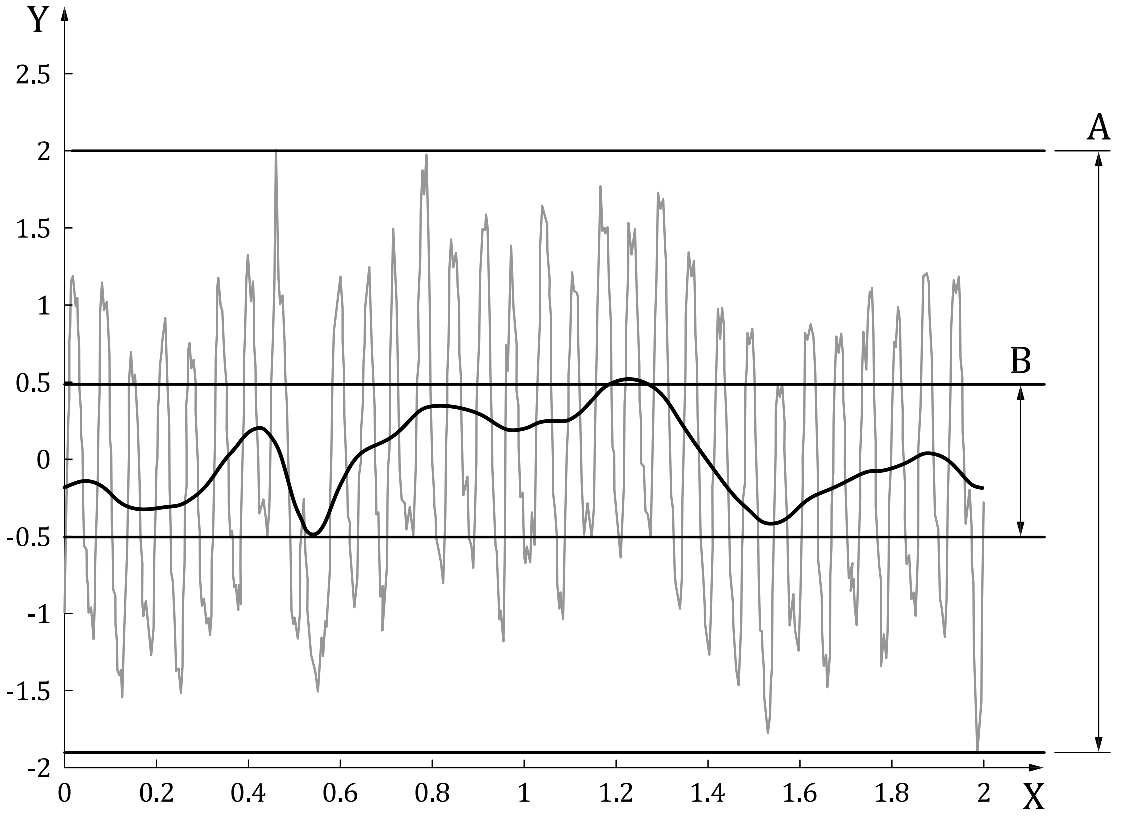
Specification Indication	Description
a) 	A roundness and straightness tolerance are applied to the same feature. These apply to a closed and open profile respectively. The roundness tolerance applies to an extracted feature filtered by a Gaussian long-wave pass filter with a nesting index of 5 UPR. The straightness tolerance applies to an extracted feature filtered by a Gaussian long-wave pass filter with a nesting index of 0,8 mm. Figure G.12 and Figure G.13 show how the unfiltered extracted profiles would not conform to this specification, but the filtered profiles would conform.
b) 	Two roundness tolerances are applied to the same feature. The upper tolerance uses a morphological closing disc filter with a radius of 0,5, eliminating any valleys in the surface which a Ø1 disc would not be able to contact. The lower tolerance has a tighter value and uses a Fourier filter applied to the third harmonic to further limit “tri-lobing” effects.
c) 	A cylindricity tolerance is applied, so the toleranced feature is an open feature in the axial direction and a closed feature in the circumferential direction. The tolerance applies to an extracted feature filtered by a Gaussian long-wave pass filter with a nesting index of 8 mm in the axial direction and a Spline long-wave pass filter with a nesting index of 15 UPR in the circumferential direction.
d) 	A closing ball morphological filter is applied to the feature with a structuring element radius of 2 mm. This is like scanning the surface with a spherical probe with a diameter of 4 mm. Any valleys which cannot be contacted by a Ø4 probe are eliminated from the extracted toleranced feature.

Figure G.12 — *Extracted profile for straightness measurement*



Key

X Distance x, mm

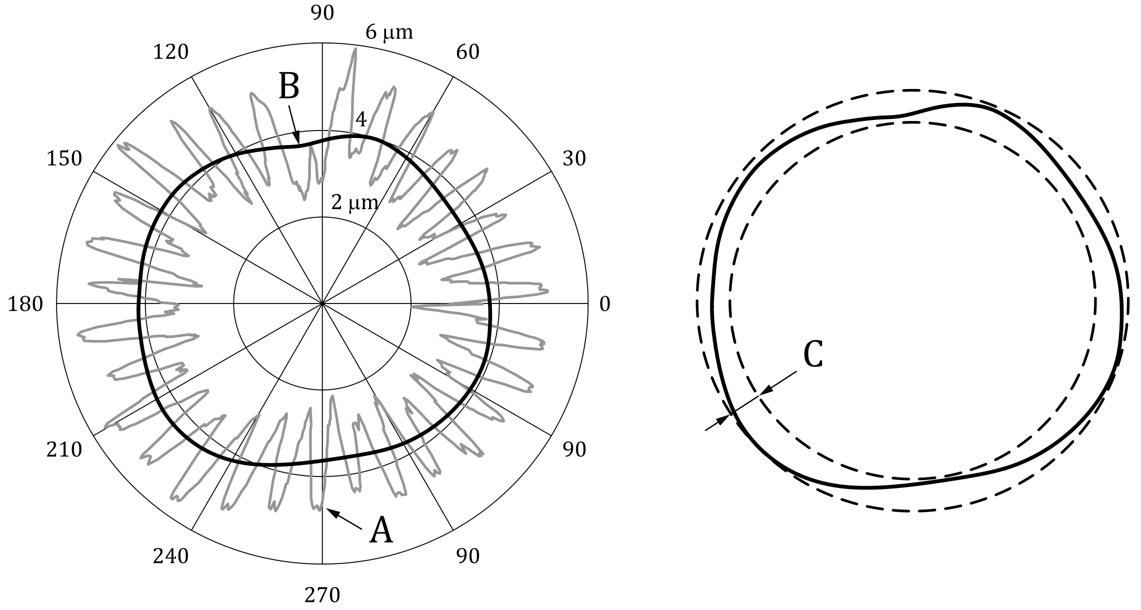
Y Height y, µm

A Unfiltered profile, Straightness = 4 µm

B Filtered profile, Straightness = 1 µm

NOTE The unfiltered straightness profile does not conform to the 2 µm straightness tolerance in Figure G.10a), but the filtered profile does. Open Gaussian filter with λc = 0,8 mm used.

Figure G.13 — *Extracted profile for roundness measurement*



Key

A Unfiltered profile. Roundness = 4 µm

B Filtered profile

C Isolated Filtered profile. Roundness = 0.7 µm

NOTE The unfiltered roundness profile does not conform to the 2 µm roundness tolerance in Figure G.11a), but the filtered profile does. Closed Gaussian filter with fc = 5 UPR used.

G.8 Datums

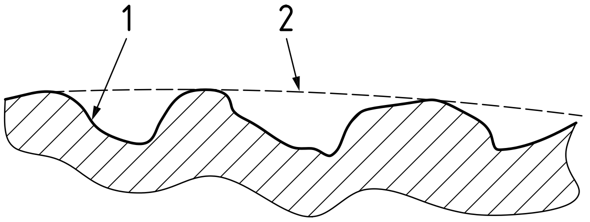
G.8.1 General

Prior to performing association methods when establishing datums, filtration is performed on the partitioned, extracted datum feature.

The filtration retains the highest points in the real integral feature. For a flat or convex nominal feature, such as a shaft, the filtration results in a convex feature (see Figure G.14). For other types of nominal feature, such as a hole, voids in the surface are similarly removed (see Figure G.15).

NOTE The default filtration is not mathematically completely defined in BS EN ISO 5459:2024, except for the case of a single plane.

Figure G.14 — *Illustration of filtration applied to a nominally flat surface*

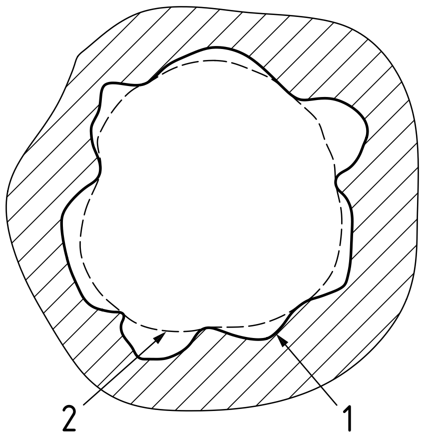


Key

1 Real integral feature

2 Filtered feature

Figure G.15 — *Illustration of filtration applied to a nominally cylindrical surface*



Key

1 Real integral feature

2 Filtered feature

G.8.2 Default filters

BS EN ISO 5459:2024 (datums and datum systems), defines the default filtration when the nominal datum feature is a plane used to establish a single datum. For this geometry, the default filtration is the Envelope Morphological (EM) filter.

G.9 Advantages and disadvantages of different filter types

Table G.7 summarizes some of the advantages and disadvantages of the various filter types. Robust versions of the Gaussian and Spline filters are less sensitive to outliers than the standard Gaussian filter.

Table G.7 — *Advantages and disadvantages of various filter types*

Filter type	Advantages	Disadvantages
Gaussian filter (BS EN ISO 16610-21)	• Well known • Well defined • Easy to compute • Closed profile: no end effects • Easy to interpret • Defined by cut-off wavelength • Closed profile: no end effects	• Outlier sensitive • Distorts skewed surfaces • Form needs to be removed (i.e. not suitable as a filter for the extraction of form portions for complex profiles, excluding cylinders, cones and spheres). • Open profile: End effects
Spline filter (BS EN ISO 16610-22)	• End effects easier to handle • No need to remove form • Does not distort skewed surfaces • Easy to compute • Closed profile: no end effects • Defined by cut-off wavelengths • Faster than Gaussian • Applicable to any surface	Currently range of application not fully established
Spline wavelet (BS EN ISO 16610-29)	• Locates and identifies outliers • Filter individual features • Form removal not necessary • Easy to compute • Closed profiles: no end effects • Defined similar to cut-off wavelengths • Faster than Gaussian • Can be used on short profiles • Can be used for surfaces	• Difficult to interpret • Currently range of application not fully established
Morphological filter (BS EN ISO 16610-41)	• Definition of mechanical surface • Simulates contact phenomena (e.g. spherical probe) • Closed profiles: no end effects • No need to remove form • Faster than Gaussian	• Range of application not fully established • Outlier sensitive
Alternating sequence filter (BS EN ISO 16610-49)	• Well defined • Naturally robust • Easy to compute • End effects easy to handle • Form removal not necessary • Defined similar to cut-off wavelengths	• Range of application not fully established • Published algorithms slower than Gaussian

Annex H (informative) Dimensioning of slots

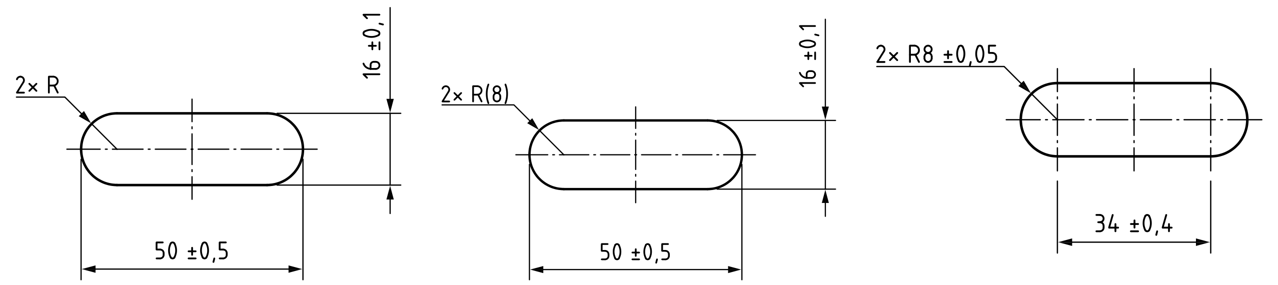
COMMENTARY ON ANNEX H

This annex describes how slots can be dimensioned and toleranced. The guidance is applicable in the case of a protrusion with the same profile.

H.1 Defining the size of the slot

When dimensioning slots with rounded ends, it is acceptable to define the length of the slot with a dimension tangent to the ends [see Figure H.1a) and Figure H.1b)], or with a dimension between the centres of the two end radii [see Figure H.1c)].

Figure H.1 — *Dimensioning slots with rounded ends*



a) Radius dimension not specified b) Auxiliary radius dimension c) Specified radius dimension

The width of the slot can be defined with a dimension defining the distance between the parallel sides of the slot, or with a dimension defining the radius of the ends.

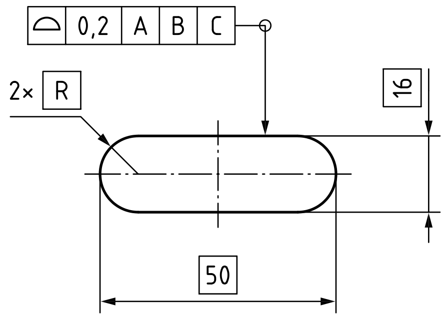
When the width of the slot is defined with a dimension between the parallel sides of the slot, the radius of the end can be indicated with a radius arrow and the symbol R without a dimension value [see Figure H.1a)], or alternatively as an auxiliary dimension, e.g. R(8) [see Figure H.1b)].

When the width of the slot is defined with a dimension defining the radius of the ends of the slot [see Figure H.1c)], the tolerance on the width of the slot is double the tolerance on the radius.

The dimensioning scheme selected should be the one that best represents the functional requirements for the feature.

In Figure H.1a) and Figure H.1b), the radius has no dimension or tolerance defined. It is assumed that the radius adjusts to match the width of the slot, but there is no direct control over the radius geometry. If the geometrical properties of the radii are functionally significant, TEDs and a profile tolerance should be used (see Figure H.2).

Figure H.2 — *Profile tolerance*

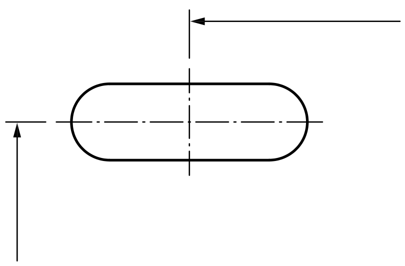


NOTE In this example, the surface profile tolerance is controlling the size of the slot and the location of the slot at the same time.

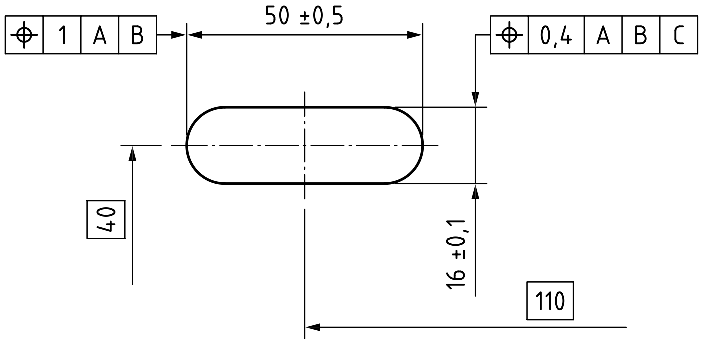
H.2 Defining the location of the slot

When dimensions are used to locate the slot, they can be applied to the centre of the slot as shown in Figure H.3a) and Figure H.3b). Dimensions with +/- tolerances can be used to locate the slot but the shortcomings of this approach are described in BS EN ISO 14405-2. The use of TEDs and surface profile tolerances (see Figure H.2), or TEDs and position tolerances (see Figure H.3) provides a less ambiguous definition.

Figure H.3 — *Defining the location of a slot*



a) Slot centre dimensions



b) TEDs and tolerances specified

Annex I (informative) Guidance for parts produced by additive manufacture (AM)

I.1 General

Additive manufacturing (AM) processes are in a state of rapid and extensive development, so this guidance might be superseded in the near future by the development of capabilities in both hardware process technology as well as the complementary software technology that enables AM. At the time of writing, many different AM processes are available for a wide range of materials, including but not exclusively:

- engineering polymers and a wide variety of metals;
- resins;
- ceramics;
- cementitious concrete;
- clay; and
- biopolymers.

The characteristics of AM processes present a range of considerations when expressing the allowable deviation in dimensional size and tolerance in geometric form of functionally critical features on a part or parts in an assembly using GPS standards. These are categorized in **I.2** to **I.7**.

I.2 Dimensional size

Fundamentally, the additive nature of material deposition or material formation in most AM processes generally leads to a near net shape (NNS) that can be larger than nominal, particularly for external features of a part. Each specific AM process also presents limitations in the spatial resolution of material deposition or material formation. Taken together, the implications of these two properties have implications for the achievable variability in dimensional size of features on a part. In particular, the dimensional size in the build direction generally behaves differently from degrees of freedom that are perpendicular to it where the build direction is towards the Earth's nadir at 1 G, generally coaxial with the Z-axis given in ISO 841. Geometry can scale non-uniformly, specifically large parts made using AM can be susceptible to greater variability in dimensional size and geometric distortion than the same geometry produced at a smaller scale.

Depending on the maturity of CAM software used to control AM processes, and its ability to compensate for differences based on a nominal CAD geometry, further steps might not be necessary if the resulting parts fall within acceptable variability for functional performance. However, several AM processes have very large differences so BS EN ISO 17295:2023, Clause **7** gives guidance, if build direction needs to be specified.

I.3 Geometric form

AM processes generally involve either a physical state change or chemical reaction, or both; the physical processes that the material composition(s) undergoes generally lead to changes in geometric form. The effects of physical thermal-distortion, chemical curing and volumetric gas diffusion contribute to some extent to geometric distortion from the CAD nominal geometry. CAM software might be limited in its capability to compensate. Such limitations might not scale well, especially if the part geometry is especially large when either the cost and/or the efficacy of software simulation to predict geometric form can impose further limitations. Particular features affected include close tolerance profiles, shafts, holes and thread forms. At the time of writing, many AM processes cannot achieve close tolerance shafts, holes or small thread forms that would pass quality control inspection and would need one or more post-processing step(s) to achieve exacting specifications.

I.4 Surface finish

There are limitations to the quality of surfaces that can be achieved with most AM processes, especially where high-quality finishes are required. As with casting, parts initially produced by an AM process to a NNS but with exacting technical requirements for the surface(s) of features, for example at the interface to an assembly, might need to include hybrid post-processing. Interestingly, some AM processes offer selective quality of finish that can be achieved on both internal and external surfaces.

I.5 Unique geometry

AM processes offer unique functionally critical features, for example achieving complicated internal lattice structures with a high iso-perimeter quotient within a closed boundary, which could not be achieved with existing casting, milling and turning processes. Assessing the conformity to specification by testable inspection of such geometries presents challenges as those surfaces are inaccessible by conventional quality control methods such as CMM probing, line-of-sight photogrammetry or scanning with structured-light. Continued development in advanced inspection methods is expected to address this in due course.

I.6 Functional grading and four-dimensional shape transformation

AM processes can achieve arrangements of two or more materials for specific functional requirements. There are considerations for specifying dimensional size and geometric form of the functionally critical features if the part undergoes transformation(s) from one shape to another. Standards in development that are complementary to GPS can support the design engineer in the specification of dimensional size and geometric form requirements in these cases.

I.7 Hybrid manufacturing

For technical specifications that exceed the capabilities of a given AM process after achieving the NNS, hybrid manufacturing can be employed. This involves complementing AM with conventional processes such as CNC drilling, milling, turning, broaching, surface grinding or others. How the build plate started in the AM process can be used as part of the datum system to coordinate GPS in each subsequent process(es). These considerations might be familiar to the challenges of applying GPS to a cast part, where technical requirements for precision that cannot be achieved in the cast are projected where required using a subsequent process(es) with datum system coordination with the NNS geometry.

When preparing documentation, referencing PD ISO/TS 21619:2018 can help designers distinguish between the document types "FUN-SPEC-COM" and "MAN-SPEC-COM". This distinction is useful when creating drawings that separate functional requirements from those related to downstream processes, such as milling, turning, drilling, boring, broaching or surface grinding and preserving the datum system between processes.

Annex J (informative) Castings

J.1 Background

The subject of general tolerances is wide. The manufacture of components by casting can at times appear more an artform than a reliable, repeatable production operation.

Originally published in the UK as BS 6615 (withdrawn), ISO 8062 was originally introduced in the 1980s and provided a framework for applying general tolerances to metal and metal alloy castings.

The 1984 version of the standard presented a series of 16 tolerance grades as "Casting Tolerance, CT" grades from CT1 to CT16, with CT 16 being the coarsest grade of tolerance values. Each of these CT grades were presented in a range of dimensional sizes, in matrix form, such that as the size of the dimensioned feature (referred to as "basic dimension") increases, the relative tolerance value increases in blocks of size.

The standard also introduced the concepts of standard machining allowance, standard permissible surface mismatch between mould closing halves, standard wall thickness and standard mould taper, along with how each of these items was to be applied to a drawing or TPD in a standardized manner.

Perhaps most importantly, the standard also guided users to the appropriate CT grades that could be achieved by different casting techniques, showing the relative accuracies of those processes and how each of them could perform and, therefore, what tolerance grade values to expect.

The idea behind the general tolerance matrices specified in the ISO 8062 series and the withdrawn BS 6615 was that the doubt and guesswork could be taken out of the design process of tolerancing cast components by making a cross-reference to the standard and in so-doing, all non-toleranced size dimensions specified on the drawing would be complemented by the appropriate size tolerance stated in Table 1, the values for which had been gathered from the casting industry trade bodies and foundries themselves over a long period of time.

When specified with appropriate care, the dimensional tolerances referred from the standards, along with the standard surface mismatch, machining allowance and taper tolerances, can complement a well dimensioned and specified casting drawing, particularly where that cast part is then to be followed by finish machining in such items as gearbox casings, pump housings and even engine cylinder blocks.

As with any general tolerancing standard document, if carelessly referred and applied from a drawing or TPD the user can obtain unintended tolerances applied to the cast component, making finish machining more difficult or more expensive, or both.

BS EN ISO 8062-1 and BS 6615 (withdrawn) were revised in the 1990s while remaining identical in all but identifier. The revised standards kept the same principle and included the values for CT1 and CT2; introduced the possibility to define taper plus or taper minus for tapered features and required machining allowance (RMA) grades with tolerance values; and broadened the range of cast raw materials for which guidance and general tolerances can be obtained.

In 2007 the ISO standard received a more major upgrade, taking account of more recent development in the field of geometrical specifications and GPS and separating the technical drawing/TPD requirements from the functional requirements defined within the TPD. As a result, the standard was split initially into two parts ISO 8062-1 and ISO 8062-3, later followed by ISO/TS 8062-2.

- BS EN ISO 8062-1:2007, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 1: Vocabulary*.

- PD CEN ISO/TS 8062-2:2013, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 2: Rules*.
- BS EN ISO 8062-3:2007, *Geometrical product specifications (GPS) – Dimensional and geometrical tolerances for moulded parts – Part 3: General dimensional and geometrical tolerances and machining allowances for castings using ± tolerances for indicated dimensions*.

BS EN ISO 8062-1 presents a vocabulary used throughout the series of standards and in general use in the moulding and casting industries.

PD CEN ISO/TS 8062-2 defines the rules to be used in the application of the series. It includes methods for identifying different types of TPD. The standard supports general tolerancing for cast/moulded parts across different manufacturing stages – whether they are to be machined, at the intermediate machined stage, at the final machined stage. Where machining is not required, provision can be in the form of one combined TPD, using different symbology to identify the applicable stage, or on separate TPDs dedicated to the specific, individual manufacturing stage.

BS EN ISO 8062-3 led to the introduction of general geometrical specifications to complement the general dimensional tolerances, RMAs, etc. The result was effectively too many general tolerances, meaning that the old, familiar “CT” casting tolerance grades had to evolve into dimensional casting tolerance grades (DCTG), providing the same general tolerances for linear dimensions. This new abbreviation then creates “space” for the introduction of the geometrical casting tolerance grades (GCTG), which provide general tolerancing of straightness, flatness, roundness, parallelism, perpendicularity, symmetry and coaxiality. The required machining allowance grades (RMAG) have also been included.

While the CT grades have changed their identity to become DCTG, the values in the tables remain unchanged. The RMAG are new, and the range of choices vary both with manufacturing process and the allowance that is deemed necessary for the component to function appropriately.

BS ISO 8062-4 introduced a whole new and potentially simpler manner of tolerancing cast/moulded components, using geometrical specifications to tolerance the surface profile of a component. For the profile tolerance to perform correctly and provide the tolerance envelope within which the surface of the component has to fall, the tolerance zone is appropriately constrained by a datum system according to BS EN ISO 5459.

Using BS ISO 8062-4 appears much simpler, in-principle; it only requires reference to an appropriate datum system establishing conforming to BS EN ISO 5459, and the workpiece is potentially completely defined within predictable limits.

In 2023, the modified versions of both BS EN ISO 8062-3 and BS ISO 8062-4 were published and are already being considered for further revision.

J.2 Whether to use BS EN ISO 8062-3 or BS ISO 8062-4

Although the ISO 8062 series is identified as a GPS standard and under the direction of the ISO Committee for dimensional and geometrical product specifications and verification (ISO/TC213), the Committee itself accepts that Parts 3 and 4 are not entirely in line with the GPS principles laid down in the other set of GPS standards.

BS EN ISO 8062-3 is currently further from the GPS set of standards than BS ISO 8062-4 and there are plans to further develop both documents along with the linking standards, BS EN ISO 8062-1, PD CEN ISO/TS 8062-2 and BS EN ISO 10135, for example.

The choice of whether to use BS EN ISO 8062-3 or BS ISO 8062-4, or neither, is for discussion and agreement between the design authority and the manufacturing foundry as to preference, level of understanding and appropriate levels of process capability versus design/functional requirements.

Annex K (informative) Examples of profile and areal surface texture specifications, and summary of changes from BS EN ISO 1302

K.1 Examples of profile surface texture specification

Figure K.1 and Figure K.2 give examples, while Table K.1 and Table K.2 give explanations for the elements of the figures.

Figure K.1 — *Minimal indication for an R-parameter with defined defaults by using all defaults*

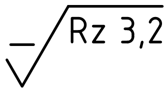


Table K.1 — *Explanation and valid defaults for the example in Figure K.1*

Specification elements	**Explanation**
Graphical symbol	Profile surface texture requirement No manufacturing requirements
Rz	R-parameter: Rz
3,2	Tolerance limit value: 3,2 µm, leads to the valid default settings (see BS EN ISO 21920-3:2022, Table 3)
	Defaults applicable to the example (see BS EN ISO 21920-3:2022, Tables 1 and 3)
Tolerance type	Upper tolerance limit
Tolerance acceptance rule	Tmax
Profile S-filter type	Gaussian
Profile S-filter nesting index N_{is}	0,002 5 mm (requires a maximum sampling distance of 0,5 µm)
Profile L-filter type	Gaussian
Profile L-filter nesting index N_{ic}	0,8 mm
Section length l_{sc}	0,8 mm
Number of sections n_{sc}	5
Evaluation length l_e	4,0 mm
Profile F-operator association method and element	Association and removal of the specified form element with total least square
Profile F-operator nesting index	Not needed
Method of profile extraction	Mechanical profile
Other requirements (OR(n))	No other requirements are valid
Manufacturing process	No requirement
Surface lay and direction of lay	No requirement
Profile direction	Orthogonal to the surface lay

Figure K.2 — *Indication for an evaluation length R-parameter including all definable specification elements by using no default*

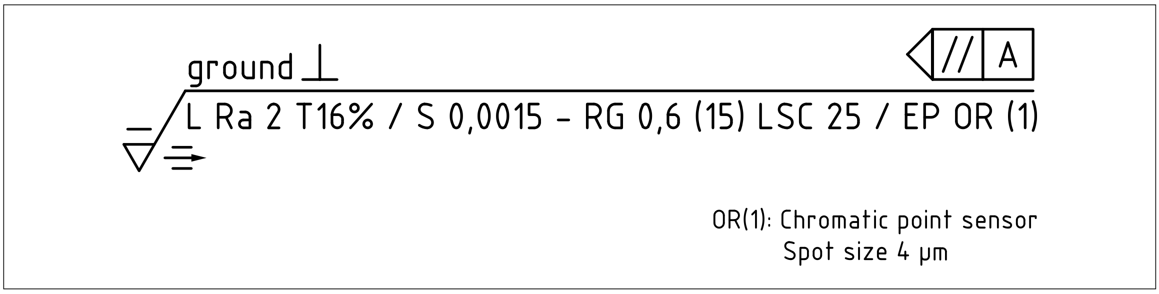


Table K.2 — *Explanation for the example in Figure K.2*

Specification elements	Explanation
Graphical symbol	Profile surface texture requirement Material shall be removed
L	Tolerance type: lower tolerance limit
Ra	R-parameter: Ra
2	Tolerance limit value: 2,0 µm
T16%	Tolerance acceptance rule: 16% tolerance acceptance rule
S	Profile S-filter type: Spline
0,001 5	Profile S-filter nesting index N_{is}: 0,001 5 mm (requires a maximum sampling distance of 0,3 µm)
RG	Profile L-filter type: Robust Gaussian
0,6	Profile L-filter nesting index N_{ic}: 0,6 mm
(15)	Evaluation length l_e: 15 mm
LSC	Association and removal of a circle (C) with least square (LS)
25	Profile F-operator nesting index: circle-radius 25 mm
EP	Method of profile extraction: electromagnetic profile
OR(1)	Other requirements: chromatic point sensor, spot size 4 µm
ground	Manufacturing process: grinding
⊥ // A	Surface lay and direction of lay: perpendicular to the direction, given by the intersection plane indicator and the datum feature indicator
	Profile direction: parallel to the grinding direction

K.2 Summary of changes from BS EN ISO 1302

K.2.1 Tolerance acceptance rules

BS EN ISO 21920-1:2022 specifies three rules: Maximum (symbol "Tmax"), 16% (symbol "T16%"), and Median (symbol "Tmed"), where "Tmed" is a new rule, and symbols for the rules are updated as well. The default rule is now Tmax (in the withdrawn BS EN ISO 1302 the default was T16%).

K.2.2 Graphical symbols

Graphical symbols for profile surface texture have changed.

K.2.3 Filter type

All profile filter types from BS EN ISO 16610 series can be applied (symbols for filters are in BS EN ISO 21920-1:2022, Table B.1).

K.2.4 Indication of the profile direction

Relative to the predominant direction of the surface lay (BS EN ISO 21920-1:2022, Table 3).

K.2.5 Indication of lay pattern

The symbol used to indicate the general type of surface lay is now placed above the crossbar of the surface texture symbol, instead of to the right of the “v” part of the symbol.

K.2.6 Setting class

The symbol Sc*n* indicates which setting class is applied for the determination of the default settings. A setting class or a default profile L-filter or a default profile S-filter nesting index value is specified for parameters without defined defaults, Sc1, Sc2, Sc3, Sc4 and Sc5, see BS EN ISO 21920-3, Table 2 to Table 6.

K.2.7 Section length

The section length l_{sc} specifies the length used to calculate height parameters based on profile hills and profile dales.

K.2.8 Number of sections

The number of sections n_{sc} specifies the number of section lengths used to calculate height parameters based on profile hills and profile dales.

K.2.9 Profile F-operator association method and element

The association method and element of the profile F-operator defines how form is removed from the profile (see BS EN ISO 21920-2). The symbols for profile F-operator methods are indicated in BS EN ISO 21920-1:2022, Tables C.1 and C.2.

K.2.10 Method of profile extraction:

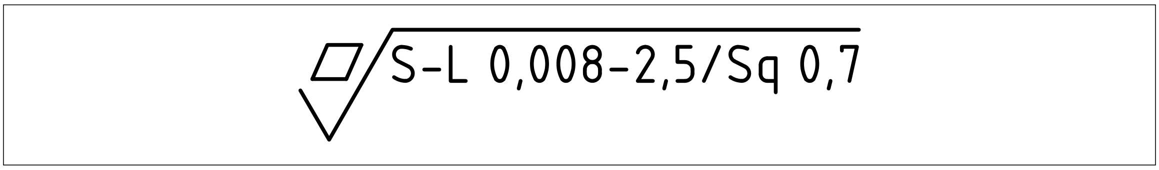
The profile can be obtained from the mechanical surface (the default) or from the electromagnetic surface.

K.3 Areal surface texture specifications

K.3.1 Examples of areal surface texture specifications

Figure K.3 and Figure K.4 give examples of areal surface texture specification.

Figure K.3 — *Example of areal surface texture specification – 1*



Interpretation: Surface with no manufacturing requirements, S – L surface, S-filter nesting index = 0,008 mm, L-filter nesting index = 2,5 mm, chosen S-parameter is root mean square height of the scale-limited surface; Sq with a maximum limit value 0,7 µm.

Implied defaults (not shown in the indication):

Upper limit tolerance “U”.

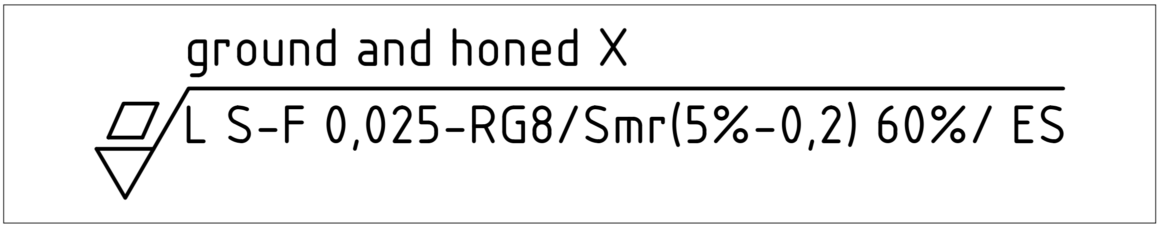
The evaluation area is equal to the definition area and is a square with a side length of 2,5 mm, the same as the value of the L-filter nesting index.

With an S-filter nesting index of 0,008 mm, the maximum sampling distance is 0,001 5 mm and the maximum sphere radius is 0,005 mm. These figures are given in BS EN ISO 25178-3:2012, Table 2 and determined independently of the size of the chosen L-filter nesting index. See also BS EN ISO 25178-3:2012, Table 1 and Note 3.

The F-operator is the total least square removal of form on S-L evaluation area.

The default unit "µm" for the Sq parameter is not indicated.

Figure K.4 — *Example of areal surface texture specification – 2*



Interpretation: Surface with a manufacturing requirement (ground and honed) and a surface lay requirement, lower limit tolerance, S-F surface, S-filter nesting index = 0,025 mm, the non-default F-operator is a Robust Gaussian filter with nesting index of 8 mm, chosen S-parameter is areal material ratio of the scale-limited surface, Smr minimum limit value at c-level 0,2 µm = 60% and measured downwards into the surface from the reference plane given by Smr = 5%. The non-default specification of extracted surface is an electromagnetic surface.

Implied defaults (not shown in the indication):

The evaluation area is equal to the definition area and is a square with a side length of 8 mm.

The S-filter is an areal Gaussian filter.

With S-filter value 0,025 mm, the maximum sampling distance is 0,008 mm and the maximum lateral period limit is 0,025 mm. These figures are given in BS EN ISO 25178-3:2012, Table 3.

NOTE The lateral period limit for an electromagnetic surface is the equivalent of the ball radius for the mechanical surface.

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